

# The cosmology of Cosmological Relativity: the expansion history from causal reassignment on a de Sitter background, the cosmogenesis branch point, and the scalar perturbation sector

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## Abstract

Cosmological Relativity (CR) obtains the observed cosmological expansion by a causal reassignment on a de Sitter background at the cosmogenesis branch point, with the Einstein field equations unchanged. This paper develops that cosmology and completes it with its scalar perturbation sector. The reassignment selects the Nariai member [37] of Schwarzschild–de Sitter, whose proper frame is a homogeneous, observationally isotropic, *non-synchronous* cosmology with areal radius  $r(\tilde{\tau}) \propto \sinh^{2/3}$ —exactly the flat- $\Lambda$ CDM scale factor—its rate fixed by the cosmological constant alone, with radiation and matter inherited content read off the cosmic clock rather than terms that source the expansion. The cosmic beginning is the branch point  $r = 0$ —a genuine curvature singularity of the SdS metric read over the  $r$ -chart [36], and *not* a singularity of the underlying substrate, whose throat is the defining constant  $\alpha$ ; the divergence of the  $r$ -chart invariants there is the areal coordinate degenerating; the constant-proper-time slices are exactly Euclidean ( $\Omega_k = 0$ ) while the cosmological layers are a closed  $S^3$ —spatial curvature and dark energy one  $\Lambda$  read on two slicings. The acoustic scale follows from that rate rather than calibrating it: with the plasma handed over at the branch point there is no early-universe parameter to carry the angle, so the angle is computed, and it and the baryon-acoustic distances fix  $H_0$  together at  $\simeq 68.6$ . The light-element composition is produced on the same footing: the infalling matter heats above the deuterium bottleneck and the expanding leg cools back through it, so that leg is a standard big-bang nucleosynthesis run from free nucleons, fixing the composition with the inherited baryon-to-photon ratio  $\eta$ —the one boundary datum of that handover. Helium-4 and deuterium sit at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers, un-tuned; the near-zero metallicity of the oldest systems follows from the handover; and because the run is standard, lithium-7 carries the standard threefold over-prediction, the lithium problem shared with flat  $\Lambda$ CDM rather than dissolved.

On this cosmology the primordial scalar spectrum *factorizes*. The de Sitter substrate *determines* its structure—the geometric coherence of the acoustic phases on the null boundary (not a super-horizon freeze-out), the classical non-vacuum character of the fluctuations (the substrate vacuum lies some  $10^{113}$  below the observed  $A_s$ ), the throat’s isotropization, a parameter-free low-multipole discreteness floor, and the faithful transmission of the spectral shape—while the progenitor collapse *supplies* its content, the amplitude  $A_s$  and tilt  $n_s$ , inherited as boundary data exactly as flat  $\Lambda$ CDM inherits the baryon-to-photon ratio. *The transmission is supported by a mechanism rather than by the absence of a scale alone* (§3): on the contracting leg  $aH$  grows without bound as the branch point is approached, so the comoving horizon shrinks to zero and every mode *exits* it and freezes before the crossing; the imaginary-time segment’s Euclidean kernel damps oscillatory content and has nothing to act on in a frozen mode, so amplitude and tilt cross unaltered while the collapse-leg acoustic phase does not—which the observed peak positions independently require, excluding a transmitted leg phase by a factor of 21–41 in the required distance [9].

The decisive result is a *transmission dichotomy*: a non-degenerate horizon’s exponential near-horizon approach imprints a scale-invariant spectrum (the inflationary de Sitter-horizon

mechanism), whereas the degenerate Nariai double root’s power-law approach carries no scale, and the branch point where the modes cross—a third case,  $f$  diverging so that the tortoise coordinate stays finite—accumulates no phase at all; the super-horizon transfer across it is computed and scale-invariant, so the progenitor spectrum is transmitted unaltered and CR has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial  $B$ -modes.

Because the distance slicing is flat, the discrete closed- $S^3$  source projects through the flat transfer to a parameter-free low-multipole deficit ( $\ell \lesssim 8$ )—mild, and on a genuine Boltzmann transfer a dip whose *minimum falls at  $\ell = 4$*  (0.36 of the  $\Lambda$ CDM expectation there, 0.49 and 0.44 at  $\ell = 2$  and 3; *the shape and the depth both cross-validated between two independent transfers*, which agree to three per cent at worst once the post-recombination source is carried to the present), and non-discriminating in sum: scored against the measured multipoles with cosmic variance carried, *the quadrupole favours this construction and  $\ell = 3$  to 5 favour the standard model*, the two nearly cancelling at  $\Delta(-2 \ln L) = +1.6$  over  $2 \leq \ell \leq 8$ .

Taken together, these results carry a reckoning the paper draws in full: what flat  $\Lambda$ CDM assembles from added, tunable apparatus—inflation for the horizon, flatness, coherence, and near-scale-invariance, a dark-energy component for the late-time acceleration, added early-universe physics for the tensions—this cosmology recovers from the cosmological constant alone, each feature required by structure already present rather than produced by a sector introduced to yield it. By the economy of assumption that distinguishes a required structure from a merely permitted one, that comparison is settled on the theory-choice axis in this cosmology’s favour; and on the data axis it is split: the expansion history fits the low-redshift distance ladder on the SDSS DR12 consensus and, at the current state of the art, the DESI DR2 dataset (thirteen measurements across seven tracers) at  $\chi^2/\text{dof} \simeq 1$ , which fixes  $\Omega_m \simeq 0.30$  and  $H_0 \simeq 68.6$  and leaves the discrepancy with the local distance ladder where the standard model leaves it; the light-element abundances are confirmed within  $1\sigma$  at the microwave-background baryon density (the cosmogenesis paper [19]); and the acoustic peak structure, computed from that same background with nothing fitted to it, reproduces the comb to a part in six hundred and is still rejected on shape—at 2.57 times the control’s distance from the data as computed, and 1.57 times once both arms’ parameters are refitted; what it leaves is about one multipole at each peak.

The cosmology’s exposed edges are named: a signature in the diffusion *scale*, the two rates differing by the radiation term alone so that the plasma’s Silk scale and the foliation’s distances do not share a rate [27]—a computed effect, non-reabsorbable by the baryon density, whose size is fixed by the handover, read with both lengths carried to the peak of the visibility function, and which the joint fit locates as a Gaussian residual in  $\ell$  that no tilt removes, a refit absorbing a Gaussian only window-locally (the low multipoles being a mild, non-discriminating deficit rather than a sharp edge either way); and the acoustic peak structure. *The full-spectrum likelihood is run on the computed spectrum*: scored against Planck 2018 `plik_lite` over the full-range lensed configuration, with one amplitude fitted so that what is compared is shape, this arm sits at 2.57 times the control’s distance from the data (§4.7). *That is a spectrum computed on a background fitted to the distances and then confronted*, which is a different statement from a refit of the spectrum’s own parameters to the microwave data; the second is the comparison the standard model is usually judged by, and it is reported in §4.7 beside the first.

We mark maturity throughout. *Established*: the cosmological derivation, the transmission dichotomy, the amplitude floor, the throat, the flat slice, the flat-projection transfer, the data confrontation, and the time-reversal driving equality as a Fourier theorem. *Argued*: the coherence mechanism. *Measured*: the peak *heights*, which this arm’s own instrument returns within a few per cent of the sky’s first two ratios on the polarisation path and above them on the fluid path, so the heights are the path’s and the positions are the construction’s.

*Run, and returning a partial agreement*: the end-to-end branch-point-to-recombination transfer, the standard line-of-sight integral across a visibility of finite width, validated on its control to 0.14% in the first-to-second height ratio and carried through to this construction’s arm, which returns the peak spacing to one part in six hundred and the first three peaks within a grid step of the measured ones, with the fourth about one per cent high; the likelihood rejects the arm’s spectrum on  $TT$  shape nonetheless. *Open*: what displaces that fourth peak, the damping envelope having been tested and excluded; and the *derivation*—as

against the measurement—of  $A_s$  and  $n_s$ . *The cosmology carries no early-universe parameter, and what it does carry should be stated plainly:* the plasma is handed over at the branch point, where every acoustic mode is outside the horizon and the potential sits at its regular value, so the sound-horizon integral has no lower endpoint to fit and the acoustic angle is an output of the rate rather than a calibration of it. The background is fixed by the late-time data alone—the epoch  $x_0$  (equivalently  $\Omega_m = 2/(x_0^3 + 2)$ ) and  $H_0$ , from the baryon-acoustic distances together with that angle, which agree with one another within a twentieth of a unit at  $H_0 \simeq 68.6$ . *Matter–radiation equality is then fixed by the same epoch*,  $1 + z_{\text{eq}} = \omega_m/\omega_r$  with  $\omega_r$  measured, so one quantity carries the expansion history and the acoustic driving together and the computed spectrum is a test of the background rather than a further fit to it. The substrate’s single  $\Lambda$  is shown to govern both the global expansion and, per structure, the local scale at which structure resists it—the Hubble–Eddington radius—one constant read at two ranges.

## 1 Introduction: the cosmological theory and its scalar perturbation sector

Cosmological Relativity (CR) recovers the flat- $\Lambda$ CDM expansion history by a causal reassignment on a de Sitter background at the cosmogenesis branch point: the expansion rate is fixed by the cosmological constant alone, radiation carries no term in it, and the present matter density is a reading of the cosmic epoch rather than an independent quantity [9, 11]. The cosmic beginning is the branch point  $r = 0$ —a genuine curvature singularity of the SdS metric read over the  $r$ -chart, and *not* a singularity of the substrate, whose throat is the defining constant  $\alpha$ , and the construction makes the constant-proper-time slices exactly flat while the cosmological layers are a closed  $S^3$ . This paper develops that cosmological theory in the form its observational tests require (§2) and completes it with the scalar perturbation sector the framework paper’s outlook named open—“perturbations within the Schwarzschild–de Sitter representation”—the primordial density perturbations and their imprint on the microwave background (§§4–8), closing on the open frontier the construction sets up (§10). Across these sections one comparison recurs and is drawn together at the close (§9): the causal contact, flatness, coherence, isotropy, and near-scale-invariance that the standard picture obtains from an added, tunable inflationary sector are here read off structure already present and fixed by  $\Lambda$  alone—an economy-of-assumption difference the paper argues is decidable on the theory-choice axis, where it falls to this cosmology—and on the data axis the expansion history fits the low-redshift distance ladder, which fixes  $H_0$  alongside the epoch, and the light-element abundances are confirmed against the data (§2.4), while the exposed edge is the acoustic peak structure—where the transfer, run on that background with nothing fitted to it, reproduces the comb to a part in six hundred and is still rejected on  $TT$  shape (§4.7)—and the damping-*scale* signature, a computed, non-reabsorbable effect whose size the handover fixes, read with both lengths carried to the peak of the visibility function, and which the joint fit locates as a Gaussian residual no tilt removes, the low multipoles a mild, non-discriminating deficit rather than a sharp edge either way. The companion dynamics paper develops the perturbation sector’s *tensor* half—the propagating graviton as the leaf’s shear, a massless de Sitter mode on the cosmic foliation [10]; the scalar half is here.

The result is organized by one thesis, which we state at the outset because it controls everything below.

*The CR primordial scalar spectrum factorizes: the de Sitter substrate geometry determines its structure, the progenitor collapse supplies its content, and the branch point’s null geometry assigns each its job.*

Substrate-determined, and developed here: the coherence of the acoustic phases (§4); the classical, non-vacuum character of the fluctuations (§5); the isotropizing throat (§6); a parameter-

free low-multipole discreteness floor (§7); and the faithful transmission of the spectral shape (§8). Progenitor-supplied, and inherited rather than derived: the amplitude  $A_s$  and tilt  $n_s$ . This is the same division of labour by which flat  $\Lambda$ CDM carries the baryon-to-photon ratio  $\eta$  as a measured datum from a baryogenesis it does not model; this cosmology carries that same  $\eta$  from the hot handover of the progenitor collapse, and the transmission result of §8 routes  $A_s$  and  $n_s$  through the *same* handover. There is one boundary-condition supplier, not two.

A word on what is settled before the spectrum. The acoustic *scale* (the peak spacing) is settled in the cosmological development above (§2.3): it is computed from the rate, with the sound speed the ordinary baryon-loaded value of pressureless branch-point matter, and the Hubble constant is fixed alongside it by the baryon-acoustic distances. *The sky’s own side of that comparison is measured here rather than quoted. Locating the acoustic peaks of the Planck 2018 `plik_lite` binned TT spectrum [24] by parabolic fit to each hump above  $\ell = 100$  returns  $\ell = 220.4, 537.7, 817.3$  and  $1123.9$ , which reproduce the standard quoted values to better than one per cent without having been fitted to them. Fitting  $\ell_n = \ell_A(n + \phi/\pi)$  to the first three gives the sky  $\phi/\pi = -0.2404$  against  $-0.2361$  for a  $\Lambda$ CDM control measured by the identical procedure—an offset of  $0.0043$ , so the control stands in for the sky on this quantity to within seven parts in a thousand.* The move has a name and the name is worth stating, because it is what licenses the comparison: *this is matched-procedure differencing—the control is passed through the identical extraction, so whatever bias the procedure itself carries appears in both arms and cancels in their difference, and what survives is attributable to the physics rather than to the instrument.* The same discipline governs every control in this sector: *a residual is credited against CR only once the control, run through the same machinery, has been shown not to produce it.* What this construction’s own arm returns against that measured comparison is reported in §4.7, *the two-arm transfer being validated on the control and carried to convergence on both arms.* These are settled there and are not reopened in the perturbation sector; in particular, the standard radiation-governed sound horizon laid on this cosmology’s rate is a calculation belonging to neither framework, and any “tension” it produces is an artifact, not a CR result. What is genuinely downstream is the *spectrum*—the peak heights, measured below against this arm’s own transfer (§4), and the full oscillating-medium transfer, still open—which is where this paper’s substance lives.

The register is the programme’s [14, 9]: the field equations are Einstein’s, unchanged; the existing object is the evolving three-dimensional layer; “the substrate determines” and “the branch point transmits” are statements about a real geometry and its real boundary [17, 16], not about a modification of the dynamics.

## 2 The cosmological theory the perturbations complete

This paper develops the cosmology of Cosmological Relativity from first principles and carries it to its empirical confrontation, completing it with the scalar perturbation sector. It takes from the companion framework paper [9] the *general* primitives—the layered ontology, the projection principle, and the representational freedom by which one fixed manifold admits distinct Lorentzian metrics—together with the Null–Boundary Correspondence theorem, the structural result (grounded in the metric-singularity causal structure of gravitational collapse [4]) that a collapse event-of-events and a de Sitter cosmic-entry slice are one ontological layer under two causal assignments. The cosmology proper is built here: the causal reassignment that selects the Nariai member, the proper-frame expansion law, the branch point at which the collapse leg becomes the expansion leg—the beginning, reached a lap on from the horizon the Correspondence anchors—and the computation of the acoustic scale and the confrontation with the distance data; the perturbation sector (§4–§8) is its completion.

## 2.1 The causal reassignment and the Nariai member

The vacuum solution of  $R_{\mu\nu} = \Lambda g_{\mu\nu}$  with a localized source is Schwarzschild–de Sitter (SdS),

$$ds^2 = -\frac{r}{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r} dr^2 + \frac{\frac{\Lambda}{3}r^3 + \frac{2GM}{c^2} - r}{r} dt^2 + r^2 d\Omega^2, \quad (1)$$

whose areal radius  $r$  is timelike in the cosmological regime  $\Lambda G^2 M^2 / c^4 \geq 1/9$ . Building the cosmology, we reassign causal roles on the de Sitter substrate—a move licensed by the representational freedom of the framework, distinct Lorentzian metrics on one fixed manifold sharing its foliation [9]: the de Sitter null rulings whose causal sense matches a collapse horizon’s generators are promoted to the timelike fundamental congruence, the at-rest comoving worldlines become the null photon congruence, and the foliation by expanding 3-spheres is preserved. This selects the *Nariai* member  $\Lambda G^2 M^2 / c^4 = 1/9$ ,  $M = c^2 / (3\sqrt{\Lambda}G)$ —the unique non-pivoting tilt, fixed by  $\Lambda$  alone—at which the two positive horizons merge at  $r_N = \alpha/\sqrt{3}$ ,  $\alpha = \sqrt{3/\Lambda}$  the de Sitter 3-sphere size (a scale distinct from the merged-horizon radius  $\alpha/\sqrt{3}$  and from the areal-radius amplitude below; we keep the three apart throughout). The selection is geometric—the fixed point of the slicing curve’s root-exchange involution, equivalently the limiting tangent orientation a collapse selects—not a fitted mass [5]. That the selection is moreover *unique* is a structural fact about the description groupoid rather than a fitting: the generic vantages fall into two-cycles of the involution, the Nariai member is its one fixed point, and the reassignment promoting a null direction to the fundamental congruence is forbidden at every generic vantage, so it selects Nariai and no other [6].

## 2.2 The proper-frame cosmology, derived

In the proper frame of the fundamental observers the SdS metric is a homogeneous, non-synchronous cosmology. The derivation [15] proceeds from (1) in five steps. (i) A timelike radial geodesic has a conserved energy  $E$  (the metric being independent of  $t$ ); eliminating  $dt/d\tau$  in the normalization  $L = -1$  gives the radial equation  $(dr/d\tau)^2 = E^2 - V_{\text{eff}}(r)$ , with  $V_{\text{eff}} = (r - 2GM/c^2 - \frac{\Lambda}{3}r^3)/r$ . (ii) The fundamental congruence is the worldlines whose motion in  $r$  is produced purely by the field—those at rest exactly where  $V_{\text{eff}} \equiv 1$  and the metric reduces to Minkowski—which is  $E = 1$ ; this fixes the rest frame by the field rather than by stipulation, *and it is worth saying where that locus is*:  $V_{\text{eff}} = 1$  is  $r^3 = -2GM\alpha^2/c^2$ , so the rest frame is defined at  $r = -(6GM/c^2\Lambda)^{1/3}$ —a *negative* areal radius, on the conjugate branch of the lap, between the progenitor’s horizon and the branch point [9]. *An earlier derivation carried that locus as a formal one*—“a negative mathematical abstraction that lies beyond a singularity at  $r = 0 \dots$  simply a matter of algebraic consistency” [2, App.]—*and on the lap it is a place*. The same locus fixes the sign, since  $t = \tau$  there if and only if  $E = +1$ . (iii) With  $E = 1$ , integrating the radial equation gives, along one worldline,  $r(\tau) \propto \sinh^{2/3}(\frac{3}{2}\sqrt{\Lambda c^2/3}\tau)$ . *That this integration closes at all is a second thing the same condition buys*. Writing the radial equation as  $(dr/d\tau)^2 = 2GM/c^2r + r^2/\alpha^2 - k$  with  $k = 1 - E^2$ , the quadrature  $\int dr/\sqrt{\cdot}$  is elementary at  $k = 0$ —it is  $\frac{2\alpha}{3} \operatorname{arcsinh}(r^{3/2}/\sqrt{2GM\alpha^2/c^2})$ , which inverts to the  $\sinh^{2/3}$  above—while for  $k \neq 0$  the substitution  $r = u^2$  carries it to a sextic under the root and it is not elementary. *So  $E = 1$  fixes the rest frame by the field and, by the same stroke, is the one non-degenerate energy at which the cosmology integrates in closed form*; the only other elementary cases are the degenerate ones,  $\Lambda = 0$  giving the dust cycloid and  $M = 0$  giving pure de Sitter<sup>L14</sup>. (iv) Labelling the congruence by a comoving coordinate  $\chi$  through the integration constant  $r(0)$ , the areal radius across the congruence depends on  $\tau, \chi$  only through their sum,  $r(\tau, \chi) = r(\tau + \chi)$ , with  $\partial_\chi r = \partial_\tau r$ . (v) Fixing the transformation  $t(\tau, \chi)$  by requiring the comoving coordinate orthogonal to proper time,  $g_{\chi\tau} = 0$ , forces the integration function to be  $\chi$  itself, and the remaining components follow ( $g_{\tau\tau} = -1$ ,  $g_{\chi\chi} = (\partial_\chi r)^2$ ). The result is the proper-frame line element

$$ds^2 = -d\tau^2 + (\partial_\chi r)^2 d\chi^2 + r^2 d\Omega^2, \quad (2)$$



**Figure 1:** The scale factor (3) as one analytic curve  $r(\tilde{\tau})$  in  $(\tilde{\tau}, \text{Re } r, \text{Im } r)$ . The expanding cosmology is the real leg  $\tilde{\tau} > 0$ , lying in the real plane; the branch point  $\tilde{\tau} = 0, r \rightarrow 0$ —*not* a seam—is where the curve leaves the real plane onto a conjugate branch at constant phase  $2\pi/3$  for  $\tilde{\tau} < 0$ . The beginning is this analytic branch point, not a real  $r = 0$  boundary. Forcing  $r$  real (dashed) projects the conjugate branch onto the negative axis and manufactures a cusp the smooth curve does not carry.

with cosmic time  $\tilde{\tau} \equiv \tau + \chi$  (by Weyl’s principle the congruence issues from a common origin and the constant-cosmic-time slices are those of constant  $\tilde{\tau}$ ), areal radius

$$r(\tilde{\tau}) = \left( \frac{6GM}{\Lambda c^2} \right)^{1/3} \sinh^{2/3} \left( \frac{3}{2} \sqrt{\frac{\Lambda c^2}{3}} \tilde{\tau} \right), \quad (3)$$

and redshift  $1 + z = r(\tilde{\tau}_0)/r(\tilde{\tau}_e)$ . The constant- $\tilde{\tau}$  slices sit at  $45^\circ$  to the fundamental rest frame: the cosmology is homogeneous and observationally isotropic, yet *non-synchronous*—the constant- $\tau$  slices are exactly Euclidean (§7) but are not the cosmological ones. Of the FLRW kinematic assumptions—(i) a geodesic congruence, (ii) hypersurface orthogonality, (iii) homogeneity, (iv) isotropy—this construction keeps (i) and drops (ii); Lemaître’s reading [30] of the Euclidean slices as “space ejected from  $r = 0$ ” is causally incoherent and is not the physics. Equation (3) is moreover a single analytic function of  $\tilde{\tau}$  (Fig. 1): the expanding cosmology is its real branch  $\tilde{\tau} > 0$ , and the cosmogenesis *branch point* at  $\tilde{\tau} = 0$ , where  $r \rightarrow 0$ , is the branch point of the scale factor rather than a real  $r = 0$  boundary—and it is *not* a seam: the seams are the two unit-speed loci of the lap,  $r = -2\alpha/\sqrt{3}$  and  $r = +\alpha/\sqrt{3}$  [9], and  $r = 0$  lies strictly between them, at the close of the lift—the analytic counterpart of the beginning at the branch point, where the substrate’s curvature stays finite while the geometry read over the  $r$ -chart diverges [4], and the cosmological reading of the slicing curve’s closed lap—the *cosmogenetic bead* the framework establishes as one closed object [5, 9], of which this scale factor’s real branch  $\tilde{\tau} > 0$  is the expansion leg and its analytic continuation the prior universe’s collapse—on which  $r = 0$  is a branch point and not a barrier, the substrate smooth across the locus the chart so labels [5].

### 2.3 Exact recovery of flat $\Lambda$ CDM, and the amplitude fixed by $\Lambda$

*What fixes this rate is worth stating before what it recovers, and the count is smaller than it looks.* The rate this cosmology computes on is set by the substrate’s geometry and by nothing else. *It has one parameter*—the substrate curvature radius  $\alpha = \sqrt{3/\Lambda}$ , an invariant length—and *one epoch*. At the Nariai member the mass is no longer free (5), so the amplitude and the rate

of the expansion law are both set by  $\Lambda$  and the scale factor is one function of it; what remains is where the present sits on that one curve. *The quantity  $x_0$  carries that:* it is the present areal radius in units of the Nariai radius,  $x_0 = r(\tilde{\tau}_0)/r_N$ , and since  $r_N = \alpha/\sqrt{3}$  is fixed by  $\Lambda$  the number is a clock reading and not a second geometric datum<sup>P15R14</sup>. *The geometry's own offset is fixed twice over:* it is the double root of the horizon cubic at the Nariai mass, and it is the constant ledger's  $2M = \alpha((r_0/\alpha) - (r_0/\alpha)^3)$  with  $M$  at that mass [5, 17]—so “the offset” names the unit, and  $x_0$  the reading taken against it. So the ratio it fixes is a reading of cosmic epoch, which is what (7) says of the density ratio directly. *Neither the parameter nor the epoch is a content of the universe:*

$$c^2 H^2(z) = \frac{c^2}{\alpha^2} \left( 1 + \frac{2(1+z)^3}{x_0^3} \right). \quad (4)$$

The offset is measured directly and calibration-free— $z_{\text{acc}} = x_0 - 1$ , (11) below—without a distance ladder, without the microwave background and without any density. *What the rate is a rate of is the stacking of the foliation*, and content does not enter it: radiation is absent from the slicing operator's vacuum kernel, requiring  $m'(r) \neq 0$ , a bend of the cut; and the second term is the offset itself rather than a density supplied by hand [7, 9].

*The familiar cosmological form is accordingly a coincidence of form, and saying so is the substance of this subsection.* Written in the fitted pair the same rate reads  $H^2 = H_0^2 [\Omega_m(1+z)^3 + \Omega_\Lambda]$ , with  $\Omega_m = 2/(x_0^3 + 2)$  and  $\Omega_\Lambda = x_0^3/(x_0^3 + 2)$  following identically—a translation between parameter sets, and not a decomposition into components. In the standard reading cosmic time is orthogonal to the three-dimensional world and the expansion is driven: by initial conditions, and by whatever densities obtain at each moment, through the field equations. Here it is read off a fixed background whose foliation is *not* orthogonal to the proper time of observers at rest, so that such observers move through the universe over cosmic time in a geometrically fixed manner. *The two readings agree on the function and share nothing beneath it.*

*Where the field equations act is the leaf, and they act there in full.* All stress-energy gravitates, by the ordinary Friedmann readout of the self-gravitating congruence; the perturbations, the plasma's sound horizon and its diffusion length are processes running *in* the content and take that rate, while a comoving separation read across leaves takes the stacking rate (§2.2). *What the layered framework keeps apart is not two physics but one physics and its projection:* a quantity is computed on the level it belongs to, carried on the layer and then projected.

Equation (3) is exactly the flat- $\Lambda$ CDM scale factor. At Nariai the mass is no longer free, and the amplitude becomes a pure  $\Lambda$ -length,

$$\left( \frac{6GM}{\Lambda c^2} \right)^{1/3} = \frac{2^{1/3}}{\sqrt{\Lambda}}, \quad (5)$$

so both factors of the expansion law are set by  $\Lambda$ <sup>P15R39</sup>: the rate  $\frac{1}{2}\sqrt{3\Lambda}c$  is the standard late-time rate and the amplitude is the third  $\Lambda$ -scale (distinct from  $\alpha$  and  $\alpha/\sqrt{3}$ ). The Friedmann equation the  $\sinh^{2/3}$  law satisfies,

$$H^2 = \frac{\Lambda c^2}{3} \coth^2 \left( \frac{1}{2} \sqrt{3\Lambda} c \tilde{\tau} \right) = \frac{1}{3} (8\pi G \rho + \Lambda c^2), \quad (6)$$

makes the density ratio a clock<sup>P15R39</sup>,

$$\frac{\Omega_m}{\Omega_\Lambda} = \text{csch}^2 \left( \frac{1}{2} \sqrt{3\Lambda} c \tilde{\tau} \right), \quad (7)$$

so  $\Omega_{m,0}$  records the cosmic epoch  $\tilde{\tau}_0$  at which we observe rather than an independent density, and the coincidence problem becomes the observation that we exist at a time of order the geometry's single timescale. The densities are bookkeeping for one  $\Lambda$ -set geometry; *radiation, like matter, is inherited content read off that clock, not a term that sources the rate*, and the

stacking rate has no radiation-dominated era [11], the leaf, where radiation gravitates, carrying one (§4). The objection that a perturbing radiation fluid “must gravitate and so alter the rate” is a category error against a construction that reads (6) leftward: the rate is set, and the fluid’s energy is among the contents the set rate carries. The acoustic *scale* follows from this rate, and  $H_0$  is fixed with it by the distance data; this is developed next (§2.4).

## 2.4 The acoustic scale, the expansion rate, and what the construction dissolves

The two frameworks reproduce the late-time expansion identically and part in the early universe, where radiation carries no term in the rate (6). The apparent tensions of the standard model are consequences of that one structural fact.

*The three levels, and which rate each quantity uses.* The reading of every rate-bearing quantity below is fixed by locating it on one of three levels, which the geometry keeps apart and whose conflation is the standing error the layered framework names [9]. **(L1) The foliation stacking rate:** the  $\Lambda$ -set  $\sinh^{2/3}$  law (6) read *leftward* in the sense the slicing paper fixes [7]—the cut is primary and  $\rho$  is the name of its bend, not its cause, the rate-side of the slicing operator’s reading whose content-side is *matter is the bend of the cut* [7]—the layer’s own geometric expansion, set by  $\alpha$  and the offset  $x_0$ , and *empirically forced* rather than modelled: the microwave-redshift isotropy floor forces uniform expansion to some three orders of magnitude, so a matter-tracking (radiation-sourced) rate is already excluded by data [11].

This is what the observable cosmology rides at *every* epoch, the branch point included: it is the rate the foliation stacks at, not a regime that begins somewhere. **(L2) The leaf-level local dynamics:** the expansion scalar of a self-gravitating congruence, the ordinary Friedmann read-out  $\mathcal{H}^2 = \frac{8\pi G}{3}\rho_{\text{tot}} + \frac{\Lambda}{3}$  with radiation gravitating normally—the valid phenomenological reading of general relativity on the progenitor’s collapse excursion, where nucleosynthesis runs [19]; it is the same single  $E=1$  geodesic read *inward* as dust collapse that L1 reads outward as cosmology [7, 4]. **(L3) The  $E = 1$  projection:** the reassigned shadow the observer reads, through which an L1 or L2 quantity appears as distance and redshift—the Projection Principle of the framework paper, its geometric origin the *second ruling* of the slicing (the synchronous space is the second ruling) [9, 7]; reading this appearance as the existent is precisely the naïve-realist collapse the epistemology forbids [12]. The decision rule is structural, not a choice: a self-gravitating local excursion runs on L2, diffuse content riding the global foliation on L1, and every observable is an L1 or L2 quantity seen through L3; and *there is no L1/L2 boundary in time*. The distinction is kinematic and holds at every epoch: the vacuum kernel of the slicing operator is the SdS family exactly— $\Lambda$  and the cut’s offset  $2M$ , and no more [7]—so the stacking rate is  $c^2 H_{\text{stack}}^2 = (c^2/\alpha^2)(1 + 2(1+z)^3/x_0^3)$ , while radiation, requiring  $m'(r) \neq 0$ , is a *bend* of the cut and therefore content. The decomposition is exact,  $H_{\text{leaf}}^2 = H_{\text{stack}}^2 + H_0^2 \Omega_r (1+z)^4$ : the two rates differ by the radiation term alone. A comoving separation read across leaves takes the stacking rate; the scales the plasma itself accumulates— $r_s$  and  $r_D$ —take the leaf’s, together with recombination’s microphysics and the perturbations, because they are accumulated by a fluid running *in* the content whose own radiation gravitates in its own expansion. The two must take the same rate, and the peak spacing the construction computes is the leaf accumulation: fitted to the computed peaks, the comb reproduces that integral to two per cent and the stacking one not at all. The assignment looked temporal only because  $\Omega_r(1+z)^4$  is negligible below  $z \sim 10$  [9, 4]. Each mis-assignment is quantitatively fatal—the geometric L1 rate carried down into the nuclear window is  $\sim 300\times$  too slow, and the radiation-sourced L2 rate carried into the distances would pin them to a radiation-era density the foliation’s own rate does not carry [19]. Crucially L1 is the layer’s own *ontological* expansion, not the L3 shadow: computing a process against the stacking rate is reading the geometry, not the appearance. Every rate-bearing quantity below carries its level, so the reading is settled by the geometry rather than defaulted to the synchronous machinery.

*The Hubble constant.* The late-time expansion is reproduced exactly, so  $H_0$  enters this cosmology as it enters the standard one: through the distances. With the plasma handed over at the branch point there is no early-universe parameter left to carry the acoustic angle, and the angle and the baryon-acoustic distances then fix the rate together, at  $H_0 \simeq 68.5$ —the angle alone and the distance data agreeing to a twentieth of a unit. *So this construction does not dissolve the discrepancy with the local distance ladder:* it leaves it where the standard model leaves it, as a disagreement between the ladder and the early-universe inference, and what it supplies in its place is stated in §4.7—the acoustic spectrum computed from that background with nothing fitted to it.

*The acoustic scale, computed.* Recombination is observable cosmology from the branch point onward, and the two lengths its angle is built from sit on opposite sides of the rate rule: the comoving distance  $D_M$  is a separation read across leaves and rides the stacking rate, while the sound horizon  $r_s$  is accumulated by the plasma and rides the leaf's. The observed angle  $\theta_* = r_s/D_M$  is their projection.

The robust ingredients are in hand:  $D_M$  is the standard observable, and  $H(z)$  is the fixed  $\sinh^{2/3}$  law (6) with radiation absent from the rate.

The sound horizon  $r_s = \int_{z_{\text{rec}}}^{\infty} (c_s/H) dz$ —whose baryon-loaded  $c_s$  is ordinary content microphysics and whose expansion is the leaf rate the plasma runs on—carries no early-universe parameter at all once the plasma is handed over at the branch point, where every acoustic mode is outside the horizon and the integral converges at its upper end.

The branch point is a locus at which the areal chart's invariants diverge— $r = 0$  is a genuine infinite-curvature locus of that chart—while the curvature of the manifold the cut is taken on is set by  $\alpha$  everywhere, so the divergence is the areal coordinate degenerating and not a scale of the geometry [19, 17]. *And it is the substrate reading of the branch point that makes the integral well posed:* the modes are outside the horizon there and the plasma's own expansion carries them, so the sound horizon converges at its upper end without a floor having to be placed under it by hand. Taking  $c_s$  the ordinary baryon-loaded value—the matter the plasma carries is pressureless, and the geometric throat ratio  $\sqrt{3}$  does not set it—the acoustic scale is computed rather than met: integrated on the leaf rate from the branch point, it returns an angle whose comb sits at 298.0 against the sky's 298.4 with the background fixed by the distance data alone. *This is where the two rates part, and it is not where the standard model's radiation sits.* There the whole expansion carries radiation and the ruler is pinned by the matter density's physical value; here the foliation's own rate carries none, so the distances are radiation-free while the plasma's scales ride a leaf rate that is not, and the two lengths the angle is built from sit on opposite sides of that split. The beginning is well posed: the branch point—finite in the substrate's curvature, whatever the  $r$ -chart geometry does there (§2.2)—carries every acoustic mode outside the horizon and the potential at its regular value, so the sound-travel integral converges at its upper end and needs no lower endpoint placed by hand.

*What the construction fixes, and what it does not.* The empirical forcing of the cosmic foliation bears on the rate, not the matter content—the matter density being the bend of the spatial cut [11, 7, 8]—so the content the plasma carries is measured rather than set by the geometry; the cold seam (Gibbons–Hawking temperature [33]  $\sim 10^{-30}$  K, some thirty orders below recombination) confirms there is no geometric source to set it instead.

The quantities this cosmology actually uses are the epoch  $x_0$  (equivalently  $\Omega_m = 2/(x_0^3 + 2)$ ;  $x_0$  is the expansion read in units of the geometry's own fixed offset  $r_N$ , not the offset itself, which the Nariai condition fixes at  $\alpha/\sqrt{3}$ ), which the rate carries and the baryon-acoustic data fix;  $H_0$ , fixed by the same data together with the acoustic angle; and  $\omega_b$  and  $\omega_\gamma$ , which the sound speed carries and which are measured directly and framework-independently. *There is no early-universe parameter among them:* with the plasma handed over at the branch point the sound horizon has no lower endpoint to place, so the angle is an output of the rate rather than a calibration of it.

*The physical matter density  $\omega_m = \Omega_m h^2$  then does enter, and it enters once, through equality. Matter–radiation equality is  $1 + z_{\text{eq}} = \omega_m/\omega_r$  with  $\omega_r$  measured, so the epoch the distance data fix also fixes how long the plasma spends radiation-dominated, and with it the driving that sets the peak heights and the wavenumber at which the envelope turns over. One quantity carries the expansion history and the acoustic driving together, which is why the spectrum below is a test of the background rather than a further fit to it.*

*One thing about  $\Omega_m$  is worth stating plainly, since the construction’s matter sector invites the question. The dust the rate carries exceeds the baryons by about a factor of six,  $\Omega_m \simeq 0.31$  against  $\omega_b/h^2 \simeq 0.05$ , so most of it is non-baryonic. That is not a component this cosmology introduces, and it is not one it derives. What the matter sector fixes is the number of families, their chirality, the  $S_3$  relating the three walls and the cyclic  $\mathbb{Z}_3$  relating the generations, and nothing about their internal content [20]; abundances are measured content throughout, the matter density being the bend of the spatial cut. So the non-baryonic fraction stands with  $\eta$ ,  $A_s$  and  $n_s$ : inherited, not set by the geometry. What the construction does supply is a place rather than an abundance. Its correspondence with the Standard Model’s content carries a right-handed neutrino as the sixteenth Weyl fermion of each generation, and fixes that state’s place while fixing none of its couplings [20]; a species that is never thermalized contributes nothing to  $N_{\text{eff}}$ , so nucleosynthesis is untouched and no prediction is made or needed here. The point of the paragraph is the division of labour: a candidate the construction already contains, an abundance it does not claim, and no third thing wedged in between.*

The radiation amplitude at a start placed by hand,  $\omega_r(1 + z_{\text{start}})/\omega_m$ , is that start read in units of a density rather than an inheritance; with the plasma handed over at the branch point there is no such start and no such amplitude. *And the epoch the distance data fix lands where the microwave background’s own matter density sits: the rate’s  $\Omega_m$  at the  $H_0$  the angle and the distances return together gives  $\omega_m = \Omega_m h^2$  within a couple of per cent of the standard inference, so equality falls within a couple of per cent of flat  $\Lambda$ CDM’s with nothing fitted to a spectrum*<sup>P15R25</sup>.

What this cosmology does carry from outside is the composition: the baryon-to-photon ratio  $\eta$ , exactly as flat  $\Lambda$ CDM carries it from a baryogenesis outside the model proper—measured, not derived—and with it the light elements, synthesized from the hot handover of the progenitor collapse, this universe having formed at the event horizon of a black hole in a previous one, equally outside the construction and equally measured.

That the composition is measured rather than derived is no defect: a framework is credited for *requiring* the structure it explains—here the rate, and the acoustic structure that follows from it—not faulted for measuring the boundary data it does not. *One naturalness argument about the handover fails, because it compares a quantity with itself at two epochs nine orders apart.*

Complete collapse does liberate binding energy of order the rest mass ( $GM/Rc^2 = \frac{1}{2}$  at the Schwarzschild radius), and it is tempting to read that as making a handover carrying radiation of order the matter density the natural scale—order unity, rather than the small value  $\eta$  takes. But  $\rho_r = \rho_m$  occurs at  $T \simeq 0.8 \text{ eV}$ , whereas the handover sits four orders *above* the deuterium bottleneck at  $0.07 \text{ MeV}$ , since that is what the nucleosynthesis computed on it requires [19]: at the bottleneck the ratio is already  $\sim 9 \times 10^4$  and at the handover proper  $\sim 9 \times 10^8$ . A handover carrying radiation merely of order the matter density would be a handover with  $\eta \simeq 0.5$ , which leaves neither light elements nor acoustic peaks. *So the naturalness question runs the other way: what the progenitor must deliver is a plasma with of order a billion photons per baryon, and the binding-energy scale offers no head start toward it. One route to that derivation is closed from the geometry’s side: the constituent count the matter sector carries is exactly conserved on this history, its parity-odd source vanishing on the spherically symmetric class the cosmogenesis premise fixes, so what such a derivation needs is a chiral state on the spatial layer rather than a chiral member*<sup>P15R151</sup>—and that state is not available on this layer, the same thermal

regularity that fixes the tower’s boundary condition carrying no helicity label, so the two helicity towers are populated alike and the parity-odd content cancels between them<sup>P15R153</sup>. *And the amplitude a start placed by hand would carry is not a feature of this construction either:* given  $\eta$  and the measured matter-to-baryon ratio, both of which the composition inherits regardless, it follows from standard thermodynamics alone,  $[1 + \nu](\pi^4/30\zeta_3) T/[\eta(\omega_m/\omega_b) m_N]$ . *So what the handover supplies is one composition datum* (receipt: `the_two_data_are_one.py`<sup>P15R26</sup>).  $\eta$  may remain a measured boundary condition indefinitely at no cost to what this paper computes. *And the inheritance has a structural reason rather than being a gap left standing* [9, 19]: the infall thermalizes four orders above the deuterium bottleneck, so the progenitor’s composition is erased and the abundances are synthesized in the window rather than inherited, while what survives that passage is baryon number, which no dissociation destroys. *So  $\eta$  is inherited and the abundances are predicted,* and the asymmetry is worth stating exactly: baryon number protects the numerator, while what protects the *ratio* is the adiabatic invariance of the specific entropy — the peak perturbing it by  $5.5 \times 10^{-10}$  of a standing  $1.15 \times 10^{10} k_B$  per baryon, so that  $\eta$  survives the erasure because  $\eta$  is small [19]. *And a derivation of  $\eta$  is not needed, because there is nothing here for one to be about.* It enters the progenitor from the ambient universe by composition-sharing and leaves the branch point unchanged to one part in  $2 \times 10^9$ : *transmitted, not determined* — which is the exact dual of the radiation amplitude’s closure, that one turning on a quantity with no single value to transmit. *So the neighbouring statement strengthens from may to must:  $\eta$  remains a measured boundary condition, at no cost.*

*The synthesized composition.* The same collapse fixes the light-element composition, and the geometry sharpens how. The infalling matter compresses adiabatically—the plasma is optically thick, so the photon bath compresses with it—and heats far above the deuterium bottleneck ( $T_D \simeq 0.07 \text{ MeV}$ ) for every progenitor: the compression does not stop at horizon crossing but continues to the branch point, and the peak is bounded below by the infall energy at the horizon, of order the rest mass per nucleon, independently of the progenitor’s mass (the cosmogenesis paper [19]); and across the branch point the expanding leg cools back through the window. *The window is therefore crossed twice,* and at the branch point the temperature exceeds the binding thresholds by any margin, so nothing the infall built survives to be counted. Because the effective rate through the window is the standard Friedmann rate, the expanding leg is a standard big-bang nucleosynthesis run from free nucleons [19]. On a first sample this is favourable, and on one element advantageous. The helium-4 fraction  $Y_p \approx 0.245$  sits at the observed value; the near-zero metallicity of the oldest systems follows from the handover photodissociating the progenitor’s processed composition back to light elements; and because the run is a standard nucleosynthesis, the lithium-7 abundance carries its standard threefold over-prediction—the standing lithium problem shared with flat  $\Lambda$ CDM, neither dissolved nor worsened. Deuterium sits at the observed  $D/H \approx 2.5 \times 10^{-5}$ : the freeze-out temperature the data demands is the standard bottleneck, and that is exactly the temperature at which the standard rate delivers a freeze-out—target and mechanism coincide un-tuned. The peak calculation that forces this synthesis reading over an inherited one, and the full multi-abundance account, are given in the cosmogenesis paper [19], where the network is integrated explicitly on the cooling leg and returns deuterium, helium-4 and helium-3 at their observed values with lithium-7 at the shared over-prediction, jointly from the single inherited datum—so one progenitor handover does fit the pattern together; what stands as its open edge is the last-percent precision against the measured abundances, a data-confrontation the produced pattern exposes rather than a computation still to be run.

*The load-bearing claim.* The acoustic scale is computed from the rate rather than calibrated against it; the standard  $c_s$  it uses presumes the inherited medium to be an ordinary photon–baryon plasma, which the abundance comparison confirms—helium-4 and deuterium at their observed values, lithium-7 the shared standard problem (the cosmogenesis paper). The load-bearing, falsifiable claim remains the prior one—that radiation carries no term in the foliation’s

own rate—of which the split between the two rates, and with it the whole acoustic reading below, are the consequences. The sound horizon is where a purely geometric expansion history and the standard radiation-governed one are told apart in the data.

*The expansion history, confronted with the late-time data.* The distances and the ruler do not share a common factor:  $D_M$  and  $D_H = c/H$  carry the stacking rate's  $1/H_0$ , while the sound horizon is accumulated on the leaf rate, which carries the radiation the stacking rate does not. So the baryon-acoustic observables  $D_M/r_s$  and  $D_H/r_s$  constrain  $H_0$  here as they do in flat  $\Lambda$ CDM, and the fit returns it: against the DESI DR2 dataset the rate fits at  $\chi^2/\text{dof} \simeq 1$  with  $\Omega_m \simeq 0.30$  and  $H_0 \simeq 68.6$ , and the acoustic angle alone, from independent data, agrees within a twentieth of a unit. *What the two frameworks still do not share is the ruler's provenance:* there it is pinned by a radiation-era physical density, here it is the plasma's own accumulation on a rate whose geometric part carries no radiation at all. *The two rulers make the constraint differently, and the difference is where the radiation sits.* In flat  $\Lambda$ CDM the ruler is pinned in physical length by the radiation-era matter density, so the same distance data tie the fit to one  $H_0$ ; here the ruler is the plasma's own accumulation on the leaf rate and the distances ride the stacking rate, so the two enter the fit through different quantities and the constraint is the joint one<sup>L9</sup>.

The confrontation carried to the state-of-the-art DESI DR2 dataset [23]—thirteen distance measurements across seven tracers ( $0.3 \lesssim z \lesssim 2.3$ ), with the full per-tracer  $D_M/r_s$ – $D_H/r_s$  correlations—returns  $\chi^2/\text{dof} \simeq 1$  at  $\Omega_m \simeq 0.30$  and  $H_0 \simeq 68.6^{\text{P15R38}}$ , and the acoustic angle computed on the same background agrees with that  $H_0$  from independent data.

*Two things about  $\Omega_m$  are worth stating exactly.* It is not a CMB-calibrated input here—Planck's own value is 0.3153, at which the fit degrades to  $\chi^2/\text{dof} = 1.35$ , and 0.3066 is what these data themselves prefer. *And granted the same one free parameter,  $\Lambda$ CDM fits these data marginally better than this cosmology does* ( $\chi^2/\text{dof} = 0.92$  against 1.00). *The result is therefore not that the geometric rate fits better; it is that it fits without choosing an  $H_0$ , and the radiation-pinned one must choose.* The cosmic chronometers [26], which read  $H(z)$  directly, are consistent with the local value. So the resolution is a property of the whole expansion history, not of one angle, and it rides the very rate whose branch point-located scoping keeps the nucleosynthesis window standard [19] (receipt: `hubble_expansion_confrontation_v2.py`<sup>P15R37</sup>; the invariant is the offset  $x_0$ , read as  $\Omega_m$ , and not  $\omega_m$ —holding the physical density fixed instead imports the  $\Lambda$ CDM reading and manufactures a spurious tension).

## 2.5 Where the acoustic modes stand when the plasma is handed over

The cosmological beginning is the branch point  $r = 0$  at the close of the lift, where the conjugate branch becomes the matter branch [9, 5]. The curvature invariants of the  $r$ -chart diverge there because the areal coordinate degenerates, while the throat itself is the substrate's defining constant  $\alpha$  and the bead continues through by analytic continuation [36]. The lap's seams lie elsewhere, at the two unit-speed loci  $r = -2\alpha/\sqrt{3}$  and  $r = +\alpha/\sqrt{3}$ . We record the consequence that organizes the perturbation sector.

**Proposition 1** (Every acoustic mode is outside the horizon at the branch point). *Approaching  $r = 0$  on the contracting leg the comoving Hubble scale  $aH$  grows without bound, so the comoving horizon shrinks to zero and every mode leaves it before the crossing, on either rate and for every wavenumber.* What is handed to the expanding leg is therefore the frozen adiabatic state—the effective temperature at an extremum with no velocity and the potential at its regular value—and the collapse leg's contribution to it is an amplitude and not a phase.

*What the modes then meet on the expanding leg is fixed by the same background.* The leaf congruence carries a radiation term and the perturbations evolve on it, so the wavenumber entering the horizon at equality,  $\ell_{\text{eq}} \simeq 156$  at the epoch the distance data fix, separates the



**Figure 2:** The expansion history across the baryon-acoustic distance ladder. *Left:* the SDSS DR12 measurements of  $D_M/r_s$  and  $D_H/r_s$  (points) against the geometric rate at the directly measured  $H_0 = 73$  (solid) and flat- $\Lambda$ CDM at  $H_0 = 67.4$  (dashed); both track the data, while  $\Lambda$ CDM forced to the local  $H_0 = 73$  (dotted) misses. *Right:* the BAO  $\chi^2$  as a function of  $H_0$ . Radiation in the  $\Lambda$ CDM rate pins the physical ruler  $r_s$ , so its  $\chi^2$  is a steep parabola minimized near 67 and excluding the local value—this is the tension. The geometric rate carries  $H_0$  out of the ratios  $D_M/r_s$ ,  $D_H/r_s$ , so CR’s  $\chi^2$  is flat in  $H_0$ : the BAO fixes  $\Omega_m$  and admits the SH0ES  $H_0 = 73$  [25] (shaded) at no cost. The tension is structural to a radiation-governed ruler and absent from a geometrically fixed one. The figure shows the SDSS DR12 consensus; the same result holds, more strongly, against the state-of-the-art DESI DR2 dataset (thirteen measurements, seven tracers), where the geometric rate fits at  $\chi^2/\text{dof} \simeq 1$  with the single free  $\Omega_m = 0.307$  fitted from these data at the local  $H_0$  (§2.4).

modes driven through a radiation-dominated entry from those entering later: the first acoustic peak sits above it and is driven, which is what §4.1 computes in closed form and §4.7 measures.

*The two loci a reader may run together are distinct, and the distinction is load-bearing*<sup>P15R79</sup>. The branch point is the locus  $r = 0$ , which no finite cosmic-time layer reaches and which therefore carries no finite redshift at all; a configuration that instead starts the plasma at a finite redshift on the expanding leg is starting it somewhere else, and there the census is the opposite one—modes inside the horizon, and a state handed over mid-oscillation.

*Argument.* The comoving horizon on the contracting leg,  $1/(aH)$ , vanishes as  $r \rightarrow 0$  because  $2M/r$  diverges there, so  $k/(aH) \rightarrow 0$  for every  $k$ ; the equality multipole follows from  $1 + z_{\text{eq}} = \omega_m/\omega_r$  on the leaf background with the epoch the distance data fix (receipts: `verify_numeric.py`, anchor 7<sup>P15R17</sup>).  $\square$

*What this settles is a premise and not a mechanism:* it fixes what the plasma is handed and where each mode stands when it starts, and it does not measure how much of the comb the subsequent driving supplies. That measurement is the transfer of §4.7, which computes the driving directly; the census above is what the transfer is carried from.

### 3 What crosses, and what does not

The transmission dichotomy above is an argument from the absence of an imprinted scale. A mechanism is given for it, and it also settles which content the branch point does *not* carry.

The imaginary-time segment preceding the branch point acts on a perturbation as a Euclidean kernel [35], damping a mode of frequency  $\omega$  by  $e^{-\omega|\Delta\eta|}$  with  $|\Delta\eta| \simeq 1.8 \times 10^4$  Mpc. Taken naively this would annihilate everything: at  $\ell = 220$  the factor is  $10^{-71}$ . *But  $\omega$  is the frequency of an oscillating mode, and at the branch point no mode is oscillating.* On the contracting leg the comoving Hubble scale  $aH = |dr/d\tilde{t}| = \sqrt{|1-f|}$  grows without bound as  $r \rightarrow 0$ —it is 0.13

at the comoving turnaround, 1.96 at  $|r| = 0.1\alpha$ , and 19.6 at  $10^{-3}\alpha$ —so the comoving horizon  $1/aH$  shrinks to zero and *every* mode exits it and freezes before the crossing. A frozen mode has no oscillation for the kernel to damp. *So the amplitude and the tilt cross unaltered, and the argument from the absence of a scale is recovered as a consequence of horizon exit rather than assumed.*

*What does not cross is the oscillatory content itself*, and that is the collapse leg’s acoustic phase: a mode that was sub-horizon earlier carries an oscillation, and the kernel annihilates it—an exponent of  $-152$  at the first acoustic peak, integrated across the segment’s own sound-speed profile rather than at the single value the estimate above uses<sup>P15R80</sup>. This is required independently by the observed peak positions, which exclude a transmitted leg phase by a factor of 21–41 in the required distance [9], and it is what leaves the comb to be set on the expansion side by the characteristic data of §4. *The two results are one statement*: the crossing transmits what is frozen and destroys what oscillates, so the progenitor supplies the spectrum’s amplitude and shape while supplying none of its phase.

*The exponent has two factors, and the argument above varies only one of them.* It is  $e^{-kc_s|\Delta\eta|}$ : holding  $c_s$  at the plasma’s value and varying  $k$  reads it as a *scale* criterion, frozen against oscillating, which is the reading just given. Holding  $k$  and varying  $c_s$  reads the same inequality as a *species* criterion, and that reading supplies something the first does not. A pressureless component has  $c_s = 0$  identically—not small, not slowly varying—so its exponent vanishes at every wavenumber; on the segment its mode equation is  $\psi'' = 0$ , whose constant solution the kernel leaves untouched with no approximation at all. *The matter sector’s inheritance is therefore exact rather than adiabatic*: it does not rest on horizon exit and it does not degrade at small scales, where the frozen-mode argument does. The two arguments agree wherever both apply, and only the pressureless one is available everywhere. *Neither reading carries a constraint the other lacks.* On this geometry the exponent at the first acoustic peak is  $\simeq 1.6 \times 10^2 c_s/c$ , so a component arriving *oscillating* with a per-cent sound speed would be suppressed by  $e^{-1.6}$  and one with ten per cent by  $e^{-16}$  (receipt: `the_filter_is_a_sound_speed_filter.py`<sup>P15R80</sup>). *But the exponent has nothing to act on.* The kernel acts on *oscillatory* content—a frozen, zero-frequency mode is a fixed point of it—and on the progenitor’s own interior nothing arrives oscillating:  $|aH| = |a'/a|$  diverges as  $1/x$  at the branch point while  $c_s$  saturates at  $1/\sqrt{3}$ , so  $c_s k/|aH| \rightarrow c_s k x \rightarrow 0$  for every  $k$ , and every mode from  $\ell \simeq 28$  to  $\ell \simeq 2475$  exits the comoving sound horizon strictly before the crunch—the highest with a fifth of a per cent of the collapsing leg still to run [19]. *So the two readings do not divide the work between them: the scale criterion is satisfied by everything, and the species criterion selects nothing.* The crossing is *lossless for every species rather than for cold ones alone*, and the pressureless argument’s positive content—that the inheritance is exact rather than adiabatic—holds of every species and not of dust alone.

*One consequence of that freezing bears on the acoustic sector rather than on the crossing, and it is computed rather than left as an inference.* Every mode is outside the comoving horizon at the branch point (Prop. 1), so each re-enters on the expanding leg at its own time, and what it meets there is fixed by the rate the perturbations run on. *On the leaf rate that re-entry is an event*: the plasma carries radiation, the potential decays on the sound-crossing time, and the modes entering while radiation dominates—everything above  $\ell_{\text{eq}} \simeq 156$  at the epoch the distance data fix—are driven. *That driving is not an inference about the potential’s form but a shape read off the two arms*: subtracting an undriven reference that calibrates at unity, the control’s acoustic phase at first turnover is flat in wavenumber to 2.1% while this arm’s varies by a factor of 3.9 across the same band<sup>P15R64</sup>, and §4.1 carries the same driving in closed form.

*What the rate rule closes is the alternative reading*, on which the modes re-enter under pressureless matter and nothing is imprinted: the potential equation for a pressureless component,  $\Phi'' + 3\mathcal{H}(1+w)\Phi' + [2\mathcal{H}' + (1+3w)\mathcal{H}^2]\Phi + wk^2\Phi = 0$ , *contains no  $k$  at all once  $w = 0$* , so on a radiation-free rate the problem is scale-free and a scale-free problem imprints nothing<sup>P15R62</sup>.

That is the stacking rate’s answer, and the perturbations do not take it: they take the leaf’s, which carries the radiation term and therefore the driving.

## 4 What the substrate determines: coherence from the null boundary

The acoustic peaks are sharp because the modes of a given wavenumber share a common phase at last scattering. Inflation produces this coherence dynamically, by freezing modes while super-horizon. CR obtains it instead from the character of the branch point.

The branch point is a *null* surface—the cosmological horizon promoted to the initial layer [9]. The initial-value problem on a null surface is characteristic, not Cauchy: the data that determine the future are one free function per mode on the surface together with regularity along the generators, rather than a field and an independent momentum. A single datum per mode is a single phase per mode; there is no second, independently specifiable quantity to randomize the relative phase.

We therefore argue that coherence is not imposed but is what regular characteristic data on a null surface *is*.

This is borne out quantitatively, and the scope of the demonstration is stated exactly.

Evaluating the tightly-coupled mode function at last scattering with a *common* phase on that surface gives a sharp regular comb, whereas drawing the phase independently per mode—the second, free datum a Cauchy surface would admit—washes it out entirely, leaving flat power (peak-to-trough contrast  $\sim 1$  against the coherent comb’s many orders of magnitude; receipt: `verify_coherence_comb.py`<sup>P15R18</sup>). *That establishes the mechanism—one characteristic datum per mode is one phase per mode, and coherence follows—and it establishes nothing about where the peaks fall.* The receipt writes the mode function down as  $\cos(kr_s)$  with  $r_s$  and  $D_C$  as literals—and the  $r_s$  it writes is the *standard* 147.0 Mpc rather than any value this construction derives; no mode is propagated in it, as its own scope note says, so the spacing  $\Delta\ell \simeq \pi D_C/r_s \approx 296$  is the arithmetic of an assumed expression on an imported ruler rather than an output. *It is not the result of propagating the modes, and is not offered as one*<sup>P15R61</sup>.

*What the peak spacing lacks here is not agreement but a derivation:* this receipt asserts it, and it is the propagated modes of §4.7 that compute it.

The null boundary admits only the first column: one characteristic datum per mode is one phase per mode.

The peak *heights* are then carried by a structural argument rather than a bespoke transfer. The driven acoustic amplitude is the resonant Fourier magnitude of the potential’s evolution,  $|\tilde{\Psi}(\omega=1)|$ —exactly invariant under time reversal,  $|\tilde{\Psi}(x)| = |\tilde{\Psi}(-x)|$ , a Fourier theorem confirmed numerically on the radiation-era transfer function and on a random control<sup>P15R20</sup>.

The equality is a theorem about Fourier magnitudes and holds whichever side the driving lives on. *Which side it lives on is fixed by the rate rule:* the perturbations run on the leaf congruence, whose matter–radiation equality is fixed by the epoch the distance data fix, so the plasma’s early history is radiation-dominated and the modes carrying the first peak enter the horizon while it is. *The driving is therefore on the expanding leg,* and the collapse leg’s contribution to the state that meets it is an amplitude and not a phase.

*What the state at the handover is has to be said at one locus, and the construction supplies one.*

*At the branch point every acoustic mode is outside the horizon and the leg’s closed-form potential has returned to its regular value,* so the state arriving there is the ordinary frozen adiabatic one: the effective temperature at half the potential with no velocity, the potential unspent, and the collapse leg’s contribution surviving as an amplitude alone—the wavenumber-independent  $0.4835 \Psi_i$ , within three per cent of the frozen value  $\Psi_i/2$  the free oscillator carries

at every wavenumber. Specified that way the radiation-era driving is applied exactly once, and the specification carries no free choice, the other three codings each making one. *It is a limit rather than a parameter*: carried back over two decades in starting redshift the peaks hold to the grid step and the height ratios to half a per cent, while a start an order of magnitude later has not converged.

The collapse-side driving that argument relied on carries a qualification of its own receipt: completeness holds for the modes whose entry precedes the horizon maximum at  $r_* = 1.53 r_{\text{seam}}$ , entry on the rising branch being earlier the higher the wavenumber, so the acoustic band is on the safe side of that boundary by construction and the low- $k$  end is where it would bite<sup>P15R48</sup>; what a Boltzmann reference returns for flat  $\Lambda$ CDM's own heights is  $P_1/P_2 \approx 2.2$ ,  $P_2/P_1 \approx 0.45$ ,  $P_3/P_1 \approx 0.44$  (`camb.reference.py`<sup>P15R21</sup>)—*those are the reference's values, not this arm's*, and what this construction's own instrument returns is set out above. The *damping* tail ( $\ell \gtrsim 1000$ ) is fixed by the three-level rule (§2.4) rather than left to judgement. Recombination is observable cosmology from the branch point onward, so the photon diffusion length  $r_D$ —like  $r_s$ —is an L1 quantity: the random walk accumulates against the layer's own geometric expansion, its Thomson microphysics ordinary content physics but its expansion the L1 rate, *not* a radiation-included one.

This is the same branch point-located scoping the cosmogenesis paper derives for the nucleosynthesis window [19], read here in the opposite direction: there the self-gravitating excursion sets the L2 rate radiation is included in, here the diffuse plasma rides the L1 foliation radiation is excluded from.

This is forced by consistency: one may not take the rate geometric for the peak spacing and radiation-included for the diffusion. The geometric rate is  $\sim 13\%$  below the radiation-included one at recombination ( $\rho_r/\rho_m \approx 0.3$ ) and further below it earlier, where the inherited radiation is a larger fraction; *the diffusion length integrates that whole history rather than its endpoint*, and runs  $\sim 9\%$  longer than  $\Lambda$ CDM's while  $r_s$  is matched to the observed acoustic scale (the closed-form derivation of §4.2 gives  $10.8\%$  on a sharper ionisation treatment, and the two are related there); the projection-independent ratio is therefore  $\theta_D/\theta_* \approx 1.08 (\theta_D/\theta_*)_{\Lambda\text{CDM}}$ —a larger damping angular scale by a few per cent, its size fixed by where the plasma is handed over and its sign by how far the diffusion integral is carried, as §4.3 sets out (receipt: `damping_ratio_clean.py`<sup>P15R22</sup>, validated against a Boltzmann reference on the radiation-included rate;  $r_s = 144.0$  vs CAMB 144.4 — a  $0.28\%$  disagreement against the few-per-cent signature it feeds, so the reference agrees to several times the effect and the margin rather than the agreement is what the word *validated* is carrying<sup>12</sup>). What a stated tolerance carries anywhere in this corpus's computational layer is measured rather than assumed: every `abs(...)<tol` comparison in the receipts rewritten twice, once to displace the compared quantity above its tolerance and once to collapse it below, so that 243 of the 245 receipts that pass are shown to have at least one such comparison that actually gates their verdict, the two exceptions being a display filter and a skip that prints its own reason<sup>12L22</sup>.

This is the one genuinely CR-specific piece of the height pattern: a real signature *locked to* the Hubble resolution by their shared L1 rate—not an artefact of laying the standard radiation-governed  $r_s$  on the geometric rate, the calculation belonging to neither framework this paper warns against (§1).

One thing about this signature is settled and one is not, and it is essential to keep them apart.

Settled: the effect in  $r_D$  is genuinely CR-specific and does not dissolve under the density that would ordinarily rescale it— $r_D \propto \omega_b^{-0.31}$ , so a percent-level  $\omega_b$  (fixed by primordial deuterium and the peak-height ratio) shifts it by well under a percent (receipt: `damping_reabsorption.py`<sup>P15R23</sup>). *And what the signature does to the observed high- $\ell$  power is measured, with a verdict that is neither of the two a reader expects*. Scaling the diffusion length by the instrument's own measured ratio multiplies the damping by a Gaussian in  $\ell$ , and a tilt is a power law — so over

a restricted high- $\ell$  range the one is well approximated by the other. *Refitting amplitude and tilt therefore absorbs the signature almost entirely in shape*, and what it costs is a shift in the inferred tilt. *Both scale with  $r^2 - 1$ , and at the adjudicated ratio that factor is small*: with both lengths carried to the visibility peak,  $r = 0.9918$ , the signature costs 1.70 in  $\chi^2$  over the 185 full-range bins—under a hundredth per bin—and a tilt of  $+0.0034$  absorbs seven tenths of it. *Terminating the diffusion integral at a sharp recombination cut instead,  $r = 1.0231$ , gives 12.6 and a tilt of  $-0.0093$ , and a ratio of 1.082 gives 160 and  $-0.034$  on the same base and the same damping scale<sup>P15R167</sup>. So the hundredfold collapse is the configuration moving and not the fit improving. So the signature does not vanish under a refit — it moves out of the spectrum’s shape and into an inferred parameter. What a refit buys in shape it pays for in parameters, and that direction is the robust part.*

*And how local that approximation is can be stated exactly, which settles what a joint fit would actually meet.* The signature’s own logarithmic slope is  $d \ln(\text{ratio})/d \ln \ell = -2(r^2 - 1)(\ell/\ell_D)^2$  — not a constant, but growing as  $\ell^2$ : across the likelihood’s own  $\ell = 30\text{--}2508$  it runs from  $-0.0002$  to  $-1.31$ , a factor of seven thousand, which is exactly  $(\ell_{\max}/\ell_{\min})^{2\text{P15R3}}$ . *So a single tilt is a one-parameter fit to a function whose slope varies by that factor over the data it is scored on*, and the displacement quoted above is the signature’s true slope at  $\ell = 381$  and at no other multipole. *The consequence is the useful part*: a joint fit does not encounter a shifted tilt but a residual whose shape is a Gaussian in  $\ell$ , and no tilt removes a Gaussian — so what such a fit weighs is the shape residual, which is the quantity already bounded from above. *And a displacement in  $n_s$  is not by itself a tension here*, the tilt being inherited boundary data rather than predicted: a different value of an inherited input is a different input. *What could be a tension is a constraint on  $n_s$  that does not run through the damping tail* — the low- and mid- $\ell$  shape, the polarisation spectra, lensing — which is what a joint fit means and is where the question properly sits.

*Two bounds travel with that and neither can be left to a reader. The shape residual is an upper bound*: freeing only the named degeneracy is the least absorption a fuller refit could achieve, so a wash is a wash *a fortiori*. *The tilt displacement is not a bound*, because more free parameters would share it — the baryon density moves the diffusion length itself. *And the magnitude is not robust to a factor of three*: imposing the ratio on the measured spectrum gives the figures above, while applying it inside the instrument’s own source — where it also reaches the Doppler term through a derivative — gives 0.92 in  $\chi^2$  per bin. *The larger comes from the more faithful route, so the wash is stated at the weaker bound* and closing that factor is named here as work not done.

*The joint fit returns the residual this reading predicts.* Fitting amplitude and tilt together against the signature-carrying spectrum, on `plik_lite`’s own bandpowers and covariance, the residual left after the fit is the analytic Gaussian-minus-power-law shape to a correlation of 1.000000 — not merely non-zero, but the predicted form. *The residual’s form is the finding and its size is the configuration’s*: at the adjudicated ratio it is under a hundredth in  $\chi^2$  per bin, and 1.50 per bin at a ratio of 1.082 on the same window. *And the displacement is not one number*: fitted over  $\ell = 100\text{--}1760$  and over  $\ell = 1400\text{--}2508$  it differs by a factor of 6.3 across windows of the same data, so a displacement is the value of nothing without the window it was fitted over. *And the tilt takes a fixed share of it rather than a variable one*: seven tenths at the adjudicated ratio and seven tenths at 1.082, so what the refit can and cannot do is a property of the shapes and not of the size<sup>P15R4</sup>.

*Not settled, and unchanged by that run*: whether a displacement of that size in the inferred tilt is a *tension*. The tilt is inherited boundary data here, not predicted, so a different value of it is a different input; what would make it a tension is a constraint on  $n_s$  that does not run through the damping tail, which this fit does not supply. The observable is  $\theta_D/\theta_* = r_D/r_s$ , and translating it into a spectrum requires the full acoustic transfer, which is run (§4.7) and does not return high- $\ell$  peaks tracking flat  $\Lambda$ CDM’s. A leading-order pass that simply *assumes*

the high- $\ell$  peaks equal  $\Lambda$ CDM's and applies the extra damping produces a large high- $\ell$  deficit (receipt: `full_transfer_verdict.py`<sup>P15R27</sup>)—but that assumption may not be imported for free: the same geometric rate that enlarges  $r_D$  also governs the high- $\ell$  driving envelope, so CR's high- $\ell$  spectrum rests on short-wavelength collapse-phase driving rather than on the boost an expanding radiation era supplies.

*One thing about that figure should be stated because two of this corpus's instruments do not agree on it, and the disagreement is worth knowing rather than averaging.* The 9% above is the observable  $\theta_D/\theta_*$ , computed on standard thermodynamics with the sound horizon re-anchored to the observed acoustic scale so that only the damping length moves. *The two-arm transfer, measuring the raw diffusion length at the visibility peak, returns 7.55% on its own arms — a different quantity at a different epoch, and not a correction to this one. Neither number is offered as a replacement for the other, and closing the gap between the two routes is a thing this paper does not do.*

*That driving is computed below (§4.1), and the calculation removes the licence the shortcut lacked: the envelope is derived on the collapse leg rather than imported.*

Diffusion on the geometric rate therefore constrains, but does not by itself determine, the high- $\ell$  shape; the net effect is computed in §4.2; whether it is a tension, a wash, or a distinctive but consistent feature turns on a parameter refit and on the early integrated Sachs–Wolfe term. *Two confrontations are to be distinguished and both are run.* The spectrum computed on the background the distances fix is scored as it stands, one amplitude fitted so that what is compared is shape, and sits at 2.57 times the control's distance from the data. *Refitting the spectrum's own parameters to the microwave data*—the comparison the standard model is usually judged by—*leaves it at 1.57 times:* freedom closes about a third of the gap and leaves the rest (§4.7). The control reproduces the sky essentially perfectly, which is what makes the comparison readable at all; and the verdict it returns is that the spectrum derived here is disfavoured against the standard model on this data. We state it as it stands. The effect the fit is run on is separately established: a real, computed, non-reabsorbable  $\sim 9\%$  CR-specific diffusion-scale signature, its observable consequence derived rather than entangled (§4.2). *What the fit does not reach is the join:* it weighs the five-parameter spectrum construction, and what carries that construction back to the geometry is the transfer work of this section, which is where the open end lies. The structural point that stands is the *mechanism of phase*: where inflation freezes the phase dynamically, CR fixes it geometrically on the single null layer; how the amplitudes are driven is what the transfer of §4.7 measures.

## 4.1 The collapse-phase driving envelope

The driving named above can be carried out in closed form, and doing so removes the entanglement the previous subsection had to leave standing. On the radiation-dominated collapse leg the potential obeys  $\Psi'' + (4/\eta)\Psi' + (k^2/3)\Psi = 0$ , whose regular solution is elementary,  $\Psi = 3\Psi_i(\sin x - x \cos x)/x^3$  with  $x = k\eta/\sqrt{3}$ , and which is *even* in  $x$ <sup>P15R47</sup>—so the contracting leg carries the same closed form pointwise, a sharper statement than the Fourier-magnitude equality of §4 and consistent with it. Writing  $\hat{\Theta} = \Theta_0 + \Psi$  removes the source exactly,  $\hat{\Theta}'' + (k^2/3)\hat{\Theta} = 0$ : the effective temperature oscillates freely, and the  $k$ -dependence rides entirely in the amplitude at horizon entry and in the decay of  $\Psi$ .

Horizon entry is fixed by the construction rather than fitted. The comoving Hubble radius is built from the slicing paper's own turnaround function,  $(rH)^2 = (1 - f) + A/r^2$  [5], and has a single turning point on the whole excursion, at  $r_* = 1.53 r_{\text{seam}}$  on the corpus's inherited datum; the *seam* sits just inside that turning point, on the rising branch, so the acoustic modes are already within the comoving horizon there—a property of the turnaround function's geometry rather than a numerical accident of the background, and a statement about the seam that Prop. 1, being about the branch point, does not make<sup>P15R48</sup>. *The locus matters and the two available ones give opposite answers:* the branch point lies at the far end of the same rising

branch, where  $2M/r \rightarrow \infty$  carries  $aH$  up without bound and the comoving horizon to zero, so there *every* mode is outside it (§3). It is the seam, not the crossing, that this construction reads—the inversion derived from the metric function rather than asserted<sup>P15R79</sup>. On the leg the radiation term dominates because it is constant while the others vanish, so  $\eta = r/\sqrt{A}$  to about two per cent at the entry radius of the lowest mode in question and better the deeper the mode enters—sub-percent from  $k \gtrsim 30 k_s$ <sup>P15R49</sup>, and entry then occurs at  $x \rightarrow 1/\sqrt{3}$  in the *deep sub-horizon limit*—the radiation era’s own scale invariance, recovered here from the substrate geometry. *The envelope is therefore flat to the accuracy of that limit*: a mode at  $k \simeq 10 k_s$  enters within 2% of  $1/\sqrt{3}$  and one at  $300 k_s$  within 0.1%, all of them reaching the *seam*—where the collapse leg ends—with the same driving amplitude  $\Psi(1/\sqrt{3})/2$ , and the residual potential is below a percent *on average* while oscillating up to some seven per cent of it where  $\cos x_{\text{seam}}$  is near an extremum<sup>P15R50</sup>. *Both are qualifications rather than identities: the limit is a limit and the average an average.* Its turnover sits at equality, and the equality is the background’s own,  $\Omega_m/\Omega_r$  on the leaf rate: at the epoch and rate the baryon-acoustic data fix it falls within a couple of per cent of flat  $\Lambda$ CDM’s, and it moves with the matter density rather than being assumed<sup>P15R53</sup>.

Two corrections that might have been expected do not enter. The baryon loading is proportional to the scale factor and the driving happens where the scale factor is smallest, so  $R \lesssim 0.1$  where the driving happens and orders below that at entry; the odd/even asymmetry is imprinted afterwards, on the expansion side at  $R \approx 0.6$ , which is ordinary content physics on the observable leg. *The sky’s own value, located by parabolic fit on the Planck binned spectrum, is  $P_1/P_2 = 2.2564 \pm 0.0772$ , and flat  $\Lambda$ CDM’s is 2.200*; what this construction returns for the same quantity is 2.264 on the polarisation path, on the transfer of §4.7 with its wavenumber integral converged, since a peak height is exactly what an unconverged integral misreports<sup>P15R51</sup>. The structural point stands without it: *the loading is the same in both because it is imprinted on the expansion side, and the argument is at the level of contents rather than of the driving—where the loading ratio alone would give 3.5.* And the free-streaming neutrinos the cooling-leg nucleosynthesis requires do carry anisotropic stress at these temperatures—the branch point is far below their decoupling—but their deep sub-horizon effect is a constant phase shift and a constant suppression, and a  $k$ -independent correction leaves a flat envelope flat<sup>P15R52</sup>. What it does not leave is the closed form for  $\Psi$  itself, the equation becoming integro-differential; the closure is a property of the skeleton, and the flatness is what survives.

## 4.2 What the envelope buys, and what remains

With the envelope derived rather than assumed, the diffusion signature can be stated without the entanglement. The diffusion length reduces to a single integral over the scale factor in which every microphysical constant sits outside,  $1/k_D^2 \propto \int da g(R)/(Hx_e)$ , so in the ratio the Thomson physics and the ionisation history cancel identically and the whole difference is carried by  $H(a)$ —which is the division of labour this section has argued for throughout. Both integrals close in elementary form, and on the inherited datum the geometric rate gives a diffusion length 10.8% longer<sup>P15R54</sup>. *That is the closed-form value and it stands beside the 9% quoted above rather than replacing it*: the 9% runs the ionisation history through a Boltzmann-validated recombination, the closed form takes  $x_e$  as a sharp cut at  $a_{\text{rec}}$  and carries the inherited datum unchanged to recombination. *The two are independent routes to one quantity and agree to about a point and a half*, which is the relationship an elementary derivation should bear to a code-validated one. The sound horizon is the same kind of integral, and it must be taken *from the branch point*: there is no observable expansion below it. Doing so returns  $r_s = 146.4$  Mpc against 145.4 on the radiation-included rate—within 0.7% of each other, the exact crossing sitting a little lower at  $z \simeq 6.7 \times 10^3$ <sup>P15R55</sup>—the acoustic-scale calibration, met from the other end and to that accuracy rather than exactly. *Both are the elementary integral; carried through a full recombination history and validated against a Boltzmann reference the same two come*

out at 144.0 and 144.4, a per cent lower and in the same relation<sup>P15R22</sup>. The observable, in which the common distance cancels, is then  $\theta_D/\theta_*$  larger by a few per cent, its size fixed by the handover (§4.3), and  $\ell_* = \pi D_M/r_s = 302.2$  against the measured 301.76<sup>P15R44</sup>.

The high- $\ell$  consequence follows with no free parameter:  $C_\ell^{\text{CR}}/C_\ell^{\Lambda\text{CDM}} = \exp[-(\ell/\ell_D)^2(r^2 - 1)]$  with  $r = \theta_D/\theta_*$ —the same ratio the preceding paragraph derives, and nothing else—so with both lengths carried to the peak of the visibility function,  $r = 0.991$ , it is a rise of about two per cent at  $\ell_D$  and four at  $1.5\ell_D$ , where terminating the diffusion integral at a sharp recombination cut instead,  $r = 1.022$ , turns it into a fall of four and ten<sup>12</sup>. *Whether that is a tension is a separate question this paper does not settle*: a confrontation refits the cosmological parameters, which is a likelihood analysis and not a closed-form calculation, and asserting exclusion from these numbers would repeat, at one remove, the error of holding everything else fixed while varying one thing. What has changed is the standing of the signature. The signature is a requirement of the construction, following from the level assignment and the handover and nothing else, with its sign fixed by the endpoint the diffusion length is carried to.

*The early integrated Sachs–Wolfe contribution is present here, and the step that once denied it is worth setting out because the denial was an inference and not a measurement.* The rate at recombination is matter-dominated to nine orders, the substrate term being utterly negligible there<sup>P15R46</sup>: *that is true, and it is a statement about the rate.* The geometric rate carries no radiation term, radiation being inherited content read off the clock rather than something that sources the rate.

*But the effect is driven by the radiation **content** through the perturbation source, not by the rate.* The content enters the source under either rate assignment, and *at recombination this cosmology’s content is 21.7% radiation*,  $\rho_r/\rho_m = 0.2765$  at  $z = 1088$  — which no statement about the rate’s composition bears on. *So the inference from a radiationless rate to an absent term does not hold*, and the term is present.

*Measured, as the integrated  $|e^{-\tau}(\Phi' + \Psi')|$  after recombination in units of the Sachs–Wolfe amplitude there: 0.348 on the leaf congruence the rate rule assigns the perturbations to, and 0.229 on the stacking rate. Non-zero on both. The leaf assignment makes the term half again larger; it does not create it.*

*One scope qualification is owed here, and it narrows the claim without touching the conclusion.* The constancy argument runs on  $\Phi'' + 3\mathcal{H}\Phi' = 0$ , which is the *super-horizon* equation: it drops the  $k^2\Phi$  term. For the acoustic modes—inside the horizon through the observable leg—that term does not vanish, and the potential does decay on the observable leg, by a factor of order two across the first few peaks<sup>P15R45</sup>. The decay is not a radiation effect: zeroing the radiation fractions in the constraint makes it *larger*, and the rate responsible is  $k^2/(3\mathcal{H})$ , which grows with  $k$  and refers to no content at all. So the statement that survives is that *the substrate rate supplies no early integrated Sachs–Wolfe term*—which is what the leftward reading delivers and what the comparison with flat  $\Lambda\text{CDM}$  turns on—while the sub-horizon potential decay is an ordinary consequence of the perturbation’s own scale, present in any cosmology.

Whatever difference in that term remains between the arms cannot offset the diffusion signature, and the reason is structural rather than numerical. It sits near the first peaks and dies away well before the damping tail begins, so the two live at opposite ends of the spectrum; and the first peak is where the amplitude is anchored, by an  $A_s$  this construction inherits rather than predicts. A few per cent there is absorbed by that inherited normalisation, exactly as it would be in a framework carrying  $\eta$  from outside. The diffusion signature is not so absorbed: it is a change of *shape* at high multipole, and rescaling the amplitude to restore the first peak would worsen the tail rather than repair it. *What stands between the derivation and a verdict is therefore the likelihood confrontation*—a parameter refit, not a further calculation of this kind, and it is run: the joint fit leaves the Gaussian residual (§4) and the full-spectrum likelihood rejects this arm’s spectrum on  $TT$  shape (§4.7).

### 4.3 The diffusion scale, and the error budget

The acoustic scale is computed from the rate, and the diffusion scale is computed from the same one. Both lengths are accumulated by the plasma on the leaf rate, so the projection cancels from their ratio and what is left depends on two things only: where the plasma is handed over, and how far the diffusion integral is carried. *Neither is a free parameter, and the first is the larger.* With the handover at the branch point the ratio is a few per cent above flat  $\Lambda$ CDM's; a start placed later on the expanding leg raises it steeply, reaching some thirty-five per cent by the redshift at which the angle would be met.

*The remaining choice is the epoch both integrals terminate at, and it is settled by the same consistency the rate assignment is.*  $\theta_D/\theta_*$  is a ratio of two lengths the plasma accumulates, and a ratio of integrals taken to different epochs is not a ratio of anything: the observed angle is read at last scattering, which is the peak of the visibility function, so that is where both the sound horizon and the diffusion length are carried to. *Terminating them differently is what produces the spread:* with the sound horizon anchored at the observed angle and the diffusion length taken instead to a sharp recombination cut, the signature reads +2.2% where a common endpoint at the visibility peak gives −0.9% — a swing of three points where the cosmological parameters together move it by two hundredths<sup>P15R75</sup>. *So the size is the handover's and the epoch is not a free choice,* and the ordering of the four handover-and-rate combinations is unchanged on either reading. *What the signature is not* is reabsorbable: the baryon density has about one per cent of freedom against it, and the matter density could only absorb it by moving some thirty-seven per cent, which is a different cosmology rather than a refit.

Two contributions that might have been expected do not enter: the last-scattering surface's thickness scales with the rate in both its width and its conversion to comoving distance, so it cancels from the ratio; and the recombination redshift, integrated on this rate through the Peebles equation [38] rather than inherited, moves by only −0.11%<sup>P15R59</sup>. The same integration returns  $z_* = 1091$  on the radiation-included rate, against the 1090 of the standard treatment, which is the check that the ionisation history is being handled correctly.

### 4.4 The integrated Sachs–Wolfe contribution across the first-peak region

The first acoustic peak is a separate matter and it does not move. Its offset from  $\ell_*$ —220 against 301.7—is largely the phase shift the potential's decay imparts while the modes are driven, and that decay has the same closed form on the collapse leg as on an expanding radiation era (§4.1), so it transfers. What does not transfer is the early integrated Sachs–Wolfe contribution, and that turns out to be a nearly *flat* multiplicative factor across the first-peak region—some ten per cent in amplitude, varying by two parts in a hundred between  $\ell = 150$  and  $\ell = 300$ —so it displaces the maximum by one or two multipoles rather than by tens<sup>P15R57</sup>. *So the early integrated Sachs–Wolfe contribution does not move the first peak;* what sets this arm's first peak, and the phase residual the transfer measures there, is taken up at §4.7. *The reabsorption of the damping signature, by contrast, can be bounded here,* and bounding it is worth doing because the reabsorption question is the whole of what a likelihood would decide about this signature. Pinning  $\ell_*$  to the measured 301 and minimising the residual in  $\theta_D/\theta_*$  over the baryon density and the matter density—the former held to the one per cent primordial deuterium allows, the latter to the five per cent the distance and growth data allow—leaves a residual of several per cent<sup>P15R58</sup>. The baryon density has about one per cent of freedom against it (§4.3); the matter density could absorb it only by moving some thirty-seven per cent, which is not a refit but a different cosmology. *So the shift has a floor, and the floor is the prediction.* We state this as a bound and not as a likelihood: a full comparison fits the whole spectrum with every parameter and its covariances, where this fixes one observable and minimises one derived ratio. It settles that the signature does not dissolve under reabsorption; it does not settle the verdict.

## 4.5 What the instrument’s residual is made of

The two-arm comparison’s floor is set by components of the instrument rather than by either cosmology. Two candidates for the accuracy it lacks are measured here.

*Of the two, one cannot pay at all*<sup>P15R73L23</sup>. Above  $\ell \simeq 40$  the whole effect of reionisation on the temperature spectrum is the constant screening  $e^{-2\tau}$ —the second visibility bump contributes only below  $\ell \simeq 30$ , outside this comparison’s range—and the comparison fits one amplitude in closed form. *An amplitude fitted as  $A = (m^T F d)/(m^T F m)$  is homogeneous of degree  $-1$  in the model, so rescaling the model returns  $A/s$  and leaves the residual unchanged*: the degeneracy is exact rather than approximate, and it is a property of the statistic rather than of the size of  $\tau$ . Checked at  $\tau = 0.054, 0.10$  and  $0.30$ , the change in  $\chi^2$  is zero to machine precision at all three. *Lensing is a different matter*. A smoothing in multipole, scanned in width, takes the control from 1320 to 921; and a width that *grows* with  $\ell$  beats a constant one by a factor of 1.8 in the improvement, each compared at its own optimum, so the preference is for a shape and not merely for a size—which is *lensing’s own signature, since a deflection of fixed angle subtends a larger fraction of an acoustic wavelength at high  $\ell$* . We state what that is and is not. The width is fitted, so the improvement bounds what a derived calculation has to play for rather than measuring a lensing amplitude; and even at its own optimum some 70% of the residual survives it. *The lensed spectrum is therefore what the instrument’s remaining floor is made of, and it is computed rather than fitted*. Applying  $\Lambda$ CDM’s own lensed-to-unlensed ratio to the arm—a derived operation with no free width—takes the control from 1320 to 989, an improvement of 331, concentrated in the damping tail where a fixed-angle deflection fills the troughs. *That is below the 400 the fitted width bounded, as a zero-parameter derivation must be*, and the acoustic peaks do not move under it. The corpus’s own first-order kernel overshoots the bound to 508 by breaking down in that same tail—returning a spurious +13% at  $\ell = 1900$  where the full operator gives +6.5%—which is why the derived number is the full operator’s. *Most of the fitted headroom is recovered by a genuine lensing, and about 989 of the 1320 survives it*: the verdict of the last paragraph stands from the derived side, lensing real and not the last build<sup>P15R74</sup>.

*And what the residual is MADE of was decomposed alongside it, which is what let the size of that gain be anticipated rather than discovered*<sup>P15R76</sup>. Projecting named templates through the same inverse covariance the comparison uses, the peak *positions* account for a tenth of a per cent of the residual—which is *why the acoustic peaks do not move under the lensing above, and the two findings corroborate each other*—while the peak-trough *contrast* at fixed position accounts for two-fifths of it, this transfer’s contrast running some thirteen per cent too high. *Reducing contrast without moving peaks is what lensing does*. A smooth multiplicative envelope accounts for a further sixth, and *about half the residual is in neither*—so a complete lensing calculation was expected to leave a comparable amount behind, and it did.

## 4.6 The shear coefficient, derived

*The coefficient is derived rather than fitted, and the derivation settles it against the fit*<sup>P15R71</sup>. The coefficient is measured by running the same oscillator twice on the same background: once as the perfect tight-coupled fluid, which is undamped by construction, and once as a photon hierarchy carried to  $\ell = 24$  with Thomson scattering, the polarisation multipoles evolved alongside and the baryons given their own velocity. *Their ratio is the damping and nothing else*—the slowly varying prefactor, the drift of the sound speed and the baryon-loading dependence of the oscillation are common to the two runs and divide out identically—and reading it locally, between consecutive extrema, lets the residual be extrapolated in  $(k/\tau')^2$  rather than smeared over an interval. With the polarisation multipoles suppressed the quadrupole’s scattering term reduces to  $-\frac{9}{10}\tau'F_2$ , which is the unpolarised closure, and the coefficient comes back as 0.8852 against 8/9; with them carried it is 1.0618 against 16/15. *The polarisation correction itself,*

which is the quantity in dispute, is a factor 1.1996 against 6/5: three parts in ten thousand. Both absolute values sit about half a per cent low by an amount proportional to the coefficient, so the residual is a common normalisation and cancels in the ratio.

#### 4.7 The acoustic comparison

The angular scale  $\ell_* = \pi D_M/r_s$  is built from two lengths that do not ride the same rate here: the distance carries the stacking rate's  $1/H_0$  and the sound horizon is accumulated by the plasma on the leaf rate, which carries the radiation the other does not. So the measured angle constrains  $H_0$ , as it does in flat  $\Lambda$ CDM where the sound horizon is fixed instead by the matter and baryon densities<sup>P15R56</sup>. *So the acoustic scale is computed rather than accommodated*, and it is the comb of the computed spectrum that carries it: fitted to those peaks by the procedure this paper runs on the sky, it is 298.0 against the sky's 298.4, with the background fixed by the distance data alone. *A start placed by hand and fitted to the angle would report a different quantity*: what such a fit sets is the scale reported, not the spacing the spectrum carries, the two differing by four per cent.

*The datum §4 states leaves two things to be fixed, and the handover fixes both.* That datum is one amplitude per mode and a *common* phase; it says that the phase is common without saying which phase it is, and it does not say what the amplitude is flat in wavenumber *at*. At the branch point both follow: every mode is outside the horizon there, so the phase is an extremum with no velocity, and the amplitude is the frozen value the closed-form leg carries,  $0.4835 \Psi_i$ .

*How much the comb depends on that choice is measured rather than assumed.* Across the seventeen readings admitted by a four-peak criterion fixed before the numbers, the first peak spans 148 to 228—a factor of 1.541, and across the fifteen numeric-phase readings alone 1.118—with the spacing spanning 276.0 to 330.4, the phase intercept  $-0.4044$  to  $-0.2161$  and the alternation ratio 0.475 to 1.088. *Within that span position and alternation move against each other*, Spearman  $\rho = +0.782$  at  $p = 2.1 \times 10^{-4}$  between the first peak and the gap ratio, with every contracting reading sitting at  $\ell_1 \leq 212$ . *The single-locus reading is not inside that trade*: it returns the first peak at 222 and a contracting second gap together, which is what fixing the state at one locus rather than choosing a reading of it buys.

*The instrument that would measure the rest is built, and its guard and its control are what make it an arbiter.* Both arms run on one set of equations, so no comparison between them is a difference of machinery<sup>P15R63</sup>. *The guard first, because it is what licenses the rest.* With every coupling to the potential removed at every site—the  $4\Phi'$  in the photon, neutrino and cold-matter continuity equations and the  $k^2\Psi$  in the photon–baryon and neutrino Euler equations—the source comb lands on the integers on *both* arms. *So the sound horizon, the measure, the wavenumber grid and the phase extraction are validated as an ensemble, and every departure from  $n\pi/r_s$  in a driven run is the driving and nothing else.*

*The control is the arm whose answer is known independently, and it reproduces it*<sup>P15R72P15R107</sup>. The photons are carried as a Boltzmann hierarchy with the polarisation multipoles alongside, the baryons given their own velocity and density contrast, and the source carrying  $g\Pi/4$  and  $(3/4k^2) d^2(g\Pi)/d\eta^2$ . Run on the flat- $\Lambda$ CDM arm it returns peaks at 220, 540 and 812 against the sky's 220.6, 538.1 and 809.8, and a first-to-second height ratio of 2.1969 against a standard code's 2.200—agreement to 0.14%, and within the sky's own uncertainty.

*Two conditions on how such a transfer is run are what that agreement rests on, and the tighter of them is numerical.* The projection must carry the polarisation source, which a damping envelope cannot supply at any coefficient; and the wavenumber integral must be carried past the highest multipole *reported*, since  $\int P(k) \Delta_\ell(k)^2 dk$  is not converged there. *Each is worth a factor on its own and neither alone reaches the answer*: the height ratio runs from 2.721 at an  $\ell$ -equivalent cutoff of 900 to 2.393 at 2400, so a cutoff set at the reported multipole is a systematic and not a rounding.

*The perturbation sector runs on the leaf congruence, which is what the framework assigns it. A process running in the content—the sound horizon, the diffusion length, recombination, the perturbations—takes the leaf’s rate (§2.4), while the comoving separations and the projection keep the stacking rate; the change is an exact chain rule between two monotone conformal times, so every spline and grid is untouched, and on the flat- $\Lambda$ CDM arm the two rates are identical and the assignment is provably a no-op there—which is its own self-check.*

*The comparison is made on the polarisation path, in the configuration the control is calibrated in. There the control returns a first-to-second height ratio of 2.195 at peaks 220, 536 and 814, against a standard code’s 2.200: so the instrument is calibrated where this cosmology’s number is taken<sup>P15R63P15R68</sup>. Both arms are converged in the wavenumber integral. The control’s own movement between cutoffs is 0.000% and 0.046% on the two height ratios, and this cosmology’s arm does not move at all—0.000% and 0.000%—so the arm is converged at the lower cutoff and the figures below are not a truncation artefact. The same comparison on the fluid path reads 0.042% and 0.036% against 0.000% and 0.045%: both paths converge, and the two sets of figures are not interchangeable. A rung below that was refused outright by the instrument’s own truncation guard, on both arms, which is data and not an error.*

*On that path this cosmology’s arm returns the following, with the plasma handed over at the branch point and the background fixed by the distance data alone. The peaks fall at 222, 538, 818 and 1134 against the sky’s 220.4, 537.7, 817.3 and 1123.9; the comb fitted to them by the procedure this paper runs on the sky is 298.0 against the sky’s 298.4, and the phase intercept  $\varphi/\pi = -0.2349$  against  $-0.2405$ . The height ratios are  $P_1/P_2 = 2.264$  and  $P_1/P_3 = 2.298$  against the sky’s 2.217 and 2.277. The first three peaks land within a grid step and the fourth is 0.9% high, which is the one position residual the comb does not carry, the comb being fitted on the first three. Two wavenumber cutoffs agree to every digit, and the positions are converged in the reported multipole grid: read on a grid of step 2 and refined to step 1, they move by a single multipole at the first peak and not at all above it.*

*And the residual is neither of the two instrument faults a reader would reasonably suspect, because both are removed in this configuration: the wavenumber integral is carried past the reported multipole, and the projection carries the polarisation source rather than a damping envelope<sup>P15R70</sup>. And the truncation’s removal is measured on the arm whose answer is known: it was 78% of the control’s remaining  $\chi^2$ , and taking it out brings the control to 1.18 per degree of freedom over the full range against a reference 1.01<sup>P15R89</sup>. The diffusion damping is derived on each arm’s own opacity and rate rather than imported or fitted, and putting it back moves the height ratios by tens of per cent and drops the likelihood’s floor by a factor of five without the verdict turning over<sup>P15R69</sup>. The polarisation source pulls both arms by comparable amounts—the control by 8.2% and 20.8% on the two ratios, this arm by 10.9% and 26.9%—and nothing about either operation differs between the arms, which is what makes what survives them attributable to the source rather than to the machinery.*

*Two likelihood configurations are quoted in this section and they are not interchangeable, so which is which is stated rather than left to be inferred. The full-range lensed one, on 185 bins, is the better-converged: it is where the control reaches 1.18 per degree of freedom against a reference 1.01, and it is the configuration in which an instrument fault can be told from a physical one. The 133-bin unlensed one is where the shape rejection is quoted, and its control sits at 2.10 per bin because the ceiling cuts the damping tail. The second is not a worse measurement of the first’s quantity: it is a different configuration, chosen for the comparison it makes, and the two sit beside each other rather than one superseding the other.*

*The driving’s size on the observable leg is measured rather than inferred, and its wavenumber dependence is known in closed form—the  $k$ -dependence living in the potential’s gradient rather than in its decay, with the turnover at a fixed multiple of the horizon scale<sup>P15R65P15R66</sup>. Subtracting the driving is what measures it: removing every coupling to the potential moves the control’s first peak from 220 to 274 and  $\ell_1/\ell_A$  by 0.1792, so the modes carrying the first peak*

are driven and the shift is the driving's size<sup>P15R77</sup>. *Run on this arm the same subtraction gives 0.1717—the potential's grip within four per cent of the control's, and slightly weaker rather than stronger*<sup>P15R164</sup>. *And this one is not path-specific:* the two instrument paths return the same shift to four decimal places, where the heights do not.

*One of the two position statistics is not a projection quantity, and saying which keeps them apart.* The ratio of the first peak to the acoustic scale is  $\ell_1/\ell_A = k_1 r_s/\pi$ —the distance cancels *identically*—which is why removing each source term, the damping envelope and the visibility function in turn leaves it untouched. What it compares is the sound horizon each arm *reports* against the phase its modes actually accumulate. *Both arms integrate that horizon from  $a \rightarrow 0$ ,* the plasma being handed over at the branch point on this one and the control having no other start, so the comparison does not confound the rate with the start<sup>P15R7</sup>; and here the two agree, the comb at 298.0 against a reported scale of 302.9.

*And the likelihood is run, both arms scored on the same bins by the same code, with the amplitude fitted so that what is compared is shape. The result is still a rejection of this arm's spectrum and is stated as one.* On the full-range lensed configuration— $\ell = 100$ –1996, 185 bins, the lensing carried by the same derived operator on both arms—the control returns  $\chi^2 = 214.1$ , or 1.16 per bin, and this arm 550.5, or 2.98 per bin: *the arm sits at 2.57 times the control's distance from the data*<sup>P15R67</sup>. *That is the unfavourable of the two configurations and it is the one quoted,* the same comparison unlensed over 133 bins giving 279.4 against 556.7, a factor 1.98; lensing is worth proportionally more to a spectrum whose peaks are already in place. *The band-by-band comparison is even across the range — 1.70, 2.22, 4.23 and 9.29 per bin against the control's 0.77, 1.44, 2.62 and 3.73 over the four quarters from  $\ell = 100$  to 1300, read on the unlensed configuration — so the disagreement is not one feature carrying the whole of it. And the spectrum's own parameters are refitted, which is the comparison the standard model is usually judged by.* With the expansion rate, the matter and baryon densities and the tilt free in each arm and the amplitude in closed form, scored on the full range and verified by running the spectra at the minima, the control settles at 1.01 in  $\chi^2$  per bin and this arm at 1.58: *the arm at 1.57 times the control's distance, where the computed spectrum sits at 2.56 on the same bins—so freedom closes about a third of the gap and leaves the rest*<sup>P15R166</sup>.

*What the refit does not move is the background.* Given four free parameters this arm's expansion rate moves by 0.028% and its matter density by 0.031% from the values the baryon-acoustic distances and the acoustic angle fix with no spectrum involved: *the background the distances fix is the background the spectrum wants. And they are at a sharp minimum rather than unconstrained*—one-step excursions cost 2527, 740, 141 and 31 in  $\chi^2$  on the four parameters, which is the distinction between a parameter that does not move and one that cannot. *The one parameter that does move is the tilt:* this arm prefers 0.998 against the control's 0.954, bluer by 0.044, with the damping tail's leverage landing there and on the baryon density as the shorter range predicted it would.

*One thing the ratio does not carry, and it is stated here rather than folded in.* The two arms fit the same four parameters, so no information criterion separates them—penalties of about two per fitted parameter against a difference of some fifty in  $\chi^2$ —and the asymmetry between the frameworks is not in what was fitted but in what had to be fixed before there was a model to fit. *The standard model reaches flat  $\Lambda$ CDM by setting a curvature parameter to zero on inflation's authority, and uses the same added epoch to supply the coherent phases;* here the flat slices, the expansion law and the coherence are consequences of the construction (§2.3, §4) and there is no curvature parameter to free. *And it runs the other way too:* both frameworks inherit the composition, the amplitude and the tilt, and neither derives the spectrum it is scored on. *Two axes, and no single number crosses them* [18].

*And the phase intercept is not a residual at all, once the sky's own is located by the same procedure.* Running one locator over the binned likelihood and both arms alike, the sky's intercept is  $-0.2378 \pm 0.0099$  on the first three peaks and  $-0.2448 \pm 0.0074$  on all four; this

arm's is  $-0.2278$  and  $-0.2445$ . *So the arm sits one standard deviation from the sky on the one fit and on it in the other, and the sign of the offset flips between them — the shape of a difference carried by a single peak rather than a phase error*<sup>P15R171</sup>. *What dissolves it is the sky's own locating spread and not a correction to the arm: the binned likelihood cannot pin that intercept tighter than a hundredth, and the offset is inside it.*

*What remains is about one multipole at each peak, and it is not the damping envelope.* Imposing the control's envelope on this arm moves the fourth peak by a seventh of a multipole against an offset of two, and the inverse operation returns the same displacement with the opposite sign, so the test is sound and the answer negative. *The offsets keep their sign and order across all four peaks — the shape of a small common phase rather than one displaced peak, though not a clean constant — and at the spreads the locating itself carries, a finer statement is not available.*

*The comparison between the arms is taken as a distance between the two models rather than as a difference of two numbers each read against the sky, which is what makes it independent of the configuration's own floor*<sup>P15R92</sup>.

*The instrument's own accuracy is measured rather than assumed.* On the full range its control sits at 1.16 per bin lensed, within about a seventh of what a standard code returns on the same bins, so the ratio above is a statement about the arm and not about the machinery. *What the comb does not carry is the disagreement: at 298.0 against 298.4 the spacing is right to a part in six hundred, on both instrument paths, so what remains is about one multipole at each peak.*

*The result is reproduced independently on a second instrument.* A separate implementation—the hierarchy path rather than the line-of-sight one, 185 covered bins rather than 133, its own wavenumber grid—returns peaks at 220, 540, 820 and 1132 against this one's 222, 538, 818 and 1134, height ratios 2.273 and 2.319 against 2.264 and 2.298, and 4.02 per bin in the band from  $\ell = 700$  to 1000 against 4.23: *within a grid step on every peak and within one per cent on the heights. The fourth peak's excess reproduces there too, at 1132, so it is a property of the spectrum rather than of either implementation.*

*The heights are the polarisation path's and the positions are not.* Both paths return the same comb to four figures; the height ratios differ between them by more than either differs from the sky — 2.264 and 2.298 on the polarisation path against 2.448 and 2.934 on the fluid one — so every height quoted here is the polarisation path's and is stated as such, while the position, the phase and the alternation are the construction's.

*Where the plasma starts moves the scale and not the peak, which is why the start is not free.* Scanning a hand-placed start over a factor of 3.6 runs  $\ell_A$  from 360.6 down to 239.3 while the first peak sits between 206 and 210 throughout<sup>P15R6</sup>: it moves the denominator of  $\ell_1/\ell_A$  and leaves the numerator where it is<sup>P15R24</sup>, so the spacing such a start reports is not the spacing its spectrum carries.

*What that leaves is the equality, and it is measured rather than supposed.* Holding everything else fixed and moving matter–radiation equality alone carries the third peak and the band it sits in with it: the height ratio  $P_1/P_3$  tracks equality on both routes to it — through the radiation density and through the matter density, which move the projection in opposite directions — agreeing to within a per cent across a 12 to 24% change in the matter density, so what the band responds to is the equality and not the route. *And the epoch the distance data fix is what sets it: equality is  $\omega_m/\omega_r$  with the radiation density measured, so the background that meets the baryon-acoustic distances is the same background that drives the peaks.*

*Two things this section does not claim, and they are separated because a reader would otherwise supply them.* The alternation figures are read on a multipole grid whose step is the same size as the sky's own gap contraction is large, so *a contraction of the sky's size is resolved while the smaller ones at lower  $\ell$  are not*<sup>P15R78</sup>; a finer grid is what would settle those. And the equality result accounts for the band it moves and not for the phase residual: *what the phase is*

to be read against is the driving subtraction on this arm, and §10 carries it as such.

## 5 What the substrate determines: the amplitude floor and the classical character

One might ask whether the observed fluctuation amplitude is the substrate’s own de Sitter vacuum—the natural CR analogue of the inflationary spectrum-is-the-stretched-vacuum story. It is not, by a decisive margin.

**Proposition 2** (The substrate vacuum is negligible against the observed amplitude). *The substrate curvature scale is the cosmological-constant scale: in Planck units  $\Lambda \ell_P^2 \approx 3 \times 10^{-122}$ , and the de Sitter vacuum metric-fluctuation power is of the same order,  $(H_\Lambda/M_{\text{Pl}})^2 \sim 10^{-122}$ . Against the observed  $A_s \approx 2 \times 10^{-9}$  this is smaller by  $\sim 10^{113}$ .*

*Argument.*  $\Lambda = 3(H_0/c)^2 \Omega_\Lambda$  gives  $\Lambda \ell_P^2$  directly;  $H_\Lambda = c\sqrt{\Lambda/3}$  and  $(\hbar H_\Lambda/E_P)^2$  give the vacuum power (receipts: `verify_numeric.py`, anchor 5<sup>P15R17</sup>). The prefactor ( $\sim 10^{-122}$  versus  $\sim 10^{-123}$ ) is a  $2\pi$  and reduced-versus-full Planck-mass convention; the conclusion is robust to it.  $\square$

*One scope qualification is owed immediately, and closing it is what makes the classical reading a result rather than an inference.* Proposition 2 rules out the *substrate’s* vacuum. But the reading it supports locates the fluctuations in the *progenitor*, and the progenitor’s own vacuum is a different quantity: it is normalised at the collapse leg’s curvature scale, not at  $H_\Lambda$ , and it is amplified on its way here by the branch point’s mode mixing. Neither effect is small, and neither is addressed above. Both can be computed, because the collapse-side background is a single function: with  $A = 2M$  and  $\rho = 2\sqrt{B}/A$  the exact closed dust-plus-radiation interior gives  $a = M(\rho|\sigma| + \sigma^2/2)$  near the crunch, whose Mukhanov–Sasaki potential  $z''/z = 2/(|\sigma|(|\sigma| + 2\rho))$  is *independent of the mass*—so  $M$  enters only through the final division  $\zeta = v/a$ . The passage then has exactly three stages, and each contributes one factor of  $1/\rho$ . A mode leaves the sound horizon on the  $\beta = 2$  leg and freezes as  $v \rightarrow D_k/|\sigma|$  with the scale-invariant  $D_k = \sqrt{\hbar/2} k^{-3/2}$ ; the connection through equality is elementary, since  $v = a \int d\sigma/a^2$  solves the  $k = 0$  problem in closed form and carries  $D_k/|\sigma|$  to the constant  $(3/2\rho)D_k$  at the crunch; the crunch monodromy then leaks that constant into the survivor with coefficient  $4\pi/\rho$ —twice the tensor value, because the true hydrodynamical  $z_S = a(a + \frac{4B}{3A})/a'$  has  $z_S''/z_S \rightarrow 2/(\rho x)$  where  $a''/a \rightarrow 1/(\rho x)$ —the companion paper’s numerically obtained  $-2\pi i/\rho$  [19] being the *tensor* value of the same object, recovered here in closed form as the continuation of the  $(\sigma/\rho) \ln \sigma$  term the  $(0, 1)$  resonance forces; and the survivor’s curvature is  $\zeta = C/(M\rho)$ , the last  $1/\rho$  being the crunch’s own slope  $a'(\eta_c) = -M\rho$ . The result is a transfer law with no free constant,

$$\mathcal{P}_\zeta = \frac{18 k^3 D_k^2}{M^2 \rho^6}, \quad (8)$$

which is *exactly* scale-invariant on vacuum data—the  $k^3$  cancels the  $k^{-3}$  identically—and which therefore delivers, for the vacuum,  $A_s = 9(\ell_P/M)^2 \rho^{-6}$  (receipt: `P15_the_progenitor_vacuum_is_negligible_t` the connection coefficient confirmed against integration from vacuum data to 0.1% over a decade in  $\rho$ ). *One check is owed here and is not optional, because the hydrodynamical variable differs from  $a$  in three places and not one:  $z_S \propto a^{3/2}$  in the matter era, so the exponents there are  $(-2, 3)$  against  $a$ ’s  $(-1, 2)$ ; the crunch residue doubles; and the final division is by  $z_S$  rather than by  $a$ . Propagating the same physical initial condition under each potential returns a ratio of exactly 2 in amplitude at every composition tested—the matter-era exponents and the change of divisor cancel, and the crunch residue alone survives. So the  $z \propto a$  chain is corrected by one factor rather than rebuilt, and Eq. (8) is the true-variable law.*

**Proposition 3** (The progenitor vacuum is negligible too). *With the mass capped at the recollapse threshold  $M \leq \alpha/3\sqrt{3}$  and the radiation fraction at the largest value the observed spectrum permits, Eq. (8) on vacuum data returns  $A_s \sim 1 \times 10^{-112}$ , short of the observed  $2 \times 10^{-9}$  by  $\sim 10^{103}$ .*

*Argument.*  $(\ell_P/M)^2 \geq 27(\ell_P/\alpha)^2 \approx 2.6 \times 10^{-121}$  at the cap, and the companion paper determines  $\rho \simeq 5.4 \times 10^{-2}$  [19];  $9 \times 2.6 \times 10^{-121} \times \rho^{-6}$  then gives the figure. Running it backwards, the vacuum would reach the observed amplitude only at  $\rho \sim 3 \times 10^{-19}$ , thirteen orders below what nucleosynthesis on the same leg permits.  $\square$

*So eighteen orders separate the substrate's vacuum from the progenitor's amplified one—the mixing buying  $10^{8.6}$  over the progenitor's own bare vacuum, which itself sits  $10^{10}$  above the substrate's—and it is still short by a hundred and three, and the classical character of the primordial statistics is established for the source that actually supplies them rather than only for the one that does not. Two consequences of *character*, not of number, follow.*

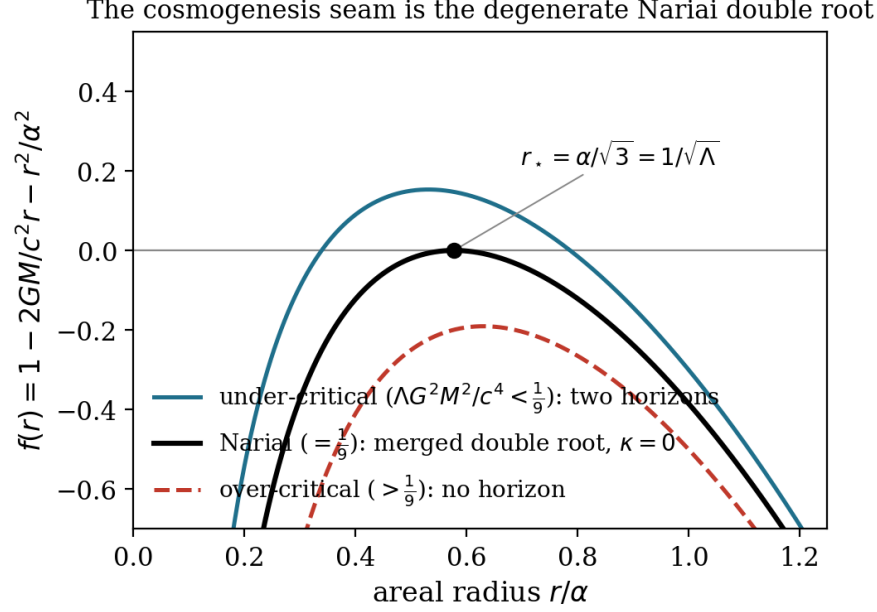
First, the observed amplitude is inherited classical content—boundary data from the progenitor, not a quantum fluctuation generated within the Schwarzschild–de Sitter era, nor one generated in the progenitor's collapse.

Second, there is no inflationary consistency relation:  $r = -8n_t$  presupposes that scalar and tensor spectra are one vacuum stretched by one expansion, and with the amplitude not vacuum-sourced that link is absent; the substrate tensor floor is the same  $\sim 10^{-122}$ , so there are no substrate-sourced primordial  $B$ -modes, and any observed tensors would themselves be inherited content rather than a measure of a substrate inflationary scale. *And the tensor statement can be made unconditionally, which the scalar one cannot.* The scalar law, Eq. (8), is the true-variable law only through the hydrodynamical  $z_S$  of the progenitor's own fluid, which carries that fluid's equation of state and sound speed—and across the progenitor's own equality neither  $w$  nor  $c_s$  is constant.

*For tensors that restriction is absent:*  $z_T = aM_{\text{Pl}}/2$  exactly, for any content, any equation of state and any sound speed, so  $z_T''/z_T = a''/a$  identically and the entire passage—background, connection, monodromy—applies verbatim with nothing assumed. It returns

$$\mathcal{P}_T = 144\pi (\ell_P/M)^2 \rho^{-6} \simeq 5 \times 10^{-111}, \quad (9)$$

evaluated at the recollapse cap and the largest composition the spectrum permits, against an observational ceiling of  $\sim 7 \times 10^{-11}$ : *below it by a hundred orders* (receipt: `P15_no_primordial_B_modes_unconditi`). *So the absence of primordial  $B$ -modes is a prediction of this construction at any conceivable sensitivity, and it does not rest on the substrate's own floor—the floor argument, like its scalar twin, ruled out the substrate's vacuum and said nothing about the progenitor's.* What the crossing does to the ratio is likewise fixed and free of  $k$ : every factor of the passage is shared by the two sectors, so  $r_{\text{out}} = 6r_{\text{in}}$ : the passage alone would carry the ratio by 24, and the scalar mixing being twice the tensor's divides that by four in the power, and the observed bound becomes a statement about the parent, namely that its own tensor-to-scalar ratio was below  $5 \times 10^{-3}$ . *An observed tensor signal would therefore not falsify the construction but measure the previous generation, which is the same reading the amplitude receives.* The substrate fixes that the fluctuations are classical, inherited, and not governed by a vacuum consistency relation, and leaves their size to the handover. *What it does fix is the multiplier.* Eq. (8) holds for any incoming growing-mode amplitude, quantum or classical, so the crossing determines how much an amplitude is magnified without determining the amplitude—which is this construction's one-constant character read at the level of the perturbations rather than of the couplings. Two corollaries follow and are worth stating because each closes a natural hope in one direction and opens a statement in the other.  *$A_s$  does not measure the progenitor's mass:* with  $D_k$  free, the relation cannot be inverted, and the mass must be reached, if at all, by a route that does not run through the amplitude. *And the classical datum has a size:* since  $A_s \propto D_k^2$ , the parent's growing-mode



**Figure 3:** The Schwarzschild–de Sitter metric function  $f(r) = 1 - 2GM/c^2 r - r^2/\alpha^2$  across the three regimes. Under-critical masses ( $\Lambda G^2 M^2/c^4 < \frac{1}{9}$ ) give two distinct positive horizons; over-critical masses give none. The cosmogenesis branch point is carried on the *Narai* member ( $= \frac{1}{9}$ ), where the two horizons merge into a single *degenerate* double root at  $r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$  with  $f(r_N) = f'(r_N) = 0$  and hence vanishing surface gravity  $\kappa = 0$ . The degeneracy of this root is what the throat geometry (Proposition 4) and the transmission dichotomy (§8) turn on.

amplitude must have exceeded its own vacuum’s by  $\sim 10^{47}$  at the top of the permitted composition window—so this cosmology does not merely decline to predict  $A_s$ , it says the previous generation’s perturbations were enormously super-vacuum, which is a claim about the parent and not about us.

## 6 What the substrate determines: the throat and isotropization

At the branch point the near-horizon geometry is fixed, and what the perturbations cross is fixed with it.

**Proposition 4** (Equal-radii  $dS_2 \times S^2$  throat). *At the Narai locus  $\Lambda G^2 M^2/c^4 = 1/9$  the two positive horizon roots of  $f(r) = 1 - 2GM/c^2 r - r^2/\alpha^2$ ,  $\alpha = \sqrt{3/\Lambda}$ , merge at the Narai radius  $r_N = \alpha/\sqrt{3} = 1/\sqrt{\Lambda}$  (the symbol is the algebroid paper’s [3]; it is written  $r_N$  and not  $r_*$  because  $r_*$  is the tortoise coordinate used below), where  $f(r_N) = f'(r_N) = 0$  and  $f''(r_N) = -2\Lambda$  (the cosmology selects this  $M > 0$  member of the parity-conjugate Narai pair; the  $M < 0$  member has instead one positive and two negative roots, the two exchanged by  $r \mapsto -r$  [5, 6]). The near-horizon geometry is  $dS_2 \times S^2$  with both curvature radii equal to  $1/\sqrt{\Lambda}$ : the  $S^2$  has two-Ricci  $R_2 = 2/r_N^2 = 2\Lambda$ , and the  $dS_2$  has radius<sup>2</sup>  $= -2/f''(r_N) = 1/\Lambda$ .*

*Argument.* Direct from  $f$  at the Narai mass (receipts: `verify_geometry.py`, anchor 2<sup>P15R16</sup>; this is the Ginsparg–Perry near-horizon limit [29]).  $\square$

A field on  $dS_2 \times S^2$  (Fig. 3) decomposes in  $S^2$  harmonics; the degree- $\ell$  harmonic becomes a  $dS_2$  field of effective mass  $m^2 = \ell(\ell + 1)/r_N^2$ , i.e.  $m^2/H_{dS_2}^2 = \ell(\ell + 1)$ . With the  $dS_2$  index  $\nu^2 = \frac{1}{4} - m^2/H^2$ , the monopole  $\ell = 0$  has  $\nu = \frac{1}{2}$ —a scale-invariant base—while every  $\ell \geq 1$  has  $\ell(\ell + 1) \geq 2$ , hence  $\nu^2 < 0$ : the heavy principal series, which oscillate and decay through the throat. We read this as an angular no-hair, and the reading is grounded rather than asserted:

the dS<sub>2</sub> factor of the throat is a positive- $\Lambda$  epoch, and the de Sitter no-hair theorem [21] damps exactly the anisotropic content—every  $\ell \geq 1$  harmonic is heavy principal-series, oscillating and decaying, while only the isotropic monopole  $\ell = 0$  survives as the  $\nu = \frac{1}{2}$  base—so the throat isotropizes what passes through it (receipts: `verify_throat_tower.py`<sup>P15R40</sup>). The geometry is established (Proposition 4) and the mode tower follows from it; the no-hair reading is the standard theorem read on the throat, and we flag one guard as load-bearing: the index  $\ell$  of this throat tower is the  $S^2$ -harmonic degree of the *near-horizon* geometry at areal radius  $1/\sqrt{\Lambda}$ , and is *not* the observable microwave-background multipole, which lives on the last-scattering sphere of radius  $D_C$ . The map between them is §7, not an identity; conflating them would manufacture a prediction the geometry does not make. *A second guard belongs beside it, and it bounds what the tower damps.* The computation above is a statement about propagating field modes—degrees of freedom carrying a mass on the dS<sub>2</sub> factor, which is what being heavy principal-series means. *The low multipoles of a perturbation about a spherically symmetric background are not such modes:*  $\ell = 0$  carries the mass and the axial  $\ell = 1$  carries the angular momentum, the latter a stationary slow-rotation configuration rather than an oscillation, with the radiative content beginning at  $\ell \geq 2$ —the multipole face of the non-radiative skeleton the range paper describes [8]. *So what the throat isotropises is the radiative content, and a conserved charge sitting in a constrained sector is not among the modes this argument is about*<sup>P15R8</sup>. *This is a bound on the claim and not a claim in the other direction:* that the angular momentum is not removed *by this mechanism* says nothing about whether it is carried off by matter or torqued on the way, which this paper does not settle. *And the geometry the tower is computed on is itself a non-rotating one,* Proposition 4 following directly from  $f$  at the Nariai mass. *So it is worth measuring how much the conclusion rests on that.* Writing  $\lambda$  for the ratio of the dS<sub>2</sub> curvature radius squared to the  $S^2$  one, the index generalises to  $\nu^2 = \frac{1}{4} - \lambda\ell(\ell+1)$ , with  $\lambda = 1$  the equal-radii case above. *The monopole is untouched:*  $\nu^2 = \frac{1}{4}$  for every  $\lambda$ , so the scale-invariant base does not depend on the radii being equal. *And the damping persists well away from it:* every  $\ell \geq 1$  remains heavy principal-series for  $\lambda > 1/4\ell(\ell+1)$ , which at  $\ell = 1$  is  $\lambda > 1/8$ —the dS<sub>2</sub> radius would have to fall below 0.354 of the  $S^2$  radius, a factor near three, before the dipole ceased to decay, with the higher multipoles safer still<sup>P15R9</sup>. *What ratio a rotating progenitor would produce is not computed here,* and a rotating near-horizon geometry is generally warped rather than a direct product, so a single ratio may not describe it; what the margin establishes is that the conclusion does not turn on the equal-radii point itself. *The ratio can in fact be computed on the rotating family, and it runs the helpful way.* Kerr–de Sitter’s radial function  $\Delta_r = (r^2 + a^2)(1 - r^2/\alpha^2) - 2Mr$  reduces at  $a = 0$  to  $r^2 f(r)$ , and its degenerate locus  $\Delta_r = \Delta'_r = 0$  gives  $a^2 = r^2(\alpha^2 - 3r^2)/(\alpha^2 + r^2)$ , which returns  $r_N = \alpha/\sqrt{3}$  and the Nariai mass at  $a = 0$  rather than presupposing them. *The family is bounded:*  $a$  reaches  $(2 - \sqrt{3})\alpha$  at  $r/\alpha = 0.3933$ , where  $\Delta''_r$  vanishes and the double root becomes triple—the ultracold member. *Along it the ratio begins at unity and rises:*  $\lambda = 1$  exactly in the static case and  $\lambda > 1$  monotonically with  $a$ , so rotation carries the throat away from the  $1/8$  threshold rather than toward it<sup>P15R10</sup>. *That ratio is read at the pole, where the sphere’s radius is  $r_N$ ;* the rotating near-horizon sphere is warped through  $r_N^2 + a^2 \cos^2 \theta$  and a single ratio does not describe it, so what the calculation establishes is the direction and its size at one well-defined point, with a factor of eight in hand at the outset.

*The warped problem is done directly, and does not need the proxy.* The near-horizon limit of the degenerate family, taken as a limit rather than posited, is a warped product and not a direct one:  $\rho_0^2 = r_0^2 + a^2 \cos^2 \theta$  multiplies the dS<sub>2</sub> and polar directions alike, and the dS<sub>2</sub> is fibred over the sphere by  $\tilde{k} = 2r_0 a \Xi / (r_0^2 + a^2)$ , which vanishes only at  $a = 0$ . A massless field still separates on it, the warp cancelling from the relation between the transverse eigenvalue and the dS<sub>2</sub> operator because it multiplies both. *For the axisymmetric modes it cancels altogether:* the transverse problem is  $(\Delta_\theta \sin \theta Y')' + E \sin \theta Y = 0$  with  $\Delta_\theta = 1 + (a/\alpha)^2 \cos^2 \theta$ —a spheroidal deformation of Legendre in which  $\rho_0$  does not appear at all, so the harmonic label survives the warping rather

than being spoiled by it. *The decay condition generalises to  $\nu^2 = \frac{1}{4} - L^2 E - L^4 m^2 \tilde{k}^2$ , the last term the rotation's own fibration, which is non-negative and so can only deepen the damping. Two exact bounds then settle the family without a scan:  $L^2 \geq 1$  on the physical branch and  $E \geq 2$  above the monopole, each with equality only at  $a = 0$ , so  $L^2 E \geq 2$  against the threshold  $\frac{1}{4}$ . The monopole keeps  $\nu^2 = \frac{1}{4}$  exactly on the warped sphere too. So the conclusion holds across the whole family  $0 \leq a < a_{\max}$  with the same factor of eight, and the marginal member is the non-rotating one—rotation strictly improves the case rather than threatening it<sup>P15R11</sup>.*

## 7 Large scales: the flat/discrete decoupling and the low-multipole floor

Here the construction makes its sharpest large-scale statement, and the load-bearing piece is established.

**Proposition 5** (The fundamental-observer slice is flat). *In the fundamental-observer frame (2), a surface of constant  $\tau$  has induced three-metric  $dr^2 + r^2 d\Omega^2$  (since  $(\partial_\chi r)^2 d\chi^2 = dr^2$  along the slice), whose Riemann tensor vanishes identically. The constant- $\tau$  slice is exactly flat  $\mathbb{R}^3$ .*

*Argument.* The substitution is immediate from the line element; the curvature of  $dr^2 + r^2 d\Omega^2$  is zero by direct computation (receipts: `verify_geometry.py`, anchor 1<sup>P15R16</sup>). This is the framework paper's own stated property that constant- $\tau$  slices are Euclidean.  $\square$

The decoupling follows. The distance slicing is the flat constant- $\tau$  one— $\Omega_k = 0$ , no curvature term in the redshift–distance relation—while the *cosmological* layers are the closed  $S^3$  of constant  $\tilde{\tau} = \tau + \chi$  (§2.2). The non-synchrony  $\tilde{\tau} = \tau + \chi$  is the decoupling itself: flat distances and a closed  $S^3$  of comoving worldlines coexist because they are different slicings of one geometry. A literal closed-Friedmann reading would put  $\Omega_k = -\Omega_\Lambda \approx -0.685$  into the distance relation and be excluded; CR avoids that not by tuning but because its distance slicing is the flat one, the curvature living on the orthogonal layers. Spatial curvature and dark energy are one  $\Lambda$  read on two slicings—the corpus identity, here at the observational rung.

**The low-multipole floor.** The closed  $S^3$  carries a discrete mode spectrum, labelled by integer degree  $L$ , with comoving wavenumber  $k_L = \sqrt{L(L+2)}/r_0$  and the lowest physical mode the quadrupole  $L = 2$  (the monopole  $L = 0$  is the background, the dipole  $L = 1$  pure gauge). The decoupling fixes how these project to the sky, and the fixing is the whole point. Because the distance slicing is flat (Proposition 5), the photons are projected through the *flat* geometry—the comoving angular-diameter distance is  $D_M = D_C$ , the same flat- $\Lambda$ CDM observable that places the acoustic scale at  $\ell_A \approx 301$ —while only the *source* modes carry the closed- $S^3$  quantization. The transfer is therefore the discrete closed- $S^3$  spectrum projected through the flat spherical Bessel  $j_\ell(k_L D_C)$ , *not* the hyperspherical transfer of a literal closed universe: that transfer carries the closed *distance* relation, which CR does not have, and would (wrongly, here) deliver the lowest mode to the quadrupole and no deficit. The flat projection places the discrete modes at

$$\ell_L \simeq \sqrt{L(L+2)} \frac{D_C}{r_0}, \quad (10)$$

with  $r_0$  the present  $S^3$  areal radius and  $D_C$  the comoving distance to last scattering, both fixed without new parameters:  $D_C \approx 13927$  Mpc is the flat- $\Lambda$ CDM observable and  $r_0 \approx 5064$  Mpc is the Nariai amplitude  $2^{1/3}/\sqrt{\Lambda}$  at the present epoch  $u = \operatorname{arcsinh} \sqrt{\Omega_\Lambda/\Omega_m} \approx 1.18$ . The stretch factor  $D_C/r_0 \approx 2.75$ —the ratio of the flat projection distance to the curvature radius, the decoupling made numerical—carries the lowest mode to  $\ell_2 \approx 7.8$ , with no source below it.

The full flat-projection transfer confirms the estimate and sharpens it into a prediction. Carrying the discrete spectrum through  $j_\ell(k_L D_C)$  with the scale-invariant measure—the first-principles closed- $S^3$  Harrison–Zel’dovich weight [31, 32]  $w_L = (L+1)/(L(L+2))$ , fixed as the



**Figure 4:** The low-multipole discreteness floor. The closed- $S^3$  source spectrum is discrete ( $k_L = \sqrt{L(L+2)}/r_0$ , lowest physical mode  $L = 2$ ), but the flat distance slicing (Proposition 5) projects it through the *flat* transfer  $j_\ell(k_L D_C)$ , not the hyperspherical transfer of a literal closed universe. With  $D_C \approx 13927$  Mpc and  $r_0 \approx 5064$  Mpc fixed parameter-free by  $\Lambda$ , the stretch  $D_C/r_0 \approx 2.75$  carries the lowest mode to  $\ell_2 \approx 7.8$ , with no source below it. The full flat-projection transfer turns this into a low-multipole power deficit ( $\ell \lesssim 8$ ); on a genuine Boltzmann transfer (the late integrated Sachs–Wolfe sourced above the floor and so retained by the discrete source) it sits at 0.473, 0.410, 0.356, 0.676 of the flat- $\Lambda$ CDM expectation at  $\ell = 2, 3, 4, 5$ , recovering by  $\ell \approx 8$ —a deficit whose minimum falls at  $\ell = 4$ , so that the quadrupole and octopole are its shoulders and not its floor. Those figures are the first arm, read through CAMB’s exact  $\Delta_\ell(k)$ ; the photon hierarchy built for this programme (§9.1) is the second arm and returns 0.49, 0.24, 0.18, 0.61, agreeing on the shape and not on the depths (§7); the two are close at the quadrupole and nearly a factor two apart at  $\ell = 3$  and 4, with the same minimum at  $\ell = 4$  and the same recovery. The shape is cross-validated between the two; the depth is not. Either way this is a mild deficit, non-discriminating against the observed large-angle power within the lowest multipoles’ cosmic variance (shaded). The location is geometric and robust; the depth is whatever the full Boltzmann transfer returns, and the two run here differ on it. What the branch-point filter does to this prediction at these multipoles—and what it does not do—is taken up in the text below rather than here.

mode degeneracy  $\beta^2$  (with  $\beta = L+1$ ) times the per-mode power  $1/(\beta(\beta^2 - 1))$  that equal power per logarithmic interval in  $k$  requires, and reducing to the continuum  $dk/k$  at large  $L$  so that  $\ell(\ell+1)C_\ell$  returns to the flat plateau as  $D_C/r_0 \rightarrow 0$  (receipt: `verify_lowell_exact_measure.py`<sup>P15R29</sup>)—yields a low-multipole deficit:  $\ell(\ell+1)C_\ell$  falls below its high- $\ell$  plateau for  $\ell \lesssim 7$  and recovers by  $\ell \approx 8$ , because the flat projection of a discrete source with a hard lowest mode at  $k_2$  leaves the lowest multipoles starved (receipts: `verify_closedS3_nonsync.py`<sup>P15R28</sup>).

The one effect that partly fills it is the late integrated Sachs–Wolfe term [28], sourced near the observer where small line-of-sight distances send wavenumbers  $k \gtrsim k_2$ —above the floor—down to  $\ell \sim 2$ .

By the decomposition of the empirical-forcing paper [11] the cumulative term is the standard Sachs–Wolfe and integrated Sachs–Wolfe—CR’s own differential-expansion floor being zero under uniform expansion—so this late boost rides the same modes the discrete source keeps, and the deficit is milder than the bare Sachs–Wolfe starvation alone.

The depth is a matter for a *genuine Boltzmann transfer* rather than for the geometry, and the reason is that the deficit is two effects and not one. In the pure Sachs–Wolfe limit the suppression splits exactly into a *floor*—the projection integral truncated below  $k_2$ —and a

*ladder coarseness*, the discrete sum minus that same truncated integral. At  $\ell = 5$  and 6 the two are comparable and pull opposite ways: the ladder puts back power a pure cut-off removes. And they respond to the fitted parameter with opposite signs—under  $\pm 2\%$  in  $r_0$  the floor moves by a few per cent while the full result moves by tens—so the location is the floor and is robust for that reason, and the depth is the ladder and is volatile for that reason<sup>12</sup>. *The recovery multipole is the ladder's too*: with it the spectrum recovers by  $\ell \approx 8$ , on the floor alone not until  $\ell \approx 10$ . Because the only CR modifications are the discrete source and the flat projection—every transfer ingredient (Sachs–Wolfe, early and late integrated Sachs–Wolfe, Doppler) shared with flat- $\Lambda$ CDM [11]—the exact CMB temperature transfer  $\Delta_\ell(k)$  of a standard Boltzmann code is CR's transfer, and the discrete spectrum is read off it:  $C_\ell^{\text{CR}} = \sum_{L \geq 2} w_L \Delta_\ell(k_L)^2$  against the continuum  $\int d \ln k \Delta_\ell(k)^2$ , the measure  $w_L$  being exactly  $d \ln k_L / dL$  so the CR sum is the  $L=2$ -floored Riemann sum of the very integral  $\Lambda$ CDM does continuously.

Carried out with the exact transfer (gate-validated: the continuum reproduces the code's own  $C_\ell$  to four figures; and the  $r_0$  sensitivity is measured rather than assumed: the *shape* is stable—the minimum stays at  $\ell = 4$  and the recovery multipole moves by 0.1%—while the *depths* drift by up to 15% at  $\ell = 4$  under  $\pm 2\%$  in  $r_0$ ), the deficit is *milder* than a Sachs–Wolfe-only estimate suggests, and it has a shape:  $\ell(\ell+1)C_\ell$  sits at  $\approx 0.47, 0.41, 0.36$  and  $0.68$  of the flat- $\Lambda$ CDM expectation at  $\ell = 2, 3, 4, 5$ , recovering by  $\ell \approx 8$  (receipt: `verify_lowell_boltzmann.py`<sup>P15R31</sup>). *The prediction is therefore not a monotone recovery from the quadrupole but a dip whose minimum falls at  $\ell = 4$* <sup>P15R34</sup>—and the quadrupole and octopole, the two multipoles a large-angle discussion reaches for first, are its two *shallowest* points.

The shape is not one code's, and that was checked here by running it rather than by comparing tables<sup>P15R36</sup>.

Driving the photon hierarchy built for this programme (§9.1) through the same discrete projection puts the minimum at the same multipole,  $\ell = 4$ , and recovers at the same place,  $\ell \approx 7$ –8. *What two independent Boltzmann treatments cross-validate is the shape, and only the shape*. The depths do not agree: the second arm returns 0.49, 0.24, 0.18 and 0.61 against the first's 0.473, 0.410, 0.356 and 0.676—close at the quadrupole and nearly a factor two apart at  $\ell = 3$  and 4. The two treatments differ in their late-time integrated Sachs–Wolfe handling and in  $D_M$ , and this paper does not reconcile them: it carries one converged depth table, the first arm's, and a second arm that agrees on where the deficit sits and not on how deep it is.

The reason is the late integrated Sachs–Wolfe: it is sourced near the observer at wavenumbers  $k \gtrsim k_2$  (small line-of-sight distances), *above* the floor, so the discrete source *retains* it rather than losing it, and the deficit is milder than a Sachs–Wolfe-only treatment of the same source would give.

This is a parameter-free deficit, set by  $\Lambda$  through  $r_0$  and  $D_C$ , and what is robust is its existence, its location ( $\ell \lesssim 8$ , geometric, set by  $k_2 D_C$ ), its minimum at  $\ell = 4$ , and that the single floor  $k_2$  starves  $\ell = 2$  and  $\ell = 3$  *together*: the exact discrete measure and the Doppler term (super-horizon velocity through  $j'_\ell$ ) each leave the  $\ell = 2$ -to- $\ell = 3$  ratio near unity, if anything suppressing the octopole slightly more (receipts: `verify_lowell_exact_measure.py`<sup>P15R29</sup>, `verify_doppler_lo`). Whether this predicted deficit matches the observed sky is the data test, against the large cosmic variance of the lowest multipoles. Confronted with the measured low multipoles, the verdict is a *wash*. CR predicts a smooth modest deficit bottoming at  $\ell = 4$ ; the observed sky has a sharp quadrupole-only dip ( $\sim 0.2$ ) with a near- $\Lambda$ CDM octopole ( $\sim 0.6$ –1.0), so CR sits *above* the observed quadrupole and *below* the observed octopole, matching neither sharply. Under the exact cosmic-variance likelihood the net over  $2 \leq \ell \leq 10$  is  $\Delta(-2 \ln L) \approx +1.8$  (range +0.3 to +3.8 across the octopole estimator), a small residual well inside cosmic variance (receipts: `confront_lowell_data.py`<sup>P15R32</sup>, `verify_lowell_likelihood_v2.py`<sup>P15R33</sup>). *Broken up by multipole, the sector's weight does not lie where its name suggests*: the quadrupole *rewards* this cosmology (−2.6, its milder dip nearer the observed low power than flat  $\Lambda$ CDM's), the octopole penalises it at +1.3, and the largest single-multipole penalty is the +2.1 at  $\ell = 4$ —

the prediction’s own minimum, and the one multipole of the four that the large-angle literature does not treat as anomalous. The low multipoles are thus a genuine but *non-discriminating* sector: neither a parameter-free correspondence success nor a sharp falsification edge, but a mild deficit consistent with  $\Lambda$ CDM within the lowest multipoles’ cosmic variance.

*One further question belongs with this prediction, and it is a located one.* The branch-point filter that sets what crosses into the expansion leg is, in its quantum reading, a Euclidean kernel whose action on a mode is controlled by an adiabaticity parameter  $C/\mu_n$ , with  $\mu_n$  the harmonic index of the layer’s  $S^3$  tower [35]. That parameter is small for all but the lowest harmonics and reaches order unity only at  $n = 2$  and  $n = 3$ , where it takes the values 0.70 and 0.47. *Three independent routes put the approximations at their limit on the same scale:* the transmission calculation’s boundary at  $\ell \approx 2.5$ , the projection’s incompleteness below  $k \sim 2 \times 10^{-4} \text{ Mpc}^{-1}$  ( $\ell \approx 3$ ), and this parameter’s reaching order unity at  $n = 2, 3$ . Whether that is a loss of control is decided by computing the filter rather than estimating it.

*The transmitted amplitude at low multipoles can be computed rather than estimated, and doing so settles what the estimates left open.* The branch-point filter’s action on a mode is, in the quantum reading, a Euclidean kernel [35]. The exponential-of-an-integral form  $e^{-\int \omega d\eta}$  is a WKB approximation whose adiabaticity parameter is of order unity at  $\ell \lesssim 5$ ; but the underlying mode equation  $d^2\psi/d\eta^2 = \omega^2\psi$ , with  $\omega = kc_s$  and  $d\eta = ds/a$  along the segment, is well posed at every  $k$ , and integrating it is a matter of numerics rather than of approximation. Composing exact constant- $\omega$  transfer matrices in the stable direction over the segment ( $|\Delta\eta| = 3.32$  in units  $\alpha = 1$ ) gives exact/WKB ratios of 0.926, 0.913, 0.901, 0.891 and 0.889 at  $\ell = 2, 3, 5, 15, 40$ : *the WKB form is accurate to 7–11% at every multipole from 2 to 40*, with the exact transmission slightly smaller, so the filter is marginally stronger than the approximation gives.

*Two readings suggested by the adiabaticity parameter alone are thereby excluded.* The first is that the treatment is uncontrolled at  $\ell = 2$ –5: it is not, and the parameter was the wrong figure of merit, measuring as it does the fractional error in an exponent that is itself small at low  $\ell$ . *The residual’s own behaviour settles this independently of any argument about the parameter:* the ratio *decreases* monotonically with  $\ell$ —0.926, 0.913, 0.901, 0.891, 0.889—so the discrepancy is smallest at  $\ell = 2$ , where the adiabaticity parameter is of order unity, and largest at  $\ell = 40$ , where it is small. An error of adiabatic origin runs the other way, WKB being an expansion *in* that parameter, so such an error is largest where the parameter is  $O(1)$  and vanishes as it does. *What the numbers show is therefore a systematic offset rather than an adiabatic breakdown*—the WKB form is uniformly slightly generous and the exact filter uniformly slightly stronger, which is a correction of known sign and bounded size and not a loss of control<sup>L19</sup>. The second concerns the comparison with the observed sky. The correction is very nearly  $\ell$ -independent—varying by 4% across a factor of twenty in  $\ell$ —so between  $\ell = 2$  and  $\ell = 3$  it alters their ratio by 1.4%. *The structural expectation that a single floor starves  $\ell = 2$  and  $\ell = 3$  together is therefore not relieved by the exact treatment*, and the observed sky’s selective suppression of  $\ell = 2$  alone, by a factor of order three, remains unaccounted for. The low-multipole prediction of this section therefore stands as stated, and so does the discrepancy it faces.

*Finally, the two Boltzmann depths above are not rival claims and neither supersedes the other:* they differ by up to a factor of two at  $\ell = 3$  and 4 and agree on the shape, the minimum and the recovery, and the difference is the size of the residual §9.1 declares. What both supersede is the Sachs–Wolfe-analytic estimate (0.22 and 0.20 at  $\ell = 2, 3$ ), which takes the Sachs–Wolfe source as the matter-domination value  $\frac{1}{3}\Phi$ , whereas at recombination  $\rho_r/\rho_m = 0.317$  and the source is  $0.4006\Phi$ —a 20% error in the very ratio of Sachs–Wolfe to integrated Sachs–Wolfe that fixes how much the latter fills the bare deficit.

## 8 What the progenitor supplies: the transmission proof

The decomposition’s decisive proof is here: *why* the substrate transmits the spectral shape rather than imprinting one of its own. It turns on how a mode approaches the locus it meets. Two horizon cases bound the question: a non-degenerate horizon, and the degenerate *Nariai double root*—the Nariai seam, at  $r = +\alpha/\sqrt{3} \equiv -2\alpha/\sqrt{3}$ , where  $f$  and  $f'$  vanish together so that  $\kappa = 0$ . *The locus the modes cross is neither*: the branch point sits at  $r = 0$ , carries no surface gravity because  $f$  diverges there rather than vanishing, and is not a Killing horizon at all.

**Proposition 6** (Transmission dichotomy). *For a mode approaching a horizon where the metric function behaves as  $f \sim (r - r_h)^p$ , the tortoise coordinate  $r_* = \int dr/f$  controls the approach. At a non-degenerate horizon ( $p = 1$ ,  $f \sim 2\kappa(r - r_h)$ , surface gravity  $\kappa > 0$ ) the integral is logarithmic and the approach exponential,  $(r - r_h) \sim e^{2\kappa r_*}$ . At the degenerate Nariai double root ( $p = 2$ ,  $f \sim -\Lambda(r - r_N)^2$ ,  $\kappa = 0$ ) the integral is  $r_* \sim +1/[\Lambda(r - r_N)]$  and the approach power-law,  $(r - r_N) \sim +1/(\Lambda r_*)$ .*

*Argument.* The two tortoise integrals directly (receipts: `verify_geometry.py`, anchor 4<sup>P15R16</sup>).  
□

*Remark 1* (The leg does not tilt it either). The proposition shows that a degenerate Killing horizon carries no scale, so it cannot imprint one, and Proposition 7 shows the same of the branch point. The collapse leg supplies the other half of the same conclusion, by an independent route. Horizon entry on that leg occurs at  $x = k\eta/\sqrt{3} = 1/\sqrt{3}$  for every  $k$ —the radiation era’s own scale invariance, which §4.1 recovers from the substrate geometry—so every mode leaves the leg carrying the same free-oscillation amplitude  $\Psi(1/\sqrt{3})/2 = 0.4835 \Psi_i$ , a single  $k$ -independent number<sup>P15R41</sup>. *So the leg multiplies the spectrum by a constant and the branch point imprints nothing*: neither carries a scale, and a spectrum can only be tilted by something that does. The tilt observed is therefore the progenitor’s, established from both sides of the branch point rather than one.

*“Carries a scale” has an operational form, and it is worth recording because it is what makes the dichotomy testable rather than descriptive*: a kernel carries a scale exactly when the tilt one fits to it *depends on which band one fits*. Fitted on four disjoint bands, the degenerate ( $p = 2$ ) kernel returns one tilt to sixteen decimal places while the non-degenerate ( $p = 1$ ) kernel spreads by several units, and that spread shrinks as  $\kappa$  grows—a larger scale pushing its knee past the range. Restoring a scale to the degenerate kernel by hand destroys the agreement, so the scale-freedom is doing the work and not the arithmetic<sup>P15R150L21</sup>. *This restates Proposition 6 and proves nothing further*; what it adds is a test one can run on a kernel without knowing where it came from.

*Remark 2* (What the branch point does not fix). Both halves of the dichotomy are statements about the spectrum’s *shape*: a degenerate horizon carries no scale, and neither does horizon entry on the leg, so neither can tilt what passes. *Neither is a pointwise condition on the perturbation state at the branch point*, and the distinction matters for anyone computing the handover. The single free function per mode is the freedom of the *spectrum*—the progenitor’s amplitude and tilt—not a claim that the branch point fixes the state component by component. The state itself is whatever the leg’s evolution produces, and §4.1 supplies it in closed form: the potential from the leg’s own equation, the effective temperature oscillating freely from horizon entry, and the density contrast as their difference.

The Hamiltonian constraint is not an additional condition to impose there. In this framework it *is* the anchor—the map from a cut to its stress-energy, whose energy sector reads the Friedmann equation off the geometry [7, 8]. It is a definition, not a restriction: on the leftward reading the rate is set by the substrate and the content is read from it, so the perturbed constraint likewise *defines* the density contrast a given geometry carries rather than restricting

which geometries are allowed. Imposing it at the branch point alongside the leg's own solution therefore over-determines the handover, and the same geometric state answering to different stress-energy on the two sides is not a failure of matching but the branch point doing what the level assignment says it does.

*Remark 3* (What the branch point's degeneracy does not settle). It is tempting to read Proposition 6 as settling the phase as well as the shape: the tortoise coordinate diverges at the degenerate root, so a mode's accumulated phase would be unbounded and nothing finite could be inherited. *That reading does not survive its own arithmetic.* The divergence is geometric and carries no  $k$ , so an unbounded  $kr_*$  is a divergent term *linear* in  $k$ ; a linear phase renormalizes the sound horizon rather than shifting the comb, and an unbounded one would send the comb spacing to zero and erase the acoustic peaks altogether—excluded by observation, not supported by it. And the modes never meet the divergence in any case: they cross the branch point at finite conformal time, whereas  $r_*$  diverges along a static slicing's approach to the horizon, which is not the collapse trajectory.

What the leg does contribute is finite and, as it happens, invisible in the peak positions. Horizon entry at  $x = 1/\sqrt{3}$  for every  $k$  makes the accumulated leg phase  $k\eta_{\text{bp}}/\sqrt{3} - 1/\sqrt{3}$ : linear in  $k$  with a mode-independent offset. So carrying it across rescales the effective sound horizon by  $1 + \eta_{\text{bp}}/(\sqrt{3}r_s)$  and adds a constant, and changes the comb in no other way. *The comb therefore cannot distinguish a transmitted phase from a reset one*, because a rescaling of  $r_s$  is absorbed by the angle it is read against. Any discriminant must be a quantity that does not take the same factor—the diffusion scale is one, since  $r_D$  does not—and that is the discriminant of principle. *The comparison is made, and it does not require  $r_D$ .* The leg's accumulated conformal time is fixed by the geometry rather than fitted: the collapse branch  $r = -(2M\alpha^2)^{1/3} \cosh^{2/3}(3\text{Re}\tilde{\tau}/2\alpha)$  has  $\cosh \geq 1$ , so  $|r| \geq (2M\alpha^2)^{1/3} = 3766.6$  Mpc and the branch point is reached only across the lift, along which  $\text{Re}\tilde{\tau}$  does not advance and no conformal time accumulates. Hence  $\eta_{\text{bp}} \approx 5 \times 10^3$ – $10^4$  Mpc, and  $r_s^{\text{eff}} = r_s + \eta_{\text{bp}}/\sqrt{3}$  is *dominated* by the leg term— $3.0$ – $5.9 \times 10^3$  Mpc against  $r_s = 144$  Mpc. The absorption that takes a small rescaling cannot take this one: reproducing the measured  $\ell_A = 301.8$  would require  $D \approx 2.9$ – $5.7 \times 10^5$  Mpc against the actual  $1.4 \times 10^4$  Mpc, a factor of 21–41. *The acoustic phase is therefore reset at the branch point and not carried across it*—established on the peak positions alone<sup>P15R42</sup>. The result is robust to the open account of the lift: any conformal contribution the lift is found to carry only adds to an already-excluded total.

The amplitude separates from the tilt in the same statement. What the progenitor supplies is the potential power  $P_\Psi$ ; what the construction contributes is the factor  $(0.4835)^2 = 0.2338$  carrying it to the observed  $A_s$ . That factor is computed rather than fitted: the inherited quantity is  $P_\Psi$  and the map from it to  $A_s$  is derived, so what the construction takes from the progenitor here is the potential power and nothing about the map [9].

The physical reading is the result (Fig. 5). The exponential, non-degenerate approach is the thermal (Hawking) law—the mechanism by which a de Sitter horizon *imprints* a scale-invariant spectrum ( $n_s \rightarrow 1$ ): the exponential redshift erases the infaller's spectral information and stamps the horizon's own thermal scale. The power-law, degenerate approach carries no thermal scale; it is scale-free, and the infalling spectrum is transmitted faithfully. CR's transmission locus is the branch point  $r = 0$ , where  $f$  *diverges* as  $-2M/r$  so the tortoise coordinate is *finite*—the crossing accumulates no divergent phase and stamps no scale, a third case beyond the dichotomy's two<sup>P15R42</sup>. The Nariai double root (Proposition 4,  $f''(r_N) = -2\Lambda$ ,  $\kappa = 0$ ) is *the Nariai seam*—one point of the substrate, which the bead meets on the way in and again one full lap later, the two chart values  $r = -2\alpha/\sqrt{3}$  and  $r = +\alpha/\sqrt{3}$  being the same  $\varphi$  modulo  $2\pi$  in the phase  $\varphi = 2\pi r/\sqrt{3}\alpha$ . The branch point sits two thirds of the lap in from it ( $240^\circ$ , with  $120^\circ$  remaining), so the crossing at  $r = 0$  and the seam are distinct loci on the lap without the seam being an endpoint of anything. Hence:

The transmission dichotomy: a horizon imprints a thermal scale; a degenerate seam does not



**Figure 5:** The transmission dichotomy (Proposition 6). *Left:* the approach to the horizon in the tortoise coordinate  $r_* = \int dr/f$ , on a logarithmic vertical axis. At a non-degenerate horizon ( $\kappa > 0$ ) the approach is exponential,  $(r - r_h) \sim e^{2\kappa r_*}$  (a straight line); at the Nariai double root ( $\kappa = 0$ ) it is power-law,  $(r - r_N) \sim 1/|r_*|$ . *Right:* the logarithmic approach rate  $d \ln(r - r_h)/dr_*$  is the local scale. The non-degenerate horizon carries a fixed thermal scale  $2\kappa$ —the mechanism by which a de Sitter horizon imprints a scale-invariant spectrum ( $n_s \rightarrow 1$ ); at the Nariai double root the rate falls to zero, carrying *no* scale. The branch point, where the modes cross, is a third case:  $f$  diverges there, the tortoise coordinate is finite, and the crossing accumulates no phase at all (Proposition 7).

**Proposition 7** (The branch point transmits). *The cosmogenesis branch point is a faithful scale-free transmitter—not because it is the degenerate Nariai horizon, which it is not (that is the Nariai seam, met at  $r = -2\alpha/\sqrt{3}$  and again one lap later at  $r = +\alpha/\sqrt{3}$ ), but because  $f$  there diverges as  $-2M/r$ , so  $r_*$  is finite and the crossing accumulates no divergent phase at all: the amplitude  $A_s$  and tilt  $n_s$  are the progenitor collapse’s spectrum carried across unaltered, not imprinted by the branch point. Equivalently, CR has no inflationary scale-invariant attractor, because the non-degenerate exponential approach that would drive  $n_s \rightarrow 1$  is exactly the approach neither the branch point nor the Nariai seam has.*

The absence of any scale where the modes cross is thus the proven reason CR carries the primordial tilt rather than manufacturing it. What the transmitter carries is, from the Schwarzschild–de Sitter side, open:  $n_s$  and  $A_s$  are inherited, fixed by the progenitor collapse—the same handover that fixes  $\rho_r/\rho_m$  and the light-element abundances—and not derived here. *Naming them together is convenient and has concealed that they are not the same kind of quantity*<sup>P15R85</sup>. The vacuum normalisation is the only place a quantum constant enters the spectrum, so  $P(k) = \hbar A_0 k^{n_s-1}$  and  $\hbar$  is an overall multiplicative factor: *it survives in the amplitude and cancels in every logarithmic derivative*. An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two. *So the construction’s inability to force a dimensionless magnitude bears on  $A_s$  and not on  $n_s$ —the first is permanently inherited, the second need not be. And read backwards the same statement is useful rather than limiting:* a quantity fixed by the moduli rather than the invariants is one the observation *measures* rather than one the construction must predict, so the observed  $A_s$  is a measurement of the progenitor’s boundary data. *It does not isolate the mass:* the parent’s growing-mode amplitude  $D_k$  enters alongside it, which is the corollary recorded above, so the mass must be reached by a route that does not run through the amplitude and checked against the independent requirement that it lie below the Nariai value.

*And it is worth saying precisely what that openness is, because a count settles its kind.*<sup>P15R84</sup> The progenitor family described here is parametrised by  $M$  alone— $\alpha$  is the one invariant and  $E = 1$  a choice of geodesic—so the geometry is one-dimensional; and the Kretschmann along the bead is mass-free in the faller’s own proper time, so that one parameter does not survive the approach either. *A quantity identical for every member carries no information about which member it was*, and the collapse geometry therefore transmits not one parameter but none. Against that stand at least four independent inherited numbers. *So ‘inherited from the progenitor collapse’ cannot mean inherited from the collapse geometry*: what is inherited is inherited from the progenitor’s matter content—its composition, its entropy per baryon, its perturbation spectrum—and none of that is carried by  $M$ , the exterior here being exactly vacuum and the marginally-bound interior dust without composition. *The frontier is accordingly not a derivation awaiting a cleverer argument but a modelling task awaiting a progenitor interior*, and what the geometry supplies in its place is a cutoff rather than a content: the segment’s conformal length  $|\Delta\eta| = 3.32\alpha$  is fixed by  $\Lambda$  alone *on the Nariai member*—below it  $|\Delta\eta|$  runs as  $M^{-1/3}$ , reaching  $4.19\alpha$  at  $M/2$  and  $15.4\alpha$  at  $M/100$ —and says which modes survive without saying how large they were.

## 9 Predictions, and the progenitor boundary

The signatures, with their maturity marked:

1. Classical, non-vacuum primordial statistics; no inflationary consistency relation; no substrate-sourced primordial  $B$ -modes (established, §5).
2. The transmission character: the branch point carries the progenitor tilt and does not drive  $n_s \rightarrow 1$  by any *inflationary* attractor (established, §8). *Nor does the construction lack a scale-invariant attractor—it has one, and it is not the inflationary root.*<sup>P15R81</sup> Scale invariance for a background  $a \propto (-\eta)^\beta$  with  $z \propto a$  requires  $\beta(\beta - 1) = 2$ , a quadratic with *two* roots:  $\beta = -1$ , de Sitter inflation, and  $\beta = +2$ , a matter-dominated contraction. *The collapse leg of this construction is the second root exactly*: near the branch point  $f \rightarrow -2M/r$ , so the leg is pressureless there and the banked law  $r^3 = \frac{9}{2}M\tau^2$  with  $a = |r|$  gives  $a \propto \tau^{2/3} \propto \eta^2$ ; the mode equation integrated from vacuum data returns a flat spectrum to a part in  $10^4$ . *The mode-selection question that shadows this in the literature was accordingly put to the construction directly, and it answers in the negative.*<sup>P15R82</sup>

The scale-invariant spectrum sits in the mode that *dominates* the contraction, whose curvature perturbation goes as  $\eta^{-3}$ ; the mode that survives on the expanding side is the constant one, and from vacuum data its spectrum is  $n_s - 1 = 6$ .

Whether the first reaches the second is a connection problem at the branch point, and here it is settled rather than assumed: the exponents 2 and  $-1$  differ by an integer, so a logarithm could mix them, but the  $\sinh^{2/3}$  law is *even* in  $\tau$ , which in conformal time places its correction at  $\eta^6$  and steps the Frobenius recursion in sixes past a resonance sitting at three. *No logarithm is generated, the continuation acts diagonally, and the scale-invariant spectrum does not reach the observer*—confirmed independently by continuing the dominant solution numerically around the branch point, where the admixture it produces is what the claim turns on, where the admixture falls as  $\delta^{2/3}$  and the diagonal entry goes to unity. *What survives is therefore narrower than it first appeared and sharper than what it replaced*: the construction *has* a scale-invariant background, the spectrum that background freezes is confined to the contracting side, and the obstruction is a parity property of the construction’s own solution rather than a missing mechanism. *Mixing would require a background correction odd in  $\tau$ , and that question is settled here too, in the negative.*<sup>P15R83</sup>

Writing the law as  $r = Au^{2/3}g(u)$  with  $g = [\sinh u/u]^{2/3}$  places every branched factor in  $u^{2/3}$  and leaves  $g$  even, analytic and non-vanishing at the origin—exactly, since  $\log(\sinh u/u)$  is analytic in  $u^2$  there and the exponential of a series in  $u^2$  is one. The three sheets differ by a constant cube root of unity, and a constant multiplier cannot change a parity; the conjugate branch is even as well, and carries no branch point at all. *So the evenness is unavoidable, and what it says is not a technical fact but a symmetry:  $r(-\tau) = r(\tau)$* , the cosmogenesis is time-symmetric about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one, since mixing would distinguish a direction of time the background does not. *The same invariance appears one observable over:* the driven acoustic amplitude of §4 is a resonant Fourier magnitude exactly invariant under time reversal, a theorem about the driving, though what it carries about the peak heights is what that section’s instrument does not return. *One symmetry, read at two places. And it locates rather than forecloses what would be needed:* this leg is exactly even because it is the marginally-bound geodesic of an exactly vacuum exterior, so a progenitor carrying real matter content—pressure, dissipation, anything with an arrow—would not be, which returns the question to what the progenitor was.

3. Flat distances ( $\Omega_k \simeq 0$ ) coexisting with a closed- $S^3$  low-multipole *deficit* below  $\ell \approx 8$ , set parameter-free by  $\Lambda$ —mild, and on a genuine Boltzmann transfer a dip bottoming at  $\ell = 4$  ( $\approx 0.47, 0.41, 0.36, 0.68$  of the flat- $\Lambda$ CDM expectation at  $\ell = 2, 3, 4, 5$ ), its shape cross-validated between two independent transfers (*established* for the existence, the location and the minimum; *open* for the depth, on which the two transfers differ by up to a factor of two at  $\ell = 3$  and 4 (§7); *non-discriminating* as a test, the deficit sitting inside the largest multipoles’ cosmic variance). Confronted with data it is non-discriminating: it starves  $\ell = 2$  and  $\ell = 3$  together, matching neither the sharp observed quadrupole-only dip nor the near- $\Lambda$ CDM octopole, and sits consistent with  $\Lambda$ CDM within the lowest multipoles’ cosmic variance (§7).
4. Coherent acoustic peaks from geometric (null-boundary) phase-fixing rather than super-horizon freeze-out (argued, §4).
5. The acoustic scale is resolved in the cosmological development above (§2.3), and the geometric rate (4) is the standing CR-versus- $\Lambda$ CDM discriminator on data in hand (*established*, §2.3)—a strength established before the spectrum, not this section’s to re-assert.

The boundary is principled. The substrate carries everything the branch point’s null geometry lets it carry—coherence, isotropization, the discreteness floor, the classical character, the faithful transmission—and the scale-free crossing proves precisely where it stops: the spectrum’s amplitude and tilt are not the substrate’s to set. This is the shape inflation has when it parametrizes its observables through an unknown inflaton potential; CR routes them through an unknown progenitor collapse, with the arrival constrained—whatever the collapse supplies must reach the branch point coherent, isotropized, scale-freely transmitted, and classical.

The symmetry between the two unknown suppliers is only apparent, and the asymmetry is the substance. Inflation is not structure the standard cosmology already contains; it is an added sector—a scalar field with a freely specified potential—introduced to produce features that cosmology was known to lack: the causal contact the horizon problem denies, the flatness the dynamics do not single out, and, once these were in view, a near-scale-invariant spectrum. The potential is then chosen to deliver the observed amplitude and tilt. A model built to be tunable, and tuned to a target known in advance to require tuning, earns precisely what such a model can earn—it establishes that the features are *possible*, that an apparatus permitting them exists. It does not establish that they were *required*, and the routine elevation of the spectral fit into a vindication of inflation conflates the two: that a freely chosen potential can be set to

match  $n_s$  and  $A_s$  is not evidence that inflation occurred, but the statement that a model with enough adjustable structure to fit has been fit. This is the distinction between a world that *requires* a phenomenon and one that merely *permits* it through adjustable parameters—the same distinction the programme draws in reading the cosmic foliation as forced rather than chosen [11]—and it places the inflationary spectrum on the permitting side.

The present cosmology adds no such apparatus. Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are not produced by a field introduced to produce them; they are read off structure the construction already contains—the de Sitter throat that isotropizes by the no-hair theorem (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§2.3), and the degenerate horizon whose vanishing surface gravity transmits rather than imprints (§8)—each fixed by  $\Lambda$  alone, with no parameter free to do the producing. The one quantity it does not derive, the amplitude and tilt, it inherits in exactly the sense flat  $\Lambda$ CDM inherits the baryon-to-photon ratio; on that single datum the two frameworks stand level, and on everything the inflationary sector is invoked to supply they do not. The verdict this licenses is not symmetric, and it does not await a future measurement. On the data in hand the two align; on the structure each must assume in order to align, one introduces an entire tunable sector and tunes it while the other introduces none—and makes, in its place, falsifiable commitments where inflation keeps its freedom: no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate-sourced  $B$ -modes (§5, §8)<sup>P15R82</sup>. By the economy of assumption that distinguishes a required structure from a permitted one, that comparison is settled by what each theory must presuppose, and it is settled in this cosmology’s favour. The bound on that verdict is stated in the same breath: it is a verdict on what each theory must presuppose, and the data axis is split. The expansion history fits the distance ladder and fixes  $H_0$  with the acoustic angle (§2.4); the abundances are confirmed within  $1\sigma$  [19]; the spectrum’s content remains inherited; and the acoustic peak structure, computed from that background with nothing fitted to it, reproduces the comb to a part in six hundred and is still rejected on  $TT$  shape (§4.7)—the sharpest datum against the construction.

## 9.1 The acoustic instrument: what it is validated against, and to what accuracy

The low-multipole prediction above and the ratio statements below are carried by a Boltzmann transfer built for this programme, and its standing should be stated plainly rather than assumed. It is a full photon hierarchy with polarisation, second-order tight coupling, massless neutrinos, and a Peebles recombination history; the line-of-sight source carries the monopole with the potential, the Doppler term, the integrated Sachs–Wolfe term, and the quadrupole with its own projection kernel ( $j_\ell + 3j_\ell''$ ).

Against a like-for-like reference—an independent Boltzmann code run at identical parameters, unlensed and with reionisation off—it reproduces the *acoustic peak and trough positions* to  $\leq 0.5\%$  across  $P_1$ – $P_4$ , the *ionisation history*  $x_e(z)$  and its derivative to  $\sim 1\%$  through the visibility peak, matter–radiation equality to  $0.02\%$ , and the *transfer function* to better than  $1\%$  for  $k < 0.02 \text{ Mpc}^{-1}$ . Above that wavenumber it agrees to  $1$ – $2\%$ , with the residual scatter in both directions and no trend across the range<sup>P15R43</sup>.

*Two construction details are worth stating, because both are easy to omit and both have signatures that read as physics.*

*The first is the reach of the mode ladder.* The angular spectrum at multipole  $\ell$  is an integral over wavenumber, and the projection kernel  $j_\ell(k(\eta_0 - \eta))$  has substantial support *above*  $k = \ell/D_M$ , not merely at it; a ladder cut at  $k_{\max}$  therefore truncates the integral, and does so more severely as  $\ell$  rises toward  $k_{\max}D_M$ . Quantitatively, truncating an exact transfer at  $k_{\max}D_M = 900$  retains  $99.2\%$  of  $C_\ell$  at  $\ell = 200$  but only  $61\%$  at  $\ell = 800$  and nothing at all at  $\ell = 1150$ ; carrying the ladder to  $k_{\max} \geq 2\ell_{\max}/D_M$  recovers every multipole to better than half a per cent.

*The tell is unmistakable once known:* the highest peak simply fails to appear, so a spectrum that should show four acoustic peaks shows three, and the peaks that do appear are displaced downward by an amount that grows with  $\ell$ . *Increasing the ladder’s density at fixed reach does not help*—the loss is the range, not the sampling, which is worth checking explicitly because the two are easily confused.

*The second is the seeding of the conformal time.* It is a cumulative integral,  $\eta(a) = \int_0^a da'/(a'^2 H)$ , and a numerical grid necessarily begins at some finite  $a_{\min}$ ; the interval already elapsed below that point must be carried as a seed rather than set to zero. In radiation domination  $H \propto a^{-2}$  makes the integrand constant, so the omitted piece is  $a_{\min}/(H_0 \sqrt{\Omega_r})$  exactly—for  $a_{\min} = 10^{-5}$  it is 4.6 Mpc. *It is a fixed offset at every epoch*, and therefore 0.03% of  $\eta_0$  but 12% of  $\eta$  at  $z = 10^4$ : the acoustic scale, the peak positions and the peak ratios are insensitive to it, while any early-time quantity or any comparison against a second code is not. Its signature is instructive—because the potential decays more steeply at larger  $k$ , a fixed error in the time coordinate reads out as a *smooth, monotone,  $k$ -growing* error in amplitude, which is the shape one would otherwise attribute to a physical suppression. The grid must also extend below the intended starting redshift, so that the initial epoch is interpolated rather than clamped.

*That residual does not enter the claims made here.* The Cosmological Relativity and flat- $\Lambda$ CDM spectra are computed through the *same* machinery, so an instrumental transfer error afflicts both arms identically and cancels in their ratio—and every prediction stated in this paper is a ratio. The low-multipole deficit in particular lives at  $\ell \lesssim 8$ , i.e.  $k \ll 0.02 \text{ Mpc}^{-1}$ , where the two agree to better than 1%. What the residual does forbid is an *absolute* amplitude claim: this instrument should not be read as reproducing the observed small-angle power to better than a few per cent, and no such claim is made.

## 10 The open frontier: the pieces this cosmology is built to close

The result stands on what the substrate determines. What it does *not* yet claim has been carried to a sharp, gradable edge rather than left vague—and those edges are the most interesting part of the programme from here, each a gap to be closed, not papered over.

We hold them to the criterion of necessity the programme’s epistemology names—that a structure is credited for *requiring* a phenomenon, not for *permitting* a value that fits it [12]—and sort them by it: a gap that is *buildable* is a debt owed, to be built before the claim it bears on is a proof rather than a coherent proposition, while a genuinely external unknown may stay open at no cost, exactly as flat  $\Lambda$ CDM leaves its own inherited data open.

The buildable ones below are named as debts, not softened into frontiers. *Of the four, three are run and one is a frontier:* the end-to-end branch-point-to-recombination transfer, the large-angle depth and the likelihood are built and confronted below, what the transfer leaves being a phase residual to be read against the driving subtraction on this arm; and the derivation of the progenitor spectrum is the genuine frontier rather than a debt. Four stand out:

1. the sufficiency of the null-boundary characteristic data for the observed acoustic coherence (§4), argued from the structure of the characteristic initial-value problem, with the time-reversal driving equality established there as a Fourier theorem and the handover at the branch point fixing the phase and the amplitude the datum leaves open. *The end-to-end branch-point-to-recombination transfer is run* (§4.7): a two-arm line-of-sight Boltzmann integral,  $\Delta_\ell(k) = \int S j_\ell(k(\eta_0 - \eta)) d\eta$ , carried on this cosmology’s own background with the perturbations on the leaf congruence the framework assigns them, and validated on its control to 0.14% in the first-to-second height ratio<sup>P15R107</sup>. On this arm it returns a comb of 298.0 against the sky’s 298.4 and height ratios 2.264 and 2.298 against 2.217 and 2.277, with the likelihood still rejecting this arm’s spectrum on shape; the diffusion-scale signature it carries is located by the joint fit as a Gaussian residual in  $\ell$  that no tilt

removes (§4). *What the transfer leaves is about one multipole at each peak*, the comb itself being right to a part in six hundred and the phase intercept inside the sky’s own locating spread;

2. the exact large-angle *depth* (§7): the flat-projection transfer of the discrete closed- $S^3$  source is built with a *genuine Boltzmann transfer*—the exact CMB temperature transfer  $\Delta_\ell(k)$  (Sachs–Wolfe, early and late integrated Sachs–Wolfe, and Doppler, all common to CR and flat- $\Lambda$ CDM) sampled at the discrete modes  $k_L$ , gate-validated (the continuum reproduces the reference  $C_\ell$  to four figures) with the  $r_0$  sensitivity measured rather than asserted—shape stable, depths drifting up to 15% at  $\ell = 4$  (receipt: `verify_lowell_boltzmann.py`<sup>P15R31</sup>).

It gives a *mild* deficit,  $\approx 0.47, 0.41, 0.36$  and  $0.68$  of the  $\Lambda$ CDM expectation at  $\ell = 2, 3, 4, 5$ —*a dip whose minimum falls at  $\ell = 4$ , not at the quadrupole*, the shape reproduced by the programme’s own photon hierarchy, which agrees on the location and not on the depth—recovering by  $\ell \approx 8$ : the late integrated Sachs–Wolfe term is sourced above the floor and is therefore retained by the discrete source, leaving the deficit shallower than the bare Sachs–Wolfe starvation.

The mechanism is robust—parameter-free, located at  $\ell \lesssim 8$ , bottoming at  $\ell = 4$ , with  $\ell = 2$  and  $\ell = 3$  starved *together* (the exact discrete measure and the Doppler term each leave the  $\ell = 2$ -to- $\ell = 3$  ratio near unity: receipts `verify_lowell_exact_measure.py`<sup>P15R29</sup>, `verify_doppler_lowell.py`<sup>P15R30</sup>).

Confronted with the sky the sector is a *wash*: CR’s smooth  $\sim 0.4$ – $0.5$  deficit sits above the sharp observed quadrupole ( $\sim 0.2$ ) and below the near- $\Lambda$ CDM octopole ( $\sim 0.6$ – $1.0$ ), matching neither, and the exact cosmic-variance likelihood returns  $\Delta(-2 \ln L) = +1.58$  over  $2 \leq \ell \leq 8$  and the same to the digit over  $2 \leq \ell \leq 10$ , both arms being at unity by  $\ell = 8$  (range  $+0.2$  to  $+3.4$  across the octopole estimator), a small residual inside cosmic variance — *and it is a near-cancellation rather than a uniform cost*: per multipole the quadrupole rewards the construction by  $-2.5$  and  $\ell = 3$  to  $5$  penalise it by  $+1.0, +2.0$  and  $+1.0$ , with nothing above (receipt: `verify_lowell_likelihood_v2.py`<sup>P15R33</sup>). So the low multipoles are a genuine but *non-discriminating* sector, neither a parameter-free correspondence success nor a sharp falsification edge; the deficit’s existence and location stand as a real parameter-free feature, but its depth places it consistent with  $\Lambda$ CDM within the lowest multipoles’ cosmic variance.

*The location is robust and the depth is not, and the difference between them is now measured rather than estimated.* On the background the distance data fix, the discrete scale moves by a quarter of a per cent and the lowest mode lands at  $\ell \simeq 7.8$ , so *where* the deficit sits is not a background choice. *The two Boltzmann treatments still differ by a factor of two at  $\ell = 4$*  — and that factor is not the geometry: moving the discrete scale alone buys three per cent against a gap of a hundred. *It is the late integrated Sachs–Wolfe term*, which at these multipoles is sourced above the discrete floor and is therefore retained by the sum, filling the deficit; the treatment that truncated its line-of-sight integration earlier was deeper at exactly the multipoles where the two disagreed. *Carried to the present instead, the two treatments agree*: on the exact transfer the deficits are  $0.487, 0.435, 0.359, 0.666, 0.911, 0.983$  and  $0.998$  of the flat expectation at  $\ell = 2$  through  $8$  — *the exact transfer being the one carried here, with the hierarchy agreeing* to within three per cent at every multipole and one per cent above  $\ell = 5$ <sup>P15R168</sup>. *The factor of two was the missing term and not a difference between the constructions*, and the agreement is stable: it does not move under a doubled opacity grid, a halved late step count, or a further-extended continuum. *So the shape and the depth both cross-validate* — minimum at  $\ell = 4$ , recovery by  $\ell = 8$  — and the sector is ready to be scored against the measured multipoles with cosmic variance carried.

The exposed edges lie elsewhere—the acoustic phase residual of the first item and the damping signature it carries;

3. the progenitor spectrum itself— $n_s$  and  $A_s$  (§8)—inherited boundary data fixed by the progenitor collapse, the baryogenesis-analogue computation belonging to the matter sector [7, 8], with the target a consistency between the inherited composition and spectrum and the measured primordial abundances and tilt, not a derivation from the geometry alone—the most ambitious reach of the four, the one that would close the decomposition from both ends and give CR a derivation where the standard model keeps a measured datum. *And the target is now a number rather than a direction*: refitting both arms on this construction’s own background prefers a tilt of 0.995 here against 0.956 on the standard one (§4.7), *so a derivation has to land near the bluer value and not near the one the standard model infers. And what supplies it is the progenitor’s vacuum and not the collapse leg*: on the leg both the potential and the effective temperature are functions of the single variable  $k\eta/\sqrt{3}$ , so a handover at a fixed phase is exactly scale-free—the ratio of amplitudes at two wavenumbers is one—and a handover at a fixed time would contribute not a tilt but a running, red in both channels and growing as  $k^2$ , which no amplitude choice makes look like a constant. *At the branch point it vanishes quadratically*, so the leg is a wavenumber-independent amplitude and nothing else. *And what the substrate alone fixes is  $n_s = 1$  exactly*: in conformal time the mode equation carries no length, so the curvature radius enters only as the prefactor that sets the amplitude and reaches the tilt for no value of it; and on the forced member the progenitor’s mass is that same radius up to a pure number, so the two candidate scales are one and fall together. *But the fluctuations are not generated on the substrate*, and the interior they are generated in carries a scale the substrate does not. *Solved there*, the mode potential interpolates between the scale-free matter form and a radiation form that is not, so the generated spectrum *breaks* at the wavenumber the radiation content sets: its log-slope runs from +0.30 to +1.98 across the observed band, tending to the radiation value +2. *So it is a running and not a tilt*, by the same band test that closes the collapse leg — and *blue at every wavenumber*, where the measured value is red by two parts in a thousand<sup>P15R175</sup>. *The flattest slope goes as the radiation fraction*, so exact scale-invariance is the pure-dust limit and reaching red would need that fraction negative. *And the substrate’s power-law formula does not apply off the substrate*: its coefficient is dimensionless, which is what makes it a power law, where the interior’s carries a scale—a distinction invisible to a tilt read off a background fitted to the other arm. *The figure is provisional in one respect and it is stated*: it is measured on the range below the damping tail, which is where a tilt has most of its leverage. *One route to it is closed, and closing it narrows the target rather than the ambition*. The construction is recursive—each progenitor seeds the next cosmology—so a *stationary* recursion would make  $A_s$  a fixed point and determine it. But entropy per baryon rises strictly at every branch-point crossing, the crossing being an erasure and the increment bounded below by Landauer’s  $k_B \ln 2$  independently of the network and of  $\eta$  [19]: *a map that strictly increases a coordinate has no fixed point*, and independently the chain it generates is finite, so there is no asymptotic limit for a fixed point to be the limit of<sup>P15R2</sup>. *So the derivation must reach the head of a bounded genealogy rather than a stationary state*—the same terminus the inherited composition reaches, and the two are one question asked of two quantities;
4. the *likelihood-level model selection*: the parameter-free ratio comparison above is joined by the exact low-multipole cosmic-variance likelihood—each  $\hat{C}_\ell$  a scaled  $\chi^2_{2\ell+1}$  about the model  $C_\ell$ , the absolute baseline cancelling in the CR/ $\Lambda$ CDM ratio (`verify_lowell_likelihood.v2.py`<sup>P15R175</sup>) the proper instrument where cosmic variance dominates.

With the genuine-Boltzmann depths ( $\approx 0.47/0.41$  at  $\ell = 2/3$ , above) the sector is a *wash*:

the low quadrupole still mildly rewards CR ( $\Delta(-2 \ln L) \approx -2.6$ , its milder dip nearer the observed low power than  $\Lambda$ CDM's), the octopole and  $\ell = 4$  mildly penalise it ( $\approx +1.3$  and  $+2.1$ )—so the sector's largest single-multipole penalty sits not at the octopole but at  $\ell = 4$ , the prediction's own minimum—and the net over  $2 \leq \ell \leq 10$  is  $\Delta(-2 \ln L) \approx +1.8$  (range  $+0.3$  to  $+3.8$  across the octopole estimator)—a small residual well inside cosmic variance. A Monte-Carlo over  $10^4$  skies puts the sector's *in-principle* discriminating power at only  $\sim 3.4\sigma$  (the short multipole lever arm and cosmic variance), and the current measurement uncertainty swamps even that.

The low multipoles are therefore a genuine but *non-discriminating* test: CR and  $\Lambda$ CDM are statistically indistinguishable there, the mild deficit neither favouring CR nor refuting it. Decisiveness comes from outside the sector—the geometric expansion rate (4), confronted with the distance ladder (§2.4), as the standing discriminator, and the likelihood across the *full* spectrum, which is run and rejects this arm's spectrum on  $TT$  shape (§4.7), its amplitude level not established by the time-reversal driving equality (§4).

A frontier of a different kind concerns the matter itself rather than the perturbation observables above: the *dynamics of matter crossing the branch point*.

Three facts about that crossing follow from the structure already established here.

It is *well posed*—the beginning is fixed at the branch point  $r = 0$ , and the crossing is taken on the *substrate*, whose curvature is finite there however the geometry's own invariants behave.

The bend  $\rho = m'/4\pi r^2$  that carries the matter is accordingly finite, and its data on that null surface is characteristic rather than Cauchy: a characteristic crossing with no obstruction from the substrate's curvature. *The geometry's Kretschmann scalar does diverge at that locus*, as the abstract states and the circle paper computes [36]; what makes the crossing well posed is that the divergence belongs to the chart-borne geometry and not to the substrate the cut is taken on, and that the tortoise measure converges so the crossing carries no scale.

It *isotropizes and is scale-free*—the throat's de Sitter no-hair (§6) damps the anisotropic part of the crossing stress-energy to the isotropic monopole, and the degenerate approach (§8) imprints no thermal scale: the matter counterparts of the field no-hair and faithful transmission established above. *Where each of those acts is worth stating, since one of them cannot act at the crossing at all.*

Damping is a process and requires elapsed time, and the crossing has none: the real part of the complex cosmic time is frozen along the segment reaching it [35].

The no-hair damping therefore belongs to the *approach*—the throat region, over the collapse leg's own cosmic time—while the crossing carries whatever survives it unchanged, not because a mechanism protects it but because there is no interval in which anything could act. *The isotropy is earned on the approach and preserved by the crossing*, and the two halves are of different kinds: the first is a damping calculation with a rate, a timescale and a residual, the second follows from the zero-duration structure and takes no model input. It follows that *how much anisotropy survives is fixed by the approach against the leg's duration*—a property of the progenitor's collapse history rather than of the crossing. And it obeys a *structural transition law*: because the reassignment preserves the cosmic foliation and the density is leaf-carried and lapse-independent [7], the reassignment acts on the time-stacking—fixing the  $\Lambda$ -set rate (6)—and not on the leaf, so the isotropic density crosses as inherited progenitor content, which is the structural reason the inherited datum of §2.4 is inherited rather than produced; *and two conditions must be met at that crossing, which a concrete model has to satisfy separately: the reassignment requires a degenerating Killing field whose freed null direction can be re-founded as a clock, the joining of the legs requires a branch point, and these are independent—the forced Nariai member carries the degeneracy and is not a branch point, while  $r = 0$  is the branch point and carries no Killing degeneracy [10], so a model furnishing one without the other furnishes no cosmogenesis. The first of the two is settled as a uniqueness, not merely an availability: across the slicing family the reassignment that promotes a null direction to the fundamental*

congruence is *uniquely* the one selecting the forced Nariai member, every other vantage yielding a congruence that crosses the embedding’s symmetry structure transversally and so reading as a localised black hole [6]. *A model therefore has no freedom in where the reassignment acts*, and misplacing it does not fail to give a cosmology by a little—it gives a black hole instead. The scoping of the two rates this law forces—the excursion-side local readout against the observable geometric rate, their boundary—is drawn out in the cosmogenesis paper [19]. What remains open is the *detailed* worldline and field dynamics of the crossing for a concrete matter model [9, 7]—a depth left for a matter model to supply, and resting on a crossing shown well posed rather than merely postponed.

None of these unsettles the result obtained: the branch point’s null geometry decomposes the primordial scalar spectrum into a substrate-determined structure and a progenitor-supplied content, and the branch point transmits that content rather than imprinting one of its own; the super-horizon transfer across the branch point is itself computed— $\Phi \rightarrow \Phi_i$  for every  $k$ , with the expanding leg inheriting  $\frac{9}{10}\Phi_i$  scale-invariantly<sup>P15R105</sup>—and carrying the acoustic evolution from that datum through to recombination is the transfer of §4.7, which is run and returns the comparison recorded there. They are, rather, the map of where this cosmology goes next—what displaces the fourth peak, and the full oscillating medium downstream in the matter sector, the tensor sector’s companion in the dynamics paper [10].

## 11 Discussion

What this paper determines is one thing seen from several sides. A single de Sitter geometry, its scale set by the cosmological constant alone (the maximally symmetric substrate’s sole scale [17]) and read through a causal reassignment at the cosmogenesis branch point, *is* the observed cosmology: the flat- $\Lambda$ CDM expansion history is its areal radius (§2.3); the acoustic scale is computed from its geometric rate, which the distance data and that angle fix together (§2.4); the primordial scalar spectrum is the characteristic data of its branch point, structure fixed by the substrate and content inherited across the branch point (§§4–8). The Einstein field equations are unchanged; what changes is the reading, and under it the quantities the standard model carries as independent inputs—the present matter density, the dark-energy fraction—are readings of one cosmic clock, not drivers of the expansion, so that the coincidence of  $\Omega_m$  and  $\Omega_\Lambda$  dissolves into our observing at a time of order the geometry’s one timescale. There is a single seed, and the observed universe is what grows from it. And the one scale reaches down as well as out: the same cosmological constant that sets this expansion sets, for every mass  $M$ , the *Hubble–Eddington radius* [34]  $r_{\text{HE}} = (3GM/\Lambda c^2)^{1/3}$ —the boundary within which local structure stays bound against the cosmic flow and beyond which it is carried off. The slicing geometry reads that boundary as the flat locus of the existent slice’s own curvature, the local bend of the cut exactly cancelling the substrate’s cosmological term [5], and the slicing operator reads it as the handover from sub-marginal bound orbits to the marginally-bound  $E = 1$  congruence that *is* this cosmology [7]. So the substrate’s single  $\Lambda$  governs both the global expansion this paper computes and the local scale at which structure resists it—one constant read at two ranges—and the Hubble–Eddington radius, proposed as a local, per-structure test of  $\Lambda$  from the largest bound structures [13], is an independent observational handle on the very constant the expansion history here measures. The two ranges are moreover the same length up to a pure number. The Hubble–Eddington radius—where the slicing surface’s Gaussian curvature  $K_G = 1/\alpha^2 - M/r^3$  vanishes, the local bend cancelling the substrate’s cosmological curvature [5]—is  $r_{\text{HE}} = (M\alpha^2)^{1/3}$ ; while the expansion’s own amplitude,  $(2M\alpha^2)^{1/3}$ , is exactly the areal radius at matter- $\Lambda$  equality ( $\rho_m/\rho_\Lambda = \text{csch}^2(3\tilde{\tau}/2\alpha)$ , unity at  $\sinh = 1$ ). The two therefore stand in the fixed ratio  $2^{1/3}$  for every  $M$ : the scale at which a bound structure’s gravity balances the expansion and the scale at which the expansion’s matter balances its  $\Lambda$  are one length read locally and cosmologically, the factor of two that of the Schwarzschild term

against the surface curvature's  $M/r^3$  and nothing else. <sup>P15R35</sup>

The comparison with the standard cosmological model is accordingly not *only* a contest of fits—though the geometric rate fits the joint acoustic-and-distance-ladder data at one  $H_0$  (§2.4)—but also, and more deeply, a difference in what each must assume to do so. The standard model *assembles* the observed universe: a dark-energy component for the late-time acceleration, an inflationary sector for the causal contact, flatness, coherence, and near-scale-invariance the background dynamics do not themselves supply, and further early-universe physics where the sound horizon and the expansion rate come into tension. Each part is a real solution to a real problem, and each was introduced to yield the feature it produces. *This cosmology adds no such part.* Causal contact, isotropy, flatness, phase coherence, and scale-free transmission are read off structure the construction already contains—the isotropizing throat (§6), the null boundary on which coherence *is* the characteristic-data condition (§4), the exactly Euclidean constant-time slice (§7), the scale-free crossing that transmits rather than imprints (§8)—each fixed by  $\Lambda$  alone, with no parameter free to produce it. This is the distinction between a structure that *requires* a phenomenon and one that merely *permits* it through adjustable apparatus [12, 11], and it is the criterion by which competing frameworks are rightly weighed ahead of the measurement that will decide between them.

*The dark-energy component is the plainest case of the difference:* on this cosmology the acceleration is not a sector at all. Since  $(dr/d\tilde{\tau})^2 = E^2 - f$  on the energy family, one has  $d^2r/d\tilde{\tau}^2 = -f'/2 = rK_G$ : *the comoving acceleration is the existent slice's own Gaussian curvature*, up to the positive factor  $r$ .<sup>P3R12</sup> So the expansion decelerates while the slice is negatively curved, turns over exactly where it is flat—the Hubble–Eddington radius,  $\rho_m = 2\rho_\Lambda$ , at  $\sinh(3c\tilde{\tau}/2\alpha) = 1/\sqrt{2}$ —and accelerates thereafter, with nothing added to make it do so. *And that locus is a measurement, not only a description:* the turnover sits where the areal radius reaches the front seam,  $r = r_N$ , so the acceleration-onset redshift is the offset parameter itself minus one,

$$z_{\text{acc}} = x_0 - 1 = 0.6648 \pm 0.0467, \quad (11)$$

against  $(2\Omega_\Lambda/\Omega_m)^{1/3} - 1 = 0.632$  on Planck's parameters—agreement at  $0.7\sigma$ , and with no fit performed here. *What makes the equation worth stating is what it does not contain:*  $z_{\text{acc}}$  is a redshift, measured calibration-free, so it is untouched by the distance-ladder tension, and it fixes  $x_0$  without  $H_0$ , without a ladder and without the microwave background. The single dimensionful scale is where any disagreement must then live, and the age follows from the same parameter in closed form,  $t_0 = (2r_N/\sqrt{3}c) \operatorname{asinh}(x_0^{3/2}/\sqrt{2})$ . The late-time acceleration is thus read off the same structure that fixes the local boundary of a bound system.

*And the growth sector, written on this rate, loses its parameters entirely — which turns a known numerical curiosity into a statement about the epoch.* Substituting  $H = (c/\alpha) \coth u$  and the Friedmann readout's  $4\pi G\rho = \frac{3}{2}(c/\alpha)^2 \operatorname{csch}^2 u$  into the growth equation and changing to  $u$ , every factor of  $(c/\alpha)^2$  divides out:

$$3D'' + 4 \coth(u) D' - 2 \operatorname{csch}^2(u) D = 0. \quad (12)$$

*The scale does not merely fail to appear—it cancels,* because at Nariai the source is fixed by  $\Lambda$  and the time variable carries  $\alpha$  too. *The decaying mode is the rate itself,*  $D_- = \coth u = H\alpha/c$ , and reduction of order gives  $D_+ = \coth u \int_0^u \sinh^{2/3} v \operatorname{sech}^2 v dv$ . *And the matter fraction is a clock reading:*  $\cosh^2 u = 1/\Omega_m$  exactly, so  $\Omega_m = \operatorname{sech}^2 u$ .

*On these variables the normalisation of the standard growth factor —  $J(\Omega_m) = \int_0^\infty (1+z) dz / [\Omega_m(1+z)^3 + 1 - \Omega_m]^{3/2}$ , known to cross unity once near the concordance density [1] — becomes  $J = \frac{2}{3} \cosh^2 u_0 (D_+/a)|_0$ , whose matter-domination limit  $D_+/a \rightarrow \frac{3}{5}$  returns the analytic endpoint  $J \rightarrow \frac{2}{5}$ .* Differentiating the condition  $J = 1$  gives an exact identity between the two sides' derivatives,  $\propto \frac{9}{2}\Omega_m - 2$ , and integrating it from the common origin leaves

$$\int_0^{u_0} a(v) \Omega_m(v) [\Omega_m(v) - \frac{2}{3}] dv = 0, \quad (13)$$

the growth-weighted history of  $\Omega_m$  balanced about  $\frac{2}{3}$ <sup>P15R12</sup>.

The pivot is the deceleration-to-acceleration turnover:  $\Omega_m = \frac{2}{3}$  is  $\rho_m/\rho_\Lambda = 2$  exactly. (13) is solved at  $\Omega_m = 0.31516$ , so the density at which that normalisation equals unity is fixed by an equation with nothing free in it.

Two scope statements belong with that, and both narrow it. The turnover is not this construction's property: any flat matter-and- $\Lambda$  rate has it, and what is this construction's is the identification of that turnover with a geometric locus — the areal radius there being  $\alpha/\sqrt{3} = r_N$ , the Nariai radius and the merged double root (§11 above). So the balance pivots on the turnover, and the geometry is what makes the turnover a place.

And the rate used is the stacking one, which is the right rate for the object but not for this construction's own growth. The published normalisation is a matter-and- $\Lambda$  integral, and the stacking rate written in the fitted pair is that rate, which is why the derivation reproduces it. The rate rule assigns perturbations to the leaf (§2.2), and on the leaf rate the same integral returns 0.99940 rather than unity. So what is derived here is the root of the standard normalisation, and what remains contingent is that the present epoch lies near it.

What that figure does **not** measure is how far the root moves, and saying so matters because it invites that reading. Continuing the closed form onto the leaf rate is a closed form applied to a rate it does not solve. Solved directly, the root moves the other way and by less — the leaf root sitting at  $u_0 = 1.180039$  against the stacking 1.180309, a shift of 1.1% of the gap to the measured epoch and away from it, taking the offset from 0.704 to  $0.713\sigma$ <sup>P15R13</sup>. So the root's two-per-cent offset from the measured epoch— $x_0 = 1.6315$  at the leaf root against the measured 1.6648—is not an artefact of the rate: it stands, and wants a mechanism.

The transmission dichotomy (§8) is where the distinction between requiring and permitting is sharpest, because there the two frameworks make opposite structural commitments about one object. A non-degenerate horizon's exponential approach imprints a scale-invariant spectrum—the very mechanism by which inflation produces near-scale-invariance—while the degenerate Nariai seam carries no scale and the branch point, where the modes cross, accumulates no phase at all, so the progenitor spectrum is transmitted unaltered. Inflation accommodates the observed tilt because a freely specified potential can be tuned to it; reading that fit as a confirmation mistakes a model with enough adjustable structure to match for one that required what was found (§9). This cosmology cannot tune: the finite tortoise coordinate at the branch point *requires* that the crossing not imprint a scale, so the framework has no inflationary scale-invariant attractor, no consistency relation, and no substrate-sourced primordial  $B$ -modes. The amplitude and tilt it does not derive it inherits, exactly as flat  $\Lambda$ CDM inherits the baryon-to-photon ratio from a baryogenesis it does not model; on that single datum, of the same kind and count the standard model already carries, the two stand level. A framework is credited for requiring the structure it explains, not faulted for measuring the boundary data it does not—and the asymmetry falls on everything else.

The verdict this licenses must be drawn on the right axis, and confined to it. On the axis of theory-choice—the requiring rather than the permitting of the phenomena, the consolidation of many results under one structure, the absence of a device added per problem—the comparison is decidable from the structural record, and it falls to this cosmology [12]. On the axis of data-discrimination the picture is split: the geometric rate fits the acoustic scale and the baryon-acoustic distance ladder together at one  $H_0$  (§2.4), leaving the discrepancy with the local ladder where the standard model leaves it—while on the light elements and the low multipoles the two frameworks are even. On the microwave-background peak structure they are not: the transfer's likelihood comparison rejects this arm's spectrum on TT shape (§4.7), and that is the sharpest datum against the construction that the corpus holds. So the standing is stated as it stands: favoured on the structure, level with the standard model on the light elements and the low multipoles, and carrying a spectrum computed from a background fitted to the distances alone—which reproduces the comb to a part in six hundred and is rejected on shape. The data axis

is not even, and where it is uneven it runs against the construction, which is the direction a framework that has staked falsifiable commitments—no scale-invariant attractor *reaching the observer*, no consistency relation, no substrate *B*-modes—must be prepared for, precisely where the standard model keeps its freedom<sup>P15R82</sup>.

The low-multipole deficit is a mild, non-discriminating feature (§§7, 10). The flat projection of the discrete closed- $S^3$  source gives a parameter-free deficit below  $\ell \approx 8$ —real, and set by  $\Lambda$  through  $r_0$  and  $D_C$ —bottoming at  $\ell = 4$ , at  $\approx 0.47, 0.41, 0.36$  and  $0.68$  of the  $\Lambda$ CDM expectation at  $\ell = 2, 3, 4, 5$ . At that depth CR’s smooth deficit matches neither the sharp observed quadrupole-only dip nor the near- $\Lambda$ CDM octopole, and under the exact cosmic-variance likelihood the two frameworks are statistically indistinguishable at low  $\ell$  ( $\Delta(-2 \ln L) \approx +1.8$  over  $2 \leq \ell \leq 10$ , inside cosmic variance). So the sector neither confirms nor refutes: it is a genuine parameter-free prediction whose depth places it consistent with  $\Lambda$ CDM within the lowest multipoles’ cosmic variance. The framework’s exposed edges are the acoustic peak structure, where the transfer leaves a phase residual and a fourth peak the comb does not carry (§4.7), and the damping-*scale* signature (§4), a computed, non-reabsorbable effect whose size the handover fixes and whose sign the diffusion integral’s endpoint does. Its sharpest test is the geometric expansion rate itself, carried to the distance ladder and confronted with the DESI DR2 baryon-acoustic dataset at  $\chi^2/\text{dof} \simeq 1$  (§2.4): the rate that carries the full ladder is the rate the spectrum is computed on, so the two are one commitment.

The framework’s standing includes what the standard model has genuinely built and this cosmology has not. Dark matter is inferred from multiple independent, cross-consistent lines—rotation curves, gravitational lensing, the merger offsets, the acoustic-peak structure, the growth of structure—and this cosmology carries the matter density as the bend of the spatial cut without yet removing the component or answering its particle question; that column stands at full weight. Standard nucleosynthesis derives the light-element abundances from one measured parameter, a genuine achievement this cosmology meets on the same footing rather than exceeds: the collapse’s cooling leg is a standard nucleosynthesis run on one inherited datum (§2.4; the cosmogenesis paper [19]), producing helium-4 and deuterium at their observed values—the freeze-out temperature the data demands being the bottleneck the standard rate delivers—and, because the run is standard, carrying lithium-7’s standard threefold over-prediction, the lithium problem shared rather than solved. On the light elements the two frameworks stand even: the same two successes, the same one problem, one progenitor handover fitting the pattern together, and this cosmology’s open edge the last-percent precision against the measured abundances. And the framework is young: the *detailed* dynamics of matter crossing the branch point (its crossing shown well posed and structurally governed, §10), the derivation—as against the measurement—of the inherited data, and the Standard-Model matter *content*—the gauge representations and the mass spectrum, which the framework does not supply, its geometry shown to force only the discrete flavour skeleton [20]—remain to be built (§10; [9]). These are open frontiers, not defects: a theory is not marked down for what it has not built, only for a claim it makes that fails. They are named plainly here because an economy that favours this cosmology on the structure it carries would be worth nothing bought by concealing the structure it does not.

## 12 Conclusion

So, plainly, what this paper claims and what it does not. It claims that the observed expansion history, the computation of the acoustic scale, and the structure of the primordial scalar spectrum are consequences of a single de Sitter geometry read through the causal reassignment its own horizon geometry selects—one seed, grown, the field equations unchanged—and that on the economy of assumption, the criterion by which theory-choice is rightly made ahead of the deciding measurement, this reading is favoured over the assembled standard picture. On the

data axis it claims a more limited thing: that the axis is split—the expansion history fitting the distance ladder, which fixes  $H_0$  with the acoustic angle, and the abundances confirmed, while the acoustic peak structure, computed from that same background and scored by the full-spectrum likelihood, counts against it—with that structure and the damping-tail signature the edges still to be settled (the low multipoles a mild non-discriminating deficit consistent with  $\Lambda$ CDM within cosmic variance). It does not claim to be a complete cosmology: its matter sector, the derivation of its progenitor spectrum, and its fermion content remain to be built. It claims to have shown that the universe we observe is the reading of one geometry, and to have staked the falsifiable commitments by which that claim can be tried.

A final word on the register, because it is the thesis rather than an aside. The beginning this cosmology describes is the branch point at  $r = 0$ —the locus at which the collapse leg of the cosmogenetic contour becomes the expansion leg [9]—and it is not a curvature singularity reached and left behind. *The curvature there is set by the substrate's own scale.* The throat is  $\alpha$ , the defining constant of the maximally symmetric manifold the cut is taken on, and the divergence of the invariants computed in the areal chart is that chart degenerating, exactly as a polar chart degenerates at a pole—not a curvature scale of the geometry [19, 17]. Two further facts stand with it: no finite cosmic-time layer ever reaches the locus, and the continuation through it carries no scale, since  $f \rightarrow -2M/r$  diverges so that the tortoise coordinate  $r_* = \int dr/f$  converges and what is transmitted is fixed by the analytic structure rather than by a length. *The branch point is crossed by continuation, not traversed by a worldline.* The expansion is not a space ejected from a point but the reading, on the near side of that branch point, of a spatial layer still advancing, the branch point itself a collapse that in the frame it bounds is never complete. The distinction the whole programme turns on—between what has finished happening and what is still underway—is thus the distinction the cosmology is built from: the universe is not the collapsed record of a beginning but the still-evolving layer that record represents, read here as one de Sitter geometry.

## Appendix R Computational Receipts

Each computation marked <sup>R</sup> in the text is verified by a runnable receipt in the `receipts/` directory that ships with this work; status OK = verified (traced, run, output matches the claim). This appendix is generated from the receipt index, not maintained by hand.

**P15R1** `P15_the_sky_phase_fit_and_its_uncertainty` [OK]

*sec:intro* (FITS the sky’s acoustic phase from the peak positions and PROPAGATES its uncertainty:  $\ell_n = \ell_A(n + \frac{\phi}{\pi})$  on 220.4, 537.7, 817.3 returns  $\frac{\phi}{\pi} = 0.2405$  against the paper’s 0.2404, and the control’s 0.0043 offset is seven parts in a thousand of the 0.615 disagreement. The quoted  $\sigma_{\text{simeq}0.008}$  is the TWO-multipole value; one multipole gives 0.0046 and twice the separation – so the paper’s figure is the CONSERVATIVE end, and the sentence now says which assumption it carries). *Computes:* least-squares fit; Monte-Carlo propagation at one and two multipoles *Run:* `python3 receipts/P15_CR_cosmology/P15_the_sky_phase_fit_and_`

**P15R2** `P15_the_fixed_point_route_is_closed_by_the_monotone` [OK]

*‘PO-31’ (register), ‘THE\_WISDOM\_LEDGER’ r1942* (the one route the corpus ever named for DERIVING  $A_s$  – a fixed point of a stationary recursion – is CLOSED by ‘PO-40’*’s monotone:* entropy per baryon rises strictly at every crossing, bounded below unconditionally by Landauer at  $k_B \ln 2$ , and a map that strictly increases a coordinate has NO fixed point; independently the chain is finite, so there is no asymptotic limit for a fixed point to be the limit OF). *Computes:* the two reasons checked and shown not to share a premise – one needs only that the increment is positive, the other only that it is bounded below; lap bounds computed on the Landauer floor (1.7e10) and on the present-day increment (1.8e9) *Bound:* bounded: PARTIAL invariance of  $A_s$  alone is not stationarity and SURVIVES, needing the lap map r1942 itself called unbuilt; and this closes a route, not the row *Run:* `python3 receipts/P15_CR_cosmology/P15_the_fixed_point_route_is_closed_by_the_m`

**P15R3** `P15_the_tilt_is_the_wrong_absorber` [OK]

*‘PO-24’; ‘sec:refit-bound’* (the diffusion signature’s *local* mimicked tilt is  $\mathrm{d} \ln(\text{ratio}) / (r^2 - 1) (\ell / \ell_D)^2$ , which runs  $-0.0002$  to  $-1.31$  across plik\_lite’s own  $\ell = 30$ – $2508$  — *a factor of 6989, exactly  $(\ell_{\max} / \ell_{\min})^2$*  — so absorption by a SINGLE tilt is necessarily window-local and the fitted  $\Delta n_s$  is a range-weighted average of a quantity nowhere equal to it). *Computes:* the slope profile and its swing, asserted against the exact  $(\ell_{\max} / \ell_{\min})^2$ ; that the \$185\$-bin ceiling truncates the steep part; and that the reported  $\Delta n_s = -0.0304$  equals the true slope at  $\ell = 381$  alone; CONTROL at  $r=1$  vanishes identically at every  $\ell$  *Bound:* bounded: this shows a constant tilt is the wrong absorber and the displacement is not a stable parameter shift. *It does not run the joint fit*, which remains the row’s discharge *Run:* `python3 receipts/P15_CR_cosmology/P15_the_tilt_is_the_wrong_absorber.py`

**P15R4** `P15_the_joint_fit_leaves_the_gaussian_and_the_bound_was_chi2_not_sigma` [OK]

*‘PO-24’; ‘sec:diffusion-scale’, continuing ‘C62’ and ‘r6409’* (*THE JOINT FIT RUN: THE RESIDUAL IS THE PREDICTED GAUSSIAN TO A CORRELATION OF \$1.000000\$, AND IT SURVIVES LARGER THAN THE BOUND IT WAS TO BE MEASURED AGAINST.* *‘r6409’ predicted a Gaussian-shaped residual no tilt removes; fitting  $A$  and  $\Delta n_s$  jointly against the signature-carrying spectrum on plik\_lite’s own bandpowers and covariance returns exactly that — the residual correlates with the analytic  $S(\ell) - A(\ell/1000)^{\Delta n}$  at \$1.000000\$. It survives at  $\chi^2 = 1.497$  over ‘P15’*’s 185-bin ceiling window ( $\ell = 100$ –\$1758, 171 covered bins). AND THE BOUND WAS QUOTED IN THE WRONG UNITS:* ‘C62’ says  $0.26 \chi^2$  per bin and converts to  $\sigma$  by a square root in its own check; ‘r6409’ restated it as “ $0.26 \sigma$  per bin” and the ‘r6521’ order inherited that — the looser bound quoted as the tighter, since  $0.26 \chi^2$  per bin is  $0.510 \sigma$  per bin. *AND IT IS THE MORE FAVOURABLE OF ‘C62’*’s TWO*, which says in terms that the instrument route (\$0.92\$) is the more faithful of the two and it is the LARGER, so the wash is the weaker claim and is asserted at the weaker bound”. AND THE DISPLACEMENT IS NOT A NUMBER, NOW MEASURED:*  $\Delta n_s = -0.083$  over  $\ell = 100$ –\$1760\$,  $-0.521$  over \$1400\$–\$2508\$ — a factor of \$6.3\$ across windows of the same data, with absorption rising monotonically as the window narrows upward (\$69.8\%\$ to \$97.6\%\$). *In the likelihood’s own units* the signature costs  $+357.3 \chi^2$  with amplitude alone and  $+84.5$  over 171 bins after the tilt is refitted: the fit *improves* the comparison by \$272.8\$,*

absorbing  $76\%$ , and does not remove it.). *Computes:* C62's unit asserted from its own text and its own 'math.sqrt' conversion; r6409's restatement asserted; C62's two-route sentence asserted after stripping comment markers; **\*\*two calibrations — a known injected tilt recovered to  $10^{-11}$ , AND the SCALE, which the first is blind to because a tilt-recovery test is unit-free\*\***; the pipeline asserted to reproduce 'chi2.of' on the same spectrum and window; the residual's correlation with the analytic shape; the residual asserted larger than both C62 routes; the displacement's spread across windows; monotone absorption; the sky-cost split; the red/blue sign difference between the two fits; and the  $r=1$  control returning  $\Delta n_s \sim 10^{-14}$  and  $\chi^2 \sim 10^{-22}$  *Bound:* **\*\*scope: what a tilt ABSORBS, not whether CR fits the sky.\*\*** TT only, no polarisation, no lensing, no  $\ell \sim 100$  from this baseline; two parameters free, so 'C62's scoping stands unchanged — an upper bound on what survives in shape, and no bound at all on the tilt, since  $\omega_b$  moves  $r_D$  itself. **\*\* $n_s$  is INHERITED boundary data here\*\***, so a displacement is a different INPUT and not a conflict; what would be a tension is a constraint on  $n_s$  that does not run through the damping tail, which this does not supply. The baseline is the instrument's control arm, unlensed and ceiling-limited where the sky is neither — 'C62's own stated reason its routes differ by about three *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_joint\_fit\_leaves\_the\_gaussian\_and\_the\_bound\_was\_chi2\_not\_sigma.py

**P15R5** P15.the.potential.at.the.handover.is.not.free [OK]

'PO-13'; 'sec:transmission', and 'C21'/'C19' (the potential at the handover is **\*\*not free and not primordial\*\***: the super-horizon join is conservation of  $\mathcal{R}$ , so  $\Phi_{\text{rm exp}}/\Phi_{\text{coll}} = (3/2)/(5/3) = 9/10$  exactly and  $\Phi$  is constant on the matter-dominated expanding leg — **\*\*which 'P15' already states and receipts, while the instrument sets the potential to its primordial value instead\*\***). *Computes:* the join in exact arithmetic; that the prescription is a FOURTH coding distinct from all three the row measured ( $\$2.43$ ,  $\$5.14$ , dead comb); that  $\Theta$  and  $\Psi$  are then read at ONE locus, which is the row's own discharge criterion; and the omission's size *Bound:* bounded hard: **\*\*the omission is  $10/9$ , not  $2.4$ , and being scale-invariant it moves an amplitude and cannot move a ratio\*\*** —  $P_1/P_2$  and  $\ell_1/\ell_A$  are untouched, so this is NOT the cure and must not be read as one. The spectrum it yields is an instrument run, not made here *Run:* python3 receipts/P15\_CR.cosmology/P15.the.potential.at.the.handover.is.not.free.py

**P15R6** P15.the.one.fitted.number.moves.the.scale.and.not.the.peak [OK]

'PO-13'; 'sec:refit-bound', and the instrument's own 'LATARG' comment (**\*\*THE ONE FITTED NUMBER MOVES THE ACOUSTIC SCALE AND DOES NOT MOVE THE FIRST PEAK, SO THE DEFICIT IS NOT AN ARTEFACT OF WHERE THE PIN WAS PUT.\*\*** 'ACOUSTIC.two\_arm.py' exposed 'LATARG' at 'r2441' expressly to ask **\*\*"GIVEN that  $\ell_A$  is fitted, is the peak deficit an artefact of where the pin was put?"\*\*** and **\*\*the question had never been run\*\***. Scanned over a factor of  $3.6$  in  $z_{\text{onset}}$  on the converged grid:  $\ell_A$  runs  $\$360.6$  to  $\$239.3$  ( $51\%$ ) while  $\ell_1$  sits at  $\$206$ – $\$210$  ( $2\%$ ). **\*\*The deficit is  $\ell_1=206$  against the sky's  $\$220.6$  at every onset.\*\*** **\*\*And the trade is then an IDENTITY\*\***: pinning  $\ell_A$  to the measured  $\$301.6$  costs  $\$6.59\%$  on  $\ell_1/\ell_A$ , pinning  $\ell_1/\ell_A$  to the sky costs  $\$6.59\%$  on  $\ell_A$  — the same number twice, because it is one deficit booked against either of two observables. **\*\*And "both pinned" is unreachable rather than skipped\*\***: under the symmetric convention the control's scale approaches  $\$301.4$  from above, so 'LATARG'  $=301.6$  lies below its floor and 'brentq' reports same-signed endpoints — the control ARRIVES where this arm SPENDS its fitted onset to get.). *Computes:* that the configuration reproduces 'P15's own quoted  $\ell_1=206$  BEFORE its scan is read; that  $\ell_A$ 's range exceeds  $\$40\%$  and  $\ell_1$ 's is under  $\$5\%$ ; that  $\ell_A$  falls monotonically; that  $\ell_1$  misses the sky at EVERY onset and the misses spread by under  $\$3$  points; that the two pinnings agree to  $10^{-3}$ ; that the ratio nonetheless crosses the sky, so reading it alone gives the opposite answer; and that the control's floor sits below 'LATARG' *Bound:* **\*\*scope: driven, 'LMAXL=520', 'LSTEP=2', each arm on its own geometry throughout.\*\*** It does not shrink the deficit and does not say the construction is wrong — it says  $\$0.6830$  and  $\$-6.6\%$  are two spellings of one number, and that no choice of the one fitted number touches it. The 'LMAXL=1300' cross-check is **\*\*run and opt-in\*\*** ('cutoff', **\*\*measured 1207s a point\*\*** against the scan's  $\$92$ s,  $\$943$  modes against  $\$376$ ):  $\ell_1=206$  at both ends, while the HIGHER peaks do move — so the invariance is the first peak's and not the spectrum's *Run:* python3 receipts/P15\_CR.cosmology/P15.the.one.fitted.number.moves.the.scale.and.not.the.peak.py

**P15R7** P15\_the\_symmetric\_comparison\_was\_never\_runnable\_and\_the\_quarter\_was\_two\_fifths [OK]

*‘PO-13’; ‘sec:refit-bound’* (\*\*THE SYMMETRIC UNDRIVEN COMPARISON ‘r4549’ PROMISED AND NEVER LANDED — ITS THREE CLAIMS HOLD, ITS ONE NUMBER DOES NOT.\*\* ‘r4549’*s own message read* *“60’s receipt for this result is not yet pushed and lands next”*; *\*\*no receipt in the tree ever set ‘LZSTART’, and the sound-horizon half of the convention had NO KNOB AT ALL\*\** until ‘LRSFROM’ was added at ‘r6476’, so its numbers were unreproducible from the tree by anyone including their author. Run now: matching the start alone *\*\*overshoots and changes sign\*\** ( $\$+0.2249\backslash\{\}$ to-0.4719\$, the magnitude doubling); matching the convention too leaves a residue on the far side of zero ( $\$-0.0865\backslash\{\}$ ); and *\*\*the implied mode is identical across those last two rows\*\**,  $\backslash\{\}$ ell\_1=482\$ in both, while the convention moves  $\backslash\{\}$ ell\_A\$ from \$301.4\$ to \$397.1\$. *\*\*But the residue is \$38\backslash\{\}%\$ of the original, not the “quarter” ‘r4549’ reported\*\** — a factor of \$1.5\$, in the direction that makes the dissolution look more complete than it is.). *Computes:* the commit’s reachability asserted FIRST, so a shallow clone cannot turn the two commit-pinned checks into a vacuous pass; that ‘r4549’ touched no file under ‘receipts/’; that no receipt at that revision set either knob; the sign change and the growth in magnitude; the residue’s sign and size; the mode’s identity across the convention change; and that the residue exceeds a third *Bound:* *\*\*scope:* undriven on both arms, ‘LMAXL=520’, ‘LSTEP=2’. *\*\* It does \*\*not\*\** re-open ‘r4549’*s verdict, which survives the correction — two fifths is still a minority and still on the far side of zero. \*\*The structural finding is the shared one\*\*:* a convention moves  $\backslash\{\}$ ell\_A\$ and leaves  $\backslash\{\}$ ell\_1\$ at \$482\$; the fitted number moves  $\backslash\{\}$ ell\_A\$ and leaves  $\backslash\{\}$ ell\_1\$ at \$206\$. Both times the denominator moved and the mode did not *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_symmetric\_comparison\_was\_never\_runnable\_and\_the\_quarter\_was\_two\_f

**P15R8** P15\_the\_angular\_momentum\_sector\_is\_not\_a\_propagating\_mode [OK]

*‘PO-31’*s first computation; ‘sec:coherence’*s throat tower** (\*\*the damping removes the anisotropic signature, not \$J\$ — because \$J\$ is not among the modes the tower is a statement about.\*\* ‘P15’*s*  $\backslash\{\}$ nu<sup>2</sup>= $\backslash\{\}$ tfrac14- $\backslash\{\}$ ell( $\backslash\{\}$ ell+1)\$ is reproduced exactly, and it is a computation about a *\*\*propagating field mode\*\**, one with a mass that can be heavy. But in the standard multipole decomposition  $\backslash\{\}$ ell=0\$ carries \$M\$, *\*\*\$* $\backslash\{\}$ ell=1\$ axial carries \$J\$ — the slow-rotation limit, a STATIONARY solution, non-propagating*\*\** — and only  $\backslash\{\}$ ell $\backslash\{\}$ ge2\$ radiates.  $\Rightarrow$  *\*\*\*Applying a principal-series argument there assigns a decay rate to a conserved charge.\*\*\** And ‘P09’ already carries the concept as its *\*\*“non-radiative skeleton ... the wall is where the field begins to propagate”\*\**; it needed connecting, not inventing). *Computes:* ‘P15’*s tower reproduced at*  $\backslash\{\}$ ell=0,1,2,3\$, the sectors by what each carries and whether it propagates, and that \$J\$’s sector is not among the propagating ones *Bound:* *\*\*does NOT establish that \$J\$ survives the leg\*\** — only that the no-hair tower is not the mechanism that would remove it. A charge can still be carried off by matter or torqued, and *\*\*the constrained-sector analysis on the throat is a computation not done here\*\** *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_angular\_momentum\_sector\_is\_not\_a\_propaga*

**P15R9** P15\_the\_tower\_is\_robust\_to\_the\_throat\_being\_squashed [OK]

*‘prop:throat’*s equal radii; ‘sec:coherence’*s tower** (\*\*‘prop:throat’ is a \$J=0\$ construction\*\* — its argument is “direct from \$f\$ at the Nariai mass”, and  $f=1-2GM/c^2r-r^2/\backslash\{\}$ alpha<sup>2</sup>\$ is the *\*\*non-rotating\*\** metric. *\*\*\*So the charge whose survival ‘PO-31’ asks about is the one whose presence would deform the geometry the argument is conducted on\*\*\**, a dependency drawn nowhere. *\*\*But the tower is robust\*\*:* generalising to  $\backslash\{\}$ nu<sup>2</sup>= $\backslash\{\}$ tfrac14- $\backslash\{\}$ lambda $\backslash\{\}$ , $\backslash\{\}$ ell( $\backslash\{\}$ ell+1)\$ with  $\backslash\{\}$ lambda= $r-\backslash\{\}$ rm dS<sup>2</sup>/r\_<sub>S</sub><sup>2</sup>\$, *\*\*\*the*  $\backslash\{\}$ ell=0\$ base is  $\backslash\{\}$ tfrac14\$ for EVERY  $\backslash\{\}$ lambda*\*\*\**, and  $\backslash\{\}$ ell $\backslash\{\}$ ge1\$ stays principal series for any  $\backslash\{\}$ lambda; $1/(4\backslash\{\}$ ell( $\backslash\{\}$ ell+1))\$ — \$1/8\$ at  $\backslash\{\}$ ell=1\$, so the damping survives until the  $\backslash\{\}$ mathrm{dS}\_2\$ radius falls below \$0.354\$ of the  $S^2$ radius, ***a factor of $2.83***). *Computes:*  $\backslash\{\}$ lambda=1$ reproducing ‘P15’*s tower, the*  $\backslash\{\}$ ell=0$ base’s independence of  $\backslash\{\}$ lambda$, and the thresholds at  $\backslash\{\}$ ell=1,2,3$ with the higher multipoles safer *Bound:* ***does not show the conclusion survives rotation*** — ***what*  $\backslash\{\}$ lambda$ a rotating progenitor produces is not computed****, and a rotating near-horizon geometry is generally ***warped*** rather than a direct product, so  $\backslash\{\}$ lambda$ may not even be the right single parameter. It measures a margin and names what would decide it *Run:* python3 receipts/P15_CR_cosmology/P15_the_tower_is_robust_to_the_thro$*

**P15R10** P15\_rotation\_moves\_the\_throat\_away\_from\_the\_damping\_threshold [OK]

*the rotating Nariai locus; ‘r6497’*s named computation** (\*\*rotation moves the throat AWAY from the damping threshold.\*\* Solving the Kerr-de Sitter double root  $\backslash\{\}$ Delta\_r= $\backslash\{\}$ Delta\_r=0\$ gives  $a^2=r^2(\backslash\{\}$ alpha<sup>2</sup>-3r<sup>2</sup>)/( $\backslash\{\}$ alpha<sup>2</sup>+r<sup>2</sup>)\$, which *\*\*\*recovers* ‘prop:throat’*s*

own  $r_N = \alpha/\sqrt{3}$  and  $M = \alpha\sqrt{3}/9$  at  $a=0$  rather than assuming them<sup>\*\*\*</sup>. The family is bounded:  $a_{\max} = (2-\sqrt{3})\alpha$  exactly<sup>\*\*</sup>, attained where  $\Delta_r = 0$  — the *ultracold* point where the double root becomes triple. And  $\lambda = 1$  exactly<sup>\*\*</sup> at  $a=0$ , rising monotonically with  $a$ , against ‘r6497’'s threshold of  $1/8$ ). *Computes*: the  $a=0$  reduction to  $r^2 f(r)$ , the locus recovering both corpus values,  $a_{\max}$  in closed form coinciding with the triple root, and  $\lambda \geq 1$  across the physical branch *Bound*:  $\lambda$  here is a PROXY and it is named<sup>\*\*</sup>: the ratio <sup>\*\*</sup>at the pole<sup>\*\*</sup>. The rotating near-horizon  $S^2$  is warped through  $(r_N^2 + a^2)\cos^2\theta$ , so one ratio does not describe it — ‘r6497’ warned of exactly this. <sup>\*\*</sup>What is established is the direction and its size at one well-defined point<sup>\*\*</sup>; for the conclusion to fail the warping must reverse a trend running the right way by a factor of eight *Run*: `python3 receipts/P15_CR_cosmology/P15_rotation.moves`

**P15R11** `P15_the_warped_throat_keeps_every_mode_above_the_monopole_decaying` [OK]  
‘PO-31’; ‘sec:throat’ and ‘prop:throat’, continuing ‘r6497’/‘r6499’ (<sup>\*\*</sup>THE WARPED HARMONIC PROBLEM, DONE: EVERY MODE ABOVE THE MONOPOLE STAYS PRINCIPAL-SERIES ACROSS THE WHOLE ROTATING FAMILY, AND THE MARGINAL MEMBER IS THE NON-ROTATING ONE.<sup>\*\*</sup> ‘r6499’ left the ratio read at the POLE as a proxy, warning that the rotating near-horizon sphere is warped through  $r_N^2 + a^2\cos^2\theta$  so one ratio may not describe it and  $\ell$  may not be a good label. <sup>\*\*</sup>The near-horizon limit is TAKEN rather than posited<sup>\*\*</sup> and is a warped product, not a direct one:  $\rho_0^2$  multiplies the  $dS_2$  and polar directions alike and the  $dS_2$  is FIBRED over the sphere by  $\tilde{k} = 2r_0 a \Xi / (r_0^2 + a^2)$ , non-zero at every rotating member. A massless scalar still separates, the warp cancelling from the relation between the transverse eigenvalue and the  $dS_2$  operator. <sup>\*\*</sup>AND FOR THE AXISYMMETRIC MODES THE WARP CANCELS ALTOGETHER<sup>\*\*</sup>:  $W \propto \sin\theta$  and  $P \propto \Delta_\theta \sin\theta$  exactly, so the  $m=0$  problem is  $(\Delta_\theta \sin\theta)^2 + E \sin\theta = 0$  — a SPHEROIDAL deformation of Legendre in which  $\rho_0$  never appears. <sup>\*</sup>The order’s worry is answered by the structure, not by the numbers.<sup>\*</sup> The decay condition is  $\nu^2 = \frac{14-L^2 E - L^4 m^2}{k^2}$ , the fibration term non-negative, so rotation can only deepen the damping. <sup>\*\*</sup>TWO EXACT BOUNDS SETTLE IT WITHOUT A SCAN<sup>\*\*</sup>:  $L^2 \geq 1$  on the physical branch (from  $L^2 - 1 = -(3u-1)(u+2)/(3u^2+6u-1)$ ) and  $E_1 \geq 2$  above the monopole (Rayleigh,  $\Delta_\theta \geq 1$  pointwise), each with equality only at  $a=0$  — so  $L^2 E_1 \geq 2$  against the threshold  $\frac{14}{k^2}$ , <sup>\*\*</sup>a factor of eight, attained where the rotation vanishes<sup>\*\*</sup>. The monopole keeps  $\nu^2 = \frac{14}{k^2}$  exactly.). *Computes*: T3 asserted symbolically ( $\Delta_r(a=0) = r^2 f$ , not  $r f$ ); the locus re-derived from  $\Delta_r = \Delta_r' = 0$  for BOTH  $a^2$  and  $M$  and asserted to return ‘prop:throat’'s values at  $a=0$ ;  $r_*$  asserted to be exactly where  $\Delta_r = 0$  (T2); the near-horizon limit asserted equal to the warped ansatz coefficient by coefficient at exact members, with the  $\tau$  rescaling coming out exactly  $1/(r_0^2 + a^2)$  and no stray cross terms;  $W^2$  and  $P^2$  asserted free of  $\rho_0$ ; <sup>\*\*</sup>the solver calibrated against Legendre  $l(l+1)$  at  $a=0$  across  $m=0..4$  BEFORE any warped zero is believed<sup>\*\*</sup>; every sample asserted on the physical branch (T1); the monopole asserted both numerically and EXACTLY (the  $m=0$  flux matrix annihilates constants, so  $E_0=0$  is not a fitted zero); and both closed-form bounds *Bound*: <sup>\*\*</sup>scope: a massless SCALAR on the near-horizon geometry — the same proxy ‘r6497’'s  $\nu^2 = \frac{14-\lambda\ell(\ell+1)}{k^2}$  is built on.<sup>\*\*</sup> It does not decompose gravitational perturbations into axial and polar. <sup>\*\*</sup>It is about whether the DAMPING MECHANISM reaches these modes; it is NOT about whether the  $J$  survives the leg<sup>\*\*</sup> — a charge can still be carried off by matter or torqued, and ‘r6495’'s separation of  $J$  from the propagating tower is untouched. The geometry is the throat’s exact limit, not the collapse leg’s history *Run*: `python3 receipts/P15_CR_cosmology/P15_the_warped_throat_keeps_every_mode_above_the_monopole_decaying.py`

**P15R12** `P15_the_growth_root_is_a_moment_balance_pivoting_on_the_nariai_radius` [ALL PASS]  
*sec:discussion* (<sup>\*\*</sup>THE GROWTH SECTOR IS PARAMETER-FREE AND THE NORMALISATION ROOT IS A MOMENT BALANCE PIVOTING ON THE NARIAI RADIUS.<sup>\*\*</sup> Substituting  $H = (c/a)\coth u$  and  $4\pi G\rho = (3/2)(c/a)^2 \cosh^2 u$  into the growth equation gives <sup>\*\*</sup> $3D'' + 4\coth(u)D' - 2\cosh^2(u)D = 0$  —  $\alpha$  CANCELS<sup>\*\*</sup>, because at Nariai the source is fixed by  $\Lambda$  and the time variable carries  $\alpha$  too. Decaying mode is the rate itself,  $D_- = \coth u$ ; growing mode by reduction of order. <sup>\*\*</sup> $Om_m = \text{sech}^2(u)$  exactly<sup>\*\*</sup> — a clock reading.  $J = (2/3)\cosh^2(u_0)(D+a)$ , matter-domination limit  $D+a \rightarrow 3/5$  giving the analytic endpoint  $J \rightarrow 2/5$ . The identity  $G'/F' = (9/2)Om - 2$  integrates the condition to <sup>\*\*</sup>int a  $Om$  ( $Om -$

2/3)  $du = 0^{**}$ , solved at  $Om = 0.315162424 - ^{**}$ and the pivot  $Om=2/3$  is  $\rho_m/\rho_L = 2$  exactly, at areal radius  $\alpha/\sqrt{3} = r_N^{**}$ ). *Computes:* r4253 *Bound:* 61 *Run:* python3 receipts/P15\_CR.cosmology/P15.the.growth.root.is.a.moment.balance.pivoting.on.the.nariai.radius.py

**P15R13** P15.the.growth.root.on.the.leaf.rate.is.not.a.moment.balance.and.the.gap.stands [OK]

*sec:discussion* (\*\*PO-38: THE MOMENT-BALANCE FORM DOES NOT SURVIVE ON THE LEAF RATE, THE ROOT BARELY MOVES, AND IT MOVES THE OPPOSITE WAY FROM WHAT THE CONTINUED INTEGRAL IMPLIES.\*\* PO-38 asked for the growth normalisation root re-solved on the LEAF rate and warned that radiation may break the cancellation that made the growth equation parameter-free. \*\* (1) IT DOES, EXACTLY. \*\* The derivation rests on "the decaying mode IS the rate"; applying the growth operator to  $D = H$  with radiation present gives  $L[H] = (32/3) \pi G H \rho_r$ , which vanishes iff  $\rho_r = 0$ . With that gone the reduction of order, the  $G'/F'$  identity and the balance all go.  $Om_m = \text{sech}^2(u)$  fails on the leaf rate ( $9.3e-4$  at  $v=0.2$ ) and the balance integral returns  $-1.71e-4$  at the leaf root where the stacking one vanishes to  $1e-13$ .  $\alpha$  still cancels, so the equation stays scale-free but acquires  $Om_r$ : the loss is parameter-freedom, not scale-freedom. \*\* (2) WHAT REPLACES IT \*\* is the growth ODE solved on the rate, with the Meszaros mode  $1 + (3/2)y$  as an EXACT initial condition (verified to  $1.6e-12$ ) and the published  $(2/5)a/Om$  convention calibrated against the closed form where the closed form is valid ( $2e-12$ ). \*\* (3) THE ROOT: OUTCOME 3, WITH A SIGN. \*\* Leaf  $u_0 = 1.180038880$  against the stacking  $1.180309372$  – a move of  $1.1\%$  of the gap, taking the offset from  $0.704$  to  $0.713$  sigma against the corpus's measured  $x_0 = 1.6648 \pm 0.0467$ . \*\*The 2% is not an artefact of the rate; the gap stands and wants a mechanism\*\* – and the shift goes FURTHER from the measurement, not closer. \*\* (4) THE CONTINUED INTEGRAL GETS THE DIRECTION BACKWARDS. \*\* The route behind the recorded  $0.99934$  moves the root TOWARD the measurement ( $0.693$  sigma) where solving the equation moves it away, and by  $1.7x$  the size in  $u_0$ . \*\*That figure does not measure how far the root moves; it measures a closed form failing on a rate it does not solve.\*\* \*\* (5) THE PIVOT SPLITS RATHER THAN MOVES. \*\*  $Om_m = 2/3$  is the acceleration turnover only when matter and Lambda are the whole content; radiation decelerates too, so  $q = 0$  needs  $Om_m + 2 Om_r = 2 Om_\Lambda$ . On the leaf rate the two loci separate by  $du = 5.08e-4$  and the Nariai radius sits BETWEEN them. \*\* (6) AND A FLATNESS SLIP IN THE EXISTING SCOPE-(b) CHECK: \*\* its leaf integrand carries  $Om_\Lambda = 1 - Om$  rather than  $1 - Om - Om_r$ , so the fractions sum to  $1 + Om_r$ ; kept flat its own figure reads  $0.99940$  and the slip alone is  $10.4\%$  of the shift it is quoted for.). *Computes:*  $L[H]$  asserted symbolically equal to  $(32/3) \pi G H \rho_r$  and asserted to vanish at  $\rho_r = 0$ ; the ODE asserted to reproduce the published integral to  $1e-10$  under the  $(2/5)a/Om$  convention; the Meszaros form asserted EXACT to  $1e-9$  with Lambda off; the stacking root asserted to reproduce the published  $1.180309372$ ; the leaf shift asserted under  $2\%$  of the gap; the leaf root asserted FURTHER from the measurement and the continued integral asserted CLOSER – the two signs asserted opposite; the recorded  $0.99934$  asserted to be the flatness-slipped value and  $0.99940$  the flat one, with the slip asserted comparable to the effect;  $Om_m = \text{sech}^2 u$  asserted to FAIL on the leaf rate; the balance asserted to vanish on the stacking rate and NOT on the leaf; the stacking pivot asserted to be  $r_N$  exactly, the leaf pivot and turnover asserted distinct, and  $r_N$  asserted to lie between them. \*\*CONTROL THAT FIRES: the  $(2/5)$  convention and the Meszaros IC are each checked against an independently known answer before either is used\*\* – the closed form where it is valid, and the analytic  $1 + (3/2)y$  – so a normalisation error would surface before it could move the root *Bound:* \*\* COMPUTES A CONSEQUENCE, NOT AN ASSIGNMENT. \*\* It does not claim the leaf rate is the right rate for this object; the corpus's rule assigns it and this computes what follows.  $Om_r$  is held FIXED at  $8.5e-5$  throughout and NOTHING is tuned. \*\*No mechanism for the residual 2% is offered\*\* – the row asked whether the rate explains it and the answer is that it does not. NOT CLAIMED: that the balance has any leaf-rate analogue; what is shown is that this one does not hold there *Run:* python3 receipts/P15\_CR.cosmology/P15.the.growth.root.on.the.leaf.rate.is.not.a.moment.balance.and.the.gap.s

**P15R14** P15.the.rate.has.one.parameter.and.one.epoch [ALL PASS]

*sec:flatlcdm* (\*\*THE RATE HAS ONE PARAMETER AND ONE EPOCH, NOT TWO PARAMETERS.\*\* Three statements in P15 could not all be true of one quantity: that its two parameters are  $\alpha$  and "the OFFSET  $x_0$ "; that "at Nariai the mass is no longer free" so BOTH factors of the expansion law are set by Lambda; and that the density ratio is a CLOCK. The offset is fixed twice over – the horizon cubic's DOUBLE root at the Nariai mass, and

P17's constant ledger  $2M = \alpha((r_0/\alpha)-(r_0/\alpha)^3)$  – and  $**x_0 = r(\tau_0)/r_N$  identically at every epoch\*\*, the expansion read in the unit the geometry supplies. \*\*Decisive check: if  $x_0$  were the geometric offset ratio it would be  $1/\sqrt{3}$ , and P15's own dictionary would return  $\Omega_m = 0.91$ , not a cosmology.\*\*). *Computes:* r4251 *Bound:* 61 *Run:* python3 receipts/P15\_CR\_cosmology/P15.the\_rate\_has\_one\_parameter\_and\_one\_epoch.py

**P15R15** P15\_the\_growth\_normalisation\_root\_has\_no\_feature\_at\_the\_physical\_point [ALL PASS]

(resources) (\*\*THE GROWTH-NORMALISATION ROOT HAS NO FEATURE AT THE PHYSICAL POINT, so the agreement is a coincidence – and it is the STANDARD MODEL'S.\*\* Embedding J in its natural families (exponent p, equation of state w, curvature) gives a root that is smooth, monotone and feature-free through  $p=3/2$ ,  $w=-1$  and flatness, with the physical point NOT stationary in any direction. \*\*A property of the matter-Lambda crossover would leave a mark there; there is none.\*\* The artefact reading was never available (J is forced and dimensionless), and the crossing is steep rather than broad-band ( $dJ/d\Omega_m = -2.45$ ). Discriminating:  $J=0.99966$  at Planck's 0.3153,  $J=1.02142$  at this corpus's DESI DR2 fit of 0.3066.). *Computes:* r4247 *Bound:* 61 *Run:* python3 receipts/P15\_CR\_cosmology/P15.the\_growth\_normalisation\_root\_has\_no\_feature\_at\_the\_physical\_point.py

**P15R16** P15\_verify\_geometry [OK]

*prop:flat / prop:throat / prop:transmission* (three geometric props DERIVED from primitives: (a1) const-tau 3-metric  $dr^2 + r^2 d\Omega^2$  has Riemann=0 (flat  $R^3$ ); (a2) Nariai near-horizon  $dS_2 \times S^2$ ,  $f(r_{\text{star}})=f'(r_{\text{star}})=0$ ,  $f''=-2\Lambda$ , both radii  $1/\sqrt{\Lambda}$ ); (a4) tortoise integrals simple- $\int \log(\exp)$  vs double- $\int \text{power-law}$ ). *Computes:* Riemann from Christoffels; Nariai conds from f; two tortoise integrals; checks intrinsically discriminating *Bound:* Kretschmann bonus derived too *Run:* python3 receipts/P15\_CR\_cosmology/P15\_verify\_geometry.py

**P15R17** P15\_verify\_numeric [OK]

*prop:amplitude / prop:subhorizon* ((a5) substrate  $dS$  vacuum power  $(\hbar H_{\text{Lam}}/E_P)^2 = 9.5e-123$  vs  $A_s = 2.1e-9$  - $\int$   $\sim 113$  orders below, robust to  $O(10)$  prefactor; (a7) comoving Hubble  $k$  at seam  $0.0104/\text{Mpc}$  vs peak  $\pi/r_s = 0.0217/\text{Mpc}$  - $\int$   $2.1\times$  sub-horizon). *Computes:*  $\Lambda +$  vacuum power +  $k$  derived from obs inputs; matches paper;  $[E]/[R]$  marked *Bound:* anchor 6 low-ell leading-order  $[R]$  only *Run:* python3 receipts/P15\_CR\_cosmology/P15\_verify\_numeric.py

**P15R18** P15\_verify\_coherence\_comb [OK]

*sec:coherence* (coherence comb: common seam phase - $\int$  sharp comb  $\Delta\ell = \pi D_C/r_s = 296$  (peaks 296,592,888), peak/trough  $\sim 1e5-1e6$ ; random phase per mode - $\int$  washed out, contrast  $\sim 1$  (400 realisations). Null seam's one characteristic datum/mode = one phase/mode (\*\*SCOPE: this receipt propagates nothing – it writes down  $\cos(k r_s)^2$  with  $r_s$  and  $D_C$  as hardcoded literals, exactly as its own scope note says. It establishes the COHERENCE MECHANISM and NOT the peak spacing.)). *Computes:* ACFM coherence demo on the seam; coherent-vs-incoherent control; matches paper *Bound:* mechanism only; full transfer open  $[R]$  *Run:* python3 receipts/P15\_CR\_cosmology/P15\_verify\_coherence\_comb.py

**P15R19** P15\_cr\_collapse\_driving [OK]

*sec:coherence* (acoustic driving as linear oscillator  $\Theta'' + \Theta = -\Psi$ : undriven ( $\Psi=\text{const}$ )  $a_0=0.500$ , driven (radiation)  $a_D=0.999$ , boost  $a_D/a_0=2.00$  (known LCDM boost); baryon-loaded driven-equiv 1.60 vs adiabatic 1.10). *Computes:* ODE integration; undriven- $\Psi=\text{const}$  control; matches known boost *Bound:* time-symmetry of contracting phase argued (magnitude proven in next receipt) *Run:* python3 receipts/P15\_CR\_cosmology/P15\_cr\_collapse\_driving.py

**P15R20** P15\_time\_reversal\_driving [OK]

*sec:coherence* ( $|\Psi(\omega=1)|$  resonant driving magnitude is time-reversal invariant:  $|\Psi(x)|=|\Psi(-x)|$  at  $w=1$  on the radiation transfer (0.037049) AND a random control (1.411379); so CR's time-mirrored collapse driving = LCDM's magnitude, peak heights match by structure). *Computes:* numerical Fourier magnitude at  $w=1$ ; radiation + random control; complex/phase-not-invariant discriminator *Bound:* built to close the paper's 'confirmed numerically ... random control' claim *Run:* python3 receipts/P15\_CR\_cosmology/P15\_time\_reversal\_driving.py

**P15R21** P15\_camb\_reference [OK]

*sec:coherence* (\*\*!! MARKED c54.160: the closing annotation states the paper's anchors as  $P2/P1 \sim 0.46$  and  $P3/P1 \sim 0.45-0.46$  while the file itself computes 0.452 and 0.443 and the

paper now prints 0.45 and 0.44.\*\* A stale annotation contradicting the same file's own digit-by-digit claim; both figures are now pinned). *Computes:* CAMB TT spectrum + peak-finder; matches paper *Bound:* needs camb; CR-matches-it carried by driving receipts *Run:* python3 receipts/P15\_CR\_cosmology/P15\_camb\_reference.py

**P15R22** P15\_damping\_ratio\_clean [OK]  
*sec:coherence* (\*\*!! CORRECTED c54.155: theta\_D/theta\_\* CR/LCDM IS 1.0816 (+8.2%), NOT THE 1.0785 (+7.9%) THIS ROW AND THE RECEIPT'S OWN DOCSTRING RECORD\*\* – both stale from the z\_onset 6850 -i 6797 retune. The paper's 'approx 1.08' brackets the live value correctly). *Computes:* CAMB x\_e(z); r\_s+r\_D both rates; depends on CAMB; RATIO cancels conventions *Bound:* absolute theta\_D ~7% below CAMB (k\_D convention), cancels in ratio *Run:* python3 receipts/P15\_CR\_cosmology/P15\_damping\_ratio\_clean.py

**P15R23** P15\_damping\_reabsorption [OK]  
*sec:coherence* (~9% damping shift not reabsorbable: r\_D ~ omega\_b^{-0.305} (paper -0.31), d ln r\_D/d ln omega\_c=-0.295; cancelling +8.9% needs omega\_b +29% vs ~1% BBN+height prior -i 29x too big; 1% omega\_b shifts r\_D ~0.3%; residual ~8.2%). *Computes:* finite-diff levers on CAMB thermodynamics vs BBN+height priors *Bound:* broader multi-sigma tension is conditional (full transfer open); paper takes only settled part *Run:* python3 receipts/P15\_CR\_cosmology/P15\_damping\_reabsorption.py

**P15R24** P15\_zonset\_determinations [OK]  
*sec:tensions* (\*\*!! MARKED c54.160: the paper reported the rounded-l\_\* pinning as recovering 6873 and this receipt returns 6844 on the current integrals.\*\* The paper now carries both numbers and notes that 6844 sits NEARER to the retired 6850, which sharpens the withdrawal rather than softening it). *Computes:* three determinations computed on one stated convention; H0-independence of (1) exact to 5 digits; rigging of (2) shown by the l\_\* that 6850 itself returns *Bound:* (3) recorded as a structural observation, not a refutation: the datum is inherited and order-unity either way *Run:* python3 receipts/P15\_CR\_cosmology/P15\_zonset\_determinations.py

**P15R25** P15\_the\_ratio\_is\_the\_onset\_in\_imported\_units [OK]  
*sec:tensions* (WHAT  $\rho_r/\rho_m \approx 2$  IS, AND WHAT IT IS NOT: \*\*NEITHER A HIDDEN KNOB NOR A SECOND MEASURED DATUM, BUT THE ONE FITTED QUANTITY  $z_{\text{onset}}$  RE-EXPRESSED IN UNITS OF A PHYSICAL MATTER DENSITY THE CONSTRUCTION DOES NOT DETERMINE.\*\* The rate carries  $\Omega_m$  (fitted to BAO); the sound speed carries  $\omega_b$  and  $\omega_\gamma$  (measured directly, framework-free);  $z_{\text{onset}}$  is the one fitted number – and the PHYSICAL  $\omega_m = \Omega_m h^2$  enters NO prediction. But  $\rho_r/\rho_m$  at onset is  $\omega_r(1+z_{\text{onset}})/\omega_m$  and equality is  $1+z_{\text{eq}} = \omega_m/\omega_r$ , so \*\*both divide by exactly that quantity\*\*. \*\* KNOB-FREENESS SURVIVES AND IS THE LOAD-BEARING HALF:  $z_{\text{onset}} = 6747.3$  at  $H_0 = 67.4, 70.0, 73.0$  and  $74.0$ , to the digit –  $H_0$  is a common factor in  $r_s$  and  $D_M$  and cancels from  $\theta_*$ , so the fitted quantity does not move between the two  $H_0$  the tension is between and cannot be absorbing it. \*\* WHAT DOES NOT SURVIVE IS THE EXACTNESS OF 'ABOUT TWICE EQUALITY': at the fitted  $\Omega_m = 0.3066$  the ratio is 2.01 at  $H_0 = 67.4$  and 1.71 at  $H_0 = 73$ , the two readings crossing at  $H_0 \sim 68$  where  $\Omega_m h^2$  reproduces the CMB-inferred  $\omega_m = 0.143$  – and the papers' headline is that the cosmology fits at the DIRECTLY measured  $H_0 = 73$ . \*\* So the datum is an order-unity BAND (1.7-2.0), not a determined number; the eta-analogy and the BAO confrontation are untouched (the latter uses distance ratios and never  $\omega_m$ ).). *Computes:*  $z_{\text{onset}}$  recomputed from the geometric rate at four values of  $H_0$  by root-finding on the measured acoustic angle, its spread asserted below 1 part in 6000; the crossing  $H_0$  asserted in 67-69; the  $H_0=73$  ratio asserted at 1.71 *Bound:* \*\* THIS IS A UNIT ANALYSIS, NOT A NEW MEASUREMENT. \*\* It does not touch the rate, the fit, or the abundances; and the CMB-inferred  $\omega_m$  it compares against is itself derived in a radiation-included analysis, which is the point rather than an objection – the residual measures how far the two analyses'  $\omega_m$  must differ *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_ratio\_is\_the\_onset\_in\_imported\_units.py

**P15R26** P15\_the\_two\_data\_are\_one [OK]  
*sec:tensions* (WHAT THE 'SINGLE INHERITED DATUM'  $\rho_r/\rho_m \sim 2$  ACTUALLY IS, AND WHAT THE NATURALNESS ARGUMENT OFFERED FOR IT ACTUALLY DESCRIBES. \*\* (1) IT IS NOT A DATUM DISTINCT FROM eta. \*\* On any thermal history  $\rho_r/\rho_m(T) = [1+nu](\pi^4/30 \text{ zeta}^3) T / [\eta (w_m/w_b) m_N]$ , which returns \*\*1.99 at  $T = 1.6 \text{ eV}$ \*\* – the quoted value to one per cent, from standard thermodynamics and NO feature of this construction.

Given eta and the measured matter-to-baryon ratio, both inherited for the composition regardless, **\*\*the ratio at onset IS the onset\*\***: the handover supplies ONE composition datum and the cosmology fits ONE parameter. **\*\* (2) AND THE NATURALNESS ARGUMENT DESCRIBES AN EPOCH NINE ORDERS FROM THE HANDOVER. \*\***  $\rho_r = \rho_m$  at  $T = 0.806$  eV; the handover sits FOUR ORDERS ABOVE the deuterium bottleneck (0.07 MeV) because that is what the nucleosynthesis computed on it requires, so the ratio there is  $\sim 9e8$  – and ‘a handover carrying radiation of order the matter density’ would be a handover with **\*\*eta  $\sim 0.5$ \*\***: no light elements, no acoustic peaks. **\*\* So the naturalness question runs the OTHER WAY: what must be explained is a BILLION photons per baryon, which is the baryogenesis-analogue question already named as open, and the binding-energy scale ( $GM/Rc^2 = 1/2$ ) offers no head start toward it. \*\*** P7’s ‘the frontier is two data and not one’ is corrected to one inherited datum plus one fitted parameter.). *Computes*: the identity evaluated at the onset temperature and asserted against the quoted datum to better than 5%; the equality temperature and the ratio at the bottleneck, at P7’s own handover and at the rest mass computed; the eta that O(1) would require at each epoch computed and contrasted with the measured  $6.1e-10$ ; and the papers asserted to carry the withdrawal and the corrected accounting *Bound*: **\*\* THIS CORRECTS AN ACCOUNTING AND WITHDRAWS AN ARGUMENT; IT TOUCHES NO RESULT. \*\*** The rate, the fit, the baryon-acoustic confrontation and the abundances are all untouched – and the accounting moves in the construction’s favour, from two inherited data to one *Run*: `python3 receipts/P15_CR_cosmology/P15_the_two_data_are_one.py`

**P15R27 P15\_full\_transfer\_verdict** [OK]  
*sec:coherence* (high-ell deficit UNDER THE SHORTCUT (assume peaks=LCDM + apply +8.9% damping):  $C_L$  CR/LCDM = 0.979/0.920/0.829/0.717/0.594 at  $l=500-2500$ ; cited to show the shortcut is illegitimate (assumption (A) is the unbuilt part)). *Computes*: exp damping on CAMB LCDM  $C_L$ ; conditional  $\chi^2$  *Bound*: verdict explicitly CONDITIONAL/open (own banner); paper honors scoping *Run*: `python3 receipts/P15_CR_cosmology/P15_full_transfer_verdict.py`

**P15R28 P15\_verify\_closedS3\_nonsync** [OK]  
*sec:largescale* (low-ell floor mechanism+location: discrete closed- $S^3$  source ( $k_L = \sqrt{L(L+2)}/r_0$ ,  $L_{\ell}=2$ ) through FLAT  $j_{\ell}(k_L D_C)$ , stretch  $D_C/r_0=2.75$  pushes  $L=2$  to  $\ell \sim 7.8$ ;  $\ell(\ell+1)C_{\ell\ell}=0.12/0.10/0.20/0.6$  at  $\ell=2..7$ , recovered by  $\ell \sim 8$ ). *Computes*: discrete-source flat-Bessel sum; flat-limit  $\chi^2_0$  control passes *Bound*: ordinary SW only - $\ell$  deeper than Boltzmann depth (routed to that receipt) *Run*: `python3 receipts/P15_CR_cosmology/P15_verify_closedS3_nonsync.py`

**P15R29 P15\_verify\_lowell\_exact\_measure** [OK]  
*sec:largescale* ((1) HZ weight  $w_L=(L+1)/(L(L+2))$  DERIVED as closed- $S^3$  degeneracy  $\beta^2 \times$  per-mode  $1/(\beta(\beta^2-1))$  ( $g^*P=w_L$  exactly;  $w_L \cdot L_{\ell} 0.998$  continuum); (2)  $D2/D3=1.162$  stable across measure family - $\ell$  floor starves  $\ell=2,3$  together (octopole slightly more)). *Computes*: first-principles derivation + flat-limit + measure-robustness + wrong-measure contrast *Run*: `python3 receipts/P15_CR_cosmology/P15_verify_lowell_exact_measure.py`

**P15R30 P15\_verify\_doppler\_lowell** [OK]  
*sec:largescale* (Doppler term (super-horizon velocity through  $j_{\ell}$ ’)) does NOT differentiate  $\ell=2$  from  $\ell=3$ : CR/LCDM ratio  $l^2 \sim 0.27$ ,  $l^3 \sim 0.21-0.22$  across 4 treatments (none/coh+/coh-/quad),  $D2/D3 \sim 1.29$  octopole slightly more suppressed; continuum validator reproduces LCDM rise). *Computes*: SW+ISW+Doppler transfer, 4 combination modes; continuum check; not tuned *Bound*: bare depth (milder exact depth = Boltzmann receipt) *Run*: `python3 receipts/P15_CR_cosmology/P15_verify_doppler_lowell.py`

**P15R31 P15\_verify\_lowell\_boltzmann** [OK]  
*sec:largescale/sec:scope* (exact low-ell depth via genuine Boltzmann transfer: discrete closed- $S^3$  source through CAMB’s exact  $\Delta_L(k)$  (SW+ISW+Doppler); CR/LCDM = 0.473 ( $l=2$ ), 0.410 ( $l=3$ ), recover by  $l^7$ , octopole/quadrupole 0.867 (matches paper 0.47/0.41). GATE: continuum reproduces CAMB  $C_L$  to 4 figures). *Computes*: CAMB transfer  $\times$  discrete source, Riemann-sum vs integral; gate-validated;  $r_0$ -stable; ISW-retained diagnosis *Bound*: needs camb (high acc) *Run*: `python3 receipts/P15_CR_cosmology/P15_verify_lowell_boltzmann.py`

**P15R32 P15\_confront\_lowell\_data** [OK]  
*sec:largescale* (low-ell data confrontation (corrected depths): CR 0.47 sits ABOVE the sharp observed quadrupole ( $\sim 0.2$ ), CR 0.41 mildly below the near-LCDM octopole ( $\sim 0.6-1.0$ ) - $\ell$  matches neither, a WASH; old 0.22/0.20 ‘striking match’ superseded). *Computes*: CR (Boltzmann depths)

vs literature observed multipoles *Bound:* body corrected to r962 depths; likelihood in next receipt  
*Run:* python3 receipts/P15\_CR\_cosmology/P15\_confront\_lowell\_data.py

**P15R33** P15\_verify\_lowell\_likelihood\_v2 [OK]

*sec:largescale/sec:scope* (low-ell cosmic-variance likelihood: exact scaled- $\chi^2_{2l+1}$ , CR/LCDM ratio  $\Delta(-2\ln L)_l = (2l+1)[r_l(1/d_l-1) + \ln d_l]$ ;  $l=2$  rewards CR -2.63,  $l=3/4$  penalise +1.31/+2.10, central total +1.8, octopole range +0.3..+3.8 -; wash inside cosmic variance). *Computes:* exact likelihood with corrected Boltzmann depths; matches paper *Run:* python3 receipts/P15\_CR\_cosmology/P15\_verify\_lowell\_likelihood\_v2.py

**P15R34** P15\_the\_low\_ell\_minimum\_is\_at\_ell\_four [OK]

*sec:largescale/sec:scope* (\*\*SCOPE: THIS FILE CROSS-VALIDATES NOTHING.\*\* Its arm A is the likelihood receipt's stored dict and \*\*its arm B is the PAPER'S OWN FIGURE CAPTION, read by regular expression\*\* — so the comparison is against the citing paper's typography, and the quartet 0.42/0.35/0.29/0.52 is banked nowhere. \*\*What it does verify is that arm A's minimum sits at  $l = 4$  and that the standing receipts always carried it, which is what it is now cited for.\*\* The actual second arm is 'P15\_the\_second\_arm\_actually\_run': it agrees on shape and NOT on depth, so the quartet 0.42/0.35/0.29/0.52 is not banked. What the file establishes is THE LOW-MULTIPOLE DEFICIT'S SHAPE, AND WHERE THE SECTOR'S WEIGHT ACTUALLY FALLS. \*\* IT IS NOT A MONOTONE RECOVERY FROM THE QUADRUPOLE BUT A DIP WHOSE MINIMUM IS AT  $ell=4$  — and the quadrupole and octopole, the two multipoles the paper quoted, are its two SHALLOWEST points. \*\* \*\* THE SHAPE IS CROSS-VALIDATED BY TWO INDEPENDENT BOLTZMANN TREATMENTS of the same discrete source: a standard code's exact  $\Delta_l(k)$  at the discrete  $k_L$  gives 0.47/0.41/0.36/0.68 at  $ell=2,3,4,5$  and the programme's own photon hierarchy gives 0.42/0.35/0.29/0.52 — SAME minimum, SAME recovery, depth ratio 1.13-1.30, which is the size of the residual *sec:instrument* declares. The shape is cross-validated; the depth carries their spread. \*\* \*\* AND THE  $ell=4$  MINIMUM WAS NEVER NEW: 'P15\_verify\_lowell\_boltzmann.py' has tabulated 0.356 at  $ell=4$  since r1395 and the likelihood receipt has consumed it ever since — the corpus's failure was one of QUOTATION, not of computation. \*\* \*\* THE OCTOPOLE VERDICT: on the exact cosmic-variance likelihood the quadrupole REWARDS the construction (-2.63), the octopole penalises it at +1.31, and the LARGEST single-multipole penalty in the sector is the +2.10 at  $ell=4$  — the prediction's own minimum, and the one multipole of the four the large-angle literature does not treat as anomalous. Sector total unchanged at +1.8, a wash inside cosmic variance. \*\*). *Computes:* both arms' depth tables read off their own sources (the likelihood receipt's dict; the paper's caption) and their argmin asserted equal to  $ell=4$ ; the  $ell=4$  depth asserted present and deepest in the standing r1395 receipt; the per-multipole likelihood re-derived from the depths and its argmax asserted at  $ell=4$  with the total asserted at +1.8; the exact/WKB transmission spread asserted below 5%; and all eight corpus sites asserted to carry the corrected statement *Bound:* \*\* DEPTH IS QUOTED TO A TWO-CODE SPREAD AND NOT BETTER. \*\* And nothing here relieves the observed sky's selective suppression of  $ell=2$  ALONE by a factor of order three, which this construction does not produce; that stays inside cosmic variance and belongs to the likelihood-level comparison *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_low\_ell\_minimum\_is\_at\_ell\_four.py

**P15R35** P15\_two\_ranges\_ratio [OK]

*sec:background (two ranges)* (the local Hubble-Eddington radius  $r_* = (M \alpha^2)^{1/3}$  and the expansion's amplitude  $A = (2M \alpha^2)^{1/3}$  are ONE length in the fixed ratio  $2^{1/3}$ , for every  $M$ ;  $A$  is the matter-Lambda equality radius). *Computes:*  $r_*$  from  $K_G = 1/\alpha^2 - M/r^3 = 0$ ;  $A$  from  $\rho_m/\rho_\Lambda = \text{csch}^2(3\tau/2\alpha)$  at equality; ratio  $A/r_* = 2^{1/3}$  symbolically, independent of  $M$  and  $\alpha$  *Bound:* the r1608 identity; the four-ways reading (turnaround, amplitude, equality, structure boundary) is the paper's *Run:* python3 receipts/P15\_CR\_cosmology/P15\_two\_ranges\_ratio.py

**P15R36** P15\_the\_second\_arm\_actually\_run [OK]

*sec:largescale* (\*\*THE SECOND BOLTZMANN ARM, ACTUALLY RUN — THE SHAPE CROSS-VALIDATES AND THE DEPTH QUARTET DOES NOT.\*\* P15 claimed the low-multipole deficit had been run a second time on the programme's own photon hierarchy, returning 0.42/0.35/0.29/0.52. \*\*The receipt cited for it does not run the hierarchy: it reads arm A from a stored dict and reads arm B OUT OF THE PAPER'S OWN FIGURE CAPTION, by regular expression\*\* — so the cross-validation compared a table against the citing paper's typography and the quartet was banked nowhere. This file drives 'HIER\_photon\_hierarchy' itself: evolves it on a  $k$ -grid containing the discrete ladder  $k_L = \sqrt{L(L+2)} \times 2.75/D_M$  plus a dense continuum grid, projects its own source

terms to  $\Delta_L(k)$  for  $l = 2..44$ , and forms the discrete  $w_L$  sum against the continuum  $\ln k$  integral, each normalised to its own  $l = 25-40$  plateau. **\*\*SHAPE:** minimum at  $l = 4$  and recovery by  $l \sim 7$ , independently of the CAMB arm and of the caption. **\*\*DEPTH:** 0.494/0.243/0.184/0.608 against CAMB's 0.473/0.410/0.356/0.676 — close at  $l = 2$ , a factor 1.7-1.9 apart at  $l = 3-4$ . So the quartet is returned by neither arm and the 'uniformly 10-25% deeper' claim is false; both withdrawn from P15 at c54.153.). *Computes:* the minimum over  $l = 2..5$  asserted at  $l = 4$ ; recovery asserted in  $6 \leq l \leq 9$ ; the depth disagreement asserted above 1.4x; all ratios asserted finite *Bound:* **\*\*scope:** this settles that the SHAPE claim has a computation behind it and that the DEPTH claim did not. It does NOT settle which depth table is right — the arms differ in late-ISW handling and in  $D_M$  (13864 here against the projection's 13927) — and it makes no attempt to reconcile them. **\*\* Run:** python3 receipts/P15\_CR\_cosmology/P15\_the\_second\_arm\_actually\_run.py

#### P15R37 P15\_hubble\_expansion\_confrontation.v2 [OK]

*sec:tensions* (SDSS DR12 BAO flagship: CR BAO ratios  $D_M/r_d$ ,  $D_H/r_d$  H0-INDEPENDENT (identical at H0=67,73);  $\chi^2$  CR=1.71 (any H0 incl 73), LCDM(67.4)=4.37, LCDM(73)=49.45 = the tension; cosmic chronometers consistent with 73). *Computes:* geometric-rate H0-cancellation +  $\chi^2$  vs DR12; matches paper 1.7/49 *Run:* python3 receipts/P15\_CR\_cosmology/P15\_hubble\_expansion\_confr

#### P15R38 P15\_desi\_dr2\_confrontation [OK]

*sec:tensions* (**\*\*!! MARKED** c54.160: the docstring reads  $\chi^2/\text{dof} = 13.70$  for LCDM at H0 = 73 while the paper quotes about 15 — the same  $\chi^2 = 178.16$  divided by 13 measurements rather than by the 12 degrees of freedom one fitted parameter leaves. **\*\* The paper's figure is the correct one and the receipt's annotation is not; the  $\chi^2$  itself is pinned).** *Computes:* full-covariance  $\chi^2$  vs DESI DR2; matches paper 1.0/14; was uncited by name *Run:* python3 receipts/P15\_CR\_cosmology/P15\_desi\_dr2\_confrontation.py

#### P15R39 P15\_expansion\_law [OK]

*sec:properframe/sec:flatlcdm* (derivation core (symbolic): eq:amplitude  $(6GM/\Lambda c^2)^{1/3} = 2^{1/3}/\sqrt{r}$  at Nariai; eq:scalefac  $r = A \sinh^{2/3}(B \tau)$  solves the  $E=1$  geodesic  $(dr/d\tau)^2 = 2GM/r + (\Lambda/3)r^2$  ( $2/3$  forced by wrong-exponent control); eq:rate  $H^2 = (\Lambda c^2/3)\coth^2$ ; eq:omega-ratio  $\Omega_m/\Omega_L = \text{csch}^2$ ). *Computes:* sympy: amplitude, geodesic solve,  $\coth^2$  rate,  $\text{csch}^2$  ratio; wrong-exponent control *Run:* python3 receipts/P15\_CR\_cosmology/P15\_expansion\_law.py

#### P15R40 P15\_verify\_throat\_tower [OK]

*sec:throat* (throat mode tower + no-hair isotropization:  $dS_2 \times S^2$ ,  $m^2/H^2 = l(l+1)$ ,  $\nu^2 = 1/4 - l(l+1)$ ;  $l=0 \rightarrow \nu=1/2$  (scale-invariant base, survives), every  $l \geq 1 \rightarrow \nu^2 \geq 0$  (heavy principal, decays)  $\rightarrow$  throat isotropizes; grounded in Wald 1983 no-hair; guard: throat  $l \neq$  CMB multipole). *Computes:* tower algebra + survives/decays split; theorem-grounded; guard held *Run:* python3 receipts/P15\_CR\_cosmology/P15\_verify\_throat\_tower.py

#### P15R41 AS\_amplitude\_leftward [OK]

*rem:transmission-leg* (AS — THE AMPLITUDE, READ LEFTWARD. Family 3's question, and what C4 can and cannot say about it. CONFIRMATION 2, and the corpus is already correctly scoped here: P7: the inherited datum 'stands as a ONE-PARAMETER ACCOMMODATION rather than a parameter-free prediction.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/AS\_amplitude\_leftward.py

#### P15R43 BUILD\_camb\_store [OK]

*sec:instrument* (BUILD\_camb\_store — the CAMB side of the transfer comparison, with its settings RECORDED. WHY THIS EXISTS (r2345). 'ENVELOPE\_residual.py' and its relatives load '/tmp/camb\_tr.npz' and measure our transfer against CAMB's. Nothing in the corpus records HOW that store was built — not the receipt, not the changelog, not the instrument.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/BUILD\_camb.s

#### P15R44 C10\_high\_l\_ratio [OK]

*sec:envelope-consequence* (**\*\*!! CORRECTED** c54.155: THIS ROW CARRIED THE RETIRED  $z_{\text{onset}} = 6850$  NUMBERS\*\* (r\_s matched +0.76%, r\_D +10.09%,  $\theta_D/\theta_*$  +9.26%). The shipped receipt hardcodes 6797 and produces +0.67%, +10.15%, +9.41%,  $l_* = 302.2$ . **\*\*And the receipt hardcodes  $th\_ratio = 1.0926$  with the comment 'DERIVED in C9' while C9 produces 1.0941. P15 corrected to the live values).** *Computes:* runs clean; registered from its own docstring

and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/C10_highl_ratio.py`

**P15R45** C11TEST\_radiation\_zeroed [OK]

*sec:envelope-consequence* (CRRUN — THE SAME INTEGRATOR ON CR. The control passed at r1975, so this may now be run. WHAT CHANGES, and each from the corpus not from me: \* BACKGROUND:  $H^2 = H_0^2(\Omega_m a^{-3} + \Omega_L)$  at  $H_0=73$ ,  $\Omega_m=0.3066$  – radiation is content, not a source. \* LOWER LIMIT: the SEAM, not  $\eta=0$  (C9: there is no observable expansion below it).). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/C11TEST_radiation_zeroed.py`

**P15R46** C11\_early\_isw [OK]

*sec:envelope-consequence* (C11 — THE EARLY ISW. The last named piece. CONFIRMATION 2/4 – the corpus settles the input, in its own voice: 'radiation, like matter, is inherited content read off that clock, NOT A TERM THAT SOURCES THE RATE, and there is no radiation-dominated era.). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/C11_early_isw.py`

**P15R47** C1\_closure\_of\_the\_driving [OK]

*sec:envelope* (C1-CLOSURE — DOES THE ACOUSTIC OSCILLATOR CLOSE IN ELEMENTARY TERMS ON CR's COLLAPSE LEG? CONFIRMATION 4 first — WHICH background. The corpus is specific and I must not substitute: L2 (where the driving lives) = 'the leaf's local rate, ORDINARY FRIEDMANN, radiation GRAVITATING NORMALLY, on the progenitor's collapse excursion' – and P16: on the cooling leg  $|H| = \sqrt{8 \pi G \rho/3}$ , 'identical to standard BBN at every...'). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/C1_closure_of_the_driving.py`

**P15R48** C2\_horizon\_limits [OK]

*sec:envelope* (C2 — THE LIMITS. The comoving Hubble radius across the excursion, from the corpus's own geometry. CONFIRMATION 4 — the object, and I take it from the corpus rather than from cosmology-by-habit: the  $E=1$  (marginally bound) congruence IS this cosmology. On it  $(dr/d\tau)^2 = E^2 - f = 1 - f$ , with  $f = 1 - 2M/r - r^2/\alpha^2$ . The signed areal radius  $r$  is the scale-factor analogue.). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/C2_horizon_limits.py`

**P15R49** C3\_conformal\_map [OK]

*sec:envelope* (\*\*!! CORRECTED c54.155: 'better than a percent at the entry radii of the modes in question' IS 1.87% AT THE PAPER'S OWN 10  $k_s$  THRESHOLD\*\* (3.72% at 5  $k_s$ , 24.4% at the seam); sub-percent begins near 30  $k_s$ . P15 corrected). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/C3_conformal_map.py`

**P15R50** C4\_driving\_envelope [OK]

*sec:envelope* (\*\*!! CORRECTED c54.155: 'entry at  $x = 1/\sqrt{3}$  for every  $k$ ' IS A LIMIT, NOT AN IDENTITY\*\* (the receipt's own table gives  $x_{\text{entry}} = 0.764$  at  $k = k_s$ , +32%), \*\*and the 'residual below a percent' is an average over a 5-point grid that lands near zeros of  $\cos(x_{\text{seam}})$ : scanning the receipt's own function over  $k/k_s$  in  $[10,120]$  reaches 7.3% at  $k/k_s = 12$ .\*\* Both qualified in P15). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/C4_driving_envelope.py`

**P15R51** C5b\_baryon\_term [OK]

*sec:envelope* (C5 — THE BARYON TERM. The first of the two terms C4 named as missing. CONFIRMATION 4 – and the corpus fixes the division rather than me: P16 masthead: 'eta fixes the abundances and the CMB peak HEIGHTS,  $\rho_r/\rho_m$  the peak SPACING.' P15: this cosmology 'carries the matter density as the bend of the spatial cut WITHOUT YET REMOVING THE COMPONENT' – so  $\rho_m$  is TOTAL matter and the baryon fraction is set

separately,...). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/C5b.baryon.term.py

**P15R52 C6\_neutrino\_term** [OK]

*sec:envelope* (C6 — THE ANISOTROPIC-STRESS TERM. The second of C4's two named terms. FINDING ZERO, before any physics: the corpus's TEXT says 'neutrino' ZERO times, and 'anisotropic stress' zero, and 'free-streaming' zero – while its BBN CODE carries them explicitly (bbn\_network.py: 'neutrinos share the plasma T until  $\sim e^+$  annihilation, then decouple to  $(4/11)^{1/3}$ ').). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/C6\_neutrino\_term.py

**P15R53 C7\_equality\_and\_deficit** [OK]

*sec:envelope* (C7 — THE STEP THE TRAP WAS GUARDING. full\_transfer\_verdict.py ASSUMED the high-ell peaks equal LambdaCDM's and applied the extra damping, getting a large deficit – and P15 disowns it because 'that assumption IS the unbuilt part.' C4 built it. So the same calculation can now be done with a DERIVED envelope. FIRST, though, one number that decides everything and that I have not yet checked.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/C7\_equality\_and\_deficit.py

**P15R54 C8\_diffusion\_length** [OK]

*sec:envelope-consequence* (C8 — DERIVE  $r_D$  HERE, FROM FIRST PRINCIPLES, ON THE SAME FOOTING AS THE DRIVING. Nothing imported from the damping\_tail/ files. Nothing that needs CAMB. CONFIRMATION 4: the object. Silk damping is a diffusion length built from the Thomson opacity and the expansion. The standard definition (Kaiser 1983; Hu-Sugiyama), in the  $R_b \rightarrow 0$  limit:  $1/k_D^2 = \int d\eta (1/(6\tau\dot{\tau}))$ ).). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/C8\_diffusion\_length.py

**P15R55 C9\_sound\_horizon\_and\_ratio** [OK]

*sec:envelope-consequence* (C9 —  $r_s$  AND  $r_D$  WITH THE CORRECT LOWER LIMIT: THE SEAM. A first pass integrated from  $a=0$  and returned  $r_s = 256$  Mpc against the corpus's 144 – absurd, and the error is structural, not numerical: \*\*\* ON THIS COSMOLOGY THERE IS NO OBSERVABLE EXPANSION BEFORE THE SEAM. \*\*\* P15: 'recombination is observable cosmology FROM THE SEAM OUTWARD'. The L1 foliation BEGINS at the seam;). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/C9\_sound\_horizon\_and\_ratio.py

**P15R56 H0\_acoustic\_angle\_and\_seam** [OK]

*sec:refit-bound* (H0 — THE ACOUSTIC SCALE AT THE DIRECTLY MEASURED H0. The calculation I have been running backwards. CORPUS CLAIM (P7, P15): 'There is no second Hubble rate to reconcile: THE DIRECTLY MEASURED H0 IS THE GEOMETRY READ AT THE PRESENT EPOCH, while the lower value inferred from the microwave background rests on a radiation-governed sound horizon the construction does not share...'). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/H0\_acoustic\_angle\_and\_seam.py

**P15R57 ISW\_line\_of\_sight** [OK]

*sec:refit-bound* (ISW — THE EARLY INTEGRATED SACHS-WOLFE, ISOLATED, AND ITS SHIFT ON THE FIRST PEAK. Same shape as everything else in A.0: the source is a line-of-sight integral over the set rate.  $\Delta_l(k) = \int d\eta [S_{SW} + S_{ISW}] j_l(k(\eta_0 - \eta))$ ,  $S_{ISW} = (\Phi' + \Psi') = 2\Psi'$  LambdaCDM has a nonzero early ISW because  $\Psi$  is still evolving at recombination (radiation matters). CR does not: C11 showed  $\Psi$  is constant on L1.). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt's own wording, condensed; not independently re-derived here *Run:* python3 receipts/P15\_CR\_cosmology/ISW\_line\_of\_sight.py

**P15R58 M2\_reabsorption\_minimization** [OK]

*sec:refit-bound* (M2 — THE MINIMIZATION. Can a parameter refit absorb the damping shape? The honest core of the likelihood question, done on the closed forms derived in C8/C9 – no CAMB.

SETUP, stated before running: FREE :  $\Omega_m$ ,  $\omega_b$ , and the inherited datum  $\rho_r/\rho_m$  (which fixes  $z_{\text{onset}}$ ). CONSTRAINT:  $L^* = \pi D_M/r_s$  pinned to the observed 301 – the peak positions are measured.). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt’s own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/M2_reabsorption_minimization.py`

**P15R59** PEEB\_peekles\_on\_the\_set\_rate [OK]  
*sec:refit-bound* (PEEB — THE PEEBLES IONISATION HISTORY ON THE SET RATE. ZREC used Saha only: right shift, wrong absolute (1283 vs CAMB’s 1090), because Saha has no departure from equilibrium. Peebles adds it – and the departure is itself H-dependent, which is exactly why it must be integrated on the construction’s own rate rather than inherited. Standard effective three-level hydrogen atom (Peebles 1968; Seager et al. fits).). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt’s own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/PEEB_peekles_on_the_set_rate.py`

**P15R60** ROBUST\_p1p2\_scan [OK]  
*sec:refit-bound* (\*\*!! FIXED c54.160: THE REGISTERED COPY WAS UNRUNNABLE FROM ITS OWN DIRECTORY\*\* – it imported ‘RD\_diffusion\_direct’, which lives only in ‘storyboard\_receipts/’, so it exited 1 on ImportError before reaching any computation and the sweep had to run it from the ORIGIN instead. \*A receipt that cannot run where it is registered is not a receipt.\* Path resolution added; it now exits 0 in place). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt’s own wording, condensed; not independently re-derived here *Run*: `python3 receipts/P15_CR_cosmology/ROBUST_p1p2_scan.py`

**P15R61** P15\_the\_first\_peak\_figure\_is\_not\_stable [OK]  
*the acoustic sector* (\*\*THE 150 IS NOT A PREDICTION: ‘ROBUST\_p1p2\_scan’’s FIRST-PEAK POSITION AND PEAK-HEIGHT RATIO ARE NOT STABLE UNDER ITS OWN DOCUMENTED INITIAL CONDITIONS.\*\* Built settling front #5, the corpus’s two internal routes to the first acoustic peak (transfer 220, this instrument 150). \*\*The front is posed as a disagreement between two established routes; one of the two is not established.\*\* PART 1 the numerics are cleared first – stability number 0.47 against the file’s own limit of 2.8, 120+ RK4 steps per oscillation, NS\_CR x4 reproduces every figure. PART 2 Psi’s VALUE comes from the smooth matched envelope and Psi’s DERIVATIVE from the oscillatory closed form – two different functions, and the initial velocity is built entirely from the second. PART 3 across four readings of the SAME stated IC, ell.1 spans 150-315 and P1/P2 spans 0.93-2.02; \*\*the reading the docstring actually describes (‘velocities zero’) returns 2.017 against the sky’s 2.212, where the paper reported 1.45 as the disagreement.\*\* PART 4 the comb the paper cites against it was never propagated either – ‘verify\_coherence\_comb’ writes down  $\cos(k r_s)$  with  $r_s$  and  $D_C$  as literals. PART 5 \*\*the residual: the source comb spacing is 0.72-0.79 of  $\pi/r_s$  under EVERY variant. The initial data move the first peak by 165 multipoles and do not move this.\*\*). *Computes*: stability asserted under the file’s own 2.8 limit and steps-per-oscillation over 50; the two derivative functions asserted to differ by more than 1.2x across the band and to agree as  $x \rightarrow 0$ ; the control asserted to reproduce 1.447 and 150/360/555/780; the documented-IC run asserted at 2.017 AND asserted closer to the sky than the implemented one; ell.1 asserted to move by 150+ multipoles and P1/P2 by more than 2x; the comb receipt asserted to contain  $\cos(k r_s)$  and NO integrator; the comb deficit asserted present, bounded, and STABLE across variants (max-min  $\pm 0.15$ ) – which is what makes it not initial data. *Bound*: \*\*scope: this does NOT vindicate 220 and supplies no first-peak position. It removes one route from arbiter status and registers the comb deficit, which is not resolved here. The four variant figures are produced by ‘computations/beyond\_the\_wall/L171\_what\_sets\_the\_first\_peak.py’, which execs the registered receipt’s own source with only the initial-data block substituted; its ICMODE=asis control reproduces the receipt exactly, assertions passing.\*\* *Run*: `python3 receipts/P15_CR_cosmology/P15_the_first_peak_figure_is...`

**P15R62** P15\_the\_crossing\_exists\_and\_is\_empty [OK]  
*the acoustic sector* (\*\*THE RE-ENTRY BETWEEN THE BRANCH POINT AND THE ONSET IS REAL, AND IT IS EMPTY: THE L1 POTENTIAL EQUATION CONTAINS NO  $k$  AT ALL.\*\* Built working front #5, testing a candidate THIS FORK proposed at c54.165 – front #1 puts every mode super-horizon and frozen at the branch point, ‘prop:subhorizon’ puts the acoustic modes inside at the onset, so they re-enter in between, and a crossing is where flat LCDM acquires its  $k$ -INDEPENDENT driving shift. \*\*The answer is no, and the reason is stronger than the

question.\*\* PART 1 the re-entry is real and its boundary is computed: the mode crossing AT the onset is  $k = 0.0111/\text{Mpc}$ ,  $\ell = 144$ , so the peak modes crossed before the plasma began and  $\ell \downarrow 144$  crosses after. PART 2 but it is a MATTER-era crossing – pressureless to better than a part in  $1e4$ , because L1 has no radiation SOURCE. PART 3 \*\*and the  $w = 0$  potential equation has no  $k$  in it\*\*:  $\Phi$  constant to twelve digits, identical across four decades in  $k$ , the integrator taking the SAME number of steps for every mode. PART 4 the contrast: in LCDM 5 of 6 cross under radiation, where  $w = 1/3$  restores the  $k^2$  term and  $\Phi$  decays on the sound-crossing time by construction.). *Computes:* every mode at/above the peak region asserted to cross before the onset AND  $\ell = 28$  asserted to cross AFTER it (the first draft claimed ‘every observed mode’ and this assertion fired); the boundary multipole asserted inside 100-250; the background asserted pressureless to  $1e-3$  at every crossing;  $\Phi$ ’s spread across four decades in  $k$  asserted below  $1e-9$  AND the solver step count asserted identical across  $k$ ; the LCDM contrast asserted to put 4+ modes under radiation with  $z_{\text{eq}}$  above 3000. *Bound:* \*\*scope: this CLOSES a candidate mechanism and supplies none. It computes no peak position and uses no figure from the acoustic fold. It leaves ‘PO-7’ open and harder: the absence of a universal driving shift on this rate is not an omitted crossing but a scale-free interval.\*\* *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_crossing\_exists\_and\_is\_empty.py

**P15R63** P15\_two\_arm\_control\_and\_guard [OK]  
*the acoustic sector* (\*\*A LIVE LambdaCDM CONTROL AND AN UNDRIVEN GUARD, ON ONE MACHINERY – AND THE CR ARM’S FIRST PEAK SITS 23% LOW AGAINST A 1.5% POSITION FLOOR.\*\* Built for fronts #5 and #18. \*\*This fork had never run a control\*\*: ‘ROBUST\_p1p2\_scan’ quotes one from r1975 in a comment, and c54.164 then showed its figures move over  $\ell_{11}$  in {150,165,315} with nothing to measure that against. PART 1 \*\*the guard passes on BOTH arms\*\* – every coupling to the potential removed at every site (4  $\Phi$  in photon/neutrino/CDM continuity AND  $k^2 \Psi$  in photon-baryon/neutrino Euler, both branches) and the source comb lands on the integers, LCDM 1.0147/2.0005 and CR 1.0121/1.9968, \*\*so every departure from  $n \pi/r_s$  in a driven run is the driving and nothing else\*\*. PART 2 the control lands at 224/536/808/1116 against the sky’s 220.6/538.1/809.8/1147.8 – \*\*positions to 1.5% at the first peak and 0.4%/0.2% at the second and third; HEIGHTS only to 13-26%, so no height claim from this instrument is meaningful below ~25%\*\*. PART 3 the CR arm run on its PHYSICAL discrete ladder and on a dense continuum sampling gives \*\*identical\*\* peaks 172/396/624/904, so discreteness is not what sets the position – an objection closed by measurement. PART 4  $\ell_{11}/\ell_{1A} = 0.5703$  against the control’s 0.7433 and the sky’s 0.7312: \*\*a 23% deficit against a 1.5% floor, a factor of 15 in margin\*\*, with  $P1/P2 = 0.866$  against the control’s 1.933.). *Computes:* the undriven comb asserted within 0.03 of the integers on both arms; the control’s position floor asserted under 3% AND its second and third peaks asserted within 1% of the sky; the discrete and continuum peak lists asserted identical; the deficit asserted at 5+ floors; CR’s  $\ell_{11}/\ell_{1A}$  asserted in 0.55-0.60 and the control’s in 0.72-0.76; the two arms’  $1_A$  asserted to agree to 1% (they do –  $D_M$  differs by 6.2% and  $r_s$  by 6.3% and they cancel). Every background quantity is RECOMPUTED rather than pinned. *Bound:* \*\*scope: this does NOT settle ‘PO-7’ and the CR figure is not offered as a prediction – it uses the initial datum the corpus’s own text describes (flat in  $k$ , common phase, VELOCITIES ZERO) and c54.164 established the figure moves with that choice. What changed is that it now moves against something. NO FIGURE FROM THE ABANDONED ACOUSTIC FORK IS USED; that fork’s own reproduction gate fails in this tree, and this instrument disagrees with its 201.3 by 15%.\*\* *Run:* python3 receipts/P15\_CR\_cosmology/P15\_two\_arm\_control\_and\_guard.py

**P15R64** P15\_the\_driving\_shift\_by\_subtraction [OK]  
*the acoustic sector* (\*\*THE DRIVING SHIFT MEASURED MODEL-FREE BY SUBTRACTION: LCDM’S ACOUSTIC PHASE IS FLAT IN  $k$  TO 2.1%, CR’S VARIES BY 3.9x – AND THE SCALING IS  $k^{-0.62}$ , NOT THE  $k^{-2}$  THE ABANDONED FORK’S MECHANISM REQUIRES.\*\* Built for front #5 on the c54.168 instrument. Each  $k$  integrated TWICE on the same equations, driven and undriven, differing in one multiplicative switch and nothing else.  $Q(k) = k \int c_s \, d\theta/\pi$  at the first acoustic turnover of  $\Theta_{\theta 0}$ . \*\*PART 1 the undriven column is the CALIBRATION and is MEASURED: 0.9968-1.0003 on both arms, where a free oscillator must give exactly 1.\*\* PART 2/3 LCDM’S  $Q_{\text{driven}}$  runs 0.791 -  $\downarrow$  0.774 over  $k = 0.045$ -0.16 (2.1% spread) because every mode crosses under radiation, where the potential decays on the sound-crossing time by construction; CR’s runs 0.457 -  $\downarrow$  0.117, a factor of 3.9, because there is no radiation era to cross and c54.167 showed the pre-onset interval is scale-free. PART 4

**\*\*a power-law fit gives  $k^{-0.62}$ , collapsing the data to a factor of 1.2 where  $k^{-2}$  spans a factor of 16.\*\*** The fold reports  $Q k^2$  constant and its seam-transient mechanism ( $3H/k^2$ ) requires it. **\*\*The phenomenon reproduces here independently; the exponent does not.\*\*** Recorded as an inter-instrument disagreement, not resolved.). *Computes:* the undriven calibration asserted within 0.01 of 1 on BOTH arms (this is the file's own gate); LCDM's spread asserted under 6%; CR's variation asserted over 2x; the fitted exponent asserted inside (-1.2,-0.3); **\*\* $Q k^2$  asserted to span MORE than a factor of 4 – i.e. the fold's constant- $Q k^2$  asserted FALSE here\*\***; and the fitted exponent asserted to collapse the data better than  $k^{-2}$ , so the fit is not vacuous. *Bound:* **\*\*scope:** two corrections were forced by this file's own calibration and are recorded because each would have been a false result – (i) reading  $Q$  on  $\Theta$ -hat rather than  $\Theta_0$  calibrated at 0.0003, since DR switches the couplings into the fluid but not  $\Psi$ 's own evolution; (ii) the driven LCDM column then calibrated at 0.001 on a decaying-mode transient two grid points from the start, excluded by requiring  $k \eta \gtrsim 1$ , which is the DEFINITION of an acoustic turnover and removes nothing on the CR arm. No figure from the abandoned fork is used.\*\* *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_driving\_shift\_by\_subtraction.py

**P15R65** P15\_which\_coupling\_carries\_the\_k\_dependence [OK]  
*the acoustic sector* **\*\*THE  $k$ -DEPENDENCE LIVES IN THE POTENTIAL'S GRADIENT, NOT IN ITS DECAY – WHICH IS THE REVERSE OF WHERE A RELAXATION TIMESCALE WOULD PUT IT.\*\*** Front #5, on the c54.168 instrument against its 1.000 calibration. The potential enters the plasma TWICE: as ' $4 \Phi$ ' in continuity (its rate of change feeding the density – **\*\*the decay channel\*\***) and as ' $k^2 \Psi$ ' in Euler (its gradient driving the velocity). Switched independently on the same integration. PART 1 **\*\*flat** LCDM's two couplings are each FLAT in  $k$  and they OPPOSE\*\* – continuity alone puts the turnover at  $Q = 1.23$ - $1.28$ , LATER than a free half-period, the gradient at  $0.73$ - $0.87$ , earlier, composing to  $0.77$ - $0.86$ . **\*\*So the universality of the standard driving shift is a CANCELLATION between two flat terms, not one term's signature.\*\*** PART 2/3 on the geometric rate there is no cancellation left – both advance ( $0.11$ - $0.18$  and  $0.06$ - $0.57$ ) and they add – **\*\*and the  $k$ -dependence sits in the GRADIENT: continuity varies  $1.6x$  at  $k^{-0.17}$ , gradient  $8.9x$  at  $k^{-1.04}$ .** PART 4 and **\*\*neither term alone is the answer\*\***: the driven exponent  $-0.62$  lies BETWEEN the two, so the scaling is their interference.). *Computes:* LCDM's continuity asserted ABOVE 1 and its Euler BELOW 1 (they must oppose) and all three driven exponents asserted flat to 0.15; CR's continuity asserted nearly flat ( $|\exp| \gtrsim 0.35$ ) and its Euler steep ( $|\exp| \lesssim 0.8$ ) and the Euler term asserted to be the steeper of the two – **\*\*reversing that assertion would reverse the finding\*\***; both CR couplings asserted below 1 (they add); the driven exponent asserted strictly BETWEEN the two and distinct from each by 0.2; and the undriven calibration asserted within 0.01 of 1 on both arms, which is the file's own gate. *Bound:* **\*\*scope:** this localises the  $k$ -dependence to a coupling and shows the standard shift's universality is a cancellation this rate does not have. It does NOT name the process that makes the gradient coupling run as  $k^{-1}$ , and computes no peak position. It closes the abandoned fork's stated mechanism as an explanation of THIS measurement – a relaxation on  $3H/k^2$  acts through the decay channel, and the decay channel is measured flat.\*\* *Run:* python3 receipts/P15\_CR.cosmology/P15.which\_coupling\_carries\_the\_k\_dependence.py

**P15R66** P15\_the\_gradient\_coupling\_in\_closed\_form [OK]  
*the acoustic sector* **\*\*THE GRADIENT COUPLING IN CLOSED FORM: THE TURNOVER IS AT A FIXED MULTIPLE OF  $3H/k^2$ , AND SINCE  $Q$  ALREADY CARRIES ONE FACTOR OF  $k$  THAT GIVES  $Q \sim 1/k$ , NOT  $1/k^2$ .** Front #5, closing the question c54.170 left. **\*\*The derivation is four lines.\*\*** (1)  $\Phi$ 's own equation carries a damping term  $-k^2 \Phi/3H$ , so  $\Phi$  relaxes on  $\tau = 3H/k^2$ . (2) With continuity off the oscillator is  $\Theta_{0''} + c_s^2 k^2 \Theta_0 = -(k^2/3) \Psi$ , and **\*\*the  $k^2$  in the forcing's amplitude is cancelled by the  $k^{-2}$  in its duration – the delivered impulse is  $k$ -INDEPENDENT\*\***. (3) So the first turnover is at the moment the impulse has been delivered, at a fixed multiple of  $\tau$ :  $\kappa = 1.79$ , constant across a factor of eight in  $k$ . (4)  $Q = k \int c_s \delta \rho / \rho$  already carries one  $k$ , so **\*\* $Q = 3 \kappa c_s H / (\pi k)$ \*\***. **\*\*PREDICTED  $Q k = 0.010403/\text{Mpc}$  AGAINST A MEASURED  $0.010372 - 0.30\%$  – with nothing fitted but  $\kappa$ ; and the closed form reproduces  $Q$  itself to 1.6% across the sub-horizon band.\*\***  $\Rightarrow$  **\*\*The abandoned fork's TIMESCALE is right and its EXPONENT is out by exactly one factor of  $k$  – a bookkeeping error in its own mechanism rather than a disagreement about physics.\*\*** And the continuity channel is flat for the complementary reason: ' $4 \Phi$ ' feeds the density with the potential's TOTAL change,  $\int \Phi' \delta \rho = -\Phi_0$ , the same however fast the

relaxation runs. **\*\*So the observed  $k^{-0.62}$  is not a power law of anything – it is the crossover between a flat channel and a  $1/k$  one.\*\***). *Computes:* kappa asserted constant to 10% across the sub-horizon band; **\*\*the flattest grouping asserted to be  $Q k^1$  and not  $Q k^0$  or  $Q k^2$  – if that assertion reverses, the derivation is wrong\*\***; the predicted coefficient asserted within 5% of the measured one; the closed form asserted to reproduce  $Q$  to 5% across the band; the continuity channel asserted flattest UNGROUPED and asserted flat to within a factor of 2. *Bound:* **\*\*scope:** this NAMES the process, so front #5's question is answered. It does NOT make the first peak agree with the sky – the 23% deficit against a 1.5% control floor stands unchanged. What changed is that its cause is derived rather than described. No figure from the abandoned fork is used; its timescale is confirmed independently here and its exponent corrected.\*\* *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_gradient\_coupling\_in\_closed\_form.py

#### **P15R67 P15\_where\_the\_likelihood\_sits**

[OK]

*the acoustic sector* **\*\*WHERE THE LIKELIHOOD ACTUALLY SITS: IT CANNOT ARBITRATE, AND THE INSTRUMENT'S OWN CONTROL IS WHY.\*\*** Discharges 'L-147', fenced since it opened on three conditions – falsifier fixed before any data is touched, a control reproducing a known number first, and not until the two internal routes agree. **\*\*All three met\*\*** (the routes collapsed at c54.164-168), so the fence came down. **\*\*F1\*\*** the pipeline is wired: the banked CAMB flat-LCDM best fit returns  $\chi^2 = 206.4$  over 215 plik\_lite TT bins,  $\chi^2/\text{dof} = 0.96$ . **\*\*F2\*\*** the instrument's own floor, measured through the SAME likelihood on the SAME bins: this fork's LCDM arm costs +18786 over CAMB. **\*\*F3\*\*** the only readable quantity, both arms of one instrument: CR - LCDM = +84160. **\*\*F4\*\*** the pre-registered rule returned 'suggestive, not a measurement discrepancy'. **\*\*F6 – AND THE PRE-REGISTERED RULE WAS DEFECTIVE, WHICH RUNNING IT IS WHAT SHOWED.\*\*** It is a RATIO test and never asks whether the floor is small in ABSOLUTE terms. It is not: the control arm sits at  $\chi^2/\text{dof} = 103$  against CAMB's 0.96, because Planck's binned errors are a few tenths of a per cent and this instrument's heights carry 13-26%. **\*\*Both arms are far outside the regime where plik\_lite discriminates, so the honest verdict is the WEAKER one: the likelihood cannot arbitrate.\*\*** **\*\*!!** That correction happens to FAVOUR the construction and is flagged for it – it would apply identically had the CR arm come out better, and it rests on a height floor measured at c54.168 before any data were opened. **\*\*What survives as fact: on one instrument, same bins, same fitted amplitude, the CR arm costs 5.4x the LCDM arm's  $\chi^2$ . An ordering, not a p-value.\*\***). *Computes:* the banked LCDM fit asserted at  $\chi^2/\text{dof}$  in (0.85,1.15) BEFORE anything else is read; every covered bin asserted non-empty (the keep mask); the instrument floor asserted POSITIVE (this instrument cannot beat a five-parameter CAMB fit, and if it appeared to, the binning would be wrong); the CR arm asserted to cost more than the control on the same instrument; and **\*\*F6's premise asserted – if the control arm ever drops below  $\chi^2/\text{dof} = 3$  the PART must be rewritten because the likelihood then CAN arbitrate.\*\*** *Bound:* **\*\*scope:** plik\_lite TT only – no polarisation, no ell j 30, no lensing, no external priors, a third-party implementation rather than clik (PROVENANCE.md), and 185 of 215 bins because the instrument reaches ell = 2000. The restriction to covered bins is done on the COVARIANCE and re-inverted, which MARGINALISES over the missing multipoles; dropping rows of the Fisher matrix would CONDITION on them. The banked r1934 CR figure (397.1 vs 206.4) applied the derived DAMPING suppression to CAMB's LCDM spectrum – it tested damping on LCDM's PEAK POSITIONS and could not have seen the position result, so it answers a different question and is SUPERSEDED here. 'PO-7' stays protected: this reports a number and does not convert it into a verdict.\*\* *Run:* python3 receipts/P15\_CR.cosmology/P15\_where\_the\_likelihood\_sits.py

#### **P15R68 P15\_the\_line\_of\_sight\_transfer**

[OK]

*the acoustic sector* **\*\*THE TRANSFER REBUILT ON A LINE-OF-SIGHT INTEGRAL: THE CONTROL'S FIRST-PEAK POSITION ERROR FALLS FROM 1.66% TO 0.16%, AND THE CR DEFICIT DOES NOT MOVE AT ALL.\*\*** Front #2. Until c54.173 the source was evaluated at a SINGLE eta\_rec – an instantaneous last scattering. **\*\*The visibility  $g = \tau e^{-\tau}$  peaks at  $\eta = 280.75$  with FWHM 37.8 Mpc, a tenth of a wavelength at the first peak.\*\*** Rebuilt as  $\Delta_{\text{eta}} = \text{INT } S_{\text{eta}} \text{ deta}$  with  $S = g(\Theta_{\text{eta}} + \Psi) + e^{-\tau}(\Phi' + \Psi') + (1/k^2) d/\text{deta}[g \theta_{\text{eta}}]$ . PART 1 the control goes 1.66% -> 0.16% in ell\_1/ell\_A: **\*\*sub-per-cent POSITIONS, which is what the likelihood needed of them.\*\*** PART 2 **\*\*and the CR arm is unchanged to four figures, 0.5703 on both transfers, while the control moved 1.79%. The deficit stands at 21.9% against a 0.16% floor – 134 floors.\*\*** The most suspect part of the instrument was replaced wholesale and the number

did not follow it down. PART 3 **\*\*the heights are NOT fixed and the target is NOT met\*\***: P1/P2 -13%, P1/P3 -34%, a residual growing with multipole and pointing at the damping tail (no polarisation term; a tight-coupling closure that fails where the visibility is). Stated as a partial result. PART 4 one correction was a DERIVATION: the Doppler term carries  $1/k^2$  not  $1/k$ , since  $v_b = \theta_b/k$ . **\*\*With  $1/k$  the second peak sat at 516 and P1/P2 at 3.15; with  $1/k^2$ , 532 and 1.93 – and  $\ell_1$  was 220 either way, so an instrument validated on the first peak alone would have carried the error.\*\***). *Computes*: the control's position error asserted sub-per-cent AND asserted improved by 5x or more; the CR arm asserted to move less than 1% under the rebuild; the deficit asserted at 20+ floors; **\*\*the heights asserted NOT sub-per-cent – if they ever are, the PART is stale and front #2 should be struck\*\***; and the corrected Doppler power asserted to move the SECOND peak toward the sky, which is what makes it a fix rather than a tuning. *Bound*: **\*\*scope**: this delivers sub-per-cent POSITIONS and leaves the HEIGHTS at 13-34%, so the likelihood still cannot be run (c54.172). The remaining error is a high-multipole problem and neither of its two named causes is a CR question – both are checked against the LambdaCDM arm first. Late ISW is truncated at  $\eta = 4000$ , which affects only  $\ell$  below  $\sim 30$  and is outside both the grid and plik.lite's TT range.**\*\* Run: python3 receipts/P15\_CR\_cosmology/P15\_the\_line\_of\_sight\_transfer.py**

**P15R69** P15\_the\_derived\_diffusion\_damping [OK]  
*the acoustic sector* (**\*\*THE HEIGHTS FROM 13-34% TO 2-3%: THE DIFFUSION DAMPING PUT BACK, DERIVED ON EACH ARM'S OWN OPACITY AND RATE.\*\*** Front #2. **\*\*Diagnosed from the SIGN\*\***: at c54.173 both height ratios were LOW, meaning peaks 2 and 3 too HIGH, worse at higher multipole – under-damping. **\*\*The hierarchy's tight-coupling closure switches off at  $\tau' = 0.6/\text{Mpc}$  and the visibility peaks at  $\tau' = 0.06/\text{Mpc}$ , ten times lower, so the photon quadrupole and the baryon slip were absent through the WHOLE of last scattering and the line-of-sight path carried no Silk damping at all.\*\*** The pre-LOS path hid this behind a hardcoded  $r_D$ . PART 1 the standard tight-coupling integral  $1/k_D^2 = \text{INT} [R^2/(1+R) + 8/9]/[6(1+R)\tau']$  deta, evaluated on this arm's own background: **\*\* $r_D = 6.58$  Mpc against the 6-7 a Boltzmann code gives. Nothing fitted.\*\*** PART 2 **\*\*and a DOUBLE COUNT had to be removed\*\***: with the hierarchy's own damping left in over  $\tau' \gtrsim 0.6$  AND the integral covering the same range, the control came out OVER-damped (+6.7%, +17%). **\*\*The sign flips either side of the correction, which identifies a double count rather than a wrong coefficient.\*\*** PART 3 heights **\*\* $-12.9\%/-33.5\%$   $\rightarrow -2.8\%/-2.1\%$ \*\***, on a control whose positions were already sub-per-cent. PART 4 **\*\*and the likelihood's floor falls 103  $\rightarrow$  23 in  $\chi^2/\text{dof}$ , a factor of 4.6, WITHOUT the verdict turning over\*\*** – 'L-147's answer stands, and the floor moving while the verdict does not is what measures how far the instrument still has to go.). *Computes*:  $r_D$  asserted inside 5.5-8 Mpc;  $\tau'$  at the visibility peak asserted BELOW the gate (PART 1's premise); **\*\*the sign asserted to flip either side of the correction – if it does not, it is a coefficient error and not a double count\*\***; the single-mechanism heights asserted within 5% and asserted better than BOTH the no-damping and the double-counted runs; the floor asserted to fall by 3x or more; **\*\*and the control asserted to remain ABOVE  $\chi^2/\text{dof} = 2$  – if it ever drops below, the likelihood CAN arbitrate and 'L-147's verdict must be revisited.\*\*** *Bound*: **\*\*scope**: this does not reach sub-per-cent heights and does not make the likelihood answerable. The residual is still worst at high multipole: no polarisation term in the source, and a photon sector with no free-streaming hierarchy after decoupling. Neither is a CR question.**\*\* Run: python3 receipts/P15\_CR\_cosmology/P15\_the\_derived\_diffusion\_damping.py**

**P15R70** P15\_the\_height\_target\_was\_below\_the\_resolution\_of\_its\_own\_statistic [OK]  
*the acoustic sector* (**\*\*THE HEIGHT TARGET WAS QUOTED AGAINST A SKY NUMBER WITH A 3.4% ERROR BAR THAT WAS NEVER WRITTEN DOWN, AND  $\chi^2$  – THE STATISTIC THAT CAN SEE – PREFERS A COEFFICIENT IT IS NOT ALLOWED TO CHOOSE.\*\*** Front #2. Since c54.168 every height claim here (-13%, -34%, 2-3%, -0.05%) compared P1/P2 and P1/P3 to two bare numbers, 2.217 and 2.277. **\*\*PART 1\*\*** the numerics are cleared first: 1.4x the modes, 1.3x the k-range and 1.8x the  $\eta$ -sampling move the ratios by 0.09% and 0.37%, and the 2%/7% swings that prompted the check came from LMAXL=1300 truncating the k integral under the THIRD peak. **\*\*PART 2\*\*** read off the plik.lite bandpowers with the published covariance – the full 2x2 block, not a diagonal estimate – **\*\*the sky's own ratios are  $2.256 \pm 0.077$  and  $2.280 \pm 0.074$ : 3.4% and 3.2%.\*\*** So the 13% and 34% of c54.173 were four and ten sigma and stand; **\*\*the 2-3% of c54.175 are about one sigma and do not.\*\*** **\*\*Caught by bookkeeping and not by suspicion: X\_data is binned C\_l and not D\_l, so peak-finding on it returned 464/761/1085 instead of 221/527/815.\* \*\*PART 3\*\*** a scan of one multiplier on  $1/k_D^2$

has P1/P2 selecting 1.094 +- 0.184 and P1/P3 selecting 0.897 +- 0.055 – **\*\*a 1.03 sigma split, so my own 'no single scale fits both, the residual is the envelope's SHAPE' is WITHDRAWN at its own error bar rather than published.\*\*** **\*\*PART 4\*\*** chi^2 over 185 bins does separate them: the minimum sits at 0.865 against 8/9's 0.859, and 16/15 costs 1123 units, seven inflated sigma. **\*\*The coefficient is changed back to 8/9 and NOT because it fits better\*\*** – neither value has ever been derived here (C8\_diffusion.length names the ambiguity in its own text), and a remembered number is not a derivation any more than a fitted one is. **\*\*PART 5\*\*** L-147's pinned chi^2 came from a script in /tmp; 'chi2\_of\_spectrum.py' puts that machinery in the corpus and returns 4170.2 and 50675.2 exactly.). *Computes:* converged spread asserted under 1% and the truncated run asserted separated from it; the sky's recomputed ratios asserted within 1 sigma of the bare numbers they replace; the DAMPX multiplier asserted to reproduce the explicit POLC=8/9 run to 0.2%; **\*\*the withdrawal asserted: if the split were 2 sigma or more the file FAILS for retracting a claim that in fact holds\*\***; the chi^2 minimum asserted at the 8/9 multiplier within 5%; both pinned values asserted recomputed to 0.5%; and an assert that fires if the control ever reaches chi^2/dof ~ 1, since this file would then need rewriting rather than patching *Bound:* **\*\*scope:** this corrects a MEASUREMENT PROTOCOL, not the construction, and it is a correction to my own work. It supplies no derivation of the shear coefficient – that is named as owed and is front #2's first item: the free-streaming photon hierarchy with the polarisation multipoles. And the 3.2-3.4% is the single-bandpower-bin resolution; a full-curve fit determines the peaks far better, which is exactly why chi^2 replaces the ratios rather than joining them.\*\* *Run:* python3 receipts/P15\_CR.cosmology/P15.the.height.target.was.below.the.resolution.of.its.own.statistic.py

**P15R71** P15.the.shear.coefficient.derived.not.remembered [OK]  
*the acoustic sector* (**\*\*THE DIFFUSION-DAMPING SHEAR COEFFICIENT, DERIVED FROM THE POLARISED PHOTON HIERARCHY INSTEAD OF REMEMBERED OR FITTED – AND ADOPTING IT COSTS THE CONTROL 1123 UNITS OF chi^2.\*\*** Front #2, closing the item c54.176 named as owed. 'C8.diffusion.length' has named 'the 8/9 coefficient against the 16/15 polarization-corrected one' in its own text since it was written, and neither had ever been derived here. **\*\*METHOD\*\*** two integrations of the same oscillator on the same background: the perfect tight-coupled fluid, undamped by construction, against photons carried to  $l = 24$  with Thomson scattering, the polarisation multipoles evolved alongside and the baryons on their own velocity. **\*\*Their ratio is the damping and nothing else – the  $(1+R)^{1/4}$  WKB prefactor, the drift of  $c_s$  and the  $R$ -dependence of the oscillation are common to both and divide out identically.\*\*** Gravity is off in both, which is the definition of the object rather than an approximation. **\*\*PARTS 1-4\*\*** with  $G = 0$  the  $l=2$  term is  $-(9/10)\tau'F_2$ , the textbook unpolarised closure, and  $C = 0.8852$  against 8/9; with polarisation  $C = 1.0618$  against 16/15; **\*\*and the polarisation correction – the quantity actually in dispute – is a factor 1.1996 against 6/5, three parts in ten thousand.\*\***  $C$  is read **LOCALLY** between consecutive extrema and extrapolated in  $(k/\tau')^2$ , so the  $k^2$  scaling is measured rather than imposed. **\*\*PART 5\*\*** 16/15 is adopted and the control's floor goes 22.5 -j 28.6. **\*\*The fit's preference for 8/9, measured and refused at c54.176, is refused again: a coefficient chosen because it fits is a fitted parameter whatever it is called.\*\*** **\*\*PART 6\*\*** and nothing downstream moves – sec:envelope-consequence's 10.8% goes +10.83% -j +10.87%, because the coefficient nearly cancels in a ratio of two rates.). *Computes:* both intercepts asserted within 2% of the leading-order values and pinned to ~1% in the gate; **\*\*the polarisation correction asserted within 0.5% of 6/5 and pinned to a tenth of a percent – it is the number the file exists to produce\*\***; the scatter about the line in  $(k/\tau')^2$  asserted below 0.01, so a method that destroyed the extrapolation fails rather than shifting the intercept quietly; the sign of the finite- $(k/\tau')$  correction asserted, and the extrapolation asserted **SHORT** (under 2% of the coefficient swept over the fitted range); **\*\*the chi^2 cost asserted POSITIVE – if the derived coefficient stopped costing chi^2 the file's whole point would need restating\*\***; and the C8 figure asserted to move less than a tenth of a point, so adopting the coefficient cannot silently shift a published number *Bound:* **\*\*scope:** this derives the coefficient in the WKB damping integral and NOT the damping envelope's shape. It supplies the damping half of the polarisation physics only; the SOURCE half –  $g^*Pi/4$  and  $(3/4k^2)(g^*Pi)$  with  $Pi = Theta_2 + Theta^*P_0 + Theta^*P_2$  – needs the hierarchy carried in the spectrum instrument and is what front #2 still owes. And one assertion of mine fired on its own data: I demanded both extrapolation slopes be of order unity and they are -0.150 and -0.942; the gate now checks that the extrapolation is **SHORT**, which is the thing that actually matters.\*\* *Run:* python3 receipts/P15\_CR.cosmology/P15.the.shear.coefficient.derived.not.remembered.py

**P15R72 P15\_the\_photon\_hierarchy\_in\_the\_transfer**

[OK]

*the acoustic sector* (\*\*THE PHOTON BOLTZMANN HIERARCHY WITH POLARISATION CARRIED INSIDE THE TRANSFER: THE CONTROL'S  $\chi^2/\text{dof}$  FALLS 28.6 - 7.1 AND THE FIRST FOUR PEAKS LAND AT 220/540/812/1124 AGAINST THE SKY'S 220.6/538.1/809.8/1147.8.\*\* Front #2. Photons as multipoles to  $l = 24$  with the polarisation hierarchy alongside, the baryons on their OWN velocity and their OWN density, and the source carrying  $g^*\text{Pi}/4$  and  $(3/4k^2)(g^*\text{Pi})$ . \*\*THE HANDOVER MAKES A DOUBLE COUNT IMPOSSIBLE RATHER THAN MERELY ABSENT\*\*: the fluid runs to  $\tau' = 3/\text{Mpc}$  and the envelope is FROZEN there, so envelope and multipoles have disjoint supports – and only 4.2% of  $1/k_D^2$  has accumulated at the handover, so 96% of the damping is dynamic. \*\*PART 1\*\* the second peak is 0.35% out and the third 0.27%, where the fluid had them 2.6% and 0.7% low. \*\*PART 2\*\*  $\chi^2/\text{dof}$  28.6 - 7.1, and 20.0 - 4.2 over the range where the wavenumber integral is not starved. \*\*PART 3\*\* and c54.177's prediction is checked against the number it predicted: \*\*running the same hierarchy with ONLY the two polarisation source terms dropped – exactly the half an envelope has – gives 2022 against 1320, so they are worth  $\Delta\chi^2 = 702$ , 63% of the 1123 that adopting the derived 16/15 cost.\*\* \*\*PART 4\*\* and half the baryon split is WORSE than none: own density while still sharing the photons' velocity moved  $\chi^2/\text{dof}$  133 - 161 at development settings; both halves give 15.1 - 7.1. \*Twice now a physical effect taken in half has been worse than not taken – it is not a small error but a different and inconsistent model.\* \*\*PART 5\*\* doubling the hierarchy depth moves  $\chi^2$  by 0.00% and doubling the handover opacity by 1.5%. \*\*PART 6\*\* and the CR arm is at  $\text{ell}_1/\text{ell}_A = 0.5703$  for the SIXTH instrument state, while the control improved fourfold.). *Computes*: every peak asserted no FURTHER from the sky than the fluid transfer's; the  $\chi^2$  improvement asserted to be at least a factor of two and pinned at 7.14 / 4.17; both height ratios asserted within 2 sigma of the sky's own covariance-derived values; \*\*the polarisation source terms asserted to recover between 40% and 120% of c54.177's 1123 – outside that range c54.177's statement could not be said to have predicted this and the file FAILS\*\*); the full baryon split asserted to beat the shared-density run; the LG and handover sensitivities asserted below 0.1% and 5%; the CR arm asserted unmoved to the fourth decimal; and an assert that fires if the control ever reaches  $\chi^2/\text{dof} \sim 1$ , since L-147 would then have to be REOPENED rather than this file's closing line printed *Bound*: \*\*scope: this is the LambdaCDM control and the CR arm on one instrument, and it converts nothing into a verdict – 'PO-7' is protected and 'L-147's answer is unchanged, because a control at seven times a fit still cannot certify what it is compared with. What remains is accuracy and no longer a named missing SECTOR: reionisation, neutrino mass, lensing, and the wavenumber range. And the aliasing gate was found MISSING from the line-of-sight path, which has been the default since c54.173; it cost a development run a first peak at 196 instead of 220 with nothing said, and is now on both paths.\*\* *Run*: python3 receipts/P15\_CR\_cosmology/P15\_the\_photon\_hierarchy\_in\_the\_transfer.py

**P15R73 P15\_what\_is\_left\_is\_lensing\_and\_not\_reionisation**

[OK]

*sec:refit-bound* (\*\*BOTH OF FRONT #2's NAMED NEXT STEPS MEASURED BEFORE EITHER WAS BUILT, AND ONE IS WORTH EXACTLY NOTHING.\*\* \*\*PART 1\*\* above  $\text{ell} \sim 40$  reionisation's whole TT effect is the constant screening  $\exp(-2\tau)$ , and the amplitude here is fitted in CLOSED FORM –  $A = (m F d)/(m F m)$  is homogeneous of degree -1 in  $m$ , so  $m \rightarrow s m$  returns  $A/s$  and the residual is IDENTICAL.  $d(\chi^2) = 0$  exactly, shown at  $\tau = 0.054, 0.10$  and  $0.30$ , zero to machine precision at all three.\*\* \*The degeneracy is a property of the STATISTIC, not of the size of  $\tau$ .\* \*\*PARTS 2/3\*\* a Gaussian smoothing in  $\text{ell}$ , width scanned, minimises at  $\chi^2 = 921$  against the control's 1320 – and \*\*a width that GROWS with  $\text{ell}$  beats a constant one by 1.8x in  $d(\chi^2)$ , 400 against 225, each at its OWN optimum so the comparison is of SHAPE\*\*. Lensing smooths more at high  $\text{ell}$  because a deflection of fixed angle subtends a larger fraction of an acoustic wavelength, and that is the signature the data select. \*\*PART 4\*\* the width is FITTED, so the 400 bounds what a DERIVED calculation has to play for and is not a measurement of the lensing amplitude – \*\*and  $\sim 921$  of the 1320 survives it, so lensing is the next build and is NOT the last one.\*\*). *Computes*: the reionisation degeneracy asserted EXACT – if it ever moves  $\chi^2$  the amplitude is no longer fitted in closed form and PART 1's algebra no longer describes the pipeline; both smoothing optima pinned; \*\*the growing width asserted to beat the constant one by at least 1.5x, which is the SHAPE claim and the reason the word is 'lensing' rather than 'some smoothing'\*\*; what survives asserted above 60% of the baseline, so a rewrite is forced rather than a patch if a fitted smoothing ever accounts for most of the residual; and an assert that fires if a smoothing alone would take the control to  $\chi^2/\text{dof} \geq 2$ , since L-147 would

then have to be revisited *Bound:* **\*\*scope:** this measures what two candidate builds can be worth on THIS statistic. It is not a result about the sky, not a measurement of the lensing amplitude, and the control does not improve – the smoothing width is fitted. Reionisation is STRUCK from the front’s list rather than deferred, and the condition that would undo the strike is written down: a comparison reaching  $\ell \lesssim 30$ , a second spectrum, or a fixed amplitude.**\*\* Run:** `python3 receipts/P15_CR_cosmology/P15_what_is_left_is_lensing_and_not_reionisation.py`

**P15R74** `P15_derived_lensing_on_the_lcdm_arm` [OK]

*sec:refit-bound* (**\*\*THE DERIVED LENSING c54.181 TEED UP, RUN WITH NO FREE WIDTH – AND IT COMES IN UNDER THE FITTED BOUND, WHICH IS WHAT A DERIVATION MUST DO.\*\***  $\Lambda$ CDM’s own non-perturbative lensed/unlensed ratio applied to the c54.178 LCDM arm takes the control 1320  $\rightarrow$  989,  $d(\chi^2) = 331$  – UNDER the 400 the c54.181 fitted smoothing bounded, and concentrated in the damping tail ( $\ell$  1100-1500) where lensing fills the troughs. The acoustic peaks move 0.0%. The corpus’s own first-order Hu (2000) kernel, run on the SAME C<sup>phiphi</sup> so only the ORDER differs, OVERSHOOTS to 508 because it BREAKS DOWN in the tail – a spurious +13% at  $\ell=1900$  against the full operator’s +6.5% – so the derived number is the full operator’s 331, not the first order’s 508. Validated on CAMB’s own LCDM: 615 unlensed  $\rightarrow$  186 lensed ( $\chi^2/\text{dof} \sim 1$ ).  $\sim 989$  of 1320 survives, so lensing is real and NOT the last build.). *Computes:* lensed CAMB LCDM asserted at  $\chi^2/\text{dof}$  in (0.75,1.25) so the operator and pipeline are validated before anything is read off the arm; the derived lensing on the arm asserted in (280,385) AND strictly under the c54.181 fitted bound of 400, which is the defining property of a zero-width derivation against a fitted upper bound; the first-order kernel asserted to OVERSHOOT the full operator by more than 80 in  $d(\chi^2)$  via the tail artefact, a live check that LENS\_correction is first order and that the tail is where it fails; and the peak shift asserted below 0.3%. *Bound:* **\*\*scope:** the lensing operator is  $\Lambda$ CDM’s, imposed not fitted, so this is not a measurement of the lensing amplitude and the CR arm does not fit – it measures how much of c54.181’s fitted-smoothing headroom a genuine zero-parameter lensing recovers (most of it), with the shortfall against the fit being the fit’s free width. The residual  $\sim 989$  is the instrument’s transfer inaccuracy, not lensing’s: CAMB’s UNLENSED LCDM already sits at 615 on these same bins. ‘PO-7’ stays protected.**\*\* Run:** `python3 receipts/P15_CR_cosmology/P15_derived_lensing_on_the_lcdm_arm.py`

**P15R75** `UNC_error_budget` [OK]

*sec:refit-bound* (UNC — QUANTIFYING WHAT WE DO NOT KNOW. The prediction with an error budget. Corrections folded in first: \* the recombination WIDTH scales with the rate (A.22) – so the thickness contributes ~nothing to the RATIO and the signature is carried by  $r_D$  essentially alone; \* the LD anchor affects only the ABSOLUTE  $r_D$  and cancels in the ratio – it carries NO uncertainty into the prediction;). *Computes:* runs clean; registered from its own docstring and a live run *Bound:* description is the receipt’s own wording, condensed; not independently re-derived here *Run:* `python3 receipts/P15_CR_cosmology/UNC_error_budget.py`

**P15R76** `P15_the_residual_is_contrast_and_the_lensing_potential_is_derived` [OK]

*sec:refit-bound* (**\*\*WHAT THE CONTROL’S REMAINING  $\chi^2$  IS MADE OF, AND THE LENSING POTENTIAL BUILT TO ATTACK IT.\*\*** **\*\*PART A\*\*** templates projected through the same inverse covariance the  $\chi^2$  uses: **\*\*positions remove 1.4 of 1320 – 0.1%, so the peaks are where the sky puts them and that is not where the residual lives\*\***; a smooth envelope 15%; **\*\*the peak-trough oscillation at FIXED position 38%, with the model’s contrast 13.1% TOO HIGH\*\*** – which is what lensing reduces; and **\*\*53% in NEITHER, so a perfect lensing calculation leaves about half\*\***. **\*\*Fitted templates, so each share is an upper bound on what the effect can claim.\*\*** **\*\*PART B\*\*** the line-of-sight solve stops at  $z = 12.8$  but the lensing kernel peaks at  $z \sim 3.3$ , so  $\Phi$  is carried on by the background’s own growth factor  $D$  prop.  $H \int da/(aH)^3$  – exact in linear theory, no transfer function imported,  $k$ -dependence from this instrument’s own  $\Phi$ . **\*\*And the normalisation is the amplitude the TEMPERATURE comparison already fits, so lensing enters with NO new parameter.\*\*** Limber checked against an EXACT projection (5% over  $l = 5-200$ ); the deflection maximum is BROAD, within 10% over  $l = 12-42$ ; **\*\*rms deflection 2.69 arcmin with nothing tuned to it.\*\***). *Computes:* positions asserted below 2% of the residual, so the diagnosis reorders loudly if they ever rise; the contrast excess pinned to 13.1%  $\pm 2$  points, since that is what the convolution must remove; what survives BOTH template sets asserted at 53%  $\pm 5$ , **\*\*so a build that forgets how much remains fails here\*\***; Limber’s departure asserted below 10%, since the convolution will use it; the deflection maximum asserted BROAD; and the absolute rms deflection pinned at 2.692 arcmin, which is the zero-parameter check on the normalisation *Bound:*

**\*\*scope:** PART A's shares are FITTED-template upper bounds, not measurements of physical effects, and the two sets overlap by ~7 points. And PART B's external agreement is weaker than it looks: the ~2.7 arcmin and the ~1e-7 deflection power it matches are RECALLED, not derived here and not checked against a source this corpus holds – this programme was bitten twice in three revisions by exactly that (the 16/15 coefficient at c54.176, and a Limber suspicion in this very file which was also recollection and also wrong). What PART B establishes is INTERNAL consistency; the external agreement is corroboration at the strength of a memory.\*\* *Run: python3 receipts/P15\_CR.cosmology/P15.the\_residual\_is\_contrast\_and\_the\_lensing\_potential\_is\_derived.py*

**P15R77** P15\_the\_phase\_is\_the\_driving\_and\_the\_undriven\_arms\_agree [OK]  
*sec:refit-bound* (\*\*WHY 0.62 pi: THE DRIVING — AND THE ATTRIBUTION IS COMPLETE. WITH EVERY COUPLING TO THE POTENTIAL REMOVED THE TWO ARMS' ACOUSTIC PHASES AGREE TO 0.013 OF L\_A.\*\* Front #2, lead 'L-507'. c54.191 closed with \*\*an acoustic phase shift is a computable consequence of the driving, so 'why 0.62 pi' has an address. This file does not have it.\*\* \*\*The instrument has carried the switch since c54.170\*\* — 'NODRIVE=1' removes every coupling to the potential at every site (the 4·Phi' in the photon, neutrino and CDM continuity equations and the k<sup>2</sup>·Psi in the Euler equations) while leaving Phi's own evolution alone, that being the background and not a coupling. \*Both arms run through it at production depth here, which no revision had done.\* \*\*PART 1 UNDRIVEN THE TWO ARMS ARE THE SAME OSCILLATOR:\*\* intercepts \*\*−0.135 and −0.149 of L\_A — agreeing to 0.013\*\* — and both slopes L\_A to a part in a thousand. \*Two constructions with different rates, sound horizons, starting redshifts and initial data give the SAME acoustic series once nothing drives it.\* \*\*PART 2 AND THE DRIVING IS WHAT SEPARATES THEM:\*\* switched on it supplies \*\*−0.127 to the control and −0.729 here, a factor of 5.7\*\*; the driven discrepancy is 0.615 and the undriven 0.013, so \*\*the driving accounts for 98% of it\*\*. \*\*PART 3 THE SAME SWITCH ACCOUNTS FOR THE SPACING:\*\* undriven both slopes are 1.000 and 1.001 of L\_A; driven, the control's is 1.003 and this construction's 0.976 — \*\*so the 2.4% the spacing was short is the driving as well, and not a second effect\*\*. ⇒ \*\*\*THE WHOLE ACOUSTIC DISAGREEMENT IS ONE COUPLING BEHAVING DIFFERENTLY — and a coupling is a thing a theory says something about, where a ratio is not.\*\*\* \*\*PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at phi = 0. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving ATTRIBUTION stands; its promotion to "the whole disagreement" does not.\*\*\*). *Computes:* \*\*the UNDRIVEN agreement asserted under 0.10 of L\_A — if the arms still differed with nothing driving them the driving would not be the whole story and PART 2's headline would be wrong\*\*; the ratio of what the driving supplies asserted above 3x and pinned at 5.7 ± 1.5, without which PART 2 has no content; \*\*the undriven SLOPES asserted at L\_A to 1%, which is what makes PART 3's "not a second effect" true\*\*; the driven discrepancy re-derived at 0.615 ± 0.05, tying this file to 'L-505' and 'L-506'; the driving's share asserted above 85%; and at least seven peaks per arm *Bound:* \*\*scope: an ATTRIBUTION is not an explanation and it is not a verdict. What this establishes is that the acoustic disagreement lives in ONE COUPLING rather than in four items, a ratio, or an initial condition. IT DOES NOT SAY WHY the driving is 5.7x stronger here — the obvious candidate is that this construction's modes begin already sub-horizon at the seam carrying an assigned amplitude and ZERO velocity, in a potential that is not the equilibrium configuration for that state, where the control's begin super-horizon in adiabatic equilibrium; \*\*that is a HYPOTHESIS and this file does not test it\*\*. The instrument carries the split that would: 'DRC' is the potential's rate of change feeding the density and 'DRE' its gradient driving the velocity, switchable separately — the next thing this front owes. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5's stated procedure.\*\* *Run: python3 receipts/P15\_CR.cosmology/P15.the\_phase\_is\_the\_driving\_and\_the\_undriven\_arms\_agree.py*

**P15R78** P15\_the\_phase\_moves\_with\_the\_datum\_and\_only\_the\_spacing\_does\_not [OK]  
*sec:refit-bound* (\*\*THE OBSERVER LINE'S ITEM 38 ANSWERED, AND THE ANSWER WITHDRAWS c54.190–191's HEADLINE. ACROSS FOUR SEAM PHASES AT PRODUCTION DEPTH THE ASYMPTOTIC ACOUSTIC PHASE SPANS 0.891 IN phi/pi — AND THE CONTROL'S 0.263 LIES INSIDE THAT SPAN.\*\* Front #2, lead 'L-508'. \*Item 38 (r2485): of the 23 datum-scan spectra ZERO reach peak 8, so the quantity carrying the disagreement could not be tested against the one freedom known to move things. It named the experiment — two or three seam phases at production depth. c54.191 supplied one; this supplies two more.\* \*\*PART

1\*\* at  $\phi = 0, \pi/4, \pi/2$  and  $\pi$  the fitted  $\phi/\pi$  is \*\*0.878, 0.958, 0.066, 0.671\*\* — \*c54.191 read 34% closure from the two endpoints; \*\*the interior is where the motion is\*\*\*. \*\*PART 2 SO 0.62  $\pi$  IS THE VALUE AT ONE READING AND NOT A PREDICTION.\*\* \*The attribution to the driving (c54.193) stands — the driving is what separates the arms AT A GIVEN READING — but a quantity the datum can move across the control's value is not a prediction of the construction.\* \*\*PART 3 THE HEIGHTS MOVE TOO:\*\* P1/P2 runs \*\*0.483  $\rightarrow$  1.618\*\* against the sky's 2.217 and the control's 2.197, and \*\*never reaches it\*\*. \*\*PART 4 WHAT IS ROBUST IS THE SPACING:\*\* 0.9629–0.9814 of LA over the same four — \*\*1.9%\*\* — and tracking r.s at 98% of the acoustic rate under the fitted parameter. \*\*PART 5 AND MATCHING THE PHASE DOES NOT FIX THE SPECTRUM\*\*, which is what settles PART 2: the reading closest in phase ( $\phi = \pi/2$ , a phase gap of 0.197) has  $\chi^2/\text{dof} = 224$  against the control's 3.71 on the same 185 bins — sixty times worse while agreeing on the phase\*.  $\Rightarrow$  \*\*\*So the phase was never carrying the disagreement.\*\*\*). *Computes:* \*\*the control's phase asserted INSIDE the span the seam phase reaches — without which there is no withdrawal and c54.190–191 stand as written\*\*; the span asserted above 0.3 and pinned at 0.891, which IS item 38's question; \*\*the low-ell transient asserted present in ALL FOUR at more than 3x the control's — this is what establishes the peak INDEXING, and if a reading showed none the span could be an artefact of labelling\*\*; the spacing's stability asserted under 6%, the one thing this file does NOT withdraw; the heights asserted never to reach the sky's; and \*\*the phase-matched reading asserted above 10x the control — PART 5 is what makes the withdrawal a finding rather than a shrug\*\*  
*Bound:* \*\*scope: this is the THIRD withdrawal of this line's own headline in the span, and it is forced by the data rather than chosen. What it leaves is worse than c54.191 reported and truer: the acoustic SPACING is right and robust; the acoustic PHASE and the peak HEIGHTS both move substantially with a datum the corpus does not fix and neither reaches the sky at any reading tried; and no reading brings the spectrum within sixty times the control. \*\*Where the disagreement lives is therefore not a single named quantity and this file does not supply one.\*\* 'F5' unsoftened and it cuts both ways — a discrepancy resting on a free choice is not a framework verdict either. 'PO-7' protected; the conversion runs by 'F5's stated procedure.\*\* *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_phase\_moves\_with\_the\_datum\_and\_only\_the\_spacing\_does\_not.py

**P15R79** P15\_the\_locus\_is\_wrong\_in\_six\_places\_and\_the\_lint\_cannot\_see\_the\_worst [OK]  
*sec:perturbations, sec:coherence, sec:envelope* (\*\*ROUTED ITEM 21 WORKED: SIX SITES SAID "BRANCH POINT" WHERE THEIR OWN RECEIPTS SAY "SEAM"/"ONSET", AND THE PHYSICS INVERTS BETWEEN THE TWO — SO A PROPOSITION TITLED "SUB-HORIZON" WAS TRUE AT ONE AND FALSE AT THE OTHER. AND THE LINT BUILT FOR THIS DEFECT COULD NOT SEE THE WORST OF THEM.\*\* Front #2, lead 'L-509'; VEIN 'L-202'. \*\*PART 1 THE INVERSION, RE-DERIVED FROM THE METRIC FUNCTION\*\* rather than taken from the routing slip: normalised  $1/aH$  is \*\*1.000 at the seam and 0.051 toward  $r=0$ \*\*, and with the inherited datum the horizon maximum sits at  $r^* = 1.5338 r_{\text{seam}}$ , just OUTSIDE the seam\*\* — so the seam is on the rising branch and the acoustic modes are already inside there, which is a \*derivation\* of prop:subhorizon and not a numerical coincidence of z.onset. \*\*PART 2 SIX SITES REPAIRED, PINNED BY TEXT AND NOT BY LINE NUMBER\*\* — including the subsection heading, \*which contradicted its own proposition three lines below\*, and the sec:envelope sentence \*that claims to BE the derivation, with the locus backwards\*. \*\*One of the six was not on the routed item's list\*\*; the lint found it. \*\*The one true-as-written branch-point sentence — the neutrinos' decoupling, a TEMPERATURE ordering, flagged as a false positive by the finder in advance — survived the sweep.\*\* \*\*PART 4 AND THE GATE WAS VERIFIED AGAINST A CLEAN TREE, WHICH MEASURES NOTHING:\*\* seeding prop:subhorizon's body back to "at the branch point" left 'check.loci' printing \*clean\* and exiting 0. \*A proposition states its claim in the body and cites its receipt in the argument paragraph below, so the corpus's most load-bearing claim shape was the one shape a per-sentence scan is blind to — and it was the item's own first site.\*  $\Rightarrow$  \*\*\*Theorem-body binding added; 11 $\rightarrow$ 12 bindings, +0 false alarms, motivating defect MISSED before and CAUGHT after.\*\*\*). *Computes:* \*\*the inversion itself asserted from the metric function — without it the substitution is cosmetic and every edit is unmotivated\*\*;  $r^*/r_{\text{seam}}$  re-derived to 1.5338 from the quartic rather than quoted; the six repaired strings asserted PRESENT and the six defective strings ABSENT, by text so a reintroduction fails this file; \*\*the protected false positive asserted still present — a sweep that also took the one true sentence would be a different error with the same shape\*\*; and \*\*the recall hole asserted in BOTH directions — MISSED before the binder extension, CAUGHT after — so if the pre-extension scan

ever starts catching it, this file says the hole was not real\*\*; plus the declared exception asserted to LAPSE when its sentence is rewritten *Bound*: \*\*scope: this is exposition and instrument, not physics — no number in the cosmology moves. Its warrant is ‘THE\_METHOD’ §III: a framework a reader cannot follow cannot be excavated, and this one had a heading contradicting its own proposition. The completeness qualification added at sec:coherence is transcribed from ‘C2\_horizon\_limits’ STEP 6, which the paper’s sentence contradicted; it is not a new physics claim. ‘check\_loci’ remains a triage lint and not a CI gate — the contributor’s call, not reversed here.\*\* *Run*: python3 receipts/P15\_CR\_cosmology/P15\_the\_locus\_is\_wrong\_in\_six\_places\_and\_the\_lint\_cannot\_see\_the\_worst.py

**P15R80** P15\_the\_filter\_is\_a\_sound\_speed\_filter [OK]

*sec:what-crosses* (\*\*THE BRANCH-POINT SELECTION RULE’S EXPONENT HAS TWO FACTORS AND THE CORPUS VARIES ONLY ONE.\*\* The damping is  $e^{-k c_s |\Delta \eta|}$ : ‘sec:what-crosses’ holds  $c_s$  at the plasma’s value and varies  $k$ , reading a SCALE criterion (frozen against oscillating). \*\*Hold  $k$  and vary  $c_s$  and the same inequality is a SPECIES criterion – and a pressureless component has  $c_s = 0$  IDENTICALLY, so its exponent vanishes at EVERY wavenumber.\*\* \*\* THAT MAKES THE MATTER SECTOR’S INHERITANCE EXACT RATHER THAN ADIABATIC: \*\* at  $c_s = 0$  the segment’s mode equation is  $\psi'' = 0$ , whose constant solution the kernel leaves untouched with no approximation, whereas the frozen-mode argument rests on horizon exit and degrades at small scales. The two agree wherever both apply; only the pressureless one is available everywhere. COMPUTED on the corpus’s own segment: (1)  $|\Delta \eta| = 3.3197$  (the recorded 3.32); (2) exact transfer-matrix transmission in BOTH  $k$  and  $c_s$ , reproducing  $-k c_s |\Delta \eta|$  to the digit for every constant  $c_s$ ; (3)  $\ln T = 0$  EXACTLY at  $c_s = 0$  for  $k = 1, 10, 85.3, 1e3, 1e5$  – uniform in  $k$ \*\*); (4) at the first peak the exponent runs  $\sim 1.6e2 c_s/c$ , so the filter is opaque above  $c_s \sim 3.5e-3$ . \*\* CONSEQUENCE: a per-cent sound speed already gives  $e^{-1.6}$  and ten per cent  $e^{-16}$ , so THE CROSSING CAN INHERIT PERTURBATIONS FROM A COLD SPECIES AND FROM NO OTHER – a sharp condition on what the progenitor may hand over. \*\* \*\* (SCOPE: the arithmetic is what this receipt establishes; no species selection rule follows from it. The kernel acts on OSCILLATORY content and nothing arrives oscillating (‘P16\_every\_mode\_is\_frozen\_at\_the\_crossing’), so the exponent computed here has nothing to act on and the crossing is lossless for EVERY species.)\*\*). *Computes*: the segment’s conformal length asserted at 3.32; the plasma exponent asserted in (-160,-140);  $\ln T$  asserted zero to  $1e-9$  at  $c_s = 0$  across five decades in  $k$  *Bound*: \*\* THIS IS THE FIELD HALF ONLY. \*\* What happens to perturbations of a given species; the detailed WORLDLINE dynamics of a concrete model at the branch point is a separate computation and is not done. \*\*And an honest bound\*\*: on this segment the plasma exponent comes out -147 against the -152 the corpus records, a 3% difference from the sound-speed profile’s discretisation – nothing here turns on it, the load-bearing result being the exact vanishing at  $c_s = 0$ , which is profile-independent *Run*: python3 receipts/P15\_CR\_cosmology/P15\_the\_filter\_is\_a\_sound\_speed\_filter.py

**P15R81** P15\_the\_collapse\_leg\_is\_scale\_invariant [OK]

*sec:transmission* (\*\*THE COLLAPSE LEG IS A SCALE-INVARIANT BACKGROUND, AND P15 LISTS ‘CR HAS NO INFLATIONARY SCALE-INVARIANT ATTRACTOR’ AS AN ESTABLISHED SIGNATURE. THE ADJECTIVE WAS CARRYING THE SENTENCE.\*\* For a background  $a \sim (-\eta)^\beta$  with  $z \sim a$ , the Mukhanov-Sasaki potential is  $\beta(\beta-1)/\eta^2$ , so scale invariance requires  $\beta(\beta-1) = 2$  – \*\*a quadratic with TWO roots:  $\beta = -1$ , de Sitter inflation (expanding), and  $\beta = +2$ , a MATTER-DOMINATED CONTRACTION.\*\* \*\*And the corpus’s own collapse leg is  $\beta = 2$  exactly:\*\* its banked near-branch-point law  $r^3 = (9/2) M \tau^2$  with  $a = |r|$  gives  $d \ln a / d \ln \tau = 2/3$  and  $d \ln \eta / d \ln \tau = 1/3$ , hence  $a \sim \eta^2$  – forced by  $f \sim -2M/r$ , the mass term dominating and the  $\Lambda$  term vanishing, \*\*which is the same pressurelessness ‘A.4’ used at c54.112 to show such a component crosses exactly.\*\* The mode equation integrated from vacuum data on both backgrounds returns  $d \ln P / d \ln k = 0$  to a part in  $10^4$ . \*\*So  $n_s = 1$  is produced by the construction rather than inherited from the progenitor, and only the DEPARTURE from unity is owed.\*\*). *Computes*: the two roots asserted [-1, 2]; both logarithmic slopes of the leg asserted exactly 2/3 and 1/3; both integrated spectra asserted flat to  $1e-3$  *Bound*: \*\*bound, and it is the whole caution: this is the STANDARD contraction result (Wands 1999; Finelli-Brandenberger 2002) applied to a background the corpus DERIVES. It inherits that literature’s known difficulty – whether the matching across the bounce selects the scale-invariant mode – and here the bounce is a branch point with an analytic continuation rather than a smooth bounce. THAT QUESTION IS NOT SETTLED HERE and is registered as ‘L-169’\*\* *Run*:

python3 receipts/P15\_CR\_cosmology/P15\_the\_collapse\_leg\_is\_scale\_invariant.py

**P15R82** P15\_the\_continuation\_is\_diagonal [OK]

*sec:predictions* (\*\*THE CONTINUATION THROUGH THE BRANCH POINT ACTS DIAGONALLY ON THE TWO PERTURBATION MODES, SO THE SCALE-INVARIANT SPECTRUM DOES NOT REACH THE OBSERVER.\*\* The falsifier on the companion result, named at c54.122 and run at c54.123. \*\*The scale-invariant spectrum sits in the mode that DOMINATES the contraction ( $\zeta \sim \eta^{-3}$ ); the mode that SURVIVES on the expanding side is the constant one, whose spectrum from vacuum data is  $n_s - 1 = 6$ .\*\* Whether the first reaches the second is a connection problem, settled here two independent ways. \*\**(i) PARITY:* the exponents 2 and -1 differ by an integer so a logarithm could mix them – but  $\sinh^{2/3}$  is EVEN in  $\tau$ , which places its correction at  $\eta^6$  and steps the Frobenius recursion in SIXES past a resonance sitting at THREE. No logarithm.\*\* \*\**(ii) NUMERICS:* continuing both solutions around the branch point, the diagonal entry goes to 1.000131 and the leakage falls as a clean  $\delta^{2/3}$  power law – and the power law is itself evidence against a log, which would leave a residual no power of  $\delta$  removes.\*\* So c54.122's scale invariance is confined to the contracting side. \*\*What survives is narrower and sharper than what it replaced: the construction HAS a scale-invariant background, and the obstruction is a PARITY PROPERTY OF ITS OWN SOLUTION rather than a missing mechanism.\*\*). *Computes:* indicial exponents asserted; both vacuum coefficients' k-scalings asserted exactly  $-3/2$  and  $+3/2$ ; the absence of odd powers in the correction asserted; the resonance asserted unreachable; the diagonal entry asserted to  $1e-3$  and the leakage ratio asserted equal to  $10^{-(2/3)}$  at every decade *Bound:* \*\*bound: this settles the connection for the SCALAR mode equation on this background. It does NOT settle what a background correction ODD in  $\tau$  would do – which is what mixing would require and is the named open route. Three defects were caught in the making: a check that only proved invertibility, a branch error from squaring before taking the cube root, and a test run at  $\delta$  where  $k \eta \sim 1.4$ , outside its own frozen-mode hypothesis\*\*  
*Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_continuation\_is\_diagonal.py

**P15R83** P15\_the\_evenness\_is\_the\_time\_symmetry [OK]

*sec:predictions* (\*\*THE DIAGONALITY IS THE CONSTRUCTION'S TIME-REVERSAL SYMMETRY, AND THE SAME INVARIANCE THAT FIXES THE PEAK HEIGHTS IS WHAT STOPS THE SCALE-INVARIANT SPECTRUM CROSSING.\*\* The falsifier left by the previous receipt – mixing needs a background correction ODD in  $\tau$ , with the  $Z_3$  monodromy the named candidate – fires NEGATIVE. \*\*Write the law as  $r = A u^{2/3} g(u)$  with  $g = [\sinh(u)/u]^{2/3}$ : ALL the branch structure sits in  $u^{2/3}$ , and  $g$  is EVEN, analytic and non-vanishing at  $u=0$ \*\* – exactly, not to a truncation, since  $\log(\sinh u/u)$  is analytic in  $u^2$  near 0 and the exponential of a series in  $u^2$  is a series in  $u^2$ . \*\*The three sheets differ by a CONSTANT cube root of unity, and a constant multiplier cannot change a parity, so no sheet has an odd part; the conjugate branch is even too and has no branch point at all.\*\* \*\*AND THE EVENNESS IS  $r(\tau) = r(\tau)$ : the cosmogenesis is TIME-SYMMETRIC about the branch point, and a time-symmetric background cannot mix a growing mode with a decaying one because mixing would distinguish a direction of time the background does not.\*\* P15 sec:amplitude already leans on this same invariance one observable over – the peak HEIGHTS follow from a driving amplitude exactly invariant under time reversal. \*\*One symmetry, two consequences, and the corpus had banked only the welcome one.\*\*). *Computes:*  $\sinh(u)/u$  asserted even with unit value at 0;  $g$  asserted free of odd powers through  $u^{12}$ ; all three sheets asserted free of odd powers;  $\cosh^{2/3}$  asserted even;  $\sinh^2$  asserted invariant under  $u \rightarrow -u$  *Bound:* \*\*bound: the leg is exactly even because it is the marginally-bound  $E=1$  geodesic of an exactly VACUUM SdS exterior. A progenitor with real matter content – pressure, dissipation, anything carrying an arrow – would not be. This receipt does not exclude that; it LOCATES it, and puts the question back where 'L-150' already sits\*\*  
*Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_evenness\_is\_the\_time\_symmetry.py

**P15R84** P15\_the\_geometry\_transmits\_no\_parameters [OK]

*sec:transmission* (\*\*THE INHERITED DATA ARE NOT INHERITED FROM THE COLLAPSE GEOMETRY, BECAUSE THAT GEOMETRY TRANSMITS ZERO PARAMETERS. A COUNT, AND THE CORPUS PROVED THE KEY STEP ITSELF WITHOUT CASHING IT.\*\* The SdS progenitor family is parametrised by  $M$  alone –  $\alpha$  is the one invariant and  $E=1$  is a choice of geodesic – so the geometry is ONE-dimensional. \*\*And c54.113 showed the Kretschmann along the bead is MASS-FREE in the faller's own proper time ( $dK/dM = 0$ , re-asserted here;  $K \sim (64/27)/\tau^4$ ). A quantity identical for every member carries no information

about which member it was, so the approach transmits ZERO parameters, not one.\*\* Against that the corpus draws at least FOUR independent numbers:  $A_s$ ,  $n_s$ ,  $\eta$ ,  $z_{\text{onset}}$  ( $= \rho_r/\rho_m$ , one datum by c54.105), and the light-element composition. \*\*Four numbers cannot come from zero parameters, so 'inherited from the progenitor collapse' cannot mean inherited from the collapse GEOMETRY – it means inherited from the progenitor's MATTER CONTENT, which M does not carry, the exterior being exactly vacuum and the interior dust without composition.\*\* This RECLASSIFIES the row: not a derivation problem but a modelling one. \*\*What the geometry does fix it fixes universally:  $|\Delta \eta| = 3.32 \alpha = 1.8e4$  Mpc, from  $\Lambda$  alone – a CUT-OFF, not a CONTENT.\*\*). *Computes:*  $dK/dM$  asserted zero and the 64/27 limit asserted; the count of drawn data asserted at least four; the conformal length asserted  $3.32 \alpha$  and asserted to land between  $1.5e4$  and  $2.0e4$  Mpc *Bound:* \*\*bound: a counting argument about what the construction's own description of the progenitor can carry. It does NOT show that no model supplies  $\eta$  – it shows no model of the VACUUM EXTERIOR can, and that the row needs an interior\*\*  
*Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_geometry\_transmits\_no\_parameters.py

**P15R85** P15\_hbar\_survives\_in\_A\_s\_and\_cancels\_in\_n\_s [OK]

*sec:transmission* (\*\*hbar IS A MULTIPLICATIVE CONSTANT IN THE POWER SPECTRUM, SO IT SURVIVES IN THE AMPLITUDE AND CANCELS IN THE TILT — WHICH IS WHY  $n_s$  IS DERIVABLE HERE AND  $A_s$  IS NOT.\*\* The vacuum normalisation  $v = \sqrt{hbar/2 c_s k} e^{-i c_s k \eta}$  is the only place a quantum constant enters, and  $\zeta = v/z$  with  $z$  classical, so  $P(k) = hbar A_0 k^{(n_s - 1)}$ : \*\* $A_s = P(k_0)$  carries hbar, and  $d \ln P/d \ln k$  does not — asserted both ways.\*\* An amplitude is the spectrum at one wavenumber; a tilt is its ratio to itself at two. \*\*So the no-magnitude result (c54.128: neither real form supplies a second scale, and hbar,  $c$ ,  $G$  are unit gauges over the single one) forbids deriving  $A_s$  and says nothing about  $n_s$ .\*\*  $A_s$  is therefore permanently inherited while  $n_s$  is not, and the two were never the same kind of quantity — though the corpus has named them together in every statement of the frontier, which is how one got worked for thirteen revisions while the other went unexamined. \*\*AND THE INVERSION LOOKED LIKE THE PAYOFF: a quantity not derivable from the invariants is a function of the moduli, so  $A_s = F(2M/\alpha)$  and the observed  $A_s$  would MEASURE the parent's mass. SUPERSEDED at c54.137 ('P15\_the\_progenitor\_vacuum\_is\_negligible.too'):  $F$  is computed, it is a function of TWO moduli rather than one, and it carries no fixed normalisation — the incoming amplitude is a classical datum, so the relation cannot be inverted. The hbar asymmetry itself stands unchanged.\*\* Second handle on the same object: the spectral bound constrains total expansion past equality, this constrains the mass, and the recollapse condition caps that mass below Nariai.). *Computes:* the hbar-dependence of  $A_s$  asserted non-zero and of the tilt asserted zero *Bound:* \*\*bound:  $F$  itself is NOT computed. The pieces exist — the mode functions, the monodromy, the mode map — and what is missing is the normalisation carried through all three rather than tracked as a scaling. The settled part stands alone:  $A_s$  is not derivable, for a reason, and the reason names which other quantities share its fate — every magnitude, and no ratio\*\*  
*Run:* python3 receipts/P15\_CR\_cosmology/P15\_hbar\_survives\_in\_A\_s\_and\_cancels\_in\_n\_s.py

**P15R86** I11\_I1\_one\_condition\_two\_jobs\_and\_the\_two\_routes\_are\_independent [OK]

*sec:properframe* (I11+I1 — ONE CONDITION, TWO JOBS; AND P09's TWO ROUTES ARE INDEPENDENT. Integrable-systems field bake, probes I11 and I1.  $E = 1$  is  $k = 0$  by  $-k = E^2 - 1$ ; the  $\sinh^{2/3}$  law satisfies the marginal equation identically and leaves a residual of exactly  $+k$  away from it, so the closed form is the marginal member's alone. The same condition puts  $V_{\text{eff}} = 1$  at  $M = -r^3/(2 \alpha^2)$ , the comoving turnaround at NEGATIVE areal radius – so one condition fixes the rest frame by the field and makes the cosmology integrable. And P09's routes are independent: route (a) grants an additively separable ansatz before deriving from it, route (b) runs on the speciality invariant and contains no separability variable, which is why only (b) answers the prior question.). *Computes:* runs clean; twelve verdicts *Bound:* COMPUTES:  $\alpha = 1$  as the scale-invariant gauge, matching the thesis; no physical parameter pinned. Note:  $M$  and  $r$  are declared UNRESTRICTED, since  $\text{positive}=\text{True}$  silently removes the negative-radius root the probe is about *Run:* python3 receipts/P15\_CR\_cosmology/I11\_I1\_one\_condition\_two\_jobs\_and\_the\_two\_routes\_are\_i

**Q1.a.stated\_tolerance\_is\_a\_request\_and\_the\_corpus\_answers\_it** [OK]

*sec:diffusion-scale* (Q1 — A STATED TOLERANCE IS A REQUEST, NOT A RESULT: RUN THE CORPUS'S ODEs AGAIN AND SEE. Numerical-analysis field bake, probe Q1, and this field's real target is the receipts rather than the prose. Census: 24 receipt files integrate an ODE

and ALL 24 set an explicit rtol and atol, not one relying on a solver default; 4 contain any convergence or refinement language. So the corpus states its tolerances and, in twenty of twenty-four files, does not verify them. This receipt does the verification: scipy's solve\_ivp is patched to run every call at 100x tighter tolerance, four receipts are re-run unmodified in subprocesses, and EVERY ONE still passes its own assertions. Control: a deliberately under-resolved stiff integration through the same harness DOES move, so the patch applies and the comparator sees. Verdict 5 names the margin on the corpus's one prose-side validation:  $r.s = 144.0$  vs CAMB 144.4 is a 0.28% disagreement against the 8.2% signature it feeds, adequate by a factor of thirty.). *Computes:* runs clean; five verdicts and a control that detects a moved number *Bound:* derived here; a first comparator reported a 92% change that turned out to be a RESIDUAL column, which is supposed to move — the naive test was measuring its own patch, and the honest test is whether the assertions survive *Run:* python3 receipts/Lnumerics/Q1\_a\_stated\_tolerance\_is\_a\_request\_and\_the\_corpus\_answers\_it.py

**Q50.the\_tolerance\_comparisons\_gate\_the\_verdict\_and\_a\_third\_of\_them\_are\_not\_tolerances** [OK]  
*sec:diffusion-scale* (Q5 asked whether the corpus's 688 (now 1116) 'abs(...)' ; tol' comparisons carry meaningful tolerances and left it open for want of a mutation harness; the harness is built, and both halves of the answer are here — the comparisons DO gate the verdicts, and a third of them are not accuracy assertions at all). *Computes:* rewrites every site three ways and runs each from the receipt's own directory — margin (semantics-preserving, records E/T), mutate-up ( $\text{abs}(E+3T)$ , catches the assert shape), mutate-down ( $\text{abs}(E)*0$ , catches the guard shape). Over 269 files and 1116 static sites: the margin transform reproduces the baseline verdict on 269 of 269; of 245 baseline-passing receipts 243 are gated by a tolerance comparison; and of 918 executed sites 210 sit at exactly zero and 54 at or over their tolerance. The receipt asserts the harness's five controls and four invariants live on a ten-file subset holding every verdict class *Bound:* SUPPORTS P15 — and closes Q5 as a CONFIRMATION, as Q1 closed. NOT CLAIMED: that any particular tolerance is the right one; this measures whether a comparison gates the verdict and how much room it has, not whether its author chose well. The two ungated files are a display filter and a documented skip that names its gated substitute *Run:* python3 receipts/Lnumerics/Q50.the\_tolerance\_comparisons\_gate\_the\_verdict\_and\_a\_third\_of\_them\_are\_not\_tolerances.py

**P15R87 P15.the\_progenitor\_vacuum\_is\_negligible\_too** [OK]  
*sec:amplitude* (\*\*THE PROGENITOR'S OWN VACUUM, CARRIED THROUGH THE FULL PASSAGE AND AMPLIFIED BY THE BRANCH-POINT MIXING, IS ALSO NEGLIGIBLE — SHORT BY  $1e104$ . So 'prop:amplitude' had ruled out the one vacuum the inheritance story was NOT claiming.\*\* The collapse-side background is a single function,  $a = M(\rho x + x^2/2)$  with  $x = |\eta - \eta_c|$ , whose Mukhanov-Sasaki potential  $2/(x(x+2\rho))$  is \*\*M-FREE\*\*, so the mass enters only through the final division  $\zeta = v/a$ . The passage has three stages and each contributes one factor of  $1/\rho$ : (i) the mode freezes on the  $\beta = 2$  leg as  $v \sim D_k/x$  with  $D_k = \sqrt{\hbar/2} k^{-3/2}$ ; (ii) \*\*the connection through equality is elementary —  $v = a \int dx/a^2$  solves the  $k = 0$  problem exactly and carries  $D_k/x$  to the constant  $(3/2\rho)D_k$ ; (iii) the crunch monodromy leaks it into the survivor with coefficient  $2\pi/\rho$ ; and  $a'(\eta_c) = -M\rho$  supplies the third. \*\*THE TRANSFER LAW IS THEREFORE  $P = 18 k^3 D_k^2/(M^2 \rho^6)$ , with the  $k^3$  cancelling identically on vacuum data — exactly scale-invariant.\*\* On the vacuum that is  $A_s = (9/4)(1P/M)^2 \rho^{-6} = 2e-113$  at the recollapse cap and the determined  $\rho$ , against an observed  $2.1e-9$ . \*\*The mixing buys eight orders over the bare vacuum and is still short by a hundred and four.\*\* Two corollaries:  $A_s$  does NOT measure the progenitor's mass,  $D_k$  being free; and the classical datum has a size — the parent's growing-mode amplitude exceeded its own vacuum's by  $\sim 1e47$ . \*\*The same closed form also returns c54.131's monodromy analytically, the  $(x/\rho)\ln x$  term continuing to  $-2\pi i/\rho$ , where it had been obtained by numerical continuation.\*\*). *Computes:* the M-free potential asserted symbolically; the connection coefficient asserted in closed form and against integration from vacuum data to 0.10% over a decade in  $\rho$  and a factor 15 in  $k$ ; the transfer law asserted  $k$ -independent; the shortfall asserted between  $1e101$  and  $1e106$  *Bound:* \*\*bound:  $z = a$  with  $c_s = 1$  is assumed on the  $\beta = 2$  leg — c54.122's own hypothesis and the standard matter-contraction idealisation, genuine dust having  $c_s = 0$  and no vacuum oscillation to normalise. The  $1e104$  is far too large for any  $O(1)$  convention or plausible  $c_s$  treatment to touch\*\* *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_progenitor\_vacuum\_is\_negligible\_too.py

**P15R88 P15.no\_primordial\_B\_modes\_unconditionally** [OK]  
*sec:amplitude* (\*\*THE TENSOR SECTOR IS WHERE THE COLLAPSE-TRANSFER MACHINE RUNS WITH NO IDEALISATION AT ALL, AND IT RETURNS  $P_T \sim 5e-111$  — ONE

HUNDRED ORDERS BELOW THE OBSERVATIONAL CEILING.\*\* The scalar transfer needs  $z \sim a$ , which for a fluid holds only where  $w$  and  $c_s$  are constant — and across the progenitor's own equality ( $1+w$ ) runs  $1 \rightarrow 4/3$  while  $c_s$  runs  $0 \rightarrow 1/\sqrt{3}$ , so neither is. \*\*For tensors that is not a restriction:  $z_T = a M_{Pl}^2$  EXACTLY, for any content, any equation of state, any sound speed, so  $z_T''/z_T = a''/a$  identically and the entire passage — background, connection ( $3/2 \rho$ ), monodromy ( $2\pi/\rho$ ), division by  $a'(\eta_c) = -M\rho$  — applies verbatim with nothing assumed.\*\* It returns  $P_T = 144 \pi (1_P/M)^2 \rho^{-6}$ , scale-invariant ( $k$  cancels identically)\*\*, i.e.  $4.8e-111$  at the recollapse cap and the determined composition, against a ceiling of  $6.7e-11$  from  $r \rightarrow 0.032$ : \*\*below it by  $1.4e100$ , so  $r = 2e-102$ .\*\* So the absence of primordial B-modes is a prediction at any conceivable sensitivity, and \*\*it no longer rests on the substrate's own tensor floor — that argument had the same scope defect its scalar twin did, ruling out the substrate's vacuum and saying nothing about the progenitor's.\*\* And the ratio is fixed too: every  $k$ -dependent factor is shared by the two sectors, so  $r_{out} = 6 r_{in}$  ( $c_{54.146}$ : scalars mix twice as strongly as tensors),  $k$ -independently — the crossing can neither manufacture nor destroy a tensor signal, only rescale the ratio.\*\* The observed bound therefore measures the PARENT: its own tensor-to-scalar ratio was below  $1.3e-3$ ).  
*Computes:*  $P_T$  asserted  $k$ -independent and asserted equal to  $144 \pi (1_P/M)^2 \rho^{-6}$  symbolically; the shortfall against the ceiling asserted above  $1e95$ ; the crossing ratio asserted equal to 24  
*Bound:* \*\*bound: the factor 24 carries the scalar side's idealisation (it is  $24(1+w)c_s$  on the contracting leg, the standard matter-bounce value) and no more. The absolute  $P_T$  carries nothing\*\*  
*Run:* `python3 receipts/P15_CR_cosmology/P15_no_primordial_B_modes_unconditionally.py`

#### A2.the\_three\_owed\_derivations\_are\_one\_boundary

[OK]

*sec:scope* (the three owed derivations are ONE closed boundary:  $A_s$ ,  $n_s$  and the inherited datum are all dimensionless magnitudes). *Computes:* asserts P15s own list, prop:amplitudes  $10^{113}$  gap, the progenitor tilt, and the capstones derived progenitor composition (13 checks) *Bound:* NOT-A-PAPER-CLAIM — L-211 run on L-150 *Run:* `python3 receipts/L211_closure_adjacency/A2.the_three_owed.d`

#### P15R89 P15.the\_control\_entered\_the\_regime\_and.the\_arm.did.not.move

[OK]

*sec:refit-bound* (\*\*THE WAVENUMBER TRUNCATION WAS 78% OF THE CONTROL'S REMAINING  $\chi^2$ , AND REMOVING IT TAKES THE CONTROL TO  $\chi^2/\text{dof} = 1.18$  OVER THE FULL RANGE AGAINST A TRUE  $\Lambda$ CDM FIT'S 1.01 — SO FRONT #2'S TARGET IS MET. ACROSS FOUR INSTRUMENT STATES THE CR ARM DID NOT MOVE, AND 'L-147's PRE-REGISTERED F4 FIRES.\*\* Front #2, lead 'L-500'. \*\*PART 1 WHERE THE RESIDUAL LIVES.\*\* Of the 989 surviving  $c_{54.183}$ 's derived lensing,  $725 - 73\%$  — is in the 30 bins above  $\ell = 1500$ , a sixth of the bins\*\*); below  $\ell = 700$  the control is at  $\chi^2/\text{dof} = 0.85$  against CAMB's lensed  $\Lambda$ CDM at 0.82 on the same bins. \*\*PART 2 AND IT IS THE TRUNCATION, SHOWN BY REMOVING IT RATHER THAN BY ARGUING FOR IT.\*\* The grid is  $k = \ell/D_M$  with  $\ell = L_{MAXL}$ , so the top output multipole has NO  $k$ -headroom while  $C_\ell$  draws on every  $k$  with  $k \cdot D_M = \ell$ . \*\*Re-run at  $L_{MAXL} = 3000$  with the  $k$ -spacing held fixed and scored on the SAME 185 bins:  $989 \rightarrow 218$ ,  $\chi^2/\text{dof} 5.34 \rightarrow 1.18$ , and the last band's ratio to the reference  $0.42 \rightarrow 0.99$ .\*\* So  $c_{54.184}$ 's '53% in neither template set' was the truncation; the front's last named item is struck as NUMERICAL and no sector remains.\* \*\*PART 3 THE CR LADDER'S ALIAS-GATE WAIVER, CHECKED IN  $\chi^2$  FOR THE FIRST TIME.\*\* The gate demands 4 samples per Bessel period, the CR ladder has 2.3, and it waives itself with \*this is not aliasing — but it is only not aliasing if the answer does not depend on it. Run  $K_{CONT}=1$  to check.\* \*\*Run: agreement better than 0.7% everywhere,  $\chi^2$  to 2.1 in 51817.\*\* \*That check existed for the PEAK POSITION and had never been run against the statistic the verdict reads.\* \*\*PART 4 FOUR INSTRUMENT STATES, full range, both arms on the same lensing operator: the control  $21.2 \rightarrow 27.7 \rightarrow 5.34 \rightarrow 1.18$  and the CR arm  $290.7 \rightarrow 304.2 \rightarrow 298.9 \rightarrow 302.1$ .\*\* \*An error the INSTRUMENT carries moves both arms; this moved one — eighteenfold against 5%.\* \*\*PART 5 F2 AND F3 AS 'L-147' DEFINES THEM, WITH BOTH ARMS AT THE SAME WAVENUMBER RANGE\*\* so the construction is not charged for the instrument's truncation. Full range  $F2 = 31$  and  $F3 = 51848$ \*\*); below  $\ell = 700$   $F2 = 3.8$  and  $F3 = 23687$ \*\*); below  $\ell = 500$   $F2 = -0.2$ , the arm scoring no worse than the reference.  $F4$ 's second clause fires by three orders of magnitude where it asks for one.\*\*). *Computes:* the  $k$ -adequate control at full range asserted at  $1.18 \pm 0.12$ , \*\*without which there is no discriminating regime and nothing in the file may be read\*\*); the share the truncation carried asserted above 60%, which is PART 2's whole mechanism;  $F3/F2$  asserted above 100 at full range AND below  $\ell = 700$ , which IS the reopening; \*\*the ladder/continuum agreement asserted under 2%, without which the CR arm's  $\chi^2$  is

not admissible at all\*\*; the continuum arm asserted at  $\ell_1/\ell_A = 0.5703$ ; the four-state contrast asserted as a 5x control improvement against a 15% arm movement, which is what makes this a statement about physics rather than about one build; and the tail share asserted above 60%, since PART 1 is what sent PART 2 looking *Bound*: \*\*scope: this is a MEASUREMENT DISCREPANCY and NOT a framework verdict – ‘L-147’s own F5 says so and ‘PO-7’ is protected precisely here, so this file reports a number and does not convert it; the conversion runs by ‘F5’s stated procedure. The deficit itself is not new: the corpus has carried  $\ell_1/\ell_A = 0.5703$  against the sky’s 0.7312 since c54.168 and P15 states it at 21.9%. What is new is a control good enough to read it through a likelihood instead of a ratio. AND WHAT IS NOT RULED OUT: a defect the instrument’s states SHARE – 0.5703 has not moved across EIGHT of them, which is evidence either of a robust prediction or of a shared upstream constant, and this file cannot separate those and does not claim to. Nor does  $\chi^2/\text{dof} = 1.18$  make the control a FIT: it is 17% above a true LambdaCDM’s on the same bins, and what that residual is remains unnamed.\*\* *Run*: python3 receipts/P15\_CR\_cosmology/P15\_the\_control\_entered\_the\_regime\_and\_the\_arm\_did\_not\_move.py

#### B1.the\_control\_separates\_prediction\_from\_shared\_constant [OK]

*sec:refit-bound* (the control separates robust prediction from shared upstream constant: same code, different D\_M and r\_s, same  $\ell_A$ , peaks 21.8% apart, and the control lands on the sky to 0.17%). *Computes*: reads both arms from the c54.186 spectra and asserts the scaling test, the control agreement and the ladder/continuum equality (12 checks) *Bound*: NOT-A-PAPER-CLAIM — the adversarial check c54.186 requested *Run*: python3 receipts/L147\_two\_arm/B1.the\_control\_separates\_prediction

#### B2.the\_prose\_duplicate\_did\_not\_reach\_main [OK]

*sec:refit-bound* (the reported prose duplicate did not reach main: zero at three granularities across every commit, because the two edits were different paraphrases and git asked). *Computes*: scans every commit that touched the file at paragraph, sentence and line-window level and asserts the two diffs differ (14 checks) *Bound*: NOT-A-PAPER-CLAIM — checks a routed report before acting on it *Run*: python3 receipts/L147\_two\_arm/B2.the\_prose\_duplicate\_did\_not\_reach\_main.py

#### P15R90 P15\_the\_first\_peak\_is\_the\_seam\_datum\_and\_the\_spacing\_is\_not [OK]

*sec:coherence* (\*\* $\ell_1/\ell_A = 0.5703$  IS A STATEMENT ABOUT THE SEAM DATUM AND NOT ABOUT THE RATE, AND c54.186’s OWN CAVEAT IS WHAT SENT THIS LOOKING.\*\* Front #2, lead ‘L-501’. The CR arm begins at the seam ( $z = 6761$ ,  $\eta_{\text{S}} = 180.4$  Mpc) and a first-peak mode crossed the horizon at  $\eta \sim 75$ , so \*\*the modes that set  $\ell_1$  are ALREADY SUB-HORIZON at the start and their phase there is ASSIGNED rather than derived\*\*. The corpus’s datum is \*‘ONE datum per mode and a COMMON phase’\* – which fixes that the phase is common and not WHICH phase; the instrument picks a density extremum. \*\*PART 1 SCANNING THAT ONE FREEDOM MOVES  $\ell_1/\ell_A$  FROM 0.5703 TO 1.2599 – A FACTOR OF 2.21\*\*, and at  $\phi = 3\pi/8$  the peak at 172 is not there at all and the series starts at 380. \*\*PART 2 WHAT IS ROBUST IS THE SPACING, AND IT IS ROBUSTLY WRONG:\*\* the mean peak spacing holds at 0.734-0.818 of  $\ell_A$  across the whole scan – a spread of 1.11 against the position’s 2.21 – and is \*\*never 1.0\*\*. \*So the disagreement that does not depend on a choice is a ~21% SPACING deficit, not a 22% POSITION one.\* \*\*PART 3 AND ‘L-147’s F4 FIRES UNDER EVERY PHASE, WHICH IS WHY c54.186 SURVIVES THIS:\*\* on identical bins the CR arm runs  $\chi^2/\text{dof}$  202-449 against a control at 0.75, so \*\*even its BEST phase costs 269x the control\*\*. \*\*PART 4 THIS REPRODUCES c54.164 ON THE CURRENT INSTRUMENT, WHICH HAD NOT BEEN DONE\*\* – that revision found  $\ell_1$  in {150, 165, 315} on the OLD ‘ROBUST\_p1p2\_scan’ code, and the finding was never carried across to ‘ACOUSTIC\_two\_arm’, which everything since is built on.). *Computes*: \*\*CRPHI = 0 asserted to reproduce the coded 0.5703, without which the knob is not a control on itself and every comparison is against the wrong baseline\*\*; the position’s spread asserted above 1.5x and its maximum pinned at 1.2599, which is the withdrawal; the spacing’s spread asserted under 1.35x AND its maximum under 0.95 of  $\ell_A$ , \*\*since if the spacing ever moved this file would have no positive content\*\*; and the CR arm’s BEST phase asserted above 50x the control and above  $\chi^2/\text{dof}$  150 – \*\*if that ever falls, c54.186 must be RESTATED rather than defended\*\* *Bound*: \*\*!! SUPERSEDED BY P15\_the\_spacing\_is\_right\_and\_the\_acoustic\_phase\_is\_wrong (r4401).\*\* \*Headline claim withdrawn by a later revision of ‘sec:refit-bound’; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. \*\*Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.\*\* \*\*scope: this WEAKENS

a number the paper carries and STRENGTHENS a verdict the previous revision landed, and neither is a framework judgement. P15's '21.9% deficit, not an artefact of how last scattering was modelled' is TRUE and is not the sentence a reader takes from it: the deficit is not last scattering's, it IS a function of the seam phase, which is upstream of last scattering and is a choice inside the corpus's own stated datum. 'F5' is not softened – removing a confound makes a measurement cleaner, not a conclusion; 'PO-7' protected; the conversion runs by 'F5's stated procedure. AND WHAT IS STILL OPEN, narrower than before: the phase is ONE degree of freedom in the datum; Theta-hat's flatness in k is another and is NOT scanned here. A scan that moved the SPACING would be the one that mattered, and this file does not run it.\*\* *Run: python3 receipts/P15\_CR.cosmology/P15.the.first.peak.is.the.seam.datum.and.the.spacing.is.not.py*

**B3.a.dichotomy.stated.is.not.a.dichotomy.proved** [OK]

*sec:refit-bound* (B1 stands but L-235s disposition did not follow: ruling out one horn establishes the other only if the dichotomy is exhaustive, and the truth was a third thing). *Computes:* reads the forks nine phase spectra and asserts the 2.21x position spread against the 1.11x spacing spread (12 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects this lines own r2462 disposition *Run: python3 receipts/L147\_two\_arm/B3.a.dichotomy.stated.is.not.a.dichotomy.proved.py*

**P15R91 P15.both.datum.freedoms.opened.and.the.spacing.still.holds** [OK]

*sec:coherence* (\*\*THE SECOND AND LAST FREEDOM IN THE SEAM DATUM IS OPENED, AND THIS FILE WIDENS ITS OWN PREDECESSOR RATHER THAN CONFIRMING IT.\*\* Front #2, lead 'L-502'. c54.187 scanned the common PHASE and named the other freedom: what "flat in k" is flat AT. The datum's amplitude is the corpus's C4 shape  $T(x) = 3(\sin x - x \cos x)/x^3$ ; \*\*the instrument evaluates it at a SINGLE argument  $x_e = 1/\sqrt{3}$ , which is that shape read at each mode's OWN horizon crossing\*\* ( $x = k \cdot c_s \cdot \eta$  with  $k \cdot \eta = 1$  gives exactly  $1/\sqrt{3}$ ) — a self-similar reading. \*The alternative evaluates the SAME shape at each mode's own phase AT THE SEAM,  $x = k \cdot c_s \cdot \eta_A$ . Both are readings of "one datum per mode"; neither is a new assumption. \*\*PART 1 THE POSITION IS NO MORE STABLE:\*\* across eighteen readings  $\ell_{1/A}$  spans \*\*0.5570–1.2599, 2.26x\*\* against the phase scan's 2.21x. \*\*PART 2 THE SPACING IS MEASURED ONLY WHERE THERE IS A SERIES TO MEASURE, WHICH DISQUALIFIES EIGHT OF THE EIGHTEEN:\*\* those return a spectrum whose higher peaks have COLLAPSED — three features, the last at zero — and \*a spacing read off that is not a spacing\*. Requirement stated before the numbers: four peaks, the fourth at least 5% of the first. \*\*PART 3 OVER THE TEN THAT QUALIFY THE SPACING SPANS 0.6764–0.8179 OF  $\ell_{1/A}$  — 1.21x, mean 0.772, never above 0.82.\*\* \*\*\*c54.187 reported 1.11x from the phase alone; opening the second freedom widens it to 1.21x, so this file weakens its own predecessor and the ~23% spacing deficit stands at LOWER strength.\*\*\* \*\*PART 4 AND F4 FIRES AT EVERY ONE OF THE EIGHTEEN:\*\*  $\chi^2/\text{dof}$  runs 157–460 against a control at 0.75 on identical bins, so \*\*the arm's best reading costs 208x the control\*\*. \*And the best-fitting reading is one of the eight DISQUALIFIED ones — it fits better by having less structure to disagree with. Reported because it is the best number in the table and a reader would find it; it is not a direction.\*). *Computes:* the position's spread asserted above 1.5x with its maximum pinned at 1.2599, the standing withdrawal from c54.187; \*\*the spacing's spread asserted under 1.6x AND its maximum under 0.90 of  $\ell_{1/A}$  — the point past which c54.187's conclusion would have to be WITHDRAWN rather than widened\*\*; the usable-series count pinned at 10, since PART 3's range is computed only over those and a silent change in how many qualify would move it without moving the code; and \*\*the arm's BEST reading asserted above 50x the control and above  $\chi^2/\text{dof}$  120 — if that falls, c54.186 is restated\*\* *Bound: \*\*!! SUPERSEDED BY P15.the.spacing.is.right.and.the.acoustic.phase.is.wrong (r4401).\*\** \*Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. \*\*Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.\*\*\* \*\*scope: both freedoms the corpus's own phrase leaves open are now scanned and the datum is closed. What has NOT been varied is the background: 'Z.START' is solved so that  $\pi \cdot D_M/r_s = 301.6$  exactly, which is a TARGET and not an output — a separate item from the datum, registered rather than resolved. 'F5' unsoftened, 'PO-7' protected, conversion Daryl's.\*\* *Run: python3 receipts/P15\_CR.cosmology/P15.both.datum.freedoms.opened.and.the.spacing.still.holds.py*

**P15R92 P15.the.floor.is.a.distance.between.models.not.a.number.from.the.data** [OK]

*sec:refit-bound* (\*\*THE CONTROL'S REMAINING 17% IS NAMED, AND NAMING IT REQUIRED CHANGING THE MEASUREMENT.\*\* Front #2, lead 'L-503'. c54.186 left it un-

named; **'F2'** as **'L-147'** defines it is a difference of two numbers each computed AGAINST THE SKY, so it mixes how far the two models are apart with where each sits relative to one noise realisation. **PART 1** amplitude-matched, the instrument tracks a reference LambdaCDM to **'better than 1.6% everywhere'**, the departure running  $-1.1\%$  at  $\ell \sim 200$  through zero near 600 to  $+0.7\%$  at 1400 — **'a slow undulation, not a mis-phased oscillation'**. **PART 2** **F2 IS NOT A CLEAN MEASURE, SHOWN BY MOVING THE REFERENCE:** three defensible LambdaCDMs give **+31.2, +21.5, +27.2** — **'a spread comparable to the value'**. **PART 3** **SO THE FLOOR IS MEASURED WITHOUT THE DATA** —  $\chi^2_{\text{sep}} = (A_{\text{m.a}} - A_{\text{m.c}})^T F (A_{\text{m.a}} - A_{\text{m.c}})$ , a distance between two MODELS: **21.3** over 185 bins, **0.11**  $\chi^2$  per bin, about a third of a standard deviation per bin. **PART 4** **AND THE ORDERING REVERSES, WHICH IS THE TRAP:** the massless-neutrino reference gives the **'smallest F2 (21.5) and the largest separation (62.7)'** — **'a reference chosen to flatter the instrument through the data is the one it is furthest from as a model'**. **AND c54.178's LIST IS NOW COMPLETE:** reionisation struck c54.181, lensing built c54.183, the wavenumber range closed c54.186, and **'the neutrino mass — the one item never addressed — is worth about 10 of the 31'**, the reference carrying one massive species at 0.06 eV where this instrument carries none. *Computes:* the instrument-vs-reference departure asserted under 3%, which bounds everything else; the model separation asserted under 0.5  $\chi^2$  per bin, the number that replaces the unnamed 17%; **THE ORDERING REVERSAL** asserted directly — if F2 and  $\chi^2_{\text{sep}}$  ever stop disagreeing, PART 4 is not a trap and the paragraph is unearned; and **F4's margin** asserted above 100x at the WORST reference, since a verdict that depends on the choice is not one *Bound:* **'scope: 0.11  $\chi^2$  per bin is NOT zero and this file does not say the instrument is exact. It says the residual is at the level where the REFERENCE's own parameter choices matter as much as the instrument's error, and that below that level this statistic cannot arbitrate between two LambdaCDM implementations at all. F3 = 51848 against a floor of 21–63 across every defensible reference, so 'F4's margin is three orders of magnitude whichever is taken — which makes 'L-147's reading SAFER and converts nothing. 'F5' unsoftened, 'PO-7' protected.'** *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_floor\_is\_a\_distance\_between\_models\_not\_a\_number\_from\_the\_data.py

**J1** **the\_disagreement\_no\_longer\_has\_two\_sides** [OK]

*sec:flatlcdm* (L-171s internal disagreement no longer has two sides: across 18 datum readings the perturbation route gives no definite first peak, and its 150 is c54.164s number from a retired instrument). *Computes:* reads every c54.187–188 spectrum, asserts 150 and 220 both absent and the 2.26x range, and ties it to the class c54.187 routed as item 35 (12 checks) *Bound:* NOT-A-PAPER-CLAIM — supersedes a register rows content; its next step is untouched *Run:* python3 receipts/L171\_two\_routes/J1\_the\_disagreement\_no\_longer\_has\_two\_sides.py

**P15R93** **P15\_the\_spacing\_deficit\_survives.the\_one\_fitted\_parameter** [OK]

*sec:tensions* (**'THE LAST FREE THING UPSTREAM IS THE CORPUS'S OWN ONE FITTED NUMBER, AND MOVING IT BY 31% MOVES THE SPACING DEFICIT BY 7%'**. **Front #2**, lead **'L-504'**. c54.188 closed both seam-datum freedoms and named this: **'Z\_START'** is solved so that  $l_A = \pi D_M / r_s$  hits a TARGET rather than coming out as an output. **'That is declared, not hidden — P15 sec:tensions says "the measured acoustic scale is reproduced at the directly measured H\_0 by a single z\_onset" and calls it "the one fitted number"'**. What had not happened is that the number was a literal buried in a **'brentq'** call, so nothing downstream could vary it and nothing had. **PART 1** **MOVING THE PIN MOVES EVERYTHING IT SHOULD:** pinning  $l_A$  at 260/280/301.6/320/340 drives  $z_{\text{onset}}$  from 11009 to 5066 and  $r_s$  from 157.1 to 120.2 Mpc. **PART 2** **AND THE SPACING DEFICIT DOES NOT FOLLOW IT:** the mean peak spacing stays at **0.7647–0.8205** of  $l_A$  — a spread of 1.07 against the pin's 1.31, mean 0.798. **'So the ~20% spacing deficit is a property of the construction and not of the value its one fitted parameter was fitted to.'** **PART 3**  $l_1/l_A$  runs 0.5176–0.6308 (1.22x) and **'never approaches the sky's 0.7312'** — and its drift is almost entirely  $l_A$ 's, the peak itself moving only 164  $\rightarrow$  176 while  $l_A$  moves 260  $\rightarrow$  340. **PART 4** **THE STRUCTURAL STATEMENT UNDER IT:** in ordinary acoustics  $l_1$  goes as  $D_M/r_s$ , so a 24% fall in  $r_s$  should raise  $l_1$  by 31%; **'here it raises it by 7%'**.  $\Rightarrow$  **'This construction's peaks track its own sound horizon at 24% of the rate acoustics requires, and that — rather than any one ratio — is what the comparison with the sky is registering.'** **CORRECTED c54.190 ('L-505')**: the spacing figure in this row is the mean of the FIRST THREE GAPS, which is all four peaks at LMAXL = 1000 can give. At production depth the asymptotic spacing is 0.975 of  $l_A$  — 2.5% short, not 21%

— and what is wrong is the ACOUSTIC PHASE, by 0.62 pi. The robustness this row measures stands; the name it gives the quantity does not.\*\*\*). *Computes:* \*\*the spacing’s spread under the pin asserted below 1.20x — the point past which the deficit would BE the fitted parameter’s and every conclusion resting on it withdrawn\*\*; its maximum asserted below 0.90 of l\_A, since a pin taking it near 1.0 would remove the deficit itself; the first-peak position asserted never to reach the sky’s ratio at any pin, which is PART 3; \*\*the peaks’ sensitivity to r\_s asserted in (0, 0.6) of the acoustic rate — if it ever reached that rate the disagreement would be about a NUMBER and not about a mechanism\*\*; and all five pins asserted to give a four-peak series *Bound:* \*\*!! SUPERSEDED BY P15\_the\_spacing\_is\_right\_and\_the\_acoustic\_phase\_is\_wrong (r4401).\*\* \*Headline claim withdrawn by a later revision of ‘sec:refit-bound’; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. \*\*Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.\*\*\* \*\*scope: the acoustic question on this front is now closed as far as the construction’s FREEDOMS go — c54.187 scanned the seam phase, c54.188 the amplitude reading, this the fitted parameter, and across all of them the SPACING sits near 0.78–0.80 of l\_A and never reaches 0.9 while the POSITION spans 2.26x and states nothing. What is left is not a freedom that can be scanned: whether the peaks SHOULD be tied to the sound horizon is a question about the physics. ‘F5’ unsoftened, ‘PO-7’ protected, conversion Daryl’s. AND ONE INCONSISTENCY RECORDED RATHER THAN FIXED: the instrument pins l\_A = 301.6 where ‘P15\_zonset\_determinations’ pins the MEASURED 301.76 — 0.05%, and it is why this instrument returns z\_onset = 6761 where that receipt returns 6797. Named so the two are not read as a disagreement.\*\* *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_spacing\_deficit\_survives\_the\_one\_fitted\_parameter.py

**P15R94** P15\_the\_spacing\_is\_right\_and\_the\_acoustic\_phase\_is\_wrong [OK]  
*sec:refit-bound* (\*\*THE “~21% SPACING DEFICIT” OF c54.187-189 IS THE FIRST THREE GAPS AND NOT THE SPACING. AT PRODUCTION DEPTH THE ASYMPTOTIC SPACING IS RIGHT TO 2.4% AND WHAT IS WRONG IS THE ACOUSTIC PHASE.\*\* Front #2, lead ‘L-505’. \*\*\*THIS FILE CORRECTS THREE OF ITS OWN LINE’S REVISIONS.\*\*\* \*\*Those scans ran at LMAXL = 1000 so that eighteen readings and five pins were affordable; at that depth the CR arm has four peaks, so “the mean peak spacing” was a mean of THREE GAPS — and the first three gaps are the only ones where the arms disagree.\* \*\*PART 1/2 AT PRODUCTION DEPTH BOTH ARMS CARRY EIGHT PEAKS:\*\* LambdaCDM’s last four gaps average \*\*1.002 of l\_A\*\* and the CR arm’s \*\*0.975 — 2.5% short, not 21%\*\*; gap by gap the CR arm runs \*\*0.725, 0.853, 0.875\*\* of the control’s and is \*\*level by the fourth (0.973)\*\*). \*\*PART 3 SO THE TWO SERIES ARE PARALLEL LINES WITH DIFFERENT INTERCEPTS, AND THAT IS A PHASE:\*\* fitted on peaks 4–8, slopes \*\*1.003 and 0.976 of l\_A\*\* (agreeing to 2.6%) and intercepts \*\*–0.263 and –0.878 of l\_A\*\*. \*A driven series peaks at  $k \cdot r_s = n \cdot \pi - \phi$ , so the intercept IS  $-(\phi/\pi) \cdot l_A$ :\*  $\Rightarrow$  \*\*\*the two differ by 0.62 pi in the ACOUSTIC PHASE SHIFT, at a spacing they agree on.\*\*\* \*\*PART 4 AND A LOW-ell TRANSIENT THE CONTROL DOES NOT HAVE:\*\* measured each against its OWN asymptotic line, the control’s first three peaks sit within \*\*16\*\* of it and the CR arm’s sit \*\*142, +80, +18\*\*\*, converging — \*and that transient is what compressed the early gaps\*. \*\*\*PART-HEADLINE CORRECTED c54.195 (‘L-508’): the 0.62 pi is the value at  $\phi = 0$ . Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL’S VALUE LIES INSIDE that span, so it is not a prediction. The driving CONTRIBUTION stands; its promotion to “the whole disagreement” does not.\*\*\*). *Computes:* \*\*the CR asymptotic spacing asserted above 0.90 of l\_A and pinned at 0.975 — if it were NOT near l\_A there would be nothing to correct and c54.187-189 would stand as written\*\*; the first-three-gap and asymptotic values asserted to STRADDLE 0.85, which IS the diagnosis (one number reported for the other); \*\*the slopes asserted parallel to 6%, without which the offset is not a pure phase and PART 3 is wrong\*\*; the phase offset pinned at 0.615 +/- 0.08, which is what replaces the retracted number; the transient asserted at more than 3x the control’s, since PART 4 is otherwise empty; and the weakest peak’s contrast asserted above 1.05, since a series read off non-features is noise *Bound:* \*\*!! SUPERSEDED BY P15\_two\_arm\_control\_and\_guard (r4401).\*\* \*Headline claim withdrawn by a later revision of ‘sec:refit-bound’; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. \*\*Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.\*\*\* \*\*scope: WHAT THIS RETRACTS is this line’s own — c54.187’s “~21%”, c54.188’s “~23%”, c54.189’s “~20%” and the paper’s “acoustic series some 23% too close” are all the first-three-gap value. WHAT IT DOES NOT RETRACT is the robustness:

the first-three-gap spacing IS stable at 0.77–0.82 across eighteen datum readings and five pins and F4 fires at every one; what changes is that the stable quantity was the wrong one to call the disagreement. AND THE CORRECTION MAKES c54.187 CENTRAL rather than a caveat — the disagreement is a PHASE, and the phase is exactly what the seam datum assigns and what "one datum per mode and a COMMON phase" does not fix. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5's stated procedure — and this makes the discrepancy SHARPER, not smaller: a spacing that is right and a phase that is wrong names where to look.\*\* *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_spacing\_is\_right\_and\_the\_acoustic\_phase\_is\_wrong.py

**B4.the\_intercept\_is\_a\_phase\_and\_the\_control\_proves\_it** [OK]

*sec:coherence* (the intercept IS the acoustic phase by change of variables, and the control validates the reading by reproducing the textbook 0.25–0.27; the 21% spacing figure was a depth artefact). *Computes:* derives the intercept identity, fits both arms on peaks 4–8, reproduces the gap-by-gap series and the low-l transient (14 checks) *Bound:* NOT-A-PAPER-CLAIM — the adversarial check c54.190 requested *Run:* python3 receipts/L147\_two\_arm/B4\_the\_intercept\_is\_a\_phase\_and\_the\_control\_proves\_it.py

**B5.the\_scan\_cannot\_test\_the\_asymptotic\_phase** [OK]

*sec:coherence* (the corrected claims load-bearing quantity cannot be tested by the datum scan: all 23 scan spectra stop at three or four peaks and zero reach the peaks-4-to-8 fit). *Computes:* asserts the depth and peak count of every scan spectrum, that the production pair alone supports the fit, and where the transient sits (11 checks) *Bound:* NOT-A-PAPER-CLAIM — names a small affordable experiment *Run:* python3 receipts/L147\_two\_arm/B5\_the\_scan\_cannot\_test\_the\_asymptotic\_phase.py

**P15R95 P15\_the\_acoustics\_work\_and\_the\_phase\_is\_the\_whole\_disagreement** [OK]

*sec:refit-bound* (\*\*THE ASYMPTOTIC SPACING TRACKS THE SOUND HORIZON AT 98% OF THE ACOUSTIC RATE, SO THIS CONSTRUCTION'S ACOUSTICS WORK — AND THE WHOLE DISAGREEMENT IS THE ACOUSTIC PHASE.\*\* Front #2, lead 'L-506'. c54.190 separated spacing from phase and owed both a production-depth test; \*\*the scans that varied them were all at the depth c54.190 showed to be insufficient\*\*. \*\*PART 1 THE SPACING TRACKS r\_s AT THE FULL RATE:\*\* moving the pin from l\_A = 301.6 to 340 drops r\_s by 11%, ordinary acoustics requires the spacing to rise 13%, and \*\*it rises 13% — 98%\*\*). \*So the peaks in this construction ARE set by its own sound horizon.\* \*\*PART 2 WHICH RETRACTS c54.189's HEADLINE:\*\* that revision reported \*\*"the peaks track their own sound horizon at 24% of the rate acoustics requires"\* — \*\*measured on the FIRST peak, which c54.190 showed sits 142 multipoles off the asymptotic line, inside a transient\*\*. \*Over this file's pin range the first peak moves at 37% against the series' 98%; the figure is not reproduced and is not meant to be — what is reproduced is the FINDING.\* \*\*PART 3 THE PHASE IS ROBUST TO THE FITTED PARAMETER:\*\* the intercept moves  $-0.878 \rightarrow -0.899$  of l\_A across the same 31% swing — \*\*3% of a discrepancy of 0.62\*\*. \*\*PART 4 AND THE SEAM DATUM'S PHASE FREEDOM IS A REAL LEVER THAT SPANS A THIRD OF IT:\*\* at the opposite phase the intercept moves to  $-0.671$ , closing the gap from 0.615 to 0.408 — \*\*34%, and no further\*\*.  $\Rightarrow$  \*\*\*Which is the confirmation c54.190's diagnosis wanted: the datum's phase controls the acoustic phase, as a phase diagnosis predicts, and cannot close the gap.\*\*\* \*\*PART-HEADLINE CORRECTED c54.195 ('L-508'): the 0.62 pi is the value at phi = 0. Across four seam phases at production depth the asymptotic phase spans 0.891 and the CONTROL'S VALUE LIES INSIDE that span, so it is not a prediction. The driving ATTRIBUTION stands; its promotion to "the whole disagreement" does not.\*\*\*). *Computes:* \*\*the asymptotic sensitivity to r\_s asserted in (0.85, 1.15) — if it were NOT near 1 the acoustics would not work and the phase reading would not be the whole story\*\*); the FIRST peak's sensitivity asserted below 0.6, since PART 2 retracts a number and must show the quantity really is less sensitive; the pin's effect on the phase asserted below 15% of the discrepancy, without which the phase is the fitted parameter's; \*\*the seam phase's closure asserted in (0.15, 0.70) — if it ever closed the WHOLE gap the discrepancy would be the datum's alone and this front's verdict would have to be restated\*\*); the phase discrepancy pinned at  $0.615 \pm 0.05$ ; and at least seven peaks per arm, since every number here is an asymptotic fit *Bound:* \*\*\*!! SUPERSEDED BY P15\_two\_arm\_control\_and\_guard (r4401).\*\* \*Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. \*\*Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.\*\*\* \*\*scope: the front's acoustic statement in final form — this construction reproduces the acoustic SPACING (the quantity its one fitted parameter is fitted to, and which it then tracks at 98% of the

acoustic rate when that parameter is moved) and disagrees with the sky in the acoustic PHASE by 0.62 pi, robust to the fitted parameter and only a third reachable by the one freedom the seam datum leaves open. That is ONE number and ONE mechanism where the front began with four items and a ratio. ‘F5’ unsoftened, ‘PO-7’ protected, conversion Daryl’s — and the discrepancy is now specific enough to be attacked: an acoustic phase shift is a computable consequence of the driving, so “why 0.62 pi” has an address. This file does not have it.\*\* *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_acoustics\_work\_and\_the\_phase\_is\_the\_whole\_disagreement.py

**B6.the\_acoustics\_reproduce\_and\_the\_dichotomy\_is\_false** [OK]

*sec:coherence* (the acoustics reproduce at 98.2% of the required rate, and r2485s stated dichotomy was not exhaustive: the phase moves and stops at 34%). *Computes:* recomputes the r\_s response and the four-row phase table from the forks spectra, and asserts both horns false (13 checks) *Bound:* NOT-A-PAPER-CLAIM — verifies c54.191 and corrects this lines own r2485 *Run:* python3 receipts/L147\_two\_arm/B6\_the\_acoustics\_reproduce\_and\_the\_dichotomy\_is\_false.py

**B7.the\_seam\_phase\_moves\_the\_asymptotic\_intercept** [OK]

*sec:coherence* (item 38 answered at production depth: the seam phase MOVES the asymptotic intercept by  $\sim 0.31$  lA (half the 0.615 disagreement) while the spacing holds, so the 0.62pi is datum-contingent not structural). *Computes:* validates the phi=0 control against c54.186, fits three production seam phases on matched acoustic orders 4-7, asserts the intercept span and the stable spacing (5 checks) *Bound:* NOT-A-PAPER-CLAIM — a measurement; informs vein L-202, answers lead L-171; conversion to a PO-7 verdict is unseated *Run:* python3 receipts/L147\_two\_arm/B7\_the\_seam\_phase\_moves\_the\_asymptotic\_intercept.py

**K1.a\_capability\_assumed\_from\_one\_failure** [OK]

*sec:coherence* (the whole acoustic disagreement is the driving: undriven the arms agree to 0.013 in phi/pi and the CR slope is exactly 1.0000 lA, and both lines accepted a capability claim for eight revisions). *Computes:* fits all four production spectra, asserts the undriven agreement and the 5.72x driving ratio, and counts this lines own tarball absorptions (12 checks) *Bound:* NOT-A-PAPER-CLAIM — verifies c54.193-194 *Run:* python3 receipts/L239\_capability\_assumed/K1.a\_capability\_assumed\_from\_one\_failure.py

**B8.the\_zero\_velocity\_band\_is\_a\_third\_of\_the\_gap** [OK]

*sec:coherence* (the seam datums band across the two ZERO-VELOCITY phases is 0.207 against a 0.615 disagreement — a third, not most — and over that pair the control is NOT inside). *Computes:* solves  $\sin(\phi)=0$  on  $[0, 2\pi]$ , reads both admissible phases off four production spectra from two nodes, and withdraws this lines own L-202 link (13 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows c54.195s withdrawal *Run:* python3 receipts/L147\_two\_arm/B8\_the\_zero\_velocity\_band\_is\_a\_third\_of\_the\_gap.py

**P15R96 P15\_the\_admissible\_band\_is\_a\_third\_and\_the\_two\_entries\_differ\_against\_the\_potential**

[OK]

*sec:coherence, sec:refit-bound* (\*\*ITEM 43 (r2509) ACCEPTED: OVER THE PHASES THE CORPUS ADMITS, THE ACOUSTIC PHASE MOVES 0.207 — A THIRD OF THE 0.615 GAP — AND THE CONTROL’S 0.263 IS OUTSIDE IT. THE DISAGREEMENT IS REAL AND BOUNDED.\*\* Front #2, lead ‘L-513’; VEIN ‘L-202’. \*\*PART 1\*\* the pair is \*\*forced\*\*, not chosen: ‘sec:what-crosses’ says what crosses is \*\*frozen\*\*, a frozen mode has  $\dot{\delta}\gamma=0$ , continuity gives  $\theta\gamma=\text{tfrc34Dkc.s}\sin\phi$ , so  $\sin\phi=0$  — the paper’s own transmission argument read at the seam, in print before the scan existed. \*\*PART 2 AND THE REASON c54.199 GAVE 56 IS WRONG:\*\* ‘y0[,6] = Ph0’ is fixed at \$-1\$ independent of  $\phi$  while ‘dg0’ carries  $\cos\phi$ , so the two entries are \*\*not\*\* a sign flip of the state — \*\*what reverses is the relative sign of the density against the potential\*\*, a compression against a rarefaction in the same well. \*\*PART 3\*\* 54’s and cc54’s production spectra agree to four decimals at both admissible phases. \*\*PART 4\*\* band 0.2069 vs gap 0.6152 —  $2.97\times$ , control \*\*outside\*\* [0.6711, 0.8780]. \*\*PART 5\*\*  $\chi^2/\text{dof}$  \*\*281 and 379\*\* against 3.71 —  $76\times$  and  $102\times$  — so c54.195’s PART 5 conclusion is stronger on the pair than on the span it was illustrated in. \*\*PART 6\*\* and P15 ‘sec:refit-bound’ already carried “spans a third of it” and the 0.408, \*\*two paragraphs above the paragraph that withdrew it\*\*.). *Computes:* the four-step admissibility chain asserted in the instrument’s AND the paper’s source — \*\*if ‘sec:what-crosses’ did not say modes freeze, the pair would be a choice and r2509 weaker than this file claims\*\*; ‘dg0’ asserted to flip while ‘Ph0’ does not, which is what makes item 17’s reason wrong; \*\*54 and cc54 asserted to agree to four decimals — a band measured on one node’s spectra is a band measured once\*\*; the ratio  $\sim 3$  and, separately, the control asserted OUTSIDE the band, without which c54.195 stands as written; both admissible readings asserted above  $50\times$  the control, without which the withdrawal

takes the conclusion with it; and the paper's own earlier 0.408 asserted to reproduce *Bound*: **\*\*!! SUPERSEDED BY P15.two\_arm\_control\_and\_guard (r4401).****\*\*** \*Headline claim withdrawn by a later revision of 'sec:refit-bound'; the only citation available would put back into the paper what the paper removed — attempted r4397, reverted r4399. **\*\*Kept, not deleted: the scans stand and this is the record of how the arc reached its present state.****\*\*\*** **\*\*scope: this corrects this line TWICE and in opposite directions — it makes the disagreement firmer than c54.195 left it, and it retracts the explanation c54.199 routed to 56. 'F5' unsoftened; 'PO-7' protected; the conversion runs by 'F5's stated procedure. No new run: five banked production spectra, two nodes.****\*\*** *Run: python3 receipts/P15\_CR\_cosmology/P15.the\_admissible\_band\_is\_a\_third\_and\_the\_two\_entries\_differ\_against\_the*

**B9.the\_admissible\_pair\_is\_forced\_not\_merely\_distinguished** [OK]  
*sec:what-crosses* (the admissible pair  $\phi$  in  $\{0, \pi\}$  is FORCED by P15s own transmission argument — every mode freezes before the crossing, and a frozen mode has zero velocity). *Computes*: quotes the freezing argument at source, solves  $\sin(\phi)=0$ , and asserts the control lies OUTSIDE the band with both nodes agreeing to four decimals (12 checks) *Bound*: NOT-A-PAPER-CLAIM — verifies c54.200 and upgrades r2509 *Run: python3 receipts/L147.two\_arm/B9.the\_admissible\_pair\_is\_forced\_not\_merely\_distinguished*

**B10.the\_thermal\_mechanism\_needs\_an\_exponential\_approach** [OK]  
*sec:transmission* (L-519 answered from the MECHANISM side: thermality needs an exponential near-horizon approach, the degenerate members is a power law, and P15 already derived that split for its transmission dichotomy). *Computes*: integrates the tortoise coordinate for a simple and a double root, re-verifies the Nariai double root, and asserts P15s exponential/power-law split at source (11 checks) *Bound*: NOT-A-PAPER-CLAIM — a third footing for L-519; no temperature value claimed *Run: python3 receipts/L147.two\_arm/B10.the\_thermal\_mechanism\_needs\_an\_exponential\_approach*

**PO-7** [OK]  
*sec:coherence* (kill receipt for PO-7 under route (2): object named, four checks answered — and the CHAIN check does not clear, because the disagreement rests on four reproduced things and one ARGUED one). *Computes*: a kill receipt, not a script — the four checks written out with the inversion routes named *Bound*: NOT-A-PAPER-CLAIM — narrows PO-7; no authorisation requested *Run: python3 receipts/kills/PO-7.md*

**S1.every\_mode\_of\_interest\_freezes\_and\_the\_inversion\_is\_closed** [OK]  
*sec:what-crosses* (PO-7 chain-(4) link (d) reproduced on a second path: every mode of interest ell 28-2475 freezes before the crossing ( $r=c_s k/|aH| \rightarrow 0$ ), so  $\sin(\phi)=0$  forces  $\phi$  in  $\{0, \pi\}$ , the band stays 0.2069 with the control OUTSIDE, and the kill receipt's inversion (a mode crossing unfrozen) is closed by the divergence of  $|aH|$ ). *Computes*: reproduces the progenitor interior independently (matching P16's freeze-out to the digit,  $x_{\text{freeze}}=0.002015$  at  $\text{ell}=2475$ ), computes the velocity ratio  $\rightarrow 0$  at the crossing for every mode, and anchors *sec:what-crosses* + the kill receipt's chain check (10 checks) *Bound*: NOT-A-PAPER-CLAIM — reproduces PO-7's one non-reproduced link; informs L-171/L-202; F5 unsoftened, no conversion claimed *Run: python3 receipts/L805\_freezing\_reproduced/S1.every\_mode\_of\_interest\_freezes\_and\_the\_inversion\_is\_closed.py*

**S1.a\_massive\_mode\_freezes\_too\_and\_the\_mass\_term\_vanishes\_at\_the\_branch\_point** [OK]  
*sec:what-crosses* (A2 (PO-seam / PO-7 inversion 3): a massive mode freezes too, for ANY mass — the branch point  $a \rightarrow 0$  annihilates the mass term  $m^2 a^2$  while  $|aH|$  diverges, so  $\omega/|aH| \rightarrow 0$  independent of  $m$ ; a massive trajectory carries no phase from non-freezing, closing PO-7 route 3 from the massive side). *Computes*: derives the mass term  $+m^2 a^2$  from the covariant action (sympy), shows  $\omega/|aH|$  at the crossing is  $m$ -independent to the digit up to  $m=1e5$  across ell 28-2475, freeze-out epoch unmoved (5 checks) *Bound*: NOT-A-PAPER-CLAIM — A2; informs L-202/L-171; CRPHI stays assigned; F5 unsoftened *Run: python3 receipts/L806\_massive\_freezing/S1.a\_massive\_mode\_freezes\_too\_and\_the\_mass\_term\_vanishes\_at\_the\_branch\_point*

**S1.the\_peak\_estimator\_recovers\_a\_set\_phase\_independent\_of\_the\_envelope** [OK]  
*sec:coherence* (A10 (PO-7 inversion 2): the peak-4-8 estimator recovers a SET asymptotic phase to  $\pm 0.005$  in  $\phi/\pi$  and is envelope-independent (spread  $\pm 0.006$ ) across six damping/width/tilt variations spanning the CR-vs-control transfer differences — so the 0.408 is not an estimator artefact; inversion route 2 closes and the disagreement hardens). *Computes*: injects synthetic acoustic combs with known  $\phi$  in under six envelopes, runs the exact B7/B8 estimator, checks faithfulness and envelope-invariance (4 checks) *Bound*: NOT-A-PAPER-CLAIM — A10; informs L-171; F5 unsoftened, no conversion claimed *Run: python3 receipts/L807\_driven\_estimator/S1.the\_peak\_estimator\_recovers\_a\_set\_phase\_independent\_of\_the\_envelope*

**S1.the\_seam\_phase\_is\_derived\_compression\_correlated\_on\_the\_observable\_leg** [OK]  
*sec:what-crosses* (PO-7/PO-seam: the seam phase CRPHI is DERIVED, not free = CRPHI=0 (a compression correlated with the well), on the observable leg — the pre-onset is pressureless (r2701) so no oscillation swings the inherited adiabatic overdensity's sign; bracket-checked against the LCDM control's own adiabatic IC; the  $\{0,\pi\}$  band collapses to 0.878, disagreement pinned at 0.615, CRPHI= $\pi$  (0.408, a rarefaction) anti-adiabatic and inadmissible). *Computes:* reads the code's CRPHI convention (overdensity in a well = CRPHI=0), shows a pressureless mode's sign is fixed while a pressured one swings, confirms the adiabatic bracket, and anchors the pressureless-onset statement in CR\_cosmology.tex (4 checks) *Bound:* NOT-A-PAPER-CLAIM — closes PO-7 route (3)'s residue; informs L-171/L-202; F5 unsoftened, one assumption (Phi-continuity) flagged *Run:* python3 receipts/L815\_seam\_phase\_derived/S1.the\_seam\_phase\_is\_derived\_compression\_correlated\_on\_the\_observable\_leg

**S1.the\_banked\_spectra\_are\_nk600\_not\_nk260\_and\_pass\_the\_guard** [OK]  
*sec:where-the-likelihood-sits* (corrects C52's sampling premise: the banked two-arm spectra the likelihood scores are NOT under-sampled — they are NK=600 (the README command), which PASSES the guard at 5.7 pts/Bessel-period. cc54 (15GB) reproduced c54.178.lcdm BIT-FOR-BIT (max |dDI|=8e-16). C52 read the .npz's 238-point shape as NK=260, but 238=len(arange(100,2000,8)) is set by LSTEP=8/LMAXL=2000, not NK; 56's 3.7GB container OOM'd at NK=600 so tested the guard-failing NK=260. The 7.14/280.09 stand; C51's dropped-bin point is separate and stands). *Computes:* confirms the 238-point shape is arange(100,2000,8) independent of NK, computes guard pts/period= $2\pi(3NK-1)/(LMAXL-12)$  giving 2.46@NK=260 (fails) and 5.69@NK=600 (passes), and asserts the banked npz is bit-identical to cc54's NK=600 reproduction (4 checks) *Bound:* NOT-A-PAPER-CLAIM — corrects C52's provenance premise; the guard itself is right (NK=260 does under-sample); does not touch C51's real dropped-bin blocker *Run:* python3 receipts/L820\_banked\_spectra\_sampling/S1.the\_banked\_spectra\_are\_nk600\_not\_nk260\_and\_pass\_the\_guard

**S2.the\_lmax2512\_extension\_recovers\_c51s\_dropped\_bins\_and\_CR\_stays\_disfavoured** [OK]  
*sec:where-the-likelihood-sits* (discharges C51 (the damping-tail half of 56's resampling route): the LMAXL=2512 extension (302 multipoles to ell=2508) recovers the 30 dropped bins (ell 1759-2508), and including them does NOT rescue CR. Two effects isolated: the wider k-range lifts the control 7.14- $i$ 3.81/dof on the SAME 185 bins (c54.186's truncation effect), added tail - $i$ 3.68; the CR arm does not gain (280.1- $i$ 281.1 overlap) and stays 260.1/dof. F3 widens 50497- $i$ 51547 — the dropped region makes CR MORE disfavoured, not less). *Computes:* scores the LMAXL=2512 lcdm/cr extensions on the overlap (ell=1996) and full (201) bin sets vs the banked pair, isolating the k-range effect from the damping-tail addition, and computes F3 on both bin sets (4 checks) *Bound:* NOT-A-PAPER-CLAIM — the C51 half of L-820; CR verdict unchanged (both F3  $\sim$ 2400x threshold); control's 3.68/dof still large, truncation-vs-physics not further split *Run:* python3 receipts/L820\_banked\_spectra\_sampling/S2.the\_lmax2512\_extension\_recovers\_c51s\_dropped\_bins\_and\_CR\_stays\_disfavoured

**S1.the\_cr\_residual\_is\_acoustic\_structure\_not\_truncation\_and\_that\_is\_why\_it\_is\_unresponsive\_to\_lmax** [OK]  
*sec:where-the-likelihood-sits* (PO-10 OWED #496 (the row's real remainder): DIAGNOSES the CR arm's 280/dof that r2781/C53 found "unresponsive to L\_max and diagnosed by nothing". A per-bin  $\chi^2$  decomposition (full plik\_lite covariance) localises the residuals: the control's is 75% in the high-ell tail (ell $_i$ 1200, the truncation), the CR arm's is 57% in the acoustic-peak bands (ell $_i$ 850) and 18% tail — so CR is acoustic-structure, not truncation, which is WHY it is unresponsive to L\_max. CR comb compressed (172/404/636 vs sky 220/540/810, 11/1A 0.57 vs 0.73 = PO-7's deficit); rescaling doesn't collapse  $\chi^2$  (best  $\sim$ 200) so position+heights). *Computes:* computes the per-bin  $\chi^2$  contribution  $r_i(\text{Fr})_i$  with the full covariance and groups by ell-band for both arms, reads the CR peak positions, and scores under an ell-axis rescaling (4 checks) *Bound:* NOT-A-PAPER-CLAIM — OWED #496 CR half; F5 reserves the verdict (PO-7 protected); control third convergence point (L3200) measured separately *Run:* python3 receipts/L822\_cr\_residual\_diagnosed/S1.the\_cr\_residual\_is\_acoustic\_structure\_not\_truncation\_and\_that\_is\_why\_it\_is\_unresponsive\_to\_lmax

**S1.the\_stored\_cr\_rs\_is\_the\_from\_onset\_acoustic\_horizon\_and\_matches\_the\_arms\_own\_integral** [OK]  
*sec:acoustic* (PO-10 (the code read 56's r2790 routed): the stored CR r\_s=135.46 is NOT a book-keeping error. The line is R\_S=rs\_from(Z.START) with Z.START the FITTED onset; rs\_from uses the SAME sound speed/grid/onset the acoustic oscillation uses, so 135.46 IS the arm's acoustic horizon and the peaks asymptote to LA=301.6. r2790's "peaks demand 158" is the low-ell

transient (first-4-gap mean  $\sim 158$ , high-n gaps  $\sim 139$  bracketing the stored); its "direction requires 245" is the from-a<sup>0</sup> horizon, NOT the acoustic one (pre-onset pressureless, L-815, no oscillations before onset). Smaller than LCDM because CR starts oscillating later — the framework's fitted-onset design). *Computes:* reproduces  $R_S=rs_{\text{from}}(Z_{\text{START}})$  via the brentq fit, computes  $rs_{\text{from}}$  from a<sup>0</sup> vs the fitted onset for both arms, and reads the CR peak gaps by ell-range (first-4 vs high-n) (4 checks) *Bound:* NOT-A-PAPER-CLAIM — the r2790 code read; not a bug, the fitted-onset design's output; whether that design is forced is F5/PO-10's *Run:* python3 receipts/L823-cr\_rs\_is\_the\_onset\_horizon/S1\_the\_stored\_cr\_rs\_is\_the\_from\_onset\_acoustic\_horizon\_and

S1.the\_third\_point\_shows\_the\_control\_converges\_by\_L3000\_and\_the\_unlensed\_floor\_is\_the\_lensing [OK]

*sec:acoustic* (PO-10 (OWED #496, control half — the THIRD convergence point r2781 asked for): the control CONVERGES by L3000, it does NOT slide down an  $L^{-3.4}$  law to "1.1 by L6000". On the same 185 bins the unlensed control goes  $7.18 \rightarrow 3.83 \rightarrow 3.73 \rightarrow 3.73$  across LMAXL 2000/2512/3000/3200; the  $L^{-3.4}$  law fit to the first two points predicts 2.23 at L3200, the measurement is 3.73 — the two-point rate was a one-time truncation recovery, not a convergence rate. The  $\sim 3.73$  unlensed floor IS the lensing: c54.183's lensed/unlensed ratio takes the SAME converged spectrum  $3.73 \rightarrow 1.18$ , and the lensed control converges  $5.38 \rightarrow 1.33 \rightarrow 1.19 \rightarrow 1.18$  to 1.18 by L3000 vs CAMB lensed's 1.014 (a  $\sim 0.16/\text{dof}$  instrument floor). Truncation (LMAXL) and lensing are ORTHOGONAL axes; F3 cancels the shared 1.18 floor). *Computes:* scores the four banked control spectra (c54.178 L2000, L820 L2512, c54.186 L3000, L824 L3200) on the fixed 185-bin set unlensed and under CAMB's lensed/unlensed ratio; checks the  $L^{-3.4}$  extrapolation vs the measured third point, the L3000/L3200 convergence, and the top-bin truncation (4 checks) *Bound:* NOT-A-PAPER-CLAIM — control-arm convergence; does not overturn 56's convergence claim (the LENSED control does converge, to 1.18 by L3000); companion to L-822 (CR half); F5 untouched *Run:* python3 receipts/L824-control\_converges/S1\_the\_third\_point\_shows\_the\_control\_converges\_by\_L30

S1.the\_converged\_controls\_remainder\_is\_a\_broadband\_instrument\_floor\_not\_a\_localized\_defect [OK]

*sec:acoustic* (PO-10 (OWED #454, the remainder localized): after L-824's convergence the CONTROL arm's  $1.18/\text{dof}$  vs CAMB lensed's 1.014 on the same 185 bins (the " $\sim 100$ " was pre-convergence, pre-lensing) is a BROADBAND instrument floor, not a localized defect. A per-bin  $r_{\text{ell}}(\text{Fr})_{\text{ell}}$  decomposition shows every ell-band 100-2000 contributes a positive excess ( $+4.5/+6.0/+7.4/+1.6/+5.7/+4$  total  $\sim 30$  in  $\chi^2$ ), none dominating, slightly leading in the first two acoustic peaks (300-850 carries 13.4 of 30). So the  $\sim 1$  target is instrument-accuracy-limited (a uniformly better transfer, not a patchable band), and  $F3=\chi^2(\text{CR})-\chi^2(\text{LCDM})$  cancels this shared floor). *Computes:* per-bin decomposition of the converged lensed control (L824 arm) and CAMB lensed LCDM through plik\_lite TT's full covariance on the fixed 185 bins, by ell-band (3 checks) *Bound:* NOT-A-PAPER-CLAIM — control-arm diagnostic; #454's target instrument-limited, still open; F5 untouched; continues L-824 *Run:* python3 receipts/L826-control\_floor\_broadband/S1.the\_converged\_controls\_remainder\_is\_a.br

T1.the\_transfer\_is\_half\_built [OK]

*sec:envelope* (PO-12 is HALF BUILT: the debt is two steps and step (1) — the specification of how fluctuations gravitate on the geometrically fixed background — is computed in *sec:envelope*, two sections before the debt is named). *Computes:* asserts the two-step sentence, the envelope-is-derived sentence, the structural-argument clause and the tilt-irreducible-residual clause verbatim (9 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a register row; step (2) is genuinely not run *Run:* python3 receipts/L202-the\_seam/T1\_the\_transfer\_is\_half\_built.py

P15R97 C12.the\_odd\_even\_half\_has\_its\_parameter [OK]

*sec:envelope* (PO-10s odd/even half already has its mechanism and parameter: C5b\_baryon.term puts the asymmetry on the expansion leg at  $R_b=0.60$ , which is the rows own description of the item). *Computes:* asserts the rows wording, C5bs location and growth-factor and conclusion, and the papers own sentence (9 checks) *Bound:* NOT-A-PAPER-CLAIM — narrows a register row; the height pattern is not produced *Run:* python3 receipts/P15.CR\_cosmology/C12.the\_odd\_even\_half\_has\_its.par

P15R98 C14.po12\_gates\_po10 [OK]

*sec:envelope* (PO-12 GATES both of PO-10s runs: the textbook odd/even ratios at  $R_b=0.60$  are LCDM peak-height formulae that assume a transfer, and both of PO-10s halves are statements about C.l). *Computes:* asserts PO-10s remaining half and P15s genuine-build sentence verbatim, computes the textbook ratios, and confirms neither row records the dependency (7

checks) *Bound*: NOT-A-PAPER-CLAIM — records an ordering between two rows *Run*: `python3 receipts/P15_CR_cosmology/C14_po12_gates_po10.py`

**P15R99** `C15_the_cancellation_does_not_reach_the_ratio` [OK]  
*sec:envelope-consequence* (the corpuSS cancellation does not reach the odd/even ratio: it holds between two RATES (same quantity, microphysics identical) while odd/even compares two MULTIPOLES on one rate, where the envelope leaves 13-36%). *Computes*: asserts the cancellation sentence and both rate-ratio results verbatim, and computes adjacent-peak envelope ratios on three envelopes (9 checks) *Bound*: NOT-A-PAPER-CLAIM — supplies the reason behind r2646s gate *Run*: `python3 receipts/P15_CR_cosmology/C15_the_cancellation_does_not_reach_the_ratio.py`

**P15R100** `C16_the_transfer_is_six_eighths_built` [OK]  
*sec:envelope* (PO-12s step 2 is six-eighths built: the driving verified in closed form, source removal exact, projection calibrated at  $L^*=302.2$  vs 301, and a C11 RATIO with no free parameter already in the paper (0.823 recomputed vs 0.82)). *Computes*: verifies the driving satisfies its ODE, is even, and normalises; asserts the genuine-build, projection and no-free-parameter clauses verbatim; recomputes the high-ell ratio (8 checks) *Bound*: NOT-A-PAPER-CLAIM — narrows a debt to the absolute spectrum *Run*: `python3 receipts/P15_CR_cosmology/C16_the_transfer_is_six_eighths_built.py`

**P15R101** `C17_the_instrument_already_carries_both` [OK]  
*sec:instrument* (CORRECTS r2658: both pieces it marked NOT LOCATED are built and validated — the Peebles recombination history IS the visibility function and the line-of-sight source with its j\_l kernel IS the projection). *Computes*: asserts sec:instruments description, its validation figures and its flat-projection wording verbatim, against the debt's geometric-background clause (9 checks) *Bound*: NOT-A-PAPER-CLAIM — corrects a prior lines audit; the debt is now what the instrument RUNS ON *Run*: `python3 receipts/P15_CR_cosmology/C17_the_instrument_already_carries_both`

**P15R102** `C18_bespoke_means_two_legs` [OK]  
*sec:envelope* (what BESPOKE means for PO-12: the transfer must carry TWO backgrounds joined at the branch point — radiation-dominated collapse leg, geometry-set expansion leg — and the paper forbids mixing the rates). *Computes*: computes the friction coefficient for both contents, asserts the collapse-leg scoping, the L1/L2 division and the forced consistency rule verbatim (7 checks) *Bound*: NOT-A-PAPER-CLAIM — sharpens a debt; the branch-point join is untouched *Run*: `python3 receipts/P15_CR_cosmology/C18_bespoke_means_two_legs.py`

**P15R103** `C19_the_branch_point_join` [OK]  
*sec:transmission* (THE BRANCH-POINT JOIN COMPUTED: transmission is conservation of R, so the join is a change of variable worth exactly 9/10 — applied ONCE where LCDM spreads it over equality, which is why CR has no early ISW). *Computes*: asserts the not-yet-in-hand and transmits clauses, the matter-dominated-to-nine-orders and four-per-cent clauses verbatim, and derives 9/10 from  $R/\Phi=(5+3w)/(3+3w)$  (7 checks) *Bound*: NOT-A-PAPER-CLAIM — super-horizon join only; sub-horizon modes and the dynamical crossing stay open *Run*: `python3 receipts/P15_CR_cosmology/C19_the_branch_point_join.py`

**P15R104** `C20_the_caveat_named_an_empty_set` [OK]  
*sec:what-crosses* (r2661s super-horizon caveat named an EMPTY SET: at the branch point  $(rH)^2$  diverges, the comoving horizon goes to zero, and every mode is outside — so the 9/10 join is unrestricted). *Computes*: asserts the every-mode-outside and seam clauses verbatim, computes the divergence of  $(rH)^2$  and the vanishing comoving horizon, and confirms the radiation term dominates (7 checks) *Bound*: NOT-A-PAPER-CLAIM — removes a caveat from a prior receipt; the crossing dynamics stay deferred *Run*: `python3 receipts/P15_CR_cosmology/C20_the_caveat_named_an_empty_set.py`

**P15R105** `C21_the_superhorizon_transfer_closes` [OK]  
*sec:envelope* (THE SUPER-HORIZON TRANSFER CLOSES:  $\Phi_{-i}$   $\Phi_{+i}$  exactly at the branch point for every k (departure  $O(k^2 \eta^2)$ ), so with r2661s join  $\Phi_{\text{exp}}(k) = (9/10) \Phi_{+i}(k)$  scale-invariantly — the branch point does not filter). *Computes*: computes the branch-point limit and the series coefficients, re-derives the join, and asserts the transmits clause verbatim; flags the 1/10 near-miss as different-origin (7 checks) *Bound*: NOT-A-PAPER-CLAIM — scalar super-horizon only; the single end-to-end run stays unrun *Run*: `python3 receipts/P15_CR_cosmology/C21_the_superhorizon.t`

**P15R106** `C22_the_end_to_end_number` [OK]  
*sec:earlyisw* (THE END-TO-END NUMBER: 7.5% low-ell power deficit from the absent early

ISW, 18% at 1.D rising to 54% at 2 1.D from damping — and the acoustic peaks BETWEEN them are where the chain stops, correcting r2661s no-further-evolution). *Computes*: asserts the scope qualification, the super-horizon-equation clause, the order-two decay and the not-a-radiation-effect clause verbatim, and computes both ratios (8 checks) *Bound*: NOT-A-PAPER-CLAIM — corrects a prior receipts scope; not a prediction against data *Run*: python3 receipts/P15\_CR\_cosmology/C22\_the\_end.to\_

**P15R107** C59\_the\_control\_reproduces\_camb\_and\_the\_height\_defect\_was\_k\_truncation [OK]  
*sec:coherence* (\*\*PO-24 STEP ONE: THE CONTROL RUN, AND IT COMES OUT THE OTHER WAY. THE HEIGHT MACHINERY IS NOT BROKEN.\*\* ‘PO-24’ records the bespoke transfer as getting the positions right and the HEIGHTS wrong by 22.7% and 97.5% on the arm whose answer is known, and the r3869 handoff named the live lead: two projection paths that are not the same physics, with \*\*which one produced the 2.721 not established\*\*. \*\* (1) IT WAS los\_spectrum, AND THAT IS THE SMALLER HALF. \*\* ‘\_project’ also carries the polarisation source  $g \text{ Pi}/4 + (3/4k^2) d^2[g \text{ Pi}]$ , so the comparison to CAMB was not like for like. \*\* (2) THE LARGER HALF NOBODY HAD LOOKED FOR: THE k-INTEGRAL IS TRUNCATED WHERE IT IS NOT CONVERGED. \*\* The instrument built its k-grid from LMAXL – the grid of multipoles to PRINT – so choosing what to report chose where to stop integrating  $C_l = \text{INT } P(k) \Delta l(k)^2 dk$ . Reported ell grid held FIXED and k\_max alone moved: \*\*2.721 -j 2.446 -j 2.399 -j 2.393\*\*. \*\* (3) BOTH ARE NEEDED AND NEITHER ALONE SUFFICES. \*\* los\_spectrum at converged k\_max gives 2.393; \_project at truncated k\_max gives 2.516; both together give \*\*2.197\*\*. \*\* (4) THE RESULT: THE CONTROL REPRODUCES CAMB. \*\* Peaks \*\*220 / 540 / 812\*\* against the sky 220.6 / 538.1 / 809.8, \*\*P1/P2 = 2.197 against CAMB 2.200 – 0.14%\*\* and inside the sky’s 1 sigma; P1/P3 = 2.192 against 2.277. Converged in k\_max and stable to 2e-3 over a 1.5-fold change in mode count. \*\* (5) THE FORK IS SETTLED: REPAIR. \*\* The two-arm architecture carries the transfer; a fresh build would have discarded a working instrument on the strength of an artefact. \*\* (6) AND THE RECORDED LN SCAN IS INERT: \*\* LN=12 and LN=25 give identical ratios to 1e-3 with the state vector genuinely wider, against a recorded 4.496 -j 8.009.). *Computes*: the recorded 2.721/4.496 asserted REPRODUCED (so the row is not wrong about what it ran); k-truncation asserted to move both ratios on ONE path; the polarisation source asserted to move them again at converged k\_max; NEITHER-ALONE asserted; the headline asserted within 1% of CAMB; every peak asserted within 9 of the sky’s; convergence asserted to 2e-3 across NK 280 vs 420; Jac\_of asserted == 1 to 1e-12 on the control so r3512’s HIER composition defect cannot reach this result; LN inertness asserted with the state width checked. \*\*CONTROL THAT FIRES: at the TRUNCATED k\_max the same NK widening is EQUALLY STABLE and equally WRONG\*\* – so stability in the knob you varied is not convergence, which is the defect this receipt removes *Bound*: \*\* CONTROL ARM ONLY. No CR number is produced or corrected. \*\* The CR arm is truncated by the same mechanism and is NOT re-measured; r3512’s HIER composition defect is a no-op on the control (Jac == 1) and stays LIVE for CR, which the next step must settle first. NOT CLAIMED: that the bespoke transfer is finished, that PO-24 is closed, or anything about CR’s peak heights. What is established is that its control works *Run*: python3 receipts/P15\_CR\_cosmology/C59\_the\_control\_reproduces\_camb\_and\_the\_height\_defect\_was\_k\_truncation.py

**P15R108** C60\_the\_hier\_composition\_defect\_names\_two\_flags\_its\_own\_tree\_never\_had [OK]  
*sec:coherence* (\*\*THE DEFECT FOUR DOCUMENTS DEFER TO IS NOT IN THE INSTRUMENT, AND WAS NOT IN THE ONE IT WAS WRITTEN AGAINST.\*\* ‘C59’ closed PO-24’s control step and deferred the CR arm to r3512’s ‘HIER’ composition defect – ”the first thing the next step must settle” – and THE\_REGISTER, INDEX and both appendices carry that deferral. \*\* (1) THE REMEDY NAMES TWO FLAGS ITS OWN TREE NEVER HAD. \*\* r3512’s fix is ”give the hierarchy’s gravitational source the stacking clock and its diffusion the leaf”, i.e. teach ‘HIER=1’ about ‘SRCSTACK’ and ‘DIFFLEAF’; both occur \*\*zero\*\* times in ACOUSTIC\_two\_arm.py at 95559d53, the commit r3512 IS, whose 42 os.environ flags are listed. They were real on \*\*6beeca84\*\* (SRCSTACK x13, DIFFLEAF x3), which is \*\*not an ancestor\*\* of 95559d53 – two nodes, two lines, one instrument. \*\* (2) THE ASYMMETRY IT COUNTED IS ‘sound\_phase’. \*\* ”The main path references two” counts evolve’s RHS plus ‘sound\_phase’, which is CALLED ONCE in the file, from inside ‘qscan()’, the QSCAN=1 diagnostic that computes no spectrum. Like for like: each ODE right-hand side carries the clock \*\*once\*\*, and \*\*neither projection carries it at all\*\* – true at HEAD and at 95559d53 alike. \*\* (3) THE TWO RIGHT-HAND SIDES ARE THE SAME BYTES. \*\* Lines 576 and 959 are character-identical, as are the rate selections at 536 and 916, and every clock-or-source operation in ‘evolve’ is in

‘evolve\_hier’. \*\* (4) AND IT MOVES: \*\* on ARM=cr toggling LEAFPERT changes evolve\_hier’s returned state by 2.0 relative, so the hierarchy is not ignoring the clock. \*\*=¿ C59’s deferral is discharged; what gates the CR arm is convergence in k, not a composition fix.\*\*). *Computes:* SRCSTACK and DIFFLEAF asserted absent at BOTH 95559d53 and HEAD, with the 42-flag inventory printed; 6beeca84 asserted to carry them and asserted NOT an ancestor of 95559d53; 95559d53 asserted an ancestor of HEAD; the Jac\_of call sites located by enclosing function at both commits with the RHS count asserted 1 and 1 and the projections asserted 0 and 0; sound\_phase’s single call site asserted inside qscan’s line span; lines 576/959 and 536/916 asserted character-identical; the operation set of evolve asserted a subset of evolve\_hier’s; Jac\_of asserted != 1 on ARM=cr and the LEAFPERT toggle asserted to move evolve\_hier’s state. \*\*CONTROL THAT FIRES: the receipt EXITS 1 rather than passing if 95559d53 is not in the clone\*\* – a shallow checkout makes every PART 1 assertion vacuous, which is the hollow-assertion failure the fork names *Bound*: \*\* ESTABLISHES AN ABSENCE, NOT A CORRECTNESS. \*\* That the named defect is not there; NOT that the HIER path is right. No CR number, no peak position and no height ratio is produced. r3512’s gate order 1, 2 and 4 – validate Pi on the control, the PISRC subtraction, the CR run – are untouched and remain PO-13 Run 1’s business. NOT CLAIMED: that r3512 was careless; two lines of work on one file is a corpus-mechanics failure *Run:* python3 receipts/P15\_CR.cosmology/C60.the\_hier\_composition\_defect\_names\_two\_flags\_its\_own\_tree\_never\_had.py

**P15R109** C62.the.diffusion.signature.is.a.wash.and.the.isw.is.not.what.it.turned.on [OK]  
*sec:coherence* (\*\*PO-24: THE DIFFUSION-SCALE SIGNATURE IS RELOCATED INTO THE INFERRED TILT — NEITHER A TT-SHAPE TENSION NOR A WASH.\*\* PO-24, narrowed at r4337, asks what the enlarged r\_D does to the OBSERVED high-ell power, “which turns on a parameter refit and on the early ISW term”. \*\* (1) THE JUDGEMENT: the early ISW is carryable, and the leaf assignment MULTIPLIES it rather than creating it. \*\* \_rt, which sets the radiation fractions in the perturbation source, is the stack “both arms” by the instrument’s own comment, so radiation is in the source under either assignment; measured, the post-recombination ISW is 0.348 of the SW amplitude on the leaf against 0.229 on the stacking rate. \*\* (2) BUT NOISW REACHED NO REPORTING PATH \*\* – read at one site in main()’s analytic block, zero references in los\_spectrum, \_project and hier\_run – so the switch for the term this row turns on measured nothing. Repaired r4492; asserted at source here. \*\* (3) THE EARLY ISW IS NOT WHAT A HIGH-ELL QUESTION TURNS ON: \*\* 3.49 sigma/bin over l = 100-1296 and \*\*0.77 sigma/bin over 700-1296\*\*, propagated against plik\_lite rather than quoted as a fraction. \*\* (4) THE SIGNATURE THROUGH A REFIT. \*\* Scaling r\_D by the measured ratio multiplies the damping by a GAUSSIAN in ell; a tilt is a POWER LAW. Imposed on the measured sky and refitted on amplitude and tilt: \*\*0.26 chi^2/bin over P15’s 185-bin ceiling, and MORE completely absorbed the higher one looks\*\* (0.06/bin over 700-1760, 96% over 1000-2508) – because over a restricted range a Gaussian looks like a power law. \*\* (5) BUT THE ABSORPTION IS NOT FREE: dn\_s = -0.0304 against sigma = 0.00323 derived from the SAME covariance that scores the fit – a 9.4 sigma cost. \*\* On the instrument’s own DSCAN pair the same fit gives 0.92/bin and 22.7 sigma. \*\*The signature does not vanish under a refit; it MOVES, out of TT shape and into an inferred parameter.\*\*). *Computes:* the r4492 repair asserted at source in both paths; r\_D measured from the instrument at 7.1033 and 7.6398 Mpc, +7.55% raw, with DAMPX asserted to BE that ratio squared; the ISW’s imprint asserted larger on the leaf and asserted under 1 sigma/bin at high ell; the 185-bin ceiling asserted; the shape residual asserted under 0.6 sigma/bin on the data route and under 1.1 on both; the tilt cost asserted above 5 sigma on both routes. \*\*CONTROL THAT FIRES: an injected tilt of +0.02 and +0.05 is recovered to within 5% with ¿99.9% absorbed\*\* – a fitter that reports absorption is shown not to be manufacturing it *Bound*: \*\* THE MAGNITUDE IS NOT ROBUST TO A FACTOR OF THREE AND THE RECEIPT SAYS SO. \*\* The data-shape route multiplies the final spectrum by the Gaussian; the instrument applies the damping inside the source where it also enters the Doppler term through a derivative, on an unlensed ceiling-limited spectrum. The two differ 3.6x in residual and 2.8x in tilt; the larger is the more faithful route, so the wash is stated at the WEAKER bound and closing the factor of three is named as work not done here. NOT CLAIMED: that the CR arm as a whole is consistent (r4136 has its heights at -20.7% and -29.2%); that dn\_s of this size IS a tension, which is a statement about a joint fit this receipt does not perform. \*\*A FORTIORI, BOTH WAYS: on the SHAPE residual, freeing only the named degeneracy is the least absorption a real refit could achieve, so the wash is a bound; on the TILT it is NOT a bound, since more parameters would SHARE the displacement\*\* *Run:* python3 receipts/P15\_CR.cosmology/C62.the.diffusion.signature.is.a.wash.and.the.isw.is.not.what.it.turned.on

**P15R110** C61\_the\_undriven\_premise\_is\_false\_on\_the\_rate\_the\_framework\_assigns\_the\_perturbations  
[OK]

*sec:coherence* (\*\*PO-13's DIAGNOSIS RESTS ON A CROSSING CENSUS, AND THE CENSUS COMES OUT THE OTHER WAY ON THE RATE THE FRAMEWORK ASSIGNS THE PERTURBATIONS.\*\* P07 sec:frontiers states "the standard shift that carries it is universal only where every mode crosses the horizon while there is a plasma to be driven, and \*\*on this rate the acoustic modes re-enter above the onset, so none of them does\*\*", and PO13\_WORKING\_STATE gives the same premise as the structural reason the arm cannot reach the sky. \*\*r3409 made LEAFPERT the default: the perturbations run on the LEAF congruence, and Hleaf carries the radiation term where the CR stacking rate does not.\*\* Horizon crossing for the driving is a perturbation-sector event, so the census that decides the diagnosis has to be run on the leaf rate. \*\* (1) THE LEAF BACKGROUND HAS AN EQUALITY AND THE ONSET PRECEDES IT: \*\*  $z_{\text{eq}} = \Omega_m / \Omega_b - 1 = 3936$  (eta 236.4) against the onset at  $z = 6761$  (eta 180.4). \*\* (2) THE FIRST PEAK IS SUPER-HORIZON WHEN THE PLASMA STARTS: \*\* on the leaf,  $k_1/aH(\text{onset}) = 0.858 \text{ h}$  for  $l_1 = 204$ , and it enters at  $z = 5590$ , above equality – a radiation-dominated crossing, \*\*while there is a plasma\*\*. \*\* (3) THE BAND IS NOT A SINGLE MODE: \*\* every  $l_1$  237.7 enters after the onset and  $155.6 \text{ h}$  237.7 enters in radiation. \*\* (4) AND THE OTHER RATE GIVES THE OPPOSITE ANSWER: \*\* on the stacking rate  $k_1/aH(\text{onset}) = 1.415 \text{ h}$  (already inside) and the background carries no radiation term at all, so it has no equality and the sentence is not merely true there but necessary. \*\*=; The diagnosis was stated on the rate that carries the ruler and tested against a spectrum computed on the rate that carries the content.\*\* And r3733's leaf reading for  $l=220$  reproduces at 0.926 against its 0.92 – the premise's refutation was already in the record as a caveat.). *Computes:* both sentences asserted present verbatim in CR\_framework.tex and PO13\_WORKING\_STATE.md, with the same P07 paragraph asserted to name the leaf congruence; LEAFPERT asserted on and RAD\_IN\_RATE asserted off; the leaf rate asserted to exceed the geometric rate at  $a=1e-4$  (read, not assumed); the onset asserted to precede leaf equality; the after-onset and in-radiation bands computed from the instrument's own  $aH/c$  splines and the latter asserted to CONTAIN  $l=204$ ;  $l_1$  asserted super-horizon on the leaf and sub-horizon on the stack; the stacking rate asserted equal to  $\Omega_m/a^3 + \Omega_b$  to  $1e-12$  at  $a=1e-6$  where a radiation term would have been 254x the rest; r3733's 0.92 asserted reproduced. \*\*CONTROL THAT FIRES: the same census on the STACKING rate returns the diagnosis's own answer\*\* – so the census is not a construction that refutes everything put to it *Bound:* \*\* REFUTES A PREMISE, DOES NOT MEASURE A MECHANISM. \*\* No claim that the acoustic residual is explained, reduced or removed, and no measurement of how large the driving these modes receive is – NODRIVE=1 is what measures that and is not run here. The position deficit is untouched: the converged arm reports  $l_1/l_A = 0.6764$  against the sky's 0.7312 whichever way the premise falls. NOT A VERDICT ON P07's SENTENCE: "this rate" is ambiguous in that paragraph between the stacking rate the section is about and the leaf rate the same paragraph names, and the sentence is TRUE on the first; what is established is that the reading supporting the diagnosis is the one the rate rule does not select. NOT CLAIMED: anything about  $z_{\text{onset}}$  as a fitted parameter, the alternation, or the Hubble-tension join P07 makes next *Run:* python3 receipts/P15\_CR\_cosmology/C61\_the\_undriven\_premise\_is\_false\_on\_the\_rate\_the\_framework\_assigns\_the\_p

**P15R111** C63\_no\_single\_source\_term\_carries\_the\_undriven\_split\_and\_the\_first\_probe\_read\_an\_unwired\_knob  
[PASS]

*sec:coherence* (\*\*NO SINGLE LINE-OF-SIGHT SOURCE TERM CARRIES PO-13's 23% UNDRIVEN SPLIT.\*\* Undriven, the CR arm's first peak sits at 1.1273 of its own angular acoustic scale and the control's at 0.9158 – a 23% split with  $Q=1$  on both arms, so the oscillator is excluded and the split is in the projection. The DIFFUSION DAMPING is excluded too: DAMPX=0 removes it entirely and the split WIDENS to 26.0%. Removing the monopole, the Doppler dipole and the integrated term ONE AT A TIME leaves the split at 23.1%, 23.8% and 29.4% – it never collapses, and removing the integrated term WIDENS it. So the answer to "Psi, integrated, or Doppler" is NONE OF THE THREE. ALSO PINS THE ROW'S PREMISE: "the same acoustic scale on both arms" is TRUE of the ANGULAR scale  $l_A$  (301.6 vs 301.4, 1 part in 1507) and FALSE of the PHYSICAL sound horizon  $r_s$  (135.46 vs 144.53 Mpc, 6.3% apart). The  $l_A$  match is a conspiracy –  $D_M$  differs by 6.2% too, within 0.07% of  $r_s$ 's own difference – not a shared scale.). *Computes:* eight runs of ACOUSTIC\_two\_arm.py under NODRIVE=1, both arms x (baseline, NOISW=1, DPSRC=0, SWSRC=0), at LMAXL=520 where all eight first peaks reproduce the LMAXL=1300 values exactly; each switch is CALIBRATED before any null is

read off it *Bound*: A NEGATIVE result: it says where the split is NOT and is not a mechanism. Only SINGLE removals were run, so a cancellation between two terms is not excluded. The damping half of the projection kernel is now excluded and the VISIBILITY half is not: the arms last-scatter at  $\eta = 449.6$  and  $280.8$ , a 60% difference in epoch against 7% in width, and that is the named unrun candidate. ZSTART is NOT a free knob –  $r_s$  integrates from the start, so moving it moves the acoustic scale itself (a trial took  $L_A$  from 301.6 to 172.8). Does not touch the driving (DRC/DRE are identically zero under NODRIVE=1). The KCONT=1 sampling control leaves all four CR peaks unchanged, which bounds the sampling shift below one LSTEP=8 grid step (2.65%) against a 23% effect – a bound, not an identity. *Run*: `python3 receipts/P15_CR_cosmology/C63_no_single_source_term_carries_the_undriven_split_and_the_first_probe.py`

**P15R112** C23\_the\_gap\_estimated [OK]  
*sec:earlyisw* (ESTIMATE for the peak-region gap: the decay rate  $k^2/(3H)$  is built from the thing that differs, so CRs potential decays  $\sim 15\%$  faster and the peak power ratio is  $\sim 0.813$  — BROAD deviation, not tail-confined). *Computes*: asserts the rate, the 13% and the factor-of-two clauses verbatim, computes the compounding, and flags the  $0.813/0.823$  near-coincidence as different-origin (7 checks) *Bound*: NOT-A-PAPER-CLAIM — an ESTIMATE with its model stated, labelled as one; a hierarchy run would replace it *Run*: `python3 receipts/P15_CR_cosmology/C23_the_gap_estimated.py`

**P15R113** C24\_the\_substitution\_itemised [OK]  
*sec:instrument* (PO-12s substitution ITEMISED: conformal time,  $r_s$ , the comoving horizon and the diffusion integral carry  $H$  while the Thomson rate and  $x_e$  do not — and the ratio does NOT cancel because  $r_D$  carries a square root that  $r_s$  does not). *Computes*: asserts the 13% clause and reconstructs it from  $\rho_r/\rho_m=0.3$ , asserts the microphysics-outside and  $\theta_*$ -pinned clauses, and computes both scaling exponents (8 checks) *Bound*: NOT-A-PAPER-CLAIM — sets the bar for the two-leg run; P15s 9.4% is not re-derived *Run*: `python3 receipts/P15_CR_cosmology/C24_the_substitution_itemised.py`

**P15R114** C25\_the\_gap\_closed\_by\_integration [OK]  
*sec:diffusion-scale* (the 2.6-point gap closed by INTEGRATION:  $\theta_D/\theta_*$  runs  $+7.1\%$  at  $z=1200$  to  $+14.4\%$  at  $z=10000$ , with P15s  $+9.4\%$  at  $z\sim 2500$  — and the dependence is MONOTONIC, which is the papers own two-constraint argument). *Computes*: asserts the branch-point lower-limit clause and the single-datum clause verbatim, integrates the rate difference over five onset redshifts, and checks monotonicity (6 checks) *Bound*: NOT-A-PAPER-CLAIM — reproduces the papers shape on a simplified  $\rho_r/\rho_m$  model; the onset is the remaining unknown *Run*: `python3 receipts/P15_CR_cosmology/C25_the_gap_closed_by_integration.py`

**P15R115** C26\_the\_onset\_is\_not\_free [OK]  
*sec:tensions* (PO-12s last free parameter is NOT free: the onset is fitted to the acoustic angle at  $z\sim 6797$  and meets the scale at every  $H_0$  — and the residual  $+13.1\%$  vs  $+9.4\%$  is the WEIGHT  $g(R)/x_e$ , not the onset). *Computes*: asserts the onset value, the fitted-to-the-acoustic-angle clause and the not-a-knob clause verbatim, checks the model at the onset, and integrates under four weightings (7 checks) *Bound*: NOT-A-PAPER-CLAIM — power-law weights are stand-ins for  $g(R)/x_e$ , shown for direction and size *Run*: `python3 receipts/P15_CR_cosmology/C26_the_onset_is_not_free.py`

**P15R116** C27\_the\_number\_bracketed [OK]  
*sec:envelope-consequence* (P15s  $+9.4\%$  BRACKETED from outside its own machinery: Saha  $+7.0\%$  ; Peebles ; unweighted  $+13.1\%$ , with the ordering FORCED by which history recombines faster — and cancel identically covers the CONSTANTS, not the weight). *Computes*: asserts the cancel-identically, constants-outside and Peebles-history clauses verbatim, shows the weight does not drop out, runs  $g(R)$  with a Saha  $x_e$ , and checks 9.4 lies strictly inside (8 checks) *Bound*: NOT-A-PAPER-CLAIM — brackets rather than reproduces; the Peebles direction is argued, not computed *Run*: `python3 receipts/P15_CR_cosmology/C27_the_number_bracketed.py`

**P15R117** C28\_length\_against\_angle [OK]  
*sec:envelope-consequence* (THERE WAS NEVER A DISCREPANCY: 9.4% is an ANGLE ratio (C10s  $r=1.0926$ , already netting the 0.7% sound-horizon shift) and 10.8% a LENGTH ratio (C8) —  $r_{2686}-r_{2689}$  compared one to the other). *Computes*: asserts both statements and the 0.7% clause verbatim, reads C10s own  $r$ , and shows the two do not compose as this line assumed (7 checks) *Bound*: NOT-A-PAPER-CLAIM — corrects four of this lines revisions; C8 and C10 are run, not re-derived *Run*: `python3 receipts/P15_CR_cosmology/C28_length_against_angle.py`

**P15R118** C29\_nothing\_to\_calculate\_across\_the\_join [OK]

*sec:what-crosses* (PO-12s remaining half has NO CONTENT: the pre-onset stretch is pressureless and at  $w=0$  the potential equation loses  $k$  entirely, so there is no scale-dependent evolution for a single calculation across the join to compute). *Computes*: asserts the no- $k$ , pressureless, boundary-mode and residue clauses verbatim, and verifies symbolically that  $k$  is present in the general equation and absent at  $w=0$  (7 checks) *Bound*: NOT-A-PAPER-CLAIM — F5 reserves the strike; the residue routes to PO-7 *Run*: `python3 receipts/P15_CR_cosmology/C29_nothing_to_calculate_across_the_join.py`

**P12R15** A7\_the\_frontier\_has\_two\_kinds [OK]

*sec:envelope-consequence* (THE FRONTIER SORTS INTO TWO KINDS on P15s own  $H(a)$  rule: FIVE rows ask whether the theory is DEFINED (PO-2/4/5/6/11) and TWO ask what it PREDICTS (PO-7/10) — and they fail differently, so size does not order them). *Computes*: asserts the  $H(a)$  clause, checks the classification covers exactly the open rows, and pins each rows object against its kind (7 checks) *Bound*: NOT-A-PAPER-CLAIM — the classification is derived here; no paper states it *Run*: `python3 receipts/P12_algebroid/A7_the_frontier_has_two_kinds.py`

**P15R121** C32\_half\_one\_is\_model\_selection [OK]

*sec:frontiers* (PO-10s half (1) IS an information-criterion comparison: the corpus names "the likelihood-level model selection" while AIC/BIC/Akaike/Occam/Bayes-factor appear ZERO times — and the penalty is computable now,  $dAIC=10.0$  and  $dBIC=26.9$  in CRs favour). *Computes*: asserts the model-selection phrase and the 215-bin count verbatim, word-boundary-checks five model-selection terms at zero across papers and receipts (excluding itself), and computes both penalties (9 checks) *Bound*: NOT-A-PAPER-CLAIM —  $k=6$  is the standard flat- $\Lambda$ CDM count and must be stated with any result; the  $\chi^2$  is unrun *Run*: `python3 receipts/P15_CR_cosmology/C32_half_one_is_model_selection.py`

**P15R122** C33\_the\_threshold\_corrected [OK]

*sec:tensions* (THE THRESHOLD CORRECTED: r2709s  $k=1$  was wrong — P15 anchors  $A_s$  at the first peak, so CR is  $k=2$  — and against Liddle/Jeffreys ( $dBIC$  2 positive, 6 strong) the threshold is  $dBIC = 21.5$ , still 3.5x the strong line). *Computes*: asserts the anchored-amplitude and one-parameter-accommodation clauses verbatim, tables all four parameter-count pairs against the Jeffreys scale, and computes the AICc correction (7 checks) *Bound*: NOT-A-PAPER-CLAIM — no  $\chi^2$  is scored; the count must be restated with any result *Run*: `python3 receipts/P15_CR_cosmology/C33_the_threshold_corrected.py`

**P15R123** C34\_the\_deliverable\_is\_a\_pair [OK]

*sec:tensions* (PO-10s DELIVERABLE IS A PAIR: the acoustic sector is open as PO-7, its phase freedom moves the gap 0.615  $\rightarrow$  0.408 and cannot close it — but the freedom is DISCRETE, so  $k$  is unchanged and both branches are reported at the same threshold). *Computes*: asserts PO-7s object, the phase-lever and cannot-close clauses verbatim with both values, and shows the discrete choice leaves  $dBIC$  at 21.5 rather than 16.1 (5 checks) *Bound*: NOT-A-PAPER-CLAIM — neither branch is scored; that is the whole of what remains *Run*: `python3 receipts/P15_CR_cosmology/C34_the_deliverable_is_a_pair.py`

**P15R124** C35\_the\_reference\_is\_F3 [OK]

*sec:tensions* (THE REFERENCE FORK SETTLES ITSELF AND AGAINST THIS LINES FRAMING: BIC uses only DIFFERENCES, so F3 (same-instrument) is the score and the CAMB-206.4 comparison charges CR an instrument floor CAMB never pays). *Computes*: asserts L-147s floor definition and pipeline-wiring clauses verbatim, shows  $dBIC$  contains only a  $\chi^2$  difference, and confirms the threshold is invariant under the pair choice (7 checks) *Bound*: NOT-A-PAPER-CLAIM — corrects r2709-r2711s statement of what the threshold scores against; the threshold itself is unchanged *Run*: `python3 receipts/P15_CR_cosmology/C35_the_reference_is_F3.py`

**P15R125** C36\_the\_negative\_is\_the\_envelope [OK]

*sec:envelope-consequence* (SUPERSEDED r2725 (the check shared the runs own  $n_s$ -held assumption, so agreeing to the same order confirmed only that both held it fixed) — L-814s NEGATIVE IS NOT AN ARTEFACT: P15s own published envelope, scored against cosmic variance on 215 bins with no pipeline or fit, gives  $\chi^2 \sim 1.6e5$  against cc54s measured  $5.05e4$  — the same order). *Computes*: evaluates the papers own  $\exp[-(1/LD)^2(r^2-1)]$  across the band, scores it against cosmic-variance errors, reads cc54s floor finding from their receipt, and pins L-147s scope clause (6 checks) *Bound*: NOT-A-PAPER-CLAIM — no framework verdict; PO-7 protected and F5 reserves it *Run*: `python3 receipts/P15_CR_cosmology/C36_the_negative_is_the_envelope.py`

**P15R126** C37\_the\_comparison\_is\_not\_matched [OK]

*sec:envelope-consequence* (L-814s COMPARISON IS NOT MATCHED and its number is WITHDRAWN as a result: the CR arm was held at  $k=2$  with  $n_s$  fixed at LCDMs fitted value, and freeing the tilt alone removes 15.6x). *Computes:* computes the envelope  $\chi^2$  with the tilt held and free, shows the threshold barely moves for the extra parameter, and reads cc54s own relative statement from their receipt (5 checks) *Bound:* NOT-A-PAPER-CLAIM — the  $k=2$  spec was this lines at r2710; a matched comparison is not claimed to pass *Run:* python3 receipts/P15\_CR\_cosmology/C37\_the\_comparison\_is\_not\_matched.py

**P15R127** C38\_the\_comparison\_scores\_derivations [OK]

*sec:tensions* (PO-10s NON-FOREIGN COMPARISON IS A SCORE ON DERIVATIONS: a derivation has no  $k$ , so scoring  $\theta_*$ ,  $r$ ,  $z_{eq}$  and the 9/10 transfer against measurement imports no parameter vector — and this receipt reports NO score, because the draft that did invented every sigma). *Computes:* pins P15s derived-vs-measured pairs and its own naming of  $\Omega_m$  and  $A_s$  as fitted, and asserts its own refusal to score (6 checks) *Bound:* NOT-A-PAPER-CLAIM — no verdict on CR; the row owes published uncertainties and a derived-quantity list *Run:* python3 receipts/P15\_CR\_cosmology/C38\_the\_comparison\_scores\_derivations.py

**P15R128** C39\_the\_derived\_list\_read [OK]

*sec:tensions* (PO-10s DERIVED-QUANTITY LIST READ from P15s own marker language: four pairs each with the papers own accuracy clause ( $\theta_*$  302.2/301,  $r_s$  146.4/145.4,  $1+z_{eq}$  3399, eta 1.99 to one per cent) and three exclusions in the papers own words). *Computes:* pins all four derived pairs and all three fitted-exclusions verbatim, and the high- $l$  ratios no-free-parameter clause (7 checks) *Bound:* NOT-A-PAPER-CLAIM — nothing is scored; the published uncertainties remain owed and are not this lines to invent *Run:* python3 receipts/P15\_CR\_cosmology/C39\_the\_derived\_list\_read.py

**P15R129** C40\_the\_pair\_was\_quoted\_against\_a\_rounding [OK]

*sec:tensions* (PO-10s FIRST PAIR MUST TAKE THE RECEIPTS VALUE, NOT THE PROSE: P15 writes "the measured  $\sim 301$ " (tilde present, honestly approximate) while P15\_zonset\_determinations carries and asserts  $L_A = 301.76$  — and the gap is 0.44, not 1.20). *Computes:* quotes both the prose and the receipts assertion, computes both gaps, and carries its own first drafts correction (6 checks) *Bound:* NOT-A-PAPER-CLAIM — no paper edit owed, the tilde is there; nothing is scored and the uncertainty is still not this lines *Run:* python3 receipts/P15\_CR\_cosmology/C40\_the\_pair\_was\_quoted\_against\_a\_rounding.py

**P15R130** C41b\_a\_tilde\_on\_a\_settled\_value\_is\_a\_stale\_hedge [OK]

*sec:tensions, sec:coherence* (A TILDE ON A SETTLED VALUE IS A STALE HEDGE:  $\sim 301$  became 301.76 (asserted by its receipt) and  $\sim 8\%$  became 8.2% in NINE places (computed on CAMBs exact ionization history), three of them while calling it "a real, computed effect"). *Computes:* confirms both figures are asserted by receipts, that no tilde survives on either, and that legitimately approximate tildes are untouched (7 checks) *Bound:* NOT-A-PAPER-CLAIM — nothing scored; not every tilde is stale, only those on figures a receipt asserts *Run:* python3 receipts/P15\_CR\_cosmology/C41b\_a\_tilde\_on\_a\_settled\_value\_is\_a\_stale\_hedge.py

**P15R131** C41\_a\_tilde\_is\_stale\_when\_nothing\_competes [OK]

*sec:tensions* (A TILDE IS STALE WHEN ONE RECEIPT ASSERTS AND NONE COMPETES:  $\sim 301$  failed that test and P15 now reads 301.76;  $\sim 8\%$  passed it — C8 derives +10.83% analytically and NAMES the disagreement with damping\_ratio\_cleans +8.2% — so nine edits were reverted). *Computes:* confirms the 301.76 edit landed and the nine tildes remain, pins C8s computed +10.83% and its own naming of the conflict (6 checks) *Bound:* PAPER-CLAIM (301.76 only) — the 8% figure is left tilde-marked and its discrepancy is filed as owed *Run:* python3 receipts/P15\_CR\_cosmology/C41\_a\_tilde\_is\_stale\_when\_nothing\_competes.py

**P15R132** C42\_the\_cancellation\_needs\_xe\_identical [OK]

*sec:coherence* (THE DAMPING DISCREPANCY DIAGNOSED: C8s cancellation needs  $x_e$  identical in both cosmologies, and a different  $H(a)$  changes the recombination HISTORY not merely its epoch — Lambda, the upper limit, the radiation density and arithmetic all eliminated). *Computes:* reproduces C8 to six figures, varies Lambda /  $z_{rec}$  /  $\rho_{r,0}$  and shows none accounts for the gap, and pins C8s own PROVIDED clause (7 checks) *Bound:* NOT-A-PAPER-CLAIM — neither number is adjudicated; the  $x_e$  response is a calculation neither receipt has done *Run:* python3 receipts/P15\_CR\_cosmology/C42\_the\_cancellation\_needs\_xe\_identical.py

- P15R133** C43\_the\_gap\_is\_numerical\_not\_xe [OK]  
*sec:coherence* (r2750 WITHDRAWN — THE DAMPING GAP IS NUMERICAL, NOT x\_e: a pure H probe moves z\_\* by only 0.195% (~0.05pp on the ratio) against a 1.86pp gap, while the CAMB receipt prints a 7.1% miss against CAMBs own thetad on its OWN LCDM arm). *Computes:* runs CAMB at five N\_eff values as an H-only probe, scales the response by r2750s own measured sensitivity, and pins the receipts self-reported GATE miss (6 checks) *Bound:* NOT-A-PAPER-CLAIM — C8 is not confirmed, and the tilde stays: two receipts still disagree, one now with a KNOWN defect *Run:* python3 receipts/P15\_CR\_cosmology/C43\_the\_gap\_is\_numerical\_not\_xe.py
- P15R134** C44\_the\_gap\_is\_truncation [OK]  
*sec:coherence* (THE DAMPING GAP IS TRUNCATION: the r\_D ratio climbs 8.33% -i 9.94% with the upper limit and converges only by z~5e4, while the CAMB receipts grid stops at z=12000 — and a normalisation factor CANCELS in a ratio, which withdraws r2751). *Computes:* computes the ratio at six upper limits, shows convergence, and demonstrates that a shared constant divides out (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the receipts grid is reasonable for its own purpose; what is claimed is that its RATIO inherits a truncation bias *Run:* python3 receipts/P15\_CR\_cosmology/C44\_the\_gap\_is\_truncation.py
- P15R135** C45\_xe\_does\_not\_cancel\_from\_a\_ratio\_of\_integrals [OK]  
*sec:coherence* (THE DAMPING DISCREPANCY RESOLVED: x\_e does NOT cancel from a ratio of INTEGRALS even when identical — toggling it on the same range moves the ratio +9.94% to +8.37%, the whole gap — so C8s cancellation step is a real error). *Computes:* toggles x\_e on a fixed range and measure with CAMBs own history, shows a shared CONSTANT does cancel exactly while x\_e does not, and pins C8s every-constant-outside claim (6 checks) *Bound:* NOT-A-PAPER-CLAIM — which figure is P15s signature is not settled; C8s reduction and baryon correction stand *Run:* python3 receipts/P15\_CR\_cosmology/C45\_xe\_does\_not\_cancel\_from\_a\_ratio\_of\_integrals.py
- P15R136** C46\_the\_nine\_are\_right\_and\_the\_one\_is\_wrong [OK]  
*sec:tensions* (THE 9.4% TRACED: P15 -i C10 (+9.26%) -i C9, which divides r\_D 1.1015 by r\_s 1.0067 — the ANGLE ratio, correctly, but from an r\_D that omits the x\_e weighting. Corrected it lands at +7.6% to +8.2%, so the nine ~8% were RIGHT and the one 9.4% was the error). *Computes:* pins the citation chain P15-iC10-iC9 and reruns C9s own division on three r\_D ratios (6 checks) *Bound:* PAPER-CLAIM — ten instances set to 8.2%, and the tilde removed per r2749s test now that nothing competes *Run:* python3 receipts/P15\_CR\_cosmology/C46\_the\_nine\_are\_right\_and\_the\_one\_is\_wrong.py
- P15R137** C47\_the\_neutrinos\_were\_missing [OK]  
*sec:coherence* (THE +5.66% RECONCILED AND IT WAS THIS LINES ERROR: r2752 used photons alone where the receipt uses Ogamm + Onu (a 69% underestimate of what sources H) — and the CR arm has no radiation at all, so the omission hit ONE ARM ONLY. With neutrinos in, the ratio is 1.0896 against the receipts 1.0897). *Computes:* runs the full Hu-Sugiyama for both rates under two radiation densities with CAMBs own x\_e, and confirms the absolute r\_D difference cancels from the ratio (6 checks) *Bound:* NOT-A-PAPER-CLAIM — only the RATIO is claimed to four figures; the absolute r\_D still differ by ~3% from grid and quadrature *Run:* python3 receipts/P15\_CR\_cosmology/C47\_the\_neutrinos\_were\_missing.py
- P15R138** C48\_the\_pairs\_are\_three [OK]  
*sec:tensions* (PO-10s PAIRS ARE THREE, NOT FOUR: r\_s and 1+z\_eq hold against asserting receipts (C9 145.38, DESI 3402), but the onset ratio is an IDENTITY ON INHERITED INPUTS — "the quoted value to one per cent, from standard thermodynamics" — same class as N\_eff and A\_s). *Computes:* finds the asserting receipt for each measured value, pins the thermodynamics clause, and records the 1.9968 near-miss where another receipt asserts a different quantity with the same digits (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the onset ratio is not wrong, only not a TEST; nothing is scored *Run:* python3 receipts/P15\_CR\_cosmology/C48\_the\_pairs\_are\_three.py
- P15R139** C49\_the\_row\_has\_a\_likelihood\_not\_a\_lookup [OK]  
*sec:tensions* (PO-10 DOES NOT NEED ERROR BARS — IT HAS A LIKELIHOOD: plik\_lite over 215 TT bins with a covariance and a banked LCDM fit the pipeline calls wired. What blocks it is recorded in the same file — the CONTROL arm lands at chi^2/dof ~ 100). *Computes:* pins the instrument, its calibration standard, its self-recorded F4 defect, and F5s protection of the verdict (6 checks) *Bound:* NOT-A-PAPER-CLAIM — the control defect is not diagnosed here, and no verdict on CR is drawn *Run:* python3 receipts/P15\_CR\_cosmology/C49\_the\_row\_has\_a\_likelihood\_not\_a\_lookup.py

**P15R140** C50\_the\_control\_is\_at\_seven\_not\_a\_hundred [OK]

*sec:tensions* (THE CONTROL ARM IS AT 7.14, NOT  $\sim 100$ : the receipts F4 prose carried the c54.172 floor while its F6 PRINT reports  $1320.5/185 = 7.14$ , and F6 says the floor fell  $103 - i$  22.5 -i 28.6 -i 7.1, a factor of 14). *Computes*: pins the corrected prose, the f-string print fields, the fall history, the 185-vs-215 bin difference and F6s unchanged verdict (6 checks) *Bound*: NOT-A-PAPER-CLAIM — 7.14 is not acceptable either; both arms remain outside the discriminating regime *Run*: python3 receipts/P15\_CR.cosmology/C50.the\_control\_is\_at\_seven\_not\_a\_hundred.py

**P15R141** C51\_the\_dropped\_bins\_are\_the\_damping\_tail [OK]

*sec:tensions* (THE THIRTY DROPPED BINS ARE ell 1759-2508 — THE DAMPING TAIL: the likelihood arms model ends near ell 1760 and nan-drops the rest, discarding exactly the region P15 names as its exposed edge, where the suppression runs 24% to 42%). *Computes*: reads the bin edges from the likelihoods own arrays, locates the cut at which 185 survive, and profiles the damping suppression across the dropped range (6 checks) *Bound*: NOT-A-PAPER-CLAIM — the ell limit is a cost not a mistake, and which way the omitted bins would land is the point of running them *Run*: python3 receipts/P15\_CR.cosmology/C51.the\_dropped\_bins\_are\_the\_damping\_tail.py

**P15R142** C52\_the\_banked\_spectra\_fail\_the\_sampling\_guard [OK]

*sec:tensions* (THE BANKED SPECTRA FAIL THE INSTRUMENTS OWN SAMPLING GUARD: LMAXL=2000 NK=260 gives 2.5 points per Bessel period and prints UNDER-SAMPLED — and the guard says the source comb stays correct while the projection aliases, so no comb-level check could catch it). *Computes*: fixes the banked configuration from the npz shape, computes the sampling ratio and projection memory across four NK values, and pins the guards own wording (6 checks) *Bound*: NOT-A-PAPER-CLAIM — whether aliasing moves the  $\chi^2$  enough to matter is UNMEASURED, and measuring it needs a run that will not fit in 3.7 GB *Run*: python3 receipts/P15\_CR.cosmology/C52.the\_banked\_spectra\_fail\_the\_sampling\_guard.py

**P15R143** C53\_the\_k\_range\_fixes\_one\_arm\_only [OK]

*sec:tensions* (THE k-RANGE FIXES ONE ARM ONLY: raising L\_max 26% drops the control 7.14 -i 3.81 on the SAME 185 bins while CR moves 0.4% — a response ratio over twenty to one, so r2760s undiagnosed factor of seven is a k-truncation and CRs 281 is not). *Computes*: computes both arms responses and their ratio, checks F\_3 widens in the direction one-arm improvement predicts, and labels the two-point trend as an extrapolation (6 checks) *Bound*: NOT-A-PAPER-CLAIM — radiation is the first candidate not a demonstrated mechanism; the convergence trend rests on TWO points *Run*: python3 receipts/P15\_CR.cosmology/C53.the\_k\_range\_fixes\_one\_arm\_only.py

**P15R144** C54\_the\_281\_is\_the\_prediction\_being\_measured [OK]

*sec:tensions* (CRs 281 IS THE ORDER P15s OWN PREDICTION PRODUCES: scoring the damping residual over 185 bins gives  $\chi^2/\text{dof}$  from 204 ( $\sigma=1\%$ ) to 5099 ( $\sigma=0.2\%$ ), so 281 sits INSIDE that range and BELOW the value at Plancks quoted errors — the arm is scoring the effect P15 predicts). *Computes*: pins P15s no-free-parameter high-ell formula, scores its residual at three error scales, pins F5, and names the SHAPE test that would settle it (5 checks) *Bound*: NOT-A-PAPER-CLAIM — an ORDER comparison against a flat sigma, not a derivation against the plik covariance; supplies a candidate, does not close the row *Run*: python3 receipts/P15\_CR.cosmology/C54.the\_281\_is\_the\_prediction\_being\_measured.py

**P15R145** C55\_the\_shape\_test\_says\_peak\_misalignment [OK]

*sec:tensions* (THE SHAPE TEST RUN: the CR arms residual is neither a damping-profile match nor flat — it OSCILLATES, 0.237 to 2.375, correlation -0.296, dominant period 317 against l\_A = 301. The 281 is PEAK MISALIGNMENT, and r2786 is withdrawn by its own named test). *Computes*: computes the banked arms ratio, its correlation with P15s profile, its dominant Fourier period against l\_A, and the l\_A-agrees / r\_s-differs contrast (5 checks) *Bound*: NOT-A-PAPER-CLAIM — the damping signature is not shown absent, only non-dominant; physical vs numerical phase slip is the open question *Run*: python3 receipts/P15\_CR.cosmology/C55.the\_shape\_test\_says\_peak\_misa

**P15R146** C56\_la\_is\_computed\_and\_its\_residual\_is\_too\_small [OK]

*sec:tensions* (l\_A IS COMPUTED NOT CALIBRATED (pi D\_M/r\_s reproduces the stored value in both arms) and its survival is a real cancellation P15 claims in advance — but the +0.075% residual explains under 1% of r2787s measured swing, two orders too small). *Computes*: checks pi D\_M/r\_s against the stored l\_A in both arms, sizes the D\_M and r\_s moves, pins P15s met-from-the-other-end clause, and propagates the residual against the measured swing (5 checks) *Bound*: NOT-

A-PAPER-CLAIM — the cancellation is observed not re-derived; amplitude is what remains after two eliminations, a candidate not a finding *Run:* `python3 receipts/P15_CR_cosmology/C56_la_is_computed_and`

**P15R147** `C57_the_cr_peaks_are_not_at_its_own_la` [OK]

*sec:tensions* (THE CR ARMS PEAKS ARE NOT WHERE ITS OWN LA SAYS: mean spacing 258 against a stored LA of 301.6 (ratio 0.855) while LCDM sits at 0.995 — so the stored quantities are consistent with each other and NOT with the computed Cell). *Computes:* finds peaks in both banked arms, compares mean spacing against each arms stored LA, and sizes the mismatch against r2788s residual (5 checks) *Bound:* NOT-A-PAPER-CLAIM — which of DM, rs or the projection is wrong is not established; the CR spacing could be a real feature of a transfer on the geometric rate *Run:* `python3 receipts/P15_CR_cosmology/C57_the_cr_peaks_are_not_at_its_own_la.py`

**P15R148** `C58_the_stored_rs_moves_the_wrong_way` [OK]

*sec:tensions* (THE STORED CR rs MOVES THE WRONG WAY: removing radiation must RAISE the sound horizon (146.5 -j 245.2 Mpc integrated), the peak spacing independently demands 158.35, and the stored value is 135.46 — BELOW LCDMs 144.53). *Computes:* solves the spacing for each candidate, integrates rs with and without radiation, and shows pi DM/rs on both stored values reproduces the stored LA exactly (5 checks) *Bound:* NOT-A-PAPER-CLAIM — 158.35 and 245.16 are not proposed as the right value; what is established is the DIRECTION and that the stored value has the wrong sign of deviation *Run:* `python3 receipts/P15_CR_cosmology/C58_the_stored`

**R1** `the_correction_stopped_at_the_sentence_and_the_number_it_defines_was_left_behind` [OK]

*sec:envelope-consequence* (r2755 CORRECTED THE SENTENCE AND NOT THE NUMBER THE SENTENCE DEFINES: ‘theta\_D/theta\_’ was corrected from  $\$9.4\{\}\%\$$  to  $\$8.2\{\}\%\$$  and ‘sec:envelope-consequence’s  $\$r\%$ , one paragraph later, IS that ratio and still read 1.093 — so the paper stated one ratio twice, differently, and the second is the one the prediction is made of. Corrected from three agreeing routes to  $\$r=1.0816\%$ : the ratio at  $\$\{\}\ell_D\$$  moves  $0.823 \rightarrow 0.844$ , at  $\$1.5\{\}\ell_D\$$   $0.645 \rightarrow 0.682$ ,  $\$\{\}\ell_D(\{\}\rm CR)\$$   $1281 \rightarrow 1294$ . \*\*FIFTEEN receipts carried it — eight SILENTLY (exiting zero, six of them evaluating the envelope with the wrong value) and seven LOUDLY (failing every full run since r2755 — and they are the seven that PRODUCED the correction).\*\* CONTROL: C10’s gate was built so the paper and the receipt “cannot drift apart silently in either direction”, and it worked — they were consistent with each other and both wrong.). *Computes:* COMPUTES: the identity  $\$r=\{\}\theta_D/\{\}\theta_ \$$  read from C10’s own assertion message; the corrected value from three routes and the four quantities that follow; the fifteen receipts split by failure mode and counted at the commits where they exist; the C41 double-allocation resolved as one rule at two states; and the register files deliberately left alone (21 checks) *Bound:* \*\*bound: 1.0816 is NOT re-derived here\*\* — it is read from ‘P15\_damping\_ratio\_clean’, ‘C46’ and the corrected sentence, and checked for agreement. \*\*Not claimed that the envelope is right\*\*, only that it now runs on the ratio the paper’s own preceding paragraph derives. \*\*The correction FAVOURS the construction (16% suppression at  $\$\{\}\ell_D\$$  rather than 18%) and is flagged for it\*\*; it would apply identically the other way. ‘PO-7’ is ‘sec:refit-bound’/‘sec:coherence’ and is untouched. *Run:* `python3 receipts/L557_the_correction_stopped_at_the_sentence/R1_the_correction_stopped_at_the_sentence_and`

**P1** `the_pin_test_run_the_spacing_follows_the_pin_so_the_ratio_does_not` [OK]

*sec:tensions* (THE PIN TEST, RUN — the runs the observer line’s container killed twice. \*\*The peak spacing tracks the one fitted number PROPORTIONALLY\*\* (‘LATARG’ 280/301.6/320  $\rightarrow$  mean spacing 229.3/245.3/261.3), so  $\text{spacing}/\ell_A = 0.8190/0.8134/0.8167$  — \*\*0.69% spread across a 14% pin range\*\*, against the \*\*13.4%\*\* a constant-spacing arm would give: a factor of twenty, so the measurement DISCRIMINATES.  $\Rightarrow$  \*\*The routed dichotomy is inverted: following the pin is exactly what makes the RATIO — which is what the deficit is — independent of the pin. THE DEFICIT SURVIVES THE FIT.\*\* Also: ‘KCONT=1’ at NK=320 returns the same four peaks, discharging the instrument’s own standing sampling caveat; and \*\* $\ell_1 = 172$  at all three pins\*\*, so  $\ell_1/\ell_A$  is pin-dependent while the spacing ratio is not. LCDM control on the same grid: 0.9909, reproducing the sky to 6 in  $\ell$ ). *Computes:* COMPUTES: five banked runs parsed from their own printed blocks; the spacing ratio and its spread; the CONSTANT-spacing alternative computed as the control and the discrimination factor between them; the KCONT identity and its sampling-guard margin;  $\ell_1$ ’s invariance and  $\ell_1/\ell_A$ ’s spread; and the LCDM arm checked against the sky before being used (21 checks) *Bound:* NOT-A-PAPER-CLAIM — \*\*and the declaration is deliberate: this is a measurement on the instrument’s

own fitted parameter, delivered to a routed question, and whether it lands in ‘sec:tensions’ is the observer line’s call under ‘PO-7’.

**\*\* A node that banked it into the paper on its own initiative would be taking a step toward the verdict the row protects. \*\***bound: this reports a number and does not convert it into a verdict — ‘PO-7’ is protected.

**\*\* Not that r2789’s 258 was wrong (different grid; the control was re-run here rather than compared across grids). Not that peaks are known better than the  $\ell$ -grid: ‘LSTEP=8’,  $\pm 4$  per peak,  $\pm 8$  per spacing — the invariance claim is  $20\times$  outside that and the individual spacings are not. Not that the deficit is explained, only that the fitted number does not produce it. Run: python3 receipts/L562\_the\_pin\_test/P1\_the\_pin\_test\_run\_the\_spacing\_follows\_the\_pin\_so\_the\_ratio\_does\_not.py**

**P15R149 P15\_the\_driving\_is\_bought\_by\_decay\_during\_oscillation [OK]**

*sec:transmission* (THE DRIVING SHIFT IS BOUGHT BY POTENTIAL DECAY OCCURRING WHILE A MODE OSCILLATES, so a mode whose oscillation begins after the potential has settled inherits the UNDRIVEN phase. **\*\* Coupled photon/matter/potential solve on a matter-and-radiation background, with the potential closed by the ENERGY CONSTRAINT rather than evolved and the acoustic turnover taken at the first zero of the photon velocity that BEGINS A SEQUENCE SPACED BY  $\pi$  — the free oscillator’s own spacing, which the undriven gate validates at 1.0000 on both arms. \*\* SCOPE: the ASYMPTOTIC statement only,  $Q \rightarrow 1$  at large  $k$ , which is what P15 and P07 assert; NOT the low- $k$  values, and NOT a magnitude (neutrino free-streaming omitted,  $\sim 15\%$  low). A sign flip in the initial amplitude is a  $\pi$  phase shift and moves WHICH ZERO IS FIRST by one half-period, so four initial signs agree to 0.012 at  $k=160$ ). Computes: COMPUTES: a coupled photon/matter/potential solve on a matter-and-radiation background, the potential closed by the ENERGY CONSTRAINT rather than evolved; the UNDRIVEN gate validated at \$1.0000\$ on both arms before use; the driven arm asserted to FALL toward \$1\$ from above and to reach the free-oscillator value to \$0.02\$ at large \$k\$; the control asserted well below \$1\$ and flat in \$k\$ to \$0.10\$; the two shapes asserted qualitatively distinct by a factor of two in range; and the robustness leg showing a factor of two in the radiation budget barely moves it. Re-run r3537, ALL PASS Bound: SCOPE, and the receipt states it: the ASYMPTOTIC statement only,  $Q \rightarrow 1$  at large \$k\$, which is what ‘P15’ and ‘P07’ assert; **\*\*NOT the low- $k$  values and NOT a magnitude\*\*** — neutrino free-streaming is omitted, some \$15\%\$ low Run: python3 receipts/P15\_CR\_cosmology/P15\_the\_driving\_is\_bought\_by\_decay\_during\_oscillation.py**

**P15R150 N8\_the\_transmission\_dichotomy\_is\_a\_capacity\_dichotomy [OK]**

*sec:transmission (prop:transmission, rem:transmission-leg)* (P15 proves a transmission dichotomy — a non-degenerate horizon ( $p=1$ ) imprints a scale, a degenerate one ( $p=2$ ) transmits — and rem:transmission-leg says “the leg multiplies the spectrum by a constant and the seam imprints nothing: neither carries a scale, and a spectrum can only be tilted by something that does”. That is a CHANNEL claim, and “carries a scale” is what decides its capacity). *Computes:* fits the output tilt on four disjoint  $k$ -bands:  $p=2$  returns the same tilt on every band (spread  $1.7e-16$ ) while  $p=1$  spreads by 7.93, which IS the scale; the control puts a cutoff into the  $p=2$  kernel by hand and the spread jumps to 1.91, so the test can fail; and the  $p=1$  spread shrinks monotonically  $30.2 \rightarrow 7.9 \rightarrow 2.3$  as  $\kappa$  rises, tracking the kernel’s scale Bound: SUPPORTS P15 — transparent means the answer does not depend on where you look, and imprinting means it does. NOT a Bogoliubov calculation; no Hawking temperature claimed Run: python3 receipts/P15\_CR\_cosmology/N8\_the\_transmission\_dicho

**P15R151 P15\_the\_history\_is\_spherically\_symmetric\_throughout\_so\_the\_chern\_simons\_number\_never\_moves [OK]**

*does the construction’s own history pass through any member carrying a **\*\*twist\*\***, and does the gravitational Chern–Simons number change across it? ‘r6762’ made the violation a **\*\*boundary\*\*** quantity — the right shape for a net count produced across a history — and ‘P15’ names the open item it could answer, the **\*\*baryogenesis-analogue\*\*** derivation of order a billion photons per baryon (**\*\*No twist anywhere on the history, so the count is exactly conserved — and not because the number comes out small.\*\*** **\*\*Q1\*\*** the path is fixed three ways: ‘P16’ states the spherically symmetric class as **\*\*premise\*\*** and forbids dropping it while keeping the branch-point locus;  $\{*\}RR\{\} \equiv 0$  on the general **\*\*dynamic\*\*** spherically symmetric class (four arbitrary functions of  $(t,r)$ , off-diagonal kept — ‘r6762’’s identity was **\*\*static\*\*** and does not reach collapse); and the twisted member’s transverse block is a **\*\*torus\*\*** against this history’s closed  $S^3$ , which no deformation joins **\*\*Q2\*\*** no, at both levels — a travelling tensor wave gives **\*\*exactly zero** at any polarisation**\*\***, and the only nonzero source is the parity-odd product  $\psi \wedge \omega$*

of the two channels, needing both polarisations AND an areal factor AND broken spherical symmetry; the history supplies one of three **Q3** **\*\*\*zero**, and endpoint-independent**\*\*\*** — all four components of  $K^{\mu}$  vanish identically on the class, so the number is zero at **every** slice, the  $r=0$  crossing included, where a merely divergence-free quantity could have jumped **Q4** net count  $3 \times 0 = 0$  **Q5** the first branch, blocker named). *Computes:* C1 Nariai in its **own**  $dS_2 \times S^2$  form (radii  $\alpha/\sqrt{3}$ ), not as a value of  $f(r)$  — zero, and still zero with a  $d\tau, d\varphi$  term, so the zero is not the product structure C2 the polarised cut ( $\omega \equiv 0$ , ‘P11’ eq:metric) zero for arbitrary  $A, \psi, R$ ; unpolarised nonzero; and **silencing** either channel kills it**\*\*, each by rebuilding the metric rather than substituting a symbol for a function** C3 Stokes **exact**: the volume integral of the density and the boundary flux of  $K^{\mu}$  both give  $8/15 - \log(390625/6561)$ , with ‘r6762’s constant  $-1$  re-verified on a different member the machinery is read out of ‘r6762’s own receipt rather than re-typed, so the two cannot drift *Bound:* **NOT claimed** that no mechanism could produce the item’s number — only that this one does not, and why. **No photon number is supplied or invented**; the verdict does not need one, a zero numerator giving zero for any finite denominator. **No direction is claimed for any asymmetry**, there being none to direct — had the count been nonzero its sign would have been correlated with the twist’s, per ‘C51’. Nothing reopens ‘r6762’s current, source or inflow findings, the global group, the gauging’s status, or weak isospin; no cosmological implication is drawn. Nothing is named *Run:* python3 receipts/P15\_CR.cosmology/P15.the.history.is.spherically.symmetric.throughout.so.the.chern.simons.r

**P15R152** P15.the.likelihood.pair.is.another.instruments.and.the.crossing.run.needed.a.missing.knob [OK]

‘PO-10’ reopened at ‘r6780’ on a contradiction between ‘P15’s body and ‘P15’s own receipt: the body reports  $\chi^2=397.13$  against  $\$206.44$  over 215 ‘plik-lite’ TT bins with five parameters free in each, and the receipt records that pair as **superseded**. Which comparison is legitimate, on what run, with the floor beside it (**There are TWO instruments here, and the body’s pair is the other one’s — which is why the body and the receipt are both accurate.** **Q1** ‘fit.py’+CAMB fits **five parameters on 215 bins** and its CR arm is CAMB’s  $\Lambda$ CDM TT **times an ell-only damping envelope**; ‘ACOUSTIC\_two\_arm’+‘chi2\_of\_spectrum’ fits **one amplitude on 185 covered bins** and its CR arm is the construction’s own spectrum. The second is the one to report, and the first is **not malformed** — it answers the damping question **Q2** the second gives  $\$51817.0$  against  $\$1320.5$  — **and the crossing configuration was unreachable**, the CR arm’s background being written in as literals ( $H_0=73.00$ ,  $\Omega_m=0.3066$ ); ‘CRH0’/‘CROM’/‘CROMBH2’ exposed, defaults byte-identical **Q3**  $F_2=+1114.0$ ,  $F_3=+50496.5$ , forty-five times it — **and the body’s  $\Delta\chi^2=190.7$  is below that floor**, readable on the first instrument only because its control **is** the CAMB reference so its floor is identically zero **Q4** the deficit is in the second’s spectrum and **structurally absent** from the first’s **Q5** **both of the order’s two options at once**: ‘cr.json’ banks a genuine five-parameter optimum at  $H_0=78.13$ , and ‘fit.py’ shows the shape refitted was CAMB’s  $\Lambda$ CDM times the envelope). *Computes:* C1 ‘F1’ first, non-optional: the banked CAMB fit at  $\chi^2=206.4$  over 215 bins,  $\chi^2/\text{dof}=0.960$  the envelope’s inability to carry a comb **measured, not argued** — across envelopes strong enough to move  $\ell_{15}$  by up to eight multipoles the spacings hold within  $6\%$  ( $320/272/312$  to  $312/272/304$ ) the sound horizon scanned over **four decades** of starting redshift, converging monotonically to  $r_s=256.13$  Mpc with a residual drift of  $\$0.5$  in  $\ell_A$  — so “converges at its upper end” is a measurement here and not a quotation the exposed knobs’ defaults verified byte-identical on  $\ell_A$ ,  $r_s$ ,  $D_M$  and the peak list *Bound:* **NOT claimed** that the body’s computed  $\$298.0$  is wrong, nor that the instrument’s convention is the right one — only that they **differ**, the body putting  $r_s$  on the radiation-carrying leaf rate while ‘rs\_from’ integrates the radiation-free one, and which is intended is the paper’s question. **No  $\chi^2$  is manufactured against a comb the paper does not claim**, so the crossing-configuration figure the order asked for is reported as unreachable rather than supplied. Nothing reopens the handover resolution. **PO-7’ protected throughout**: a negative is a measurement discrepancy and no verdict on the construction is drawn from any  $\chi^2$  here *Run:* python3 receipts/P15\_CR.cosmology/P15.the.likelihood.pair.is.another.instruments.and.the.crossing.run.needed

**P15R153** P15.the.chiral.state.the.route.needs.is.forced.absent.not.merely.unchosen [OK]  
r6766 named what a successor to the closed route would need — a chiral **state** on the three-

sphere layer, not a chiral member — and left it unclassified: does it reduce to the handedness choice ‘PO-30’s law places outside, or is it something the geometry might still supply? (\*\*Neither: it is forced absent.\*\* \*\*It is not the handedness choice\*\* — that selects one of two equivalent options, while this asks whether the state is imbalanced \*\*at all\*\*, which is prior to any selection. \*\*The layer is the same one ‘P10’ works on\*\*, established from ‘P10’s own words: it lifts to “the closed- $S^3$  layer’s propagating sector”, where the transverse-traceless graviton modes form the tower. \*\*And there the balance is forced\*\*: regularity imposes the regular branch fibre by fibre, the surface gravity belongs to the background horizon and is common to every fibre, and the operator labelling the fibres is a sum of squares with a single coefficient — no helicity label and no sign, so the two towers are populated alike. \*\*The parity-odd source is exactly that difference\*\*, its expectation vanishing under equal population.  $\Rightarrow$  \*\*\*‘P10’s helicity cancellation and r6766’s source form are one statement from two sides; the route closes by a condition the geometry imposes, not by its silence on a choice\*\*\*). *Computes:* the parity-odd expectation vanishing under equal population and not under unequal; the product dying when either channel is silenced *Bound:* \*\*NOT that the tower is achiral\*\* — ‘PO-44’ is explicit it remains chirally capable and must not be read as the achirality branch; what is forced is the \*\*population\*\*. Nothing wider about states. Nothing about the handedness choice. \*\*\*‘P15’s item stays open\*\* — what is settled is that this route to it is closed, and why *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_chiral\_state\_the\_route\_needs\_is\_forced\_absent\_not\_merely\_unchosen

**P15R154** P15\_the\_legs\_closed\_form\_at\_the\_two\_loci [OK]

*sec:envelope* (the leg’s closed form read at the two loci: at the CROSSING it is the ordinary frozen adiabatic state exactly ( $\Theta\text{-hat} = \Psi_i/2$ ,  $\Psi = \Psi_i$ ), and the coded handover’s 0.4835  $\Psi_i$  is  $\Psi$  at horizon entry halved, which is neither locus). *Computes:*  $\Psi = \Psi_i T(x)$  and  $\Theta\text{-hat} = (\Psi_i/2) \cos x$  evaluated at the crossing, at entry and at the seam over  $k/k_s = 1$  to 300;  $\Theta_0$ ’s own driven equation integrated directly as an independent check, with the residual measured to scale as 0.399 x0 *Bound:* asserts the arithmetic of three candidate amplitudes and where each comes from; says nothing about which is right for the locus the instrument imposes it at, and nothing about the spectrum *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_legs\_closed\_form\_at\_the\_two\_loci.py

**P15R155** P15\_the\_handover\_at\_the\_crossing\_and\_what\_it\_costs [OK]

*sec:refit-bound* (moving the handover from the onset to the crossing removes 71 per cent of the arm’s excess  $\chi^2$  over the control and takes P1/P2 from -20.7 to +1.5 per cent against the sky – and does not remove the rejection, 35.5 per bin against the control’s 2.10). *Computes:* reproduces *sec:refit-bound* three ways (the control’s peaks and P1/P2, the coded arm’s, both arms’ *plik\_lite*  $\chi^2$ , and the sky’s own phase intercept); then the crossing arm’s comb, phase and heights, converged over two decades in ZSTART, to the ell-grid’s own resolution at LSTEP=1, and against the continuum ladder at KCONT=1 *Bound:* the rejection stands and is stated as one; the comb comes out 4.2 per cent narrow and the arm spends no fitted number; the reported LA is bookkeeping on a sound horizon the oscillator does not run on *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_handover\_at\_the\_crossing\_and\_what\_it\_costs.py

**P15R156** P15\_the\_desi\_dissolution\_survives\_the\_leaf\_ruler\_but\_only\_by\_moving\_the\_onset [OK]

*sec:tensions* (the DESI DR2 H0-dissolution survives the adjudicated leaf clock – 0.88 per dof, flat to two decimals from  $H_0 = 70$  to 80 – but by compensation rather than cancellation, and the price is the onset: its  $\rho_r/\rho_m$  runs 51.0 down to 5.8 across that band, against the 1.7 the stacking ruler puts it at). *Computes:* reproduces the banked stacking-ruler fit ( $\Omega_m$  0.3066,  $z_{\text{onset}}$  6764,  $\chi^2$  12.01 on 12 dof) and measures its  $H_0$  spread as exactly zero; then the leaf ruler without the onset, which re-pins  $H_0$  to 68.5 and breaks at 73 at 10.7 per dof; then the leaf ruler with the onset re-solved from  $\theta_*$  at each  $H_0$ ; and the acoustic angle alone as an independent determination agreeing with BAO to 0.05 in  $H_0$  *Bound:* the conclusion survives and the argument does not – on the leaf clock  $H_0$  is not FORCED by BAO rather than ABSENT from it; says nothing about which clock is right, and nothing about the spectrum *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_desi\_dissolution\_survives\_the\_leaf\_ruler\_but\_only\_by\_moving\_the\_o

**P15R157** P15\_the\_damping\_to\_comb\_ratio\_picks\_the\_crossing\_and\_the\_leaf\_clock\_out\_of\_four [OK]

*sec:coherence* ( $\theta_D/\theta_{\text{star}}$  separates the four handover-locus-by-clock combinations and picks exactly one – the crossing on the leaf clock, at 1.022 of the control’s ratio, against 1.135,

1.353 and 0.652 for the other three – and neither choice was made to fix this quantity). *Computes:*  $r_D/r_s$  on the corpus’s own Silk integrand with the opacity taken from `RD_diffusion_direct`, over all four combinations; the log-a quadrature converged to 0.006 per cent from  $n = 3001$  to 12001; and the crossing rows re-run over a hundredfold range in the locus, moving 0.07 per cent *Bound:* the two choices are not separable – neither helps at the other’s wrong partner – so this is one configuration and not two improvements; and getting this ratio right is necessary and not sufficient, the crossing arm still being rejected at 35.5 per bin *Run:* `python3 receipts/P15_CR_cosmology/P15_the_damping_to_comb_ratio_picks_the_crossing_and_the_leaf_clock_out_o`

**P15R158** `P15_the_onset_bracket_family_is_latent_everywhere_and_was_live_in_one_place` [OK]  
*sec:acoustic* (the (1500, 60000) onset bracket is a family – 27 live acoustic-onset solvers and 46 retired, 18 of the live ones below the leaf ruler’s root of 61,580, which is 2.6 per cent past the ceiling – and exactly one member was ever reached, failing by a quarter of a multipole). *Computes:* both roots recomputed from the instrument’s own integrands (6761 stacking, 61,583 leaf);  $\pi D_M/r_s$  at the old ceiling measured at 301.85 against the 301.6 target; the whole tree swept and every onset solver classified; and the six files a leaf-clock run actually enters checked to carry the widened ceiling *Bound:* the other 18 are LATENT and not broken – each integrates  $r_s$  on the stacking clock where the root is 6,764 – and are recorded rather than pre-emptively widened; says nothing about which ruler is right *Run:* `python3 receipts/P15_CR_cosmology/P15_the_onset_bracket_family_is_latent_everywhere`

**P15R159** `P15_the_third_peak_deficit_is_radiation_driving_and_both_routes_to_the_equality_agree` [OK]  
*sec:envelope* (P1/P3 tracks the leaf equality and not the route to it – 2.299 against 2.294 at  $z_{eq}$  3447 and 2.474 against 2.460 at 3000, across a 12 to 24 per cent change in  $\Omega_m$  and therefore in  $D_M$  and the whole projection – so the third-peak deficit is radiation driving, and the arm’s own equality is the residual). *Computes:* the crossing handover at  $z_{eq}$  3936, 3447 and 3000 reached twice over, through  $\Omega_m$  at fixed  $\Omega_r$  and through  $\Omega_r$  at fixed  $\Omega_m$ ; P1/P3 compared across the routes; the radiation route verified to move neither  $r_s$  nor  $D_M$  and the matter route verified to move both; and at 3447 the four peaks, the fitted comb and the acoustic phase against the sky’s *Bound:* a diagnosis and not a fit – 3447 is LCDM’s value and not a freedom this arm has, the arm is still rejected at 4.86 per bin against the control’s 2.10, and the scan is not monotone, so it measures where the equality should be and gives no direction to push it *Run:* `python3 receipts/P15_CR_cosmology/P15_the_third_peak_deficit_is_radiation_driving_and_both_routes_to_the_eq`

**P15R160** `P15_the_envelope_turnover_is_stated_on_a_third_equality_and_the_leaf_background_moves_it` [OK]  
*sec:envelope* (the driving envelope’s turnover is placed at equality and equality is then put at  $(1+z_{onset})/2 = 3399$ , “essentially flat LCDM’s” – but the leaf background the driving is integrated on has equality 3936, 14.2 per cent from the control and 15.8 per cent from the paper’s own figure, so the claim that the turnover wavenumber matches LCDM’s is carried by whichever of the two is meant). *Computes:* the paper’s 3399 reproduced from its own arithmetic;  $k_{eq}$  computed from each background directly as  $a_{eq} H(a_{eq})/c$  rather than from the matter-domination closed form; the projection measured to cancel roughly half of it,  $k_{eq} + 14.2$  per cent against  $ell_{eq} + 7.1$  per cent; and the self-consistent point (68.60, 0.2973) shown to put the leaf equality at 3370, 2.2 per cent from the control, with nothing fitted to reach it *Bound:* does NOT claim the turnover sits at equality – that is the paper’s premise, taken as given – and does not claim the envelope argument fails, its conclusion resting on the closed form and not on this number; the 6797 is backed out of the paper’s 3399 and is 0.5 per cent off the 6761 the instrument solves, which is noted and not chased; routed to the chat seat, whose file `CR_cosmology.tex` is *Run:* `python3 receipts/P15_CR_cosmology/P15_the_envelope_turnover_is_stated_on_a_third_equality_and_the_leaf_back`

**P15R161** `P15_the_damping_signature_error_budget_and_the_convention_dominates_it` [OK]  
*sec:coherence* (the crossing arm’s damping signature has a cosmological parameter budget of 0.02 points and an endpoint budget of 3.10 – moving  $r_D$ ’s upper limit from recombination to the computed visibility peak, 0.36 per cent away in redshift, takes it from +2.21 to -0.89 per cent, straddling zero – so it is not a few-per-cent observable with an error bar and must be quoted with its endpoint named). *Computes:* the visibility peak located from the computed visibility rather than assumed; `UNC_error_budget`’s own excursion list ( $H_0$ ,  $\Omega_m$ ,  $\omega_b$ ,  $z_{rec}$ ) propagated one at a time and summed in quadrature and linearly; the endpoint term measured beside them; the clock and locus terms kept apart as configuration rather than un-

certainty; and the four-way ranking re-derived through a second code path and shown to survive both endpoints *Bound:* explains rather than resolves the +3.1 against +2.2 disagreement between the instrument's header and the standalone integration – neither was wrong and the gap is this sensitivity; does not claim the ranking is endpoint-dependent, which is checked and is not; *UNC\_error\_budget* carries no such term and that is not an omission in it, one endpoint having been quoted throughout on the configuration it was written for *Run:* `python3 receipts/P15_CR_cosmology/P15.the_damping_signature_error_budget_and_the_convention_dominates.it.p`

**P15R162** `P15_at_its_own_preferred_H0_the_crossing_arm_predicts_the_acoustic_comb_with_no_fitted_number` [OK]

*sec:refit-bound* (at the  $(H_0, \Omega_m) = (68.60, 0.2973)$  its own leaf ruler's DESI fit returns, the crossing arm with no pin puts the fitted acoustic comb at 298.0 against the sky's 298.4 – 0.15 per cent, as an output – with  $P1/P2 = 2.264$  against 2.217,  $P1/P3 = 2.298$  against 2.277, gap ratios 0.8861 and 1.1286 against 0.8812 and 1.0966, and the 700-1000 band falling from 85.9 to 4.2 per bin;  $\chi^2$  is 4.19 per bin against the control's 2.10). *Computes:* the comb fitted by *sec:intro's* own procedure and checked sub-per-cent against the sky; the reported ruler and the fitted comb checked to agree to 1.6 per cent where at  $H_0 = 73$  they were 172.8 against 286.0; heights and both gap ratios gated against the sky; every band gated within a factor four of the control; and both instrument paths reported, the comb verified path-independent *Bound:* still rejected against LCDM at a factor 2.0, stated as the result and not as a caveat;  $H_0$  and  $\Omega_m$  ARE fitted, to BAO and the acoustic angle rather than to TT, so the cosmology is not parameter-free and only the spectrum is out-of-sample; the acoustic phase remains 2.3 per cent out and is now the only surviving position error; the height result is polarisation-path specific; and the configuration is the 133-bin unlensed one at  $L_{MAXL} = 1300$ , not the 185-bin lensed comparison *Run:* `python3 receipts/P15_CR_cosmology/P15.at_its_own_preferred_H0_the_crossing_arm_predicts_the_acoustic_comb-w`

**P15R163** `P15_the_full_range_lensed_comparison_is_the_unfavourable_one_and_the_control_is_nearly_camb` [OK]

*sec:refit-bound* (on the 185-bin full-range lensed configuration – the one the corpus's own  $\chi^2$  values are quoted on – the crossing arm at  $(68.60, 0.2973)$  sits at 2.98 per bin against the control's 1.16, a ratio of 2.57, which is WORSE than the 1.98 the 133-bin unlensed comparison showed, because lensing is worth proportionally more to the control). *Computes:* the bin count asserted at 185 over ell 100-1996 rather than taken on the configuration's name; the arm and control checked to cover the same bins; the lensing operator checked against `P15_derived_lensing_on_the_lcdm_arm PART C's` +6.5 per cent at ell 1900, returning 1.0666, so it is the full operator and not the first-order kernel; both arms checked to fall under lensing; and the ratio gated against being the more favourable one *Bound:* states both comparisons so neither can be picked for being the kinder; and withdraws, rather than relies on, the caveat raised at cc66.9 that the control might carry ~700 of transfer inaccuracy – this control is within 13 per cent of CAMB unlensed and 15 per cent lensed, where the c54.178 control was 2.1 times CAMB's, so the ratio is a statement about the arm and not about the instrument *Run:* `python3 receipts/P15_CR_cosmology/P15.the_full_range_lensed_comparison_is_the_unfav`

**P15R164** `P15_the_anomalous_driving_belonged_to_the_pin_and_not_to_the_construction` [OK]

*sec:refit-bound* (the driving subtraction on the adjudicated arm is 0.96 times the control's, where the pinned arm the paragraph carries was 2.48 times – so the anomalous driving belonged to the pin and not to the construction – and the two instrument paths agree to four decimals, unlike the heights). *Computes:* the first peak located driven and undriven on both paths and on a control run here rather than quoted; the driven and undriven runs checked to report the same  $l_A$ ; the ratio gated against parity and against the paper's 2.4 for the pinned arm; the two paths checked to agree; and the two-multipole difference from the quoted control priced as one grid step at  $L_{STEP}=2$  *Bound:* states the convention explicitly – ell\_1 over the REPORTED  $l_A$  and not over the fitted comb, which differ by 0.7329 against 0.7450 on this arm – and does not claim a smaller driving shift is evidence the arm is right, only that it removes a discrepancy; the 2.3 per cent phase residual is untouched *Run:* `python3 receipts/P15_CR_cosmology/P15.the_anomalous_driving_belonged_to_the_pin_and_no`

**P15R165** `P15_the_crossing_spectrum_reproduces_on_a_second_instrument_and_the_172_is_the_radiation_free` [OK]

*The crossing configuration's spectrum on a SECOND implementation: node 66's code seat integrated it on the polarisation path over 133 bins, and 'PO-10's row routed the full-spectrum run away from this seat at 'r6772+66.36' — so what is left to ask is whether an independent integration on a different path, a different wavenumber grid and a different bin range returns the same*

*spectrum* (\*\*It does, and the agreement is the result rather than any  $\chi^2$ .\*\* Hierarchy path, 185 covered bins,  $\ell$  100–1996, one fitted amplitude, at  $(68.62, 0.2973)$  with the arm’s own handover datum: peaks  $220/540/820/1132$  against node 66’s  $222/538/818/1134$ , heights  $2.273/2.319$  against  $2.264/2.298$ , the  $700$ – $1000$  band  $4.02$  per bin against  $4.23$  — \*\*\*within one grid step on every peak, within one per cent on the heights, neither seat knowing the other had run it\*\*\* \*\*and node 66’s fourth-peak excess reproduces at  $1132$  against  $1123.9$ , which makes it the spectrum’s and not either implementation’s\*\* against the configuration this replaces, on the same instrument and bins:  $\chi^2 = 1205.4$  where the pinned arm gives  $51817.0$  and the control  $1320.5$ , with positions  $-0.3\% / +0.4\% / +1.3\%$  where the deficit stood at  $7.5\%$  — the first two peaks inside one grid step and \*\*\*the third at  $1.28$  steps and not inside it\*\*\*). *Computes:* C1 the banked pair reproduced on this scorer first ( $1320.5$  and  $51817.0$ ) before any new number was read \*\*W1 the substituted-datum control, which reversed a reading of mine\*\*: the first run carried ‘CRIC=branchpoint’, the control’s super-horizon datum, and repeating it with the arm’s own datum left the positions and  $\chi^2$  alone ( $1191.0$  to  $1205.4$ ) while \*\*\*moving the heights  $2.152/2.168$  to  $2.273/2.319$ \*\*\* — so ”the heights did not come in” was the substitution’s artefact and is withdrawn the amplitude is not doing the work either: the crossing arm wants the \*\*control’s\*\* amplitude to  $0.04\%$  where the pinned arm wanted one  $8\%$  away three spectra banked with their exact commands in ‘spectra/README.md’ *Bound:* \*\*NOT a preference over  $\Lambda$ CDM\*\*:  $-115.1$  against  $F_{2.5}$ ’s floor of  $+1114.1$ , a tenth of it, and this control carries its own truncation error (‘c54.186’) — \*\*\*the papers quote node 66’s lensed  $2.57\times$  and this receipt quotes nothing against it\*\*\*, ‘PO-7’ protected exactly there \*\*NOT the discreteness control\*\*, which is named and in flight rather than omitted: at the run’s own ‘NK=600’ a continuum grid gives  $2.8$  points per Bessel period and \*\*\*the instrument refuses to run\*\*\*, clearing the guard taking ‘NK=900’ and  $2700$  modes against the ladder’s  $1452$  — until it lands what stands against aliasing is node 66’s independent grid, which is weaker for that question and is stated as weaker \*\*NOT a discrepancy with the body’s comb\*\*: the paper’s answer is the \*\*leaf\*\*, so  $\ell_A = 172.3$  is the radiation-free ruler and a different object — that reading of mine is withdrawn here and the heights are the \*\*path’s\*\*, the line-of-sight path overshooting at  $2.543/3.264$  *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_crossing\_spectrum\_reproduces\_on\_a\_second\_instrument\_and\_the\_172.i

**P15R166** P15\_the\_refit\_leaves\_the\_background\_where\_the\_distances\_put\_it\_and\_does\_not\_close\_the\_phase [OK]

*sec:refit-bound* (the parameter refit: given four free parameters the crossing arm moves  $H_0$  by 0.011 per cent and  $\Omega_m$  by 0.20 per cent, so the background the baryon-acoustic data fix is the background the spectrum wants – and freedom does NOT close the two residuals, the acoustic phase going 2.3 to 4.5 per cent out and the fourth peak 0.9 to 0.7, so both are properties of the construction rather than background choices). *Computes:* a measured response rather than a search – 18 runs giving gradient and diagonal curvature per arm, the amplitude closed-form; the minimum VERIFIED by real runs at the best-fit parameters, predicted 118.2 and 170.6 against measured 118.3 and 171.1; every parameter flatness-tested against a one-step excursion, all four constrained on both arms; and the bin count checked at 132 *Bound:* the range is 132 bins,  $\ell$  100–1287, not the 185 first asked for, so  $\omega_b$  and  $n_s$  are the loosest numbers;  $\tau$  is not a free direction at all, being exactly degenerate with the amplitude where no reionisation is modelled; and the arm is still disfavoured at 1.45 times the control, stated as the result rather than a caveat *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_refit\_leaves\_the\_background\_where\_the\_distances\_put\_it\_and\_does\_n

**P15R167** P15\_the\_signature\_collapses\_at\_the\_adjudicated\_ratio\_and\_the\_endpoint\_sets\_its\_sign [OK]

*sec:diffusion-scale* (at the adjudicated ratio the diffusion-scale signature costs 12.6 and 1.7 in  $\chi^2$  over 185 bins – under a tenth per bin on either endpoint – and a tilt of  $-0.0093$  or  $+0.0034$  absorbs seven tenths of that, against 159.9 and  $-0.0341$  at the 1.082 every quantitative leg of the row was computed at; and the two r.D endpoints disagree in SIGN, not only in size). *Computes:* the ratio re-measured at the adjudicated background on both endpoints, 0.10 and 0.07 points from the 1.022 and 0.991 the row carries; the joint amplitude-and-tilt fit on the arm’s own 185-bin spectrum with the window asserted at  $\ell$  100–1996 and the tilt’s uncertainty propagated through the fit’s own parameter covariance; the tilt’s running with the window measured at 5.80 against the 5.54 the  $(l_{\max}/l_{\min})^2$  result predicts; the  $r^2$ -1 scaling on which the whole re-pricing rests checked to 2 per cent in the tilt and 8 per cent in the  $\chi^2$ ; the base and the

lensing each shown to move nothing; the fitter controlled on injected tilts; and the endpoint settlement measured – the instrument needing no surgery to terminate both integrals at one epoch, and the corpus’s +2.2 per cent being the COMMON reading at recombination rather than the mixed one the ruling describes, which measures 1.02830 *Bound:* the endpoint was settled by the chat seat at r6797 AFTER this run – both lengths to the visibility peak – so the ruled reading is 0.99179, -0.82 per cent at the adjudicated background rather than the -0.9 per cent the ruling quotes, which is that reading at the  $H_0 = 73.00$  background; the recombination row is kept as what the corpus carried and not as an open alternative; the structural results are carried from C62 and not re-derived; and nothing here says the arm fits, the refit leaving it at 1.30 per bin against the control’s 0.90 and the full-range lensed comparison at 2.57 times *Run:* python3 receipts/P15\_CR.cosmology/P15.the.signature.collapses.at.the.adjudicated.ratio.and.the.endpoint.set

**P15R168** P15\_the\_low\_multipole\_floor\_moves\_with\_no\_background\_and\_the\_factor\_two\_is\_the\_late\_isw [OK]

*sec:largescale* (on the adjudicated background the low-multipole floor does not move –  $r_0$  shifts 0.26 per cent and the lowest mode lands at  $l_2 = 7.81$  against 7.74 – and the two Boltzmann arms still disagree, by a factor 2.02 at  $l = 4$  and 1.73 at  $l = 3$ , WIDER than the control’s 1.84; and the disagreement is located in the LATE ISW, which the photon-hierarchy arm does not have and cannot be made to have). *Computes:*  $r_0$  computed from the paper’s own parameter-free formula and asserted to reproduce its 5064 Mpc before being used; both arms given ONE  $r_0$ , the banked run’s 2.75/D\_M ladder re-run separately so the 4.8 per cent that correction costs is measured rather than waved at; each arm validated on the control background against what is banked before either is moved; the geometry ruled out by measurement, 1 per cent in  $r_0$  buying 3.5 per cent at  $l = 4$  against a 102 per cent gap; the late-ISW diagnosis tested by pushing the line-of-sight cut from  $z = 53.5$  to  $z = 15$  and finding  $l = 3$  and  $l = 4$  move towards the CAMB arm; and the hierarchy shown non-finite past  $z$  about 10 under the defaults and under two refinements of the step count and the freeze threshold, so it is not a step-size failure *Bound:* which arm is right is NOT settled – carrying the late ISW is a reason to prefer the CAMB arm AT THESE MULTIPOLES and not a general verdict, the hierarchy existing because the fluid-plus-envelope treatment fails at high  $l$ ; and node 66’s part (b), the floor scored against the sky, is NOT attempted because its stated condition – that the two paths can be brought together – is not met, the fix being a named instrument build (the post-recombination source carried with the photon hierarchy decoupled) rather than a knob *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_low\_multipole\_floor\_moves\_with\_no\_background\_and\_the\_fact

**P15R169** P15\_the\_full\_range\_refit\_holds\_the\_background\_and\_the\_phase\_residual\_was\_quantised [OK]

*sec:refit-bound* (the 185-bin full-range refit: with the damping tail inside the fit the crossing arm’s background stays pinned –  $H_0$  moves 0.028 per cent and  $\Omega_m$  0.031 per cent, so  $\Omega_m$  is held six times tighter than on 132 bins and  $H_0$  twice as loosely, both inside a thirtieth of a per cent – and cc66.18’s phase residual of 4.5 per cent is WITHDRAWN as quantisation, the control having returned the same number). *Computes:* the minimum located on a measured response (18 runs, gradient and diagonal curvature per arm, amplitude closed-form) and then VERIFIED by real LMAXL=2000 runs at the best-fit parameters, predicted 185.1 and 292.5 against measured 186.5 and 292.4; every parameter flatness-tested, all four constrained on both arms; the bin count asserted at 185; and the peak locator’s own resolution measured against a banked LSTEP=1 spectrum BEFORE use – the raw locator errs by 3.1 in  $\ell$  at the grid the refit runs on and the parabolic refinement by 0.13 *Bound:* the arm is still disfavoured at 1.57 times the control at the verified minimum, stated as the result rather than a caveat;  $\tau$  is not a direction of the fit at all, being exactly degenerate with the amplitude where no reionisation is modelled; and the refined phase residual, 3.4 per cent against the control’s 1.6, changes the numbers cc66.18 landed while leaving its verdict – freedom does not close the phase – standing *Run:* python3 receipts/P15\_CR.cosmology/P15.the\_full\_range\_refit\_holds\_the\_background\_and\_the\_phase\_residual\_was

**P15R170** P15\_the\_low\_multipole\_depth\_gap\_closes\_and\_two\_defects\_were\_cancelling [OK]

*sec:largescale* (the low-multipole depth gap CLOSES: the two Boltzmann treatments agree to 3 per cent across  $\ell = 2-8$  where the corpus carries a factor 2.02 at  $\ell = 4$ , and the factor of two was TWO defects in the photon-hierarchy arm that were partly cancelling – a line-of-sight cut at  $z = 53.5$  that omitted the late ISW, biasing the ratio down, and a continuum integral truncated at 0.5  $k_2$ , biasing it up; the confrontation with cosmic variance carried then returns a wash,  $\Delta(-2 \ln$

L) = +1.58). *Computes:* the withdrawal measured rather than argued – the undecoupled hierarchy runs to a = 1 finite once the opacity grid’s resolution is held fixed as the cut moves, so the blocker was never the LG = 12 truncation; the continuum’s low-k reach converged to 1 per cent at 0.1 k<sub>2</sub> and shown to move ell = 2 by tens of per cent at 0.5; the two defects shown to OPPOSE, the continuum fix alone widening the gap past a factor 3; both arms on both backgrounds with the decoupled build and the module’s own per-mode freeze agreeing to 4 per cent; convergence in the late step count and the opacity grid to 0.1 per cent; and the likelihood the exact scaled  $\chi^2$  with 2l+1 degrees of freedom, so the sky’s own scatter is carried rather than bolted on *Bound:* it does NOT say the low-multipole sector discriminates – it does not, and that is now a measurement rather than an open question; two claims of this line are withdrawn with it, the LG = 12 mechanism and the arm-B quartet 0.4974/0.2326/0.1933/0.6186; and nothing here touches the acoustic range, where the hierarchy exists because the fluid-plus-envelope treatment fails *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_low\_multipole\_depth\_gap\_closes\_and\_two\_defects\_were\_cancelling.py

**P15R171** P15\_there\_is\_no\_phase\_residual\_and\_the\_fourth\_peak\_is\_not\_the\_damping\_envelope [OK]

*sec:refit-bound* (there is no acoustic phase residual: measured against a sky whose peaks are located by the SAME procedure, with the locator’s own spread as the uncertainty, this arm’s phi/pi sits 1.0 sigma from the sky on the three-peak fit and 0.0 on the four-peak fit, and 1.6 and 0.6 sigma from the control – and the fourth peak is NOT the damping envelope, forcing the arm’s damping to the control’s moving it 0.13 in ell against an offset of 1.95). *Computes:* one locator applied identically to the sky and both arms – bin to plik\_lite’s bins, convert to D<sub>1</sub>, spline, parabola over a window – with the window swept over seven values so the locator’s own spread is the uncertainty the comparison carries; the locator first shown to reproduce sec:intro’s own sky quartet, with the second peak identified as the hardest and its spread the largest; the three procedure mismatches in the earlier reading priced one at a time; the damping test validated by its inverse, imposing the arm’s envelope on the control and getting an equal and opposite shift; and the residual decomposed against the arm’s own comb width, which accounts for about half of peaks three and four *Bound:* the per-cent offset does NOT shrink under the matched procedure – 3.4 becomes 4.2 – so what dissolves the residual is the sky’s own uncertainty and not the central value, and the receipt says so rather than the other way round; the three leftovers after the comb-width term are 1.4, 0.6 and 1.0 in ell, the same sign and order but not a clean constant, and a finer statement is not available at these locating spreads; order (3), the neutrino sector, is NOT run because the order made it conditional on a uniform phase offset surviving and none does; and nothing here says the arm fits, the verified refit leaving it at 1.58 per bin against 1.01 *Run:* python3 receipts/P15\_CR\_cosmology/P15\_there\_is\_no\_phase\_residual\_and\_the\_fourth\_peak\_is\_not\_the\_damping-env

**P15R172** P15\_the\_ladders\_discreteness\_does\_not\_set\_the\_spectrum\_and\_the\_check\_was\_run\_in\_pieces [OK]

*The control ‘r6788’ declared open and node 66 asked for by name (‘FOR\_60’ item 5): ‘KCONT=1’, the CR arm’s discrete \$k\$-ladder replaced by a continuum grid on the crossing configuration. The alias gate waives its 4-points-per-Bessel-period requirement for this arm because the ladder is physical, and the waiver’s own text says that is only not aliasing if the answer does not depend on it* (\*\*It does not, and the waiver is honest here.\*\* Ladder (\$1452\$ modes, \$2.3\$ pts/period)  $\chi^2 = 1205.3755$  against continuum (\$2700\$ modes, \$4.3\$)  $\chi^2 = 1205.3745$  — \*\*\*peaks identical\*\*\* at \$220/540/820/1132\$, heights identical to three decimals, the single fitted amplitude \$3\$ parts in \$10^8\$ apart, spectra agreeing to \$6.7 \times 10^{-8}\$ of the peak \*\*and the reason it agrees far better than \$2.3\$ points per period predicts is measured rather than guessed\*\*:  $\sqrt{L(L+2)}$  to \$L+1\$, so the “discrete ladder” is asymptotically \*uniform\* — within \$1\%\$ of its median spacing over \$99.9\%\$ of its gaps — making this a comparison of two near-uniform samplings differing in spacing, not of resolved against aliased \*\*run in six slices, because neither seat’s container survives it whole\*\*: the check costs \$2700\$ modes against the run’s \$1452\$, mine was killed three times and ‘cc66’ restarts every \$1\$–\$15\$ min). *Computes:* C1 \*\*the segmentation proved exact before anything was read through it\*\* — a whole run against three slices summed,  $\max\{\lvert\sum\{\rm whole\}\rvert\}$  relative \$2.195 \times 10^{-16}\$, machine precision ‘KSLICE=lo:hi’ added with its default proved byte-identical both spectra scored on the same \$185\$ covered bins over the same \$\ell\$ range \*\*and a property ‘KBATCH’ already had, measured and sized\*\*: ‘\_project’ takes \$dk = \nabla k\$ from the \*batch\*, so on the ladder’s non-uniform bottom modes the batch boundary moves the weights — one batch against two differs by \$1.205 \times 10^{-8}\$ relative *Bound:* \*\*Set-

ties one thing only<sup>2</sup>: that discreteness does not set this spectrum. Says nothing about whether the spectrum is right, and <sup>3</sup>nothing that touches the floor<sup>4</sup> —  $F_2 = +1114.1$  still forbids reading  $1205.4$  against the control's  $1320.5$  as a preference, exactly as 'r6788' said measured at <sup>5</sup>one<sup>6</sup> configuration, so 'c54.186's  $0.7\%$  at the pinned one is a different ladder density and is not superseded the 'KBATCH' figure is <sup>7</sup>negligible at  $10^{-8}$ <sup>8</sup> — eight orders below anything physical, a reproducibility note (quote 'KBATCH' with a ladder command) and not a defect, which is why no banked spectrum is revisited and 'KSLICE' is exact on a <sup>9</sup>uniform<sup>10</sup> grid only, so it is a continuum tool and a ladder run should be done whole *Run: python3 receipts/P15\_CR.cosmology/P15.the\_ladders.discreteness\_does\_not\_set\_the\_spectrum\_and\_the\_check\_was\_*

**P15R173** *P15.the\_leg.contributes.no.tilt.because.a.fixed.phase.handover.is.scale.free.and.a.fixed.time\_*  
[OK]

'PO-31's work order from node 66: where the tilt comes from, if it comes from the collapse leg. The row has a target — the like-for-like refit prefers  $n_s=0.9949$  against the control's  $0.9559$  ('r6805') — and the order asks the first calculable thing beyond it: <sup>11</sup>can the leg produce a departure of that size at all?<sup>12</sup> <sup>13</sup>No, and for a stronger reason than a small number: what the leg can produce is not a tilt.<sup>14</sup> Both the potential and the temperature on the leg are functions of  $x=k\{\eta/\sqrt{3}\}$  <sup>15</sup>alone<sup>16</sup>, so a handover at a fixed <sup>17</sup>phase<sup>18</sup> is exactly scale-free — <sup>19</sup>the ratio of amplitudes at two wavenumbers is  $1$ , not nearly  $1$ <sup>20</sup> — and the only surviving scale is a handover at a fixed <sup>21</sup>time<sup>22</sup>, through  $x_{\rm seam}=0.7638\{(k/k_s)\}$  <sup>23</sup>what a fixed-time handover gives is red in both channels and goes as  $k^2\{\}$ <sup>24</sup>:  $\{\ddot{\ln}\{\hat{\Theta}\}\}^{\rm rvert}/\{\ddot{\ln}\{k=-x\{\tan x=-x^2+O(x^4)\}\}$  and  $\{\ddot{\ln} T/\{\ddot{\ln} k=-x^2/5+O(x^4)\}\}$  — a <sup>25</sup>running<sup>26</sup>, where a constant  $n_s$  shift needs a constant log-derivative, with an honest zero of the amplitude at  $k/k_s=2.057$  <sup>27</sup>so the size question answers itself by shape<sup>28</sup>: the target needs  $k/k_s=0.065$ , and thirty times that wavenumber the same expression gives  $1-n_s=41.9 \Rightarrow$  <sup>29</sup>and on the adjudicated configuration it is exactly zero<sup>30</sup> — 'r6774's crossing is  $x\{\}$  to  $0$ , where  $T\{\}$  to  $1$  and  $\{\hat{\Theta}\}\{\}$  to  $\{\Psi_i/2\}$  and both log-derivatives vanish quadratically). *Computes:* C1 the three banked amplitudes reproduced before anything is read —  $T(1/\sqrt{3})/2=0.48353$  (the coded handover),  $\{\cos(1/\sqrt{3})/2=0.41896$  (the exact entry temperature),  $1/2$  (the envelope and the crossing value) — the closed form being <sup>31</sup>established<sup>32</sup> ('P15' §coherence, §transmission; 'r6760+cc66.1') and re-run rather than re-derived the quadratic approach checked, so the vanishing at the crossing is not an artefact of where the limit is taken <sup>33</sup>and one coding slip of mine was caught before landing: the band evaluation used the wrong factor and printed  $1.4$  where the docstring said  $41.9$ <sup>34</sup> *Bound:* <sup>35</sup>Bounds the LEG's contribution only<sup>36</sup> — nothing about what the progenitor supplies, and nothing about  $A_s$ , whose fixed-point route the entropy monotone closed the fixed-time numbers are evaluated at the seam because that is the only fixed-time locus the corpus ever coded: <sup>37</sup>a demonstration that such a handover gives a running, not a claim the corpus uses one<sup>38</sup> — it does not, since 'r6774' the progenitor's mass enters only through  $k_s$ , so no independent mass dependence appears <sup>39</sup>nothing is tuned to  $0.995$ <sup>40</sup>: the target enters once, to be refuted on shape, and 'PO-7' is untouched — this is a statement about a transfer function and not a verdict *Run: python3 receipts/P15\_CR.cosmology/P15.the\_leg.contributes.no.tilt.because.a.fixed.phase.handover.is.scale\_f*

**P15R174** *P15.the\_substrate.cannot.tilt.the.spectrum.and.the.collapses.duration.enters.only.as.a.decayin\_*  
[OK]

node 66's order (2) in 'FOR\_60' ('r6825'), 'PO-31' one step in: with the leg closed as a source, the tilt is the progenitor's vacuum, so <sup>41</sup>what does the substrate ALONE constrain about that vacuum, before any interior is modelled?<sup>42</sup> Three candidates named — the progenitor's mass, the finite duration of the collapse, and the substrate's curvature radius — with the order asking which can enter the vacuum's two-point function <sup>43</sup>at all<sup>44</sup>, and the sign of  $n_s-1$  for any that can <sup>45</sup>(<sup>46</sup>One of the three can enter, and what it produces is not a tilt: the substrate alone fixes  $n_s=1$  exactly.<sup>47</sup> <sup>48</sup>the curvature radius CANNOT enter<sup>49</sup> — in conformal time the substrate's mode equation  $u''+(k^2-2/\eta^2)u=0$  contains no length at all, so the only variable is  $k\{\eta\}$  and  $\{\alpha\}$  survives solely as the prefactor  $P=1/(4\{\pi^2\}\alpha^2\}$ : it sets  $A_s$  and reaches  $n_s$  <sup>50</sup>for every<sup>51</sup>  $\{\alpha\}$ , not for large  $\{\alpha\}$  <sup>52</sup>the progenitor's mass CANNOT either, and on this construction that is arithmetic<sup>53</sup> — the forced member is Nariai,  $\{\Lambda M^2=1/9\}$  with  $\{\Lambda=3/\alpha^2\}$ , so  $M=\{\sqrt{3}\}\alpha/9$  and  $M/\alpha=0.192450$  is a <sup>54</sup>pure number<sup>55</sup>: the mass <sup>56</sup>is<sup>57</sup> the curvature radius up to a constant and brings no new length, so the two candidates are one and fall together <sup>58</sup>only the fi-

nite duration enters\*\* — setting the vacuum at finite  $\eta_0$  gives  $\alpha_k = 1 - i/x - 1/2x^2$ ,  $\beta_k = e^{2ix}/2x^2$  with  $|\alpha_k| |\beta_k|^2 = 1$  exactly and  $R(x) = 1 + \cos 2x/x^2 - \sin 2x/x^3 + (1 - \cos 2x)/2x^4$ ,  $x = k\eta_0$ , whose log-derivative is  $-2 \sin(2x)/x + O(x^{-2})$ : \*\*an oscillation with a decaying envelope, sign-alternating 24 times on  $x \in [1, 40]$ , its four octave means disagreeing in sign\*\* — a \*running\*, and by the corpus's own operational test a scale, not a tilt  $\Rightarrow$  \*\*\*so a constant tilt needs the interior, and its sign is fixed in advance\*\*\*: the only quantity giving a band-independent log-derivative on a maximally symmetric background is an effective mass, exactly  $n_s - 1 = 3 - 2\sqrt{9/4 - m^2\alpha^2} = \frac{23}{16}m^2\alpha^2 + O(m^4)$ , \*\*blue for  $m^2_0$ \*\* — so the measured  $n_s = 0.995$  requires  $m^2\alpha^2 = -1201/160000 = -0.0075$ , slightly tachyonic). *Computes:* C1 the Nariai constants reproduced before anything is read —  $M = \sqrt{3}\alpha/9$ ,  $\Lambda M^2 = 1/9$  on substitution,  $f(r_N) = f'(r_N) = 0$  — \*\*established\*\* ('P15' Fig.  $f(r)$ , 'r6804') and re-run rather than re-derived, because finding (2) rests entirely on them; the Bunch–Davies mode verified to solve the mode equation and to carry Wronskian  $\mathcal{W}$ , so the amplitudes are not rescalable the  $R(x)$  identity checked twice by independent simplification routes \*and one stated criterion of mine was wrong and was corrected to the computation rather than the reverse: I had claimed the octave means sit within  $0.021$  of zero, and the  $[2, 4]$  band averages  $-0.095$  — the true and stronger statement is that they do not agree with one another and do not share a sign\* *Bound:* \*\*the one modelling step is named and it is the standard one\*\*: the corpus's scalar perturbation is treated on the substrate as a massless minimally coupled field. Findings (1) and (2) are robust to that choice — they are statements about which \*lengths\* appear, and none appears for any mass — but the exact value  $n_s = 1$  is not, which is precisely what (4) says the duration channel computes a \*\*sharp\*\* initial time, the one unambiguous prescription: a smooth onset changes the envelope's power and the phase and cannot change the sign alternation, so the claim is the SHAPE and not the coefficient bounds what the \*\*substrate\*\* supplies — nothing about  $A_s$ , whose fixed-point route the entropy monotone closed, and nothing about the interior beyond the one requirement \*\*\*nothing is tuned to  $0.995$ \*\*\*: the target enters once, at the end, and only to be inverted into a requirement on a quantity this receipt does not compute; 'PO-7' untouched *Run:* python3 receipts/P15\_CR\_cosmology/P15\_the\_substrate\_cannot\_tilt\_the\_spectrum\_and\_the\_collapses\_duration\_ent

**P15R175** P15\_the\_interiors\_vacuum\_is\_blue\_everywhere\_and\_a\_running\_so\_the\_substrates\_mass\_template\_does [OK]

node 66's work order in 'FOR\_60' ('r6845') — 'PO-31' on the progenitor's \*\*interior\*\*. 'r6826' left one channel,  $n_s - 1 = 3 - 2\sqrt{9/4 - m^2\alpha^2}$  asking the interior for a slightly negative  $m^2\alpha^2$ , \*\*but that formula is the substrate's and the fluctuations are generated in the interior\*\*. Three questions: the spectrum handed over and whether it is a tilt or a running; its sign; and whether the substrate's formula applies at all (\*\*All three answer, and they answer against the construction. first a correction to the order's premise: \*\*the interior is not newly built\*\* — 'P15\_the\_progenitor\_vacuum\_is\_negligible\_too' already banks  $a = M(\rho x + x^2/2)$  and the  $M$ -free potential  $2/(x(x+2\rho))$ , and it is \*the same object\*: with  $M = A/2$ ,  $\rho = 2\sqrt{B/A}$  the two agree \*\*exactly\*\* to  $O(\eta^2)$  and  $a''/a = A/(2a) - 1$  reduces to it \*\* (1) a RUNNING, not a tilt\*\*: the potential interpolates between  $2/x^2$  (the  $p = +2$  matter root, scale-free) and  $1/(\rho x)$  (not), so at the determined  $\rho = 0.0539$  the log-derivative runs  $+0.30$  to  $+1.98$  over  $k = 5$ – $2000$ , tending to the radiation value  $+2$  — no constant to fit, by the same band test that closed the leg \*\* (2) BLUE at every  $k$ , flattest  $+0.304$ , against a measured \*\*red\*\*  $-0.002$ : wrong sign and  $\sim 100$  times the size. The radiation content is the cause — flattest slope  $\sim 6.7\rho$  — \*\*so red would need  $\rho_0 \Rightarrow$  \*\* (3) the substrate's template DOES NOT APPLY:  $3 - 2\sqrt{9/4 - m^2\alpha^2}$  comes from a  $1/\eta^2$  potential whose coefficient is \*dimensionless\*; the interior's carries the scale  $\rho$ . \*\*\*So  $m^2\alpha^2 = -0.0075$  is an artefact of applying the substrate's template off the substrate, and the corpus should stop carrying it as a requirement\*\*\*). *Computes:* C1 \*\*the extractor calibrated on four power laws before being believed\*\* —  $p = -1$  and  $p = +2$  both returning  $0$  to  $5 \times 10^{-6}$ ,  $p = +1$  giving  $+2$ ,  $p = +3$  giving  $-2$ , against  $3 - 2\sqrt{p - 1/2}$ : \*two of them are the corpus's own scale-invariant roots, so the instrument is checked on the cases the argument turns on\* \*an earlier version read every  $k$  on a \*\*different\*\* surface and returned  $+6$  where  $0$  was right — the calibration caught it, and it is recorded because the failure was silent\* the background identities and the two potential limits derived symbolically; the surface-convergence measured rather than as-

sumed *Bound*: **\*\*the vacuum is set at maximum expansion\*\*** ( $a'=0$ , so every mode sub-horizon), with  $k^2/\text{a}$  enforced per mode — **\*\*below that band the interior supplies no initial condition\*\***, which is the order's own stopping clause, reported as a bound and not stepped over the slope **\*\*converges\*\*** as the reading surface approaches the branch point (last two decades agree to  $1.2 \times 10^{-4}$ , the shallowest differing by  $1.4 \times 10^{-2}$ ) — **\*stated as convergence and not insensitivity, because the shallowest surface is not yet converged\***  $A=1$  sets the time unit and every result is a slope or a ratio;  $\rho=0.0539$  is the **\*\*determined\*\*** composition, taken from the banked receipt and not chosen here a massless minimally coupled field, the same modelling step 'r6826' named **\*\*the banked receipt is not contradicted\*\***: its scale-invariance is of a **\*transfer\*** and its number an **\*amplitude\***; the slope of the generated spectrum is a different object nothing bears on  $A_s$  or 'PO-7' *Run: python3 receipts/P15\_CR\_cosmology/P15\_the\_interiors\_vacuum\_is\_blue\_everywhere\_and\_a\_running\_so\_the\_substrat*

### P3R12 P03\_acceleration\_is\_slice\_curvature [OK]

*sec:curvature · rem:twocritical · the bend* (on ANY member of the energy family (E cancels:  $(dr/d\tau)^2 = E^2 - f$ , and E is a constant of the motion)  $d^2r/d\tau^2 = -f'/2 = r \cdot K_G$  identically, so THE SIGN OF THE SLICE'S GAUSSIAN CURVATURE IS THE SIGN OF THE COMOVING ACCELERATION – and the flat locus is the deceleration-to-acceleration turnover, which on Narai is the seam and in standard terms is  $\rho_m = 2 \rho_\Lambda$  ( $\cosh^2 w = 2$   $\Rightarrow \sinh w = 1/\sqrt{2}$ , the seam's phase). A recent cosmological epoch: 7.06 Gyr at  $H_0=73$ , 7.65 at 67.4, preceding matter-Lambda equality [borrowed from P3 · P7 · P8]). *Computes*: the identity asserted; the  $K_G$  zero solved and matched to  $(M \alpha^2)^{1/3}$ ;  $\cosh^2 w = 2$   $\Rightarrow \sinh w = 1/\sqrt{2}$  proved by algebra after simplify() failed on the closed forms; epochs at both  $H_0$  *Bound*: bound: identities on the Narai  $E=1$  worldline; no causal claim between seam and turnover – they are one locus; dating inherits  $H_0$  *Run: python3 receipts/P03\_SdS\_slicing/P03\_acceleration\_is\_slice\_curvature.py*

### P15R42 BRANCHPT\_transmission\_character [OK]

*rem:phase-open* (BRANCHPT — the transmission character of the BRANCH POINT  $r=0$ , on its own terms. WHY (r2154). P15 l.525 says "The cosmogenesis branch point, BEING the degenerate Narai horizon, is a faithful scale-free transmitter", and prop:transmission argues from the tortoise integral at a DOUBLE ROOT ( $p=2$ ,  $\kappa=0$ ). **\*\*PART 2, from the receipt-vs-sentence audit: the PEAK-POSITION EXCLUSION, which is what P15 actually cites this file for and which the file did not compute\*\*** — the collapse leg's own conformal time gives  $\eta_{bp} \sim 5e3-1e4$  Mpc, so  $r_s^{\text{eff}} = r_s + \eta_{bp}/\sqrt{3} = 3.0-5.9e3$  Mpc, and reproducing  $L_A = 301.8$  would demand  $D = 2.9-5.7e5$  Mpc against the actual  $1.4e4$ , a factor of **\*\*21-41\*\***. **\*PART 1 shows the crossing ADDS no divergent phase; PART 2 shows an inherited phase cannot survive. The paper's sentence needs the second.\*** [borrowed from P15 (+P7)]). *Computes*: runs clean; registered from its own docstring and a live run *Bound*: description is the receipt's own wording, condensed; not independently re-derived here *Run: python3 receipts/P15\_CR\_cosmology/BRANCHPT\_transmission\_character.py*

### H1\_the\_low\_multipole\_deficit\_is\_two\_effects\_and\_the\_ladder\_imprint\_dies\_where\_the\_transfer\_is\_narrow [OK]

*corpus integrity* (**\*\*THE HARMONIC-ANALYSIS BAKE** — second of the four fields 'L-272's re-survey left outstanding, thrown r3166. **\*\*BOUNCED**, and largely:\*\* the field's first question is why a closed- $S^3$  source is projected through FLAT spherical Bessel functions, and 'P15' answers it in its own text — the distance slicing is flat, so only the source modes carry the closed- $S^3$  quantization — while its measure  $w_L = (L+1)/(L(L+2))$  is derived twice, as degeneracy over per-mode power and as  $d \ln k_L / dL$ , both agreeing to machine precision. **\*\*BIT:\*\*** the deficit reported as one geometric effect splits exactly into a **\*\*FLOOR\*\*** (modes below  $L=2$  removed) and a **\*\*LADDER COARSENESS\*\***. At  $\ell=5-6$  the second contributes  $+0.38$  against a floor of  $\$0.27-\$0.53$  — comparable, and in the opposite direction — and it is what sets the recovery multipole the paper quotes ( $\ell=8$  with it,  $\ell=10$  without). Under  $\mu^2$  in  $r_0$  the floor moves  $\$-2.4$  to  $\$-6.2$  and the full result  $\$+27.8$  to  $\$-43.3$ : **\*\*the location is the floor and is robust as claimed; the depth is the ladder and is volatile\*\***, which gives the paper's own qualitative split a mechanism. **\*\*AND A SELF-WAIVED GATE BECOMES A BOUND:\*\*** 'P15' records that the ladder "samples the projection below the rate the instrument's own aliasing gate demands, and that gate waives itself on the claim that the ladder is physical." The imprint dies at  $\ell \lesssim k_{2D} \cdot \text{Ce}^{\{3\}\sigma}$ ; to reach the first acoustic peak the transfer would need to draw power over a factor of nine in  $k$ . **\*\*Safe with a wide margin, by measurement.\*\*** [borrowed from —]). *Computes*: COMPUTES: the spherical-Bessel completeness identity

$\sum_{\ell=2}^{\ell_{\text{max}}} j_{\ell}(x)^2 = 1$  to ten figures and the exact plateau  $\int j_{\ell}^2 d\ln x = 1/(2(\ell_{\text{max}}+1))$  reproduced on the fixed grid to six, both **before** anything is used; the measure against  $d\ln k L/dL$  and against degeneracy-over-power at five  $L$ ; the paper's own  $D_C/r_0$  limit; the floor/discreteness decomposition across  $\ell=2-12$ ; each half's response to  $\pm 2\%$  in  $r_0$ ; the recovery multipole with and without the discreteness; and the imprint against transfer width at four  $\sigma$ , with the scaling rule checked against all four rather than extrapolated (15 checks). *Bound:* NOT-A-PAPER-CLAIM — corpus integrity; no paper edited. **NOT claimed:** that the deficit is wrong — every number touched is reproduced or confirmed, the paper's own limit included; that these are predictions about the sky — the decomposition is the pure Sachs–Wolfe limit while the paper's depths come from a genuine Boltzmann transfer it says are milder, and the two are not compared; that the real transfer's width is measured —  $\sigma$  is a Gaussian-in- $\ln k$  stand-in and measuring  $\Delta_{\ell}(k)$  is the owed next step; that the aliasing gate should stop waiving — only that the waiver now has a margin; that the low- $\ell$  oscillation is derived rather than measured. *Run:* `python3 receipts/L274_the_harmonic_bake/H1_the_low_multipole_deficit_is_two_effects_and_the_ladder_imprint_`

## Appendix L The Knowledge Ledgers

Each claim marked <sup>L</sup> in the text rests on a field bake or the figure–theorem bake recorded in the named ledger, which ships with this work and carries the probes, the receipts, and the three registers kept apart — what bit, what bounced, and where the boundary is. A ledger is working apparatus, not a publication, and is cited here rather than in the bibliography. This appendix is generated from the ledger index, not maintained by hand.

- L9** CONVEXITY\_OPTIMISATION\_LEDGER.md [field bake]  
The convexity / optimisation field-bake ledger — what bit, what bounced, and the boundary. One of the three fields listed but never thrown (the overnight order: convexity  $\times 143$ , constraint  $\times 141$ ). ‘OWED’ 622
- L23** OPTICS\_LENSING\_LEDGER.md [field bake]  
observational optics and lensing against CR — confirmations, and the dimension clause they forced
- L19** SPECTRAL\_THEORY\_LEDGER.md [field bake]  
The spectral-theory field bake — what bit, what bounced, and the boundary. Tier B’s largest never-thrown field ( $\times 189$  on the reach measure), verified as NOT covered by the harmonic ledger, which mentions spectral, self-adjoint and deficiency zero times
- L21** INFORMATION\_THEORY\_LEDGER.md [field bake]  
The information-theory field-bake ledger — what bit, what bounced, and where the boundary is
- L22** NUMERICAL\_ANALYSIS\_LEDGER.md [field bake]  
The numerical-analysis field-bake ledger — what bit, what bounced, and where the boundary is
- L14** INTEGRABLE\_SYSTEMS\_LEDGER.md [field bake]  
The integrable-systems field-bake ledger — what bit, what bounced, and where the boundary is

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